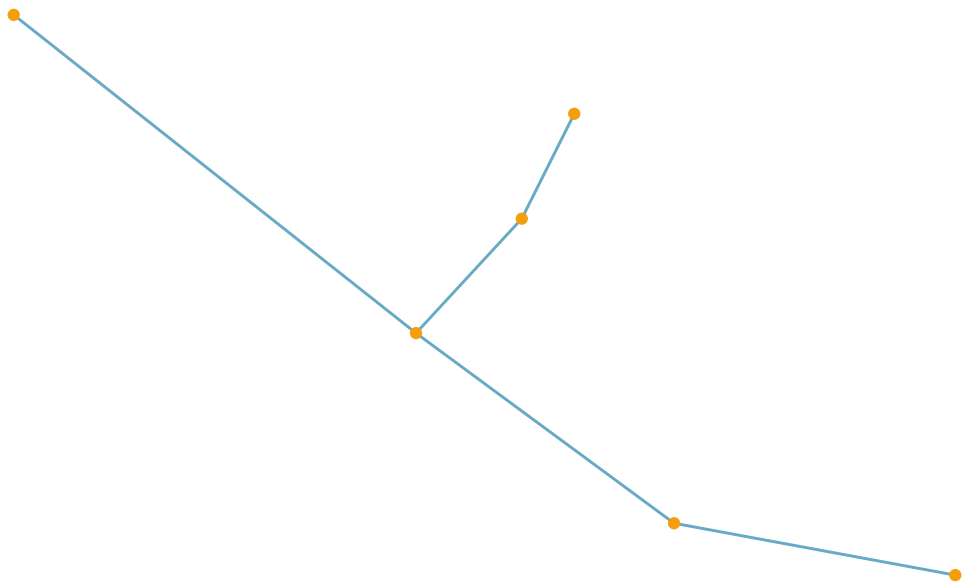
# 01/88 Andromeda (And) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

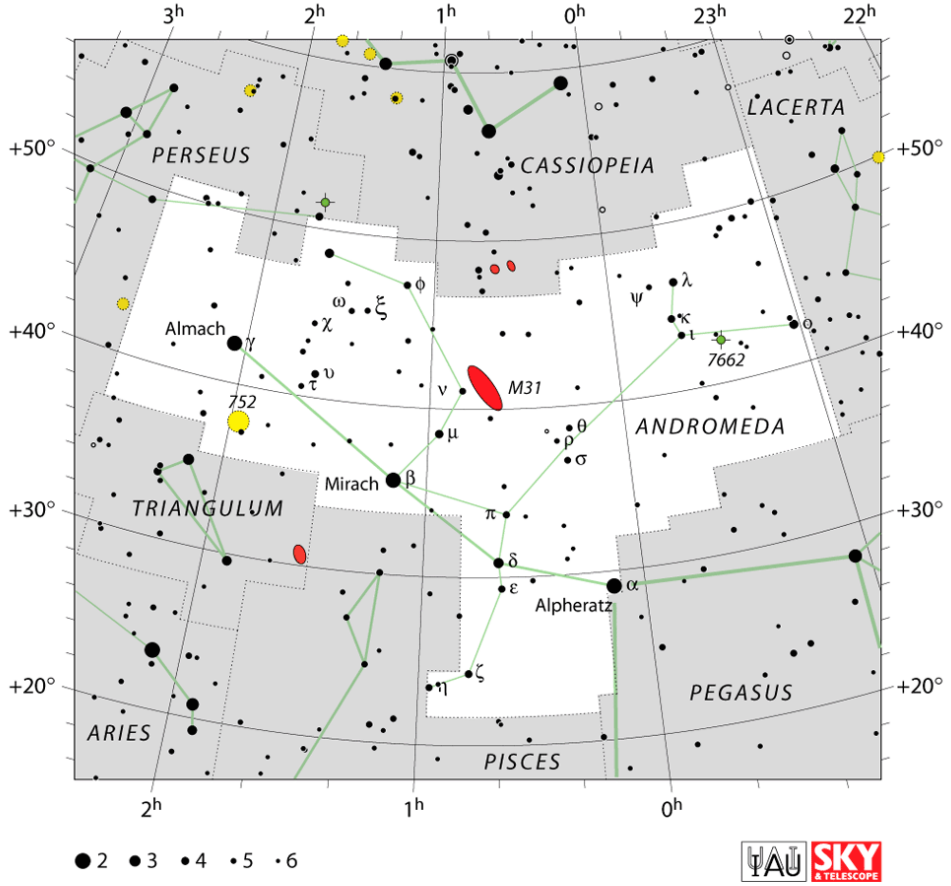
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 01/88  Andromeda (And) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 677 | 2.096542 | +29.090833 | HIP 3092 |
| HIP 3092 | 9.831667 | +30.861222 | HIP 677, HIP 5447 |
| HIP 5447 | 17.432500 | +35.620833 | HIP 3092, HIP 9640, HIP 4436 |
| HIP 9640 | 30.974667 | +42.329861 | HIP 5447 |
| HIP 4436 | 14.187917 | +38.499250 | HIP 5447, HIP 3881 |
| HIP 3881 | 12.453458 | +41.078944 | HIP 4436 |

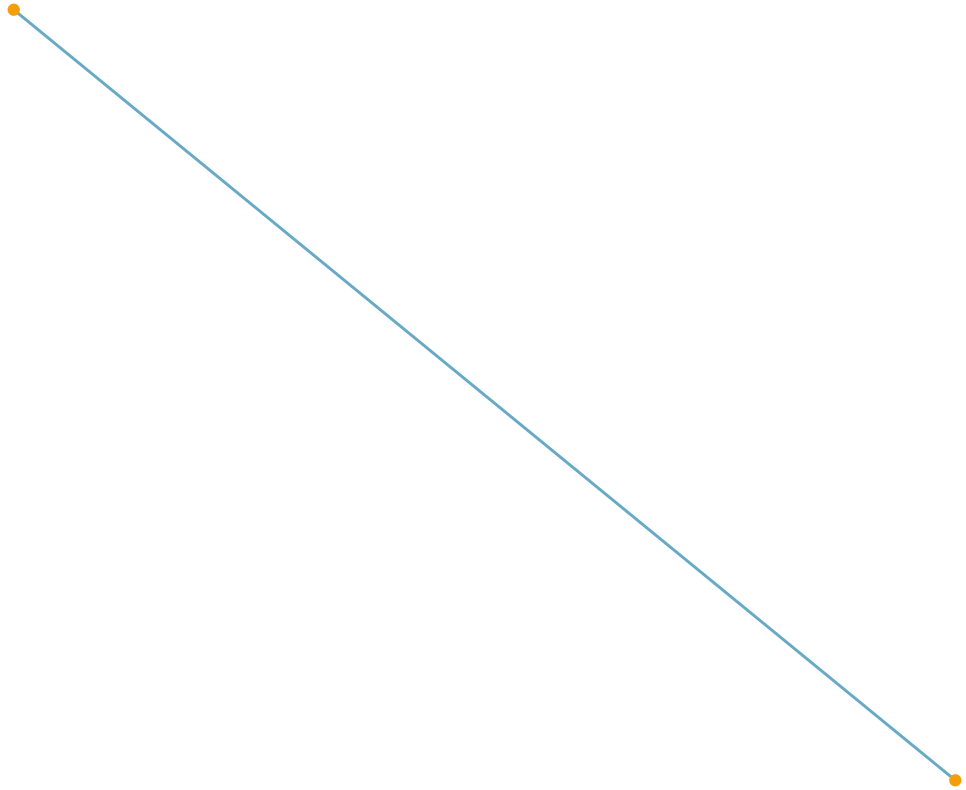
Effective release edges: 5 • unique plotted points: 6 • source: release_effective_edges.csv

# 02/88 Antlia (Ant) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

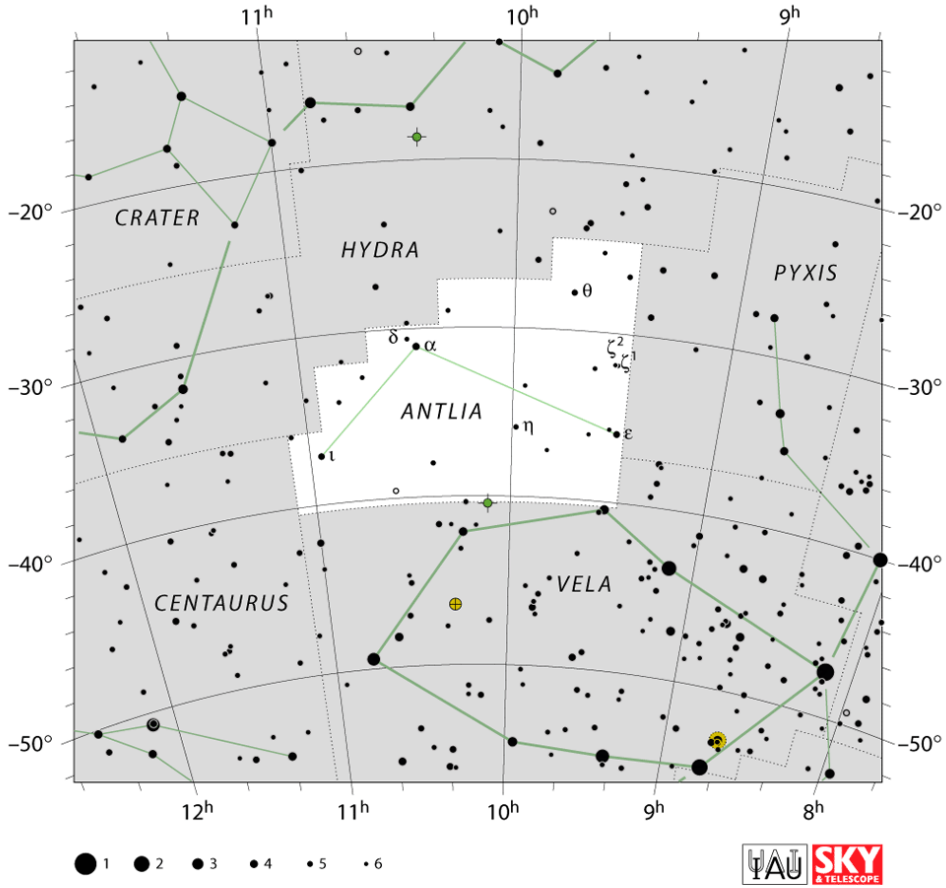
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 02/88  Antlia (Ant) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 51172 | 156.788167 | -31.067806 | HIP 48926 |
| HIP 48926 | 149.718083 | -35.890944 | HIP 51172 |

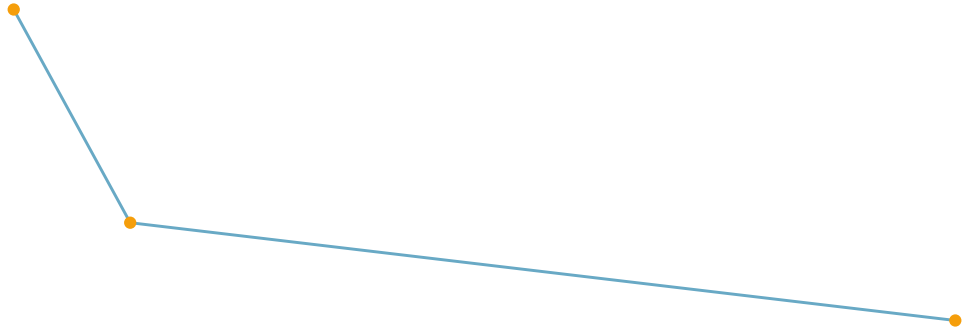
Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv

# 03/88 Apus (Aps) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

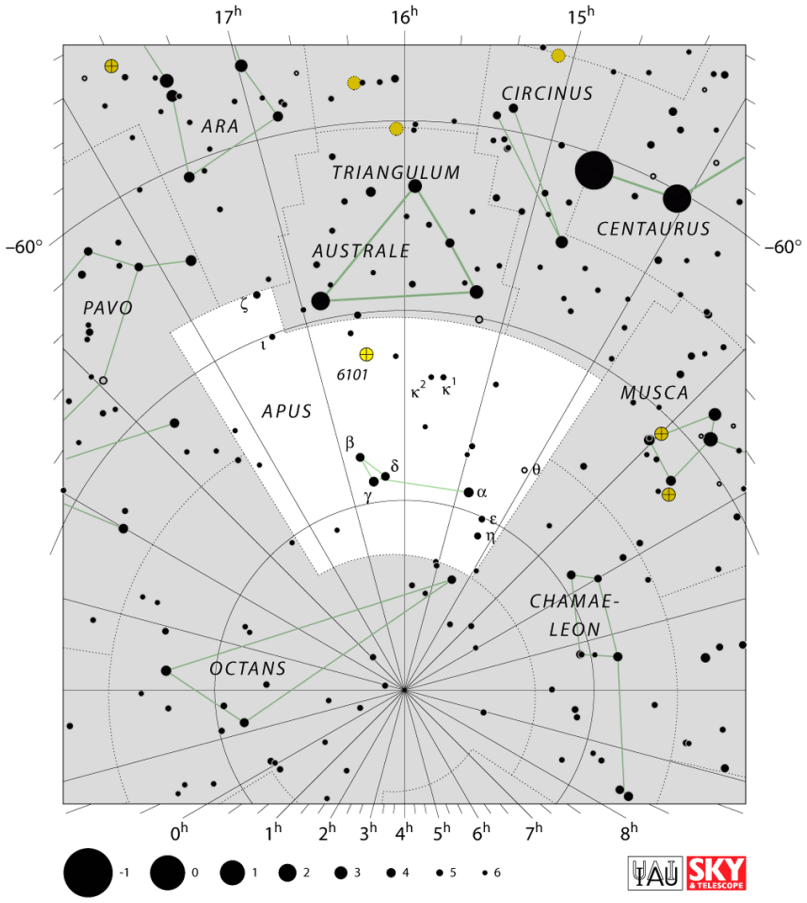
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 03/88  Apus (Aps) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 72370 | 221.965542 | -79.044722 | HIP 81065 |
| HIP 81065 | 248.364417 | -78.896972 | HIP 72370, HIP 81852 |
| HIP 81852 | 250.772583 | -77.516583 | HIP 81065 |

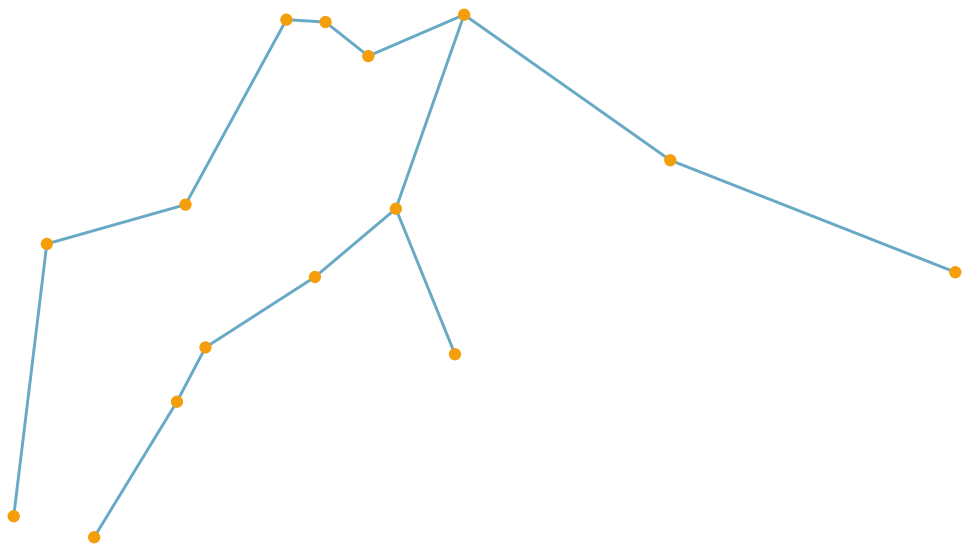
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

# 04/88 Aquarius (Aqr) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

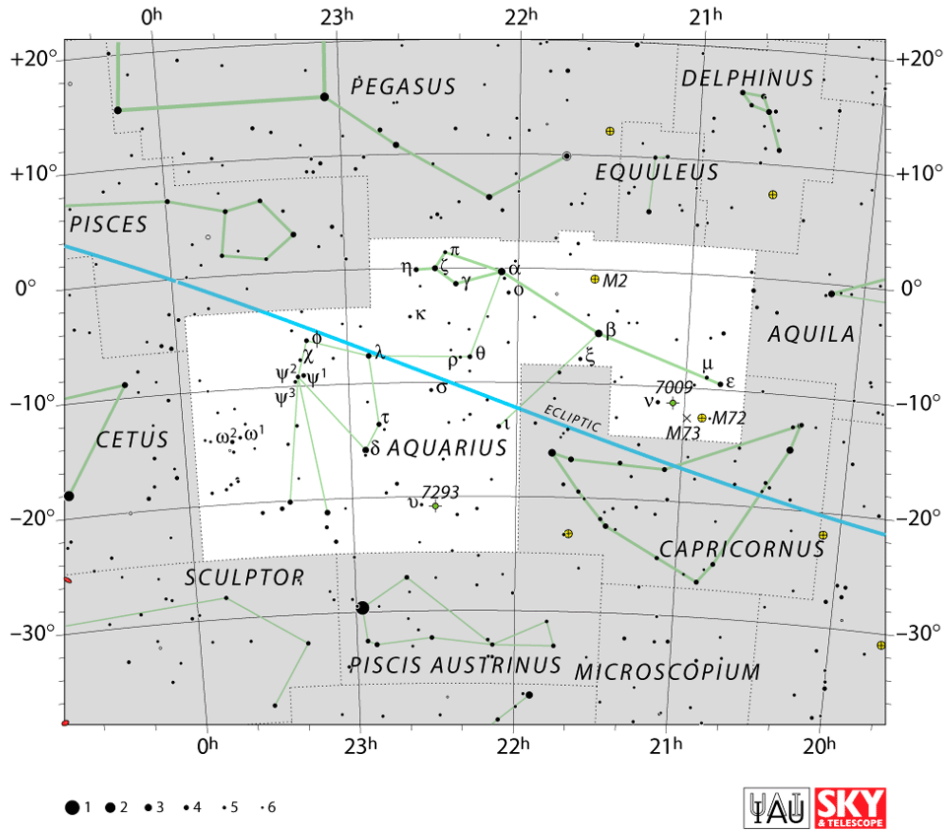
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 04/88  Aquarius (Aqr) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 106278 | 322.889667 | -5.571167 | HIP 109074, HIP 102618 |
| HIP 109074 | 331.445958 | +0.319833 | HIP 106278, HIP 110395, HIP 110003 |
| HIP 110395 | 335.413750 | -1.387361 | HIP 109074, HIP 110960 |
| HIP 110960 | 337.207500 | +0.020056 | HIP 110395, HIP 111497 |
| HIP 111497 | 338.838875 | +0.117361 | HIP 110960, HIP 112961 |
| HIP 112961 | 343.153583 | -7.579667 | HIP 111497, HIP 114855 |
| HIP 114855 | 348.972000 | -9.087694 | HIP 112961, HIP 115438 |
| HIP 115438 | 350.742917 | -20.100333 | HIP 114855 |
| HIP 110003 | 334.208208 | -7.783250 | HIP 109074, HIP 109139, HIP 111123 |
| HIP 109139 | 331.609208 | -13.869528 | HIP 110003 |
| HIP 111123 | 337.661750 | -10.677889 | HIP 110003, HIP 112716 |
| HIP 112716 | 342.397958 | -13.592528 | HIP 111123, HIP 113136 |
| HIP 113136 | 343.662667 | -15.820750 | HIP 112716, HIP 114341 |
| HIP 114341 | 347.361500 | -21.172472 | HIP 113136 |
| HIP 102618 | 311.918875 | -9.495694 | HIP 106278 |

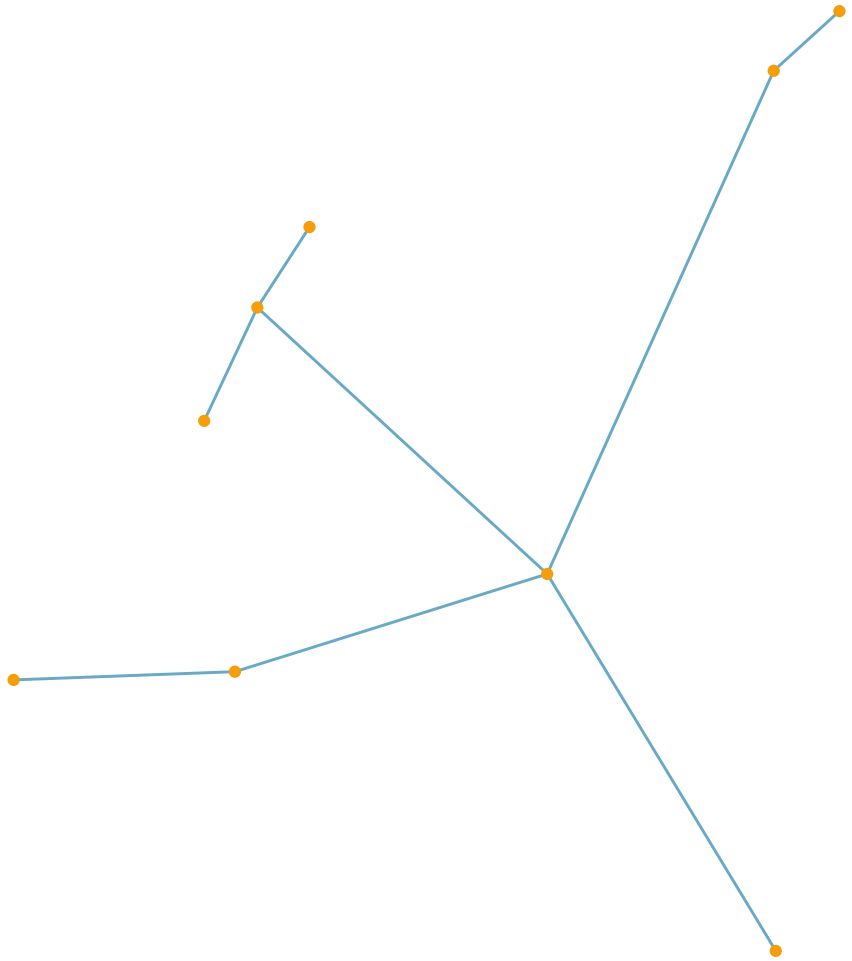
Effective release edges: 14 • unique plotted points: 15 • source: release_effective_edges.csv

# 05/88 Aquila (Aql) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

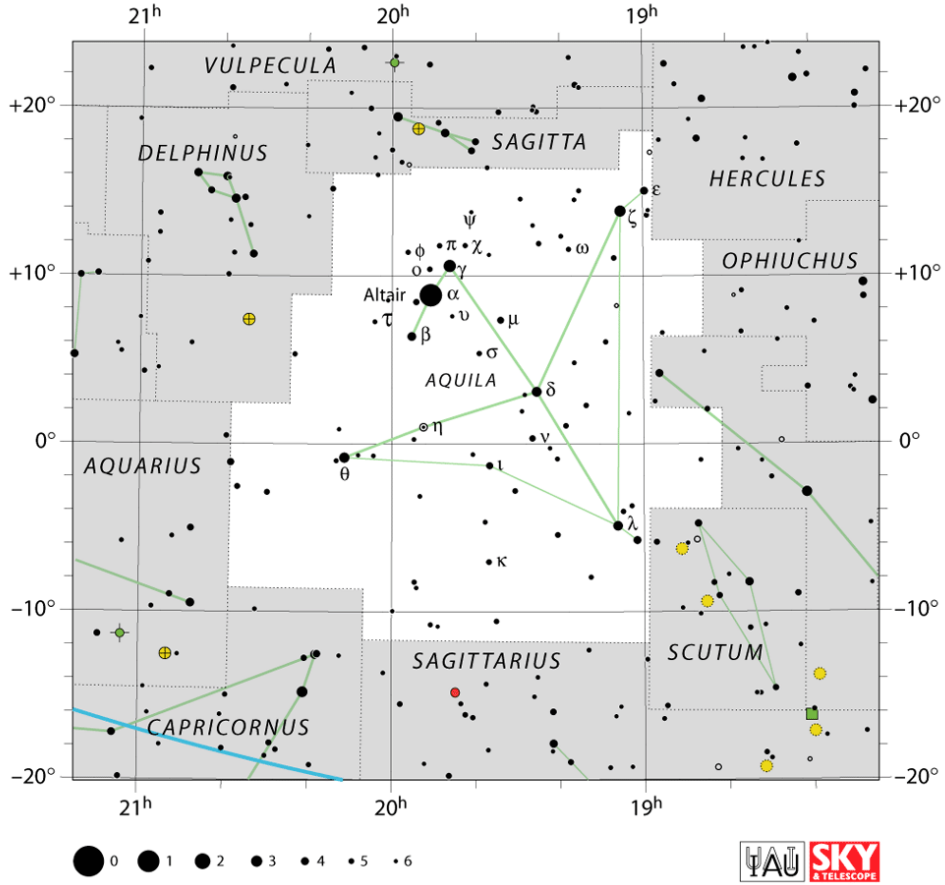
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 05/88  Aquila (Aql) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 98036 | 298.828208 | +6.407944 | HIP 97649 |
| HIP 97649 | 297.694500 | +8.867389 | HIP 98036, HIP 97278, HIP 95501 |
| HIP 97278 | 296.564875 | +10.613278 | HIP 97649 |
| HIP 95501 | 291.373958 | +3.114583 | HIP 97649, HIP 97804, HIP 93747, HIP 93805 |
| HIP 97804 | 298.118167 | +1.005667 | HIP 95501, HIP 99473 |
| HIP 99473 | 302.826083 | +0.821472 | HIP 97804 |
| HIP 93747 | 286.352542 | +13.863722 | HIP 95501, HIP 93244 |
| HIP 93244 | 284.905792 | +15.068472 | HIP 93747 |
| HIP 93805 | 286.562292 | -4.882333 | HIP 95501 |

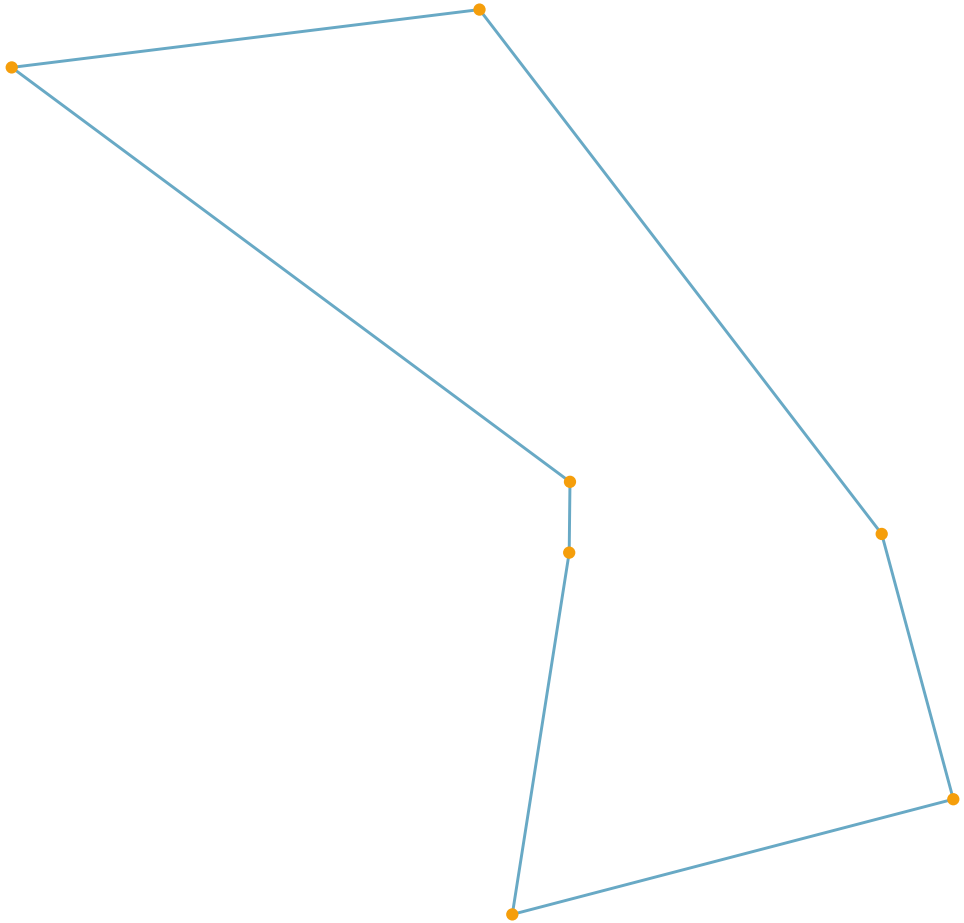
Effective release edges: 8 • unique plotted points: 9 • source: release_effective_edges.csv

# 06/88 Ara (Ara) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

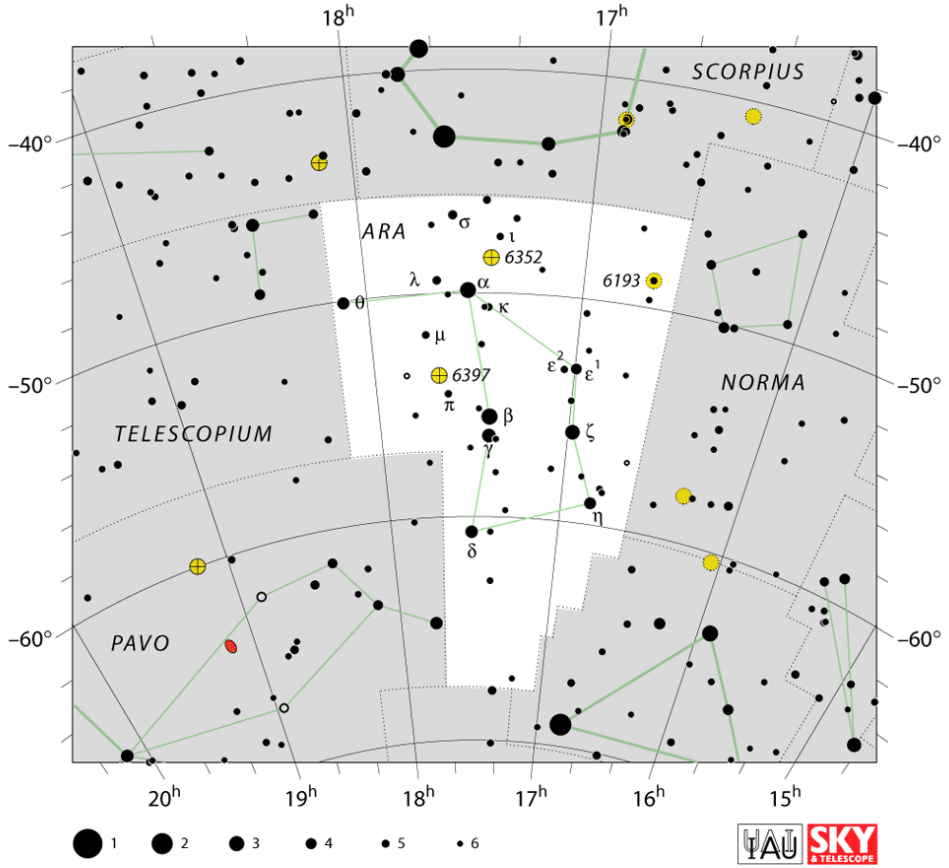
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 06/88  Ara (Ara) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 88714 | 271.657833 | -50.091444 | HIP 85792, HIP 85258 |
| HIP 85792 | 262.960500 | -49.875972 | HIP 88714, HIP 83081 |
| HIP 83081 | 254.655125 | -55.990056 | HIP 85792, HIP 82363 |
| HIP 82363 | 252.446292 | -59.041306 | HIP 83081, HIP 85727 |
| HIP 85727 | 262.774917 | -60.683611 | HIP 82363, HIP 85267 |
| HIP 85267 | 261.348583 | -56.377694 | HIP 85727, HIP 85258 |
| HIP 85258 | 261.325000 | -55.529833 | HIP 85267, HIP 88714 |

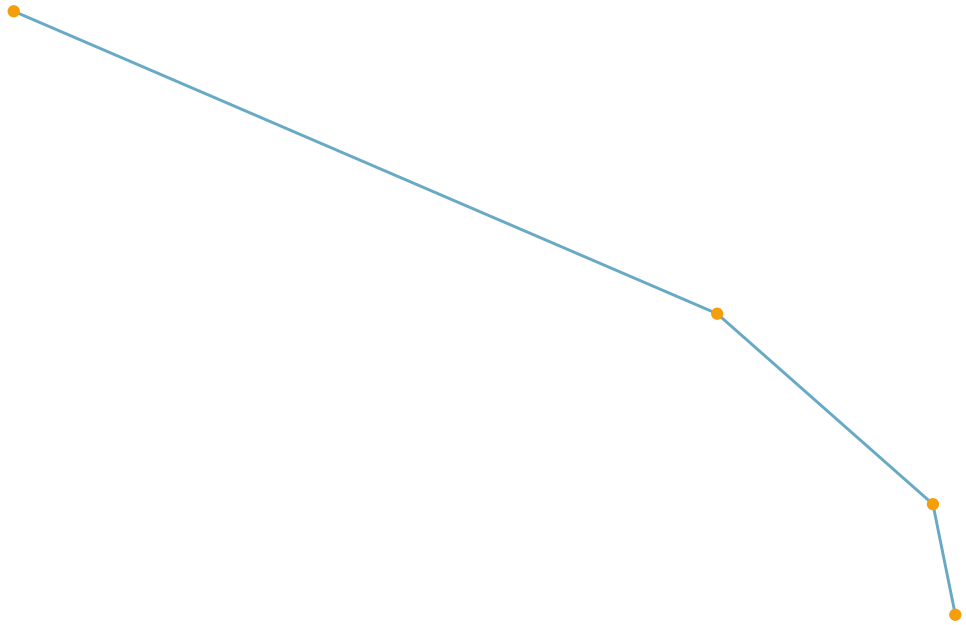
Effective release edges: 7 • unique plotted points: 7 • source: release_effective_edges.csv

# 07/88 Aries (Ari) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

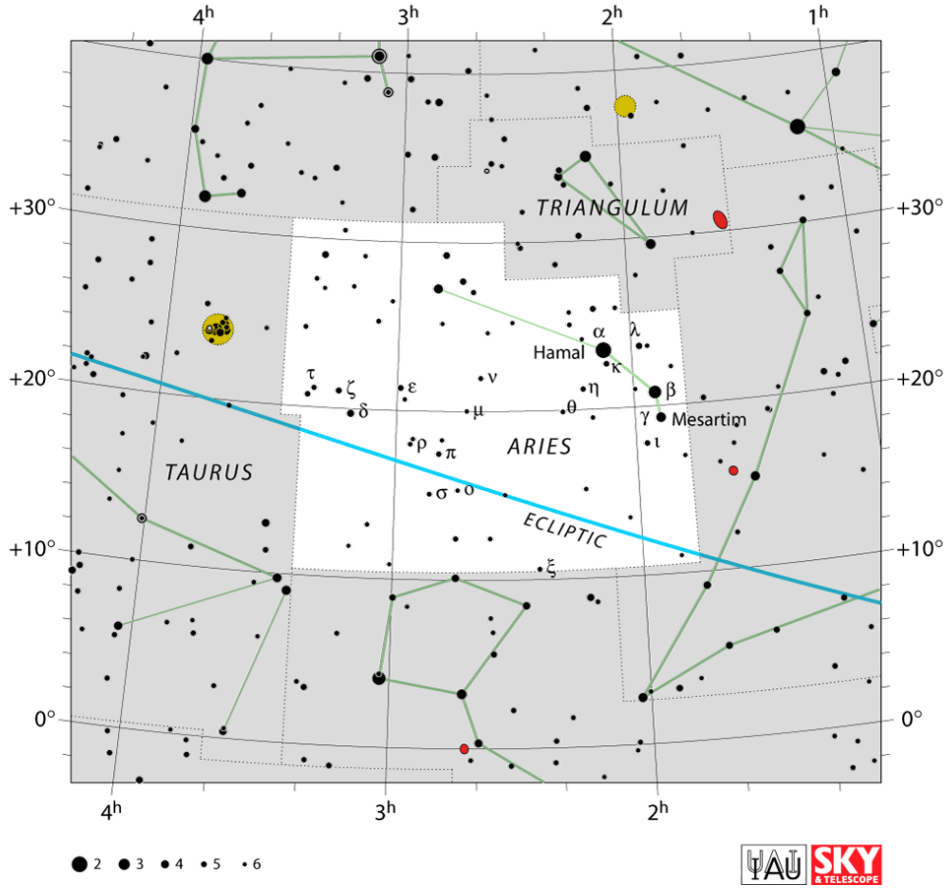
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 07/88  Aries (Ari) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 13209 | 42.495792 | +27.260778 | HIP 9884 |
| HIP 9884 | 31.792875 | +23.462778 | HIP 13209, HIP 8903 |
| HIP 8903 | 28.659792 | +20.808306 | HIP 9884, HIP 8832 |
| HIP 8832 | 28.382375 | +19.294083 | HIP 8903 |

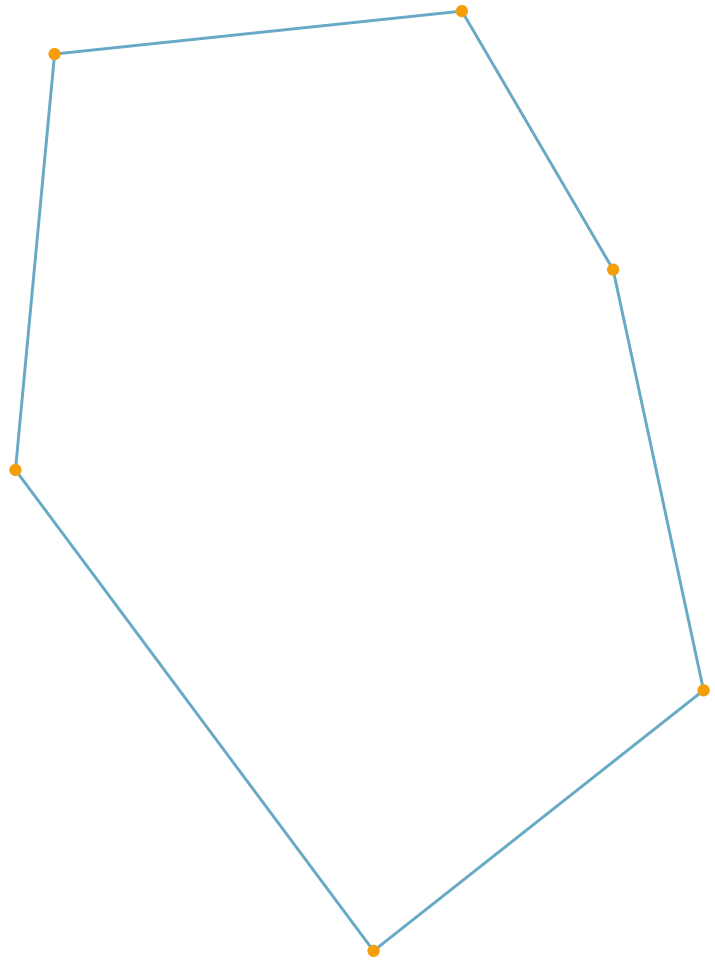
Effective release edges: 3 • unique plotted points: 4 • source: release_effective_edges.csv

# 08/88 Auriga (Aur) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

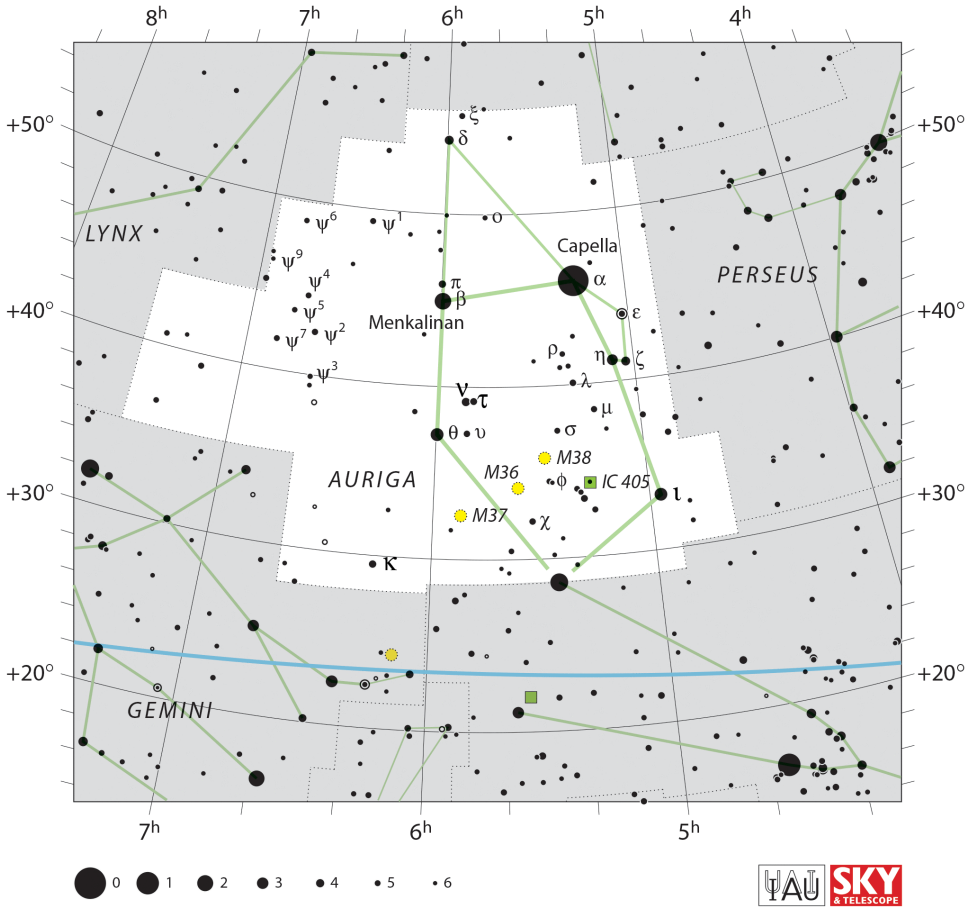
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 08/88  Auriga (Aur) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 28380 | 89.930167 | +37.212778 | HIP 28360, HIP 25428 |
| HIP 28360 | 89.882375 | +44.947444 | HIP 28380, HIP 24608 |
| HIP 24608 | 79.172083 | +45.999028 | HIP 28360, HIP 23453 |
| HIP 23453 | 75.619500 | +41.075889 | HIP 24608, HIP 23015 |
| HIP 23015 | 74.248417 | +33.166139 | HIP 23453, HIP 25428 |
| HIP 25428 | 81.572917 | +28.607861 | HIP 23015, HIP 28380 |

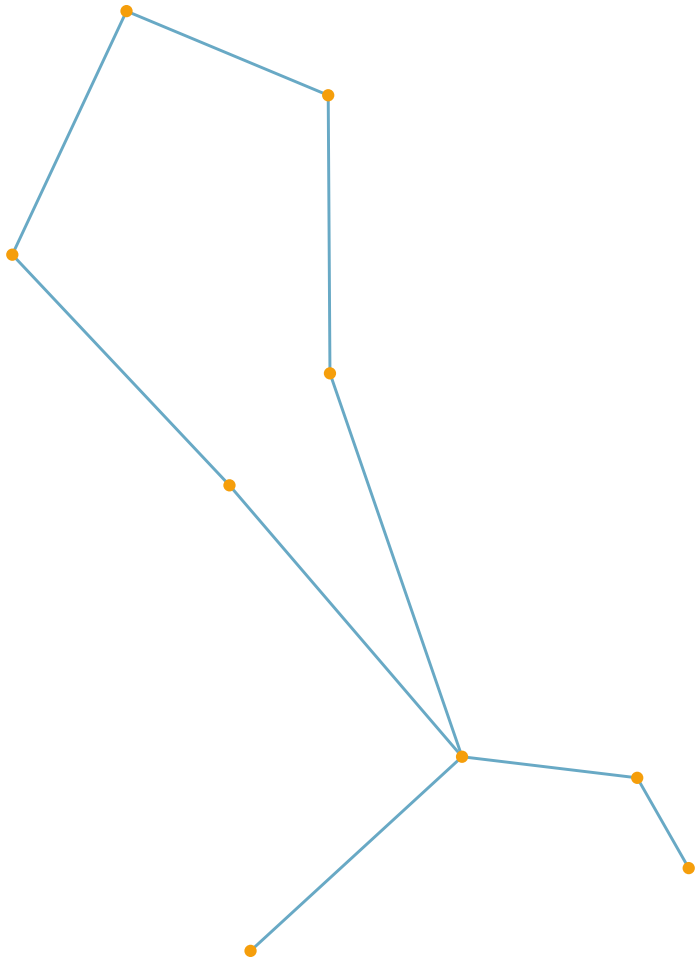
Effective release edges: 6 • unique plotted points: 6 • source: release_effective_edges.csv

# 09/88 Boötes (Boo) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

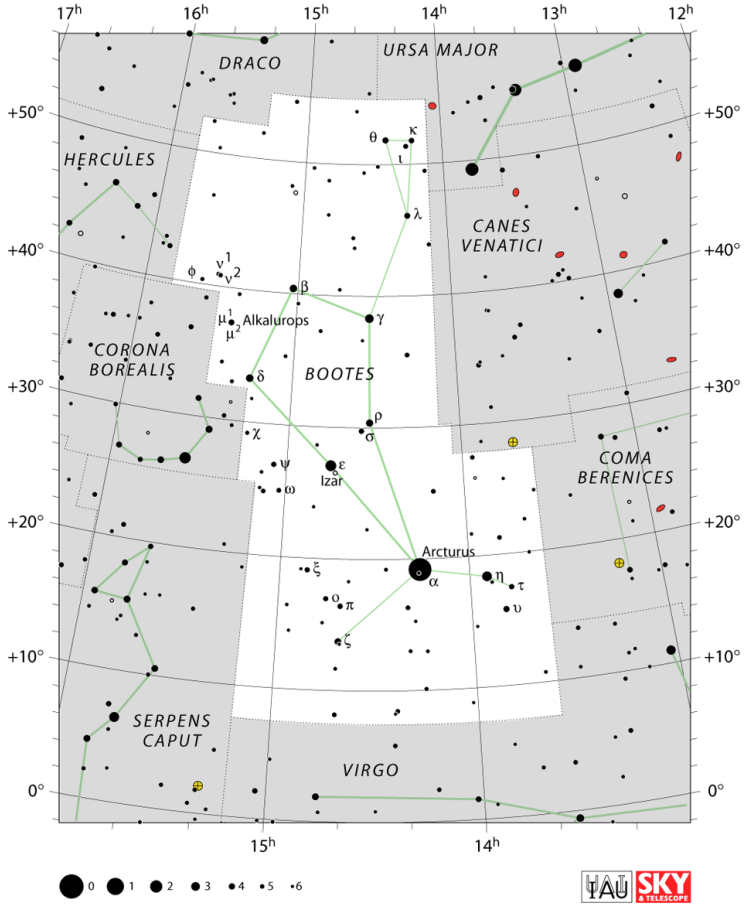
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 09/88  Boötes (Boo) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 71795 | 220.287167 | +13.728333 | HIP 69673 |
| HIP 69673 | 213.918125 | +19.187278 | HIP 71795, HIP 72105, HIP 71053, HIP 67927 |
| HIP 72105 | 221.246875 | +27.074167 | HIP 69673, HIP 74666 |
| HIP 74666 | 228.875417 | +33.315111 | HIP 72105, HIP 73555 |
| HIP 73555 | 225.486625 | +40.390639 | HIP 74666, HIP 71075 |
| HIP 71075 | 218.019833 | +38.307889 | HIP 73555, HIP 71053 |
| HIP 71053 | 217.957750 | +30.371139 | HIP 71075, HIP 69673 |
| HIP 67927 | 208.671333 | +18.398583 | HIP 69673, HIP 67459 |
| HIP 67459 | 207.369583 | +15.797806 | HIP 67927 |

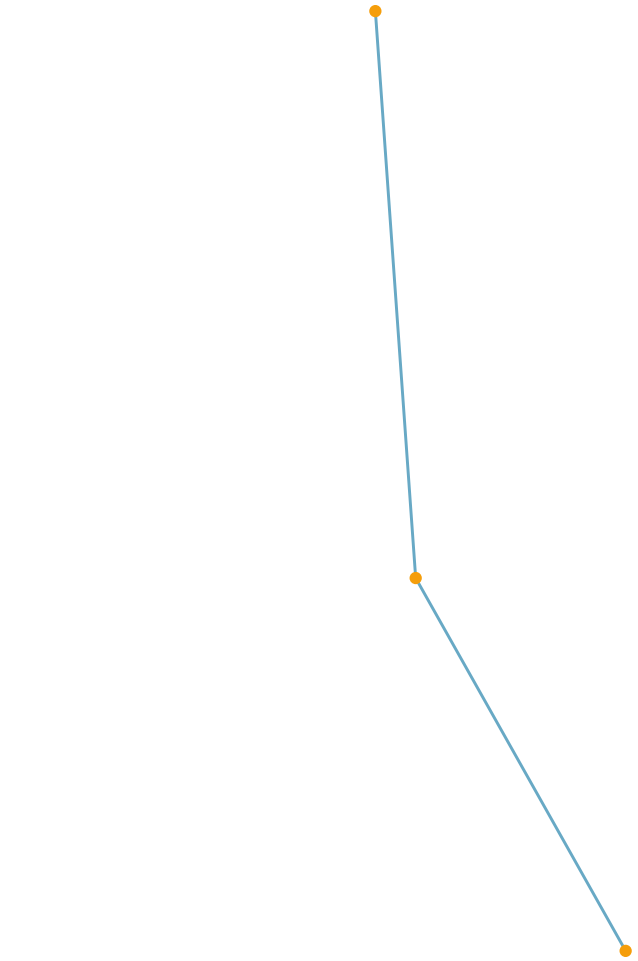
Effective release edges: 9 • unique plotted points: 9 • source: release_effective_edges.csv

# 10/88  Caelum (Cae) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

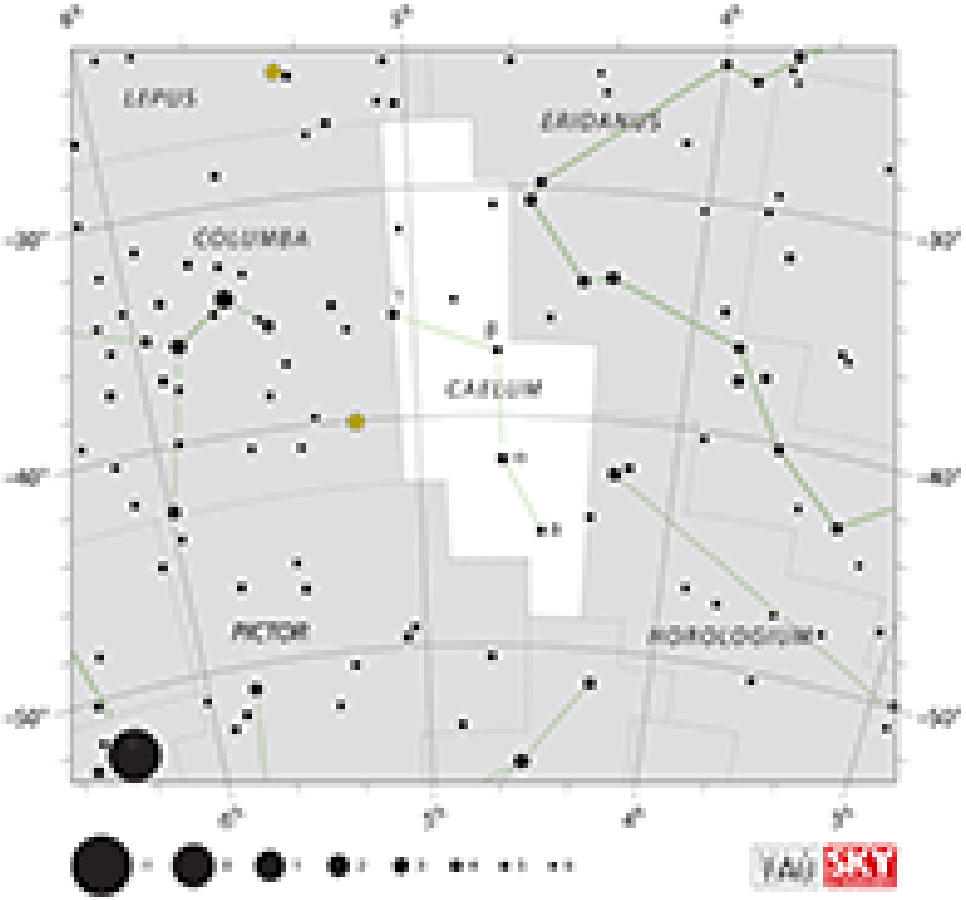
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review:  [ ] Correct  [ ] Incorrect  [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 10/88  Caelum (Cae) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 21060 | 67.708750 | -44.953750 | HIP 21770 |
| HIP 21770 | 70.140917 | -41.863583 | HIP 21060, HIP 21861 |
| HIP 21861 | 70.514375 | -37.144778 | HIP 21770 |

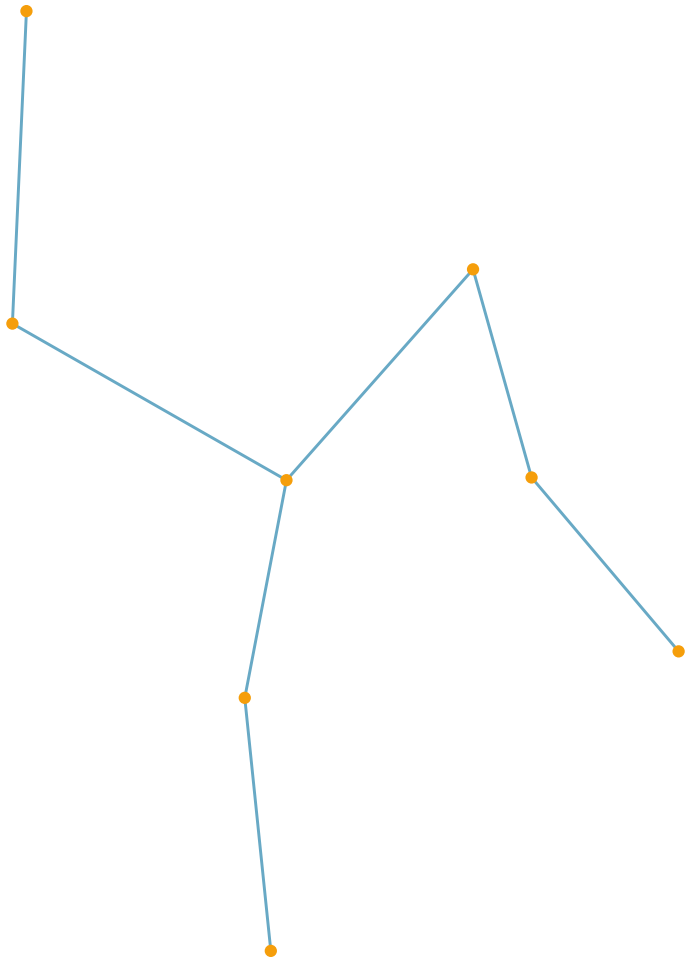
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

# 11/88 Camelopardalis (Cam) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

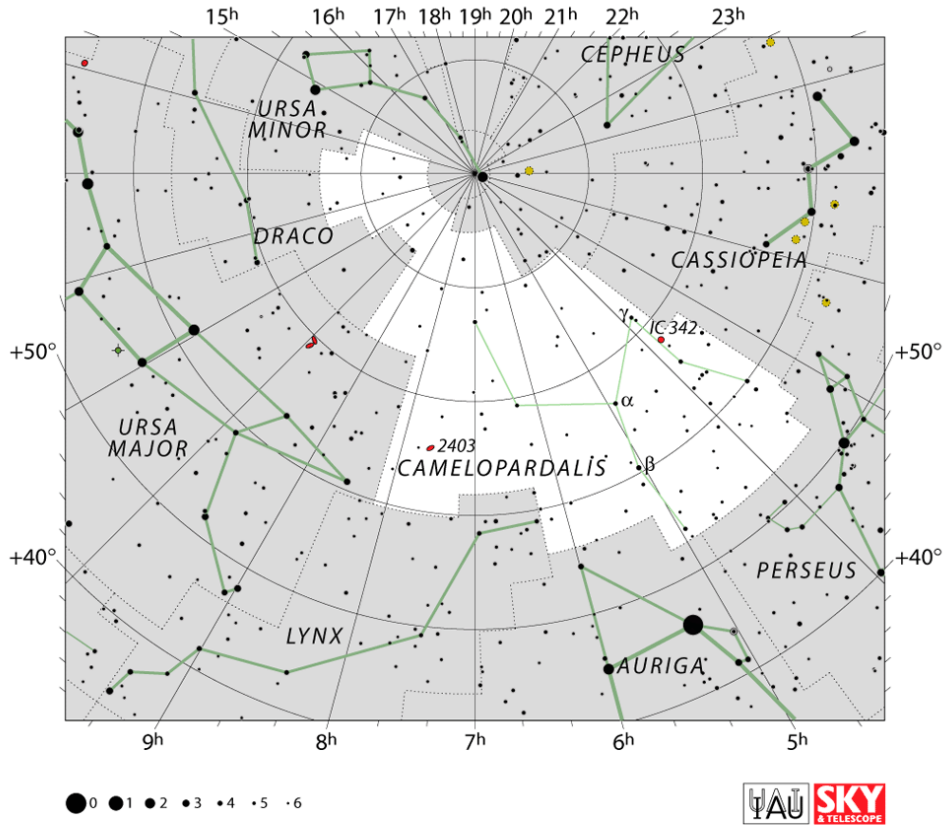
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 11/88  Camelopardalis (Cam) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 33694 | 105.016033 | +76.977442 | HIP 29997 |
| HIP 29997 | 94.711583 | +69.320034 | HIP 33694, HIP 22783 |
| HIP 22783 | 73.512544 | +66.342660 | HIP 29997, HIP 23522, HIP 17959 |
| HIP 23522 | 75.854572 | +60.442281 | HIP 22783, HIP 23040 |
| HIP 23040 | 74.321752 | +53.752083 | HIP 23522 |
| HIP 17959 | 57.589483 | +71.332368 | HIP 22783, HIP 17884 |
| HIP 17884 | 57.380356 | +65.526006 | HIP 17959, HIP 16228 |
| HIP 16228 | 52.267227 | +59.940335 | HIP 17884 |

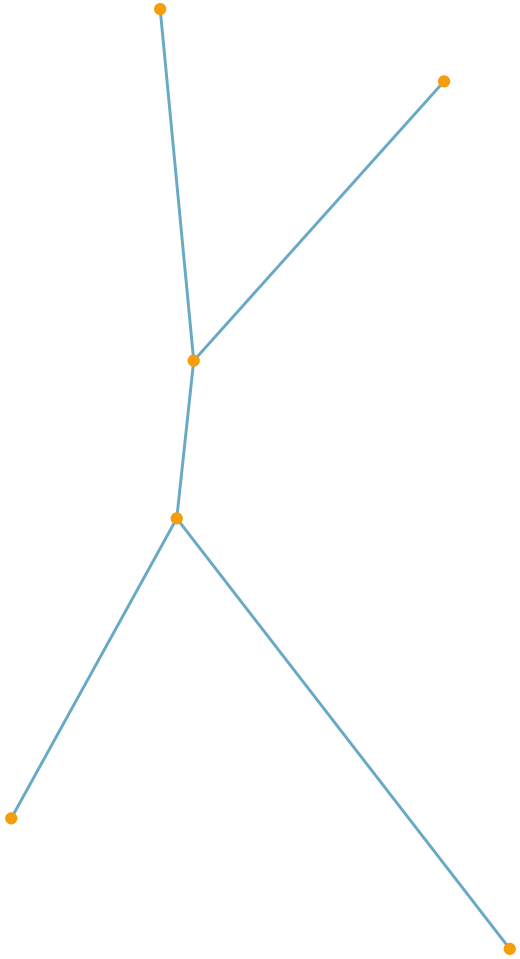
Effective release edges: 7 • unique plotted points: 8 • source: release_effective_edges.csv

# 12/88 Cancer (Cnc) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

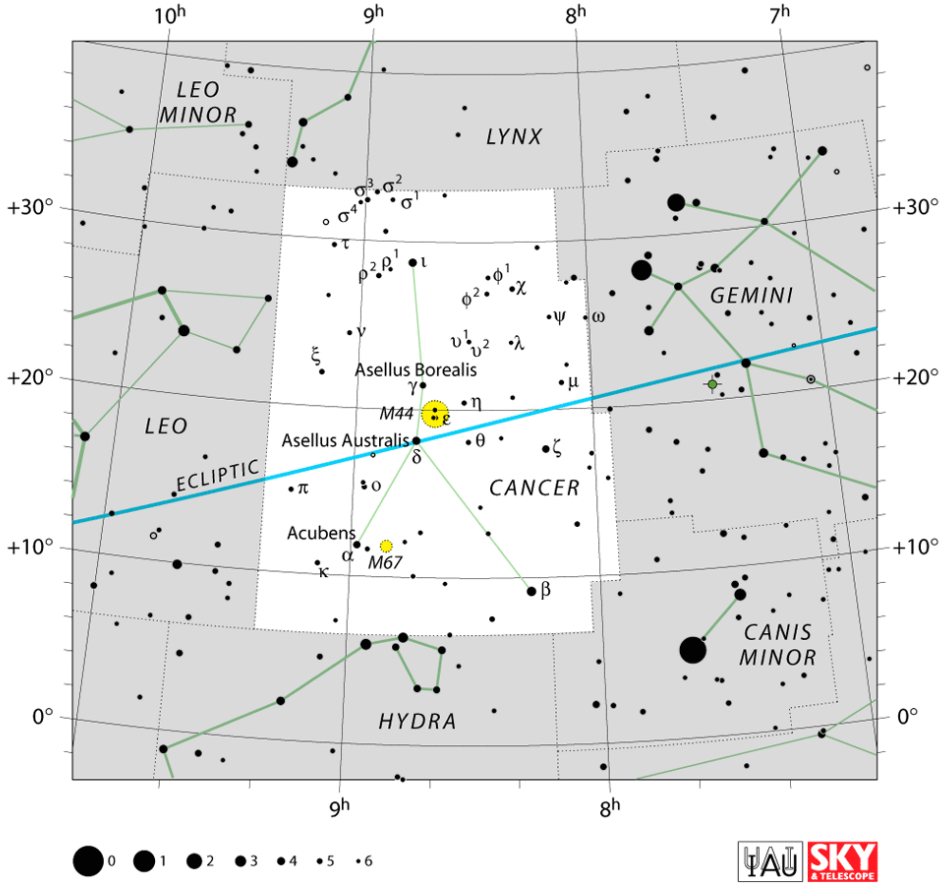
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 12/88  Cancer (Cnc) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 43103 | 131.674292 | +28.760000 | HIP 42806 |
| HIP 42806 | 130.821708 | +21.468583 | HIP 43103, HIP 40843, HIP 42911 |
| HIP 40843 | 125.016125 | +27.218611 | HIP 42806 |
| HIP 42911 | 131.171292 | +18.154861 | HIP 42806, HIP 40526, HIP 44066 |
| HIP 40526 | 124.128958 | +9.185667 | HIP 42911 |
| HIP 44066 | 134.621667 | +11.857778 | HIP 42911 |

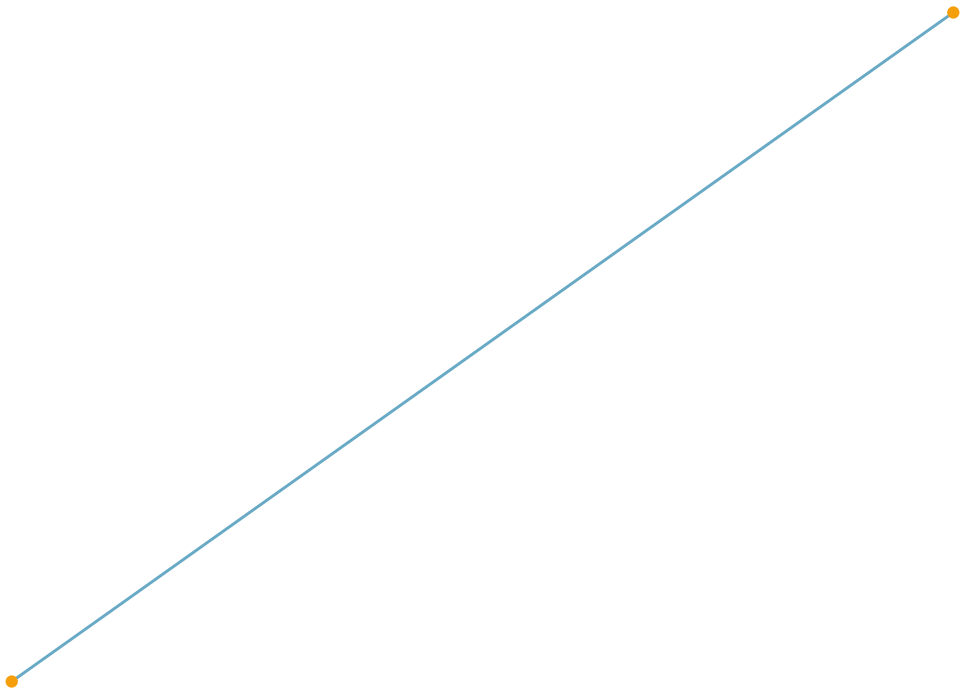
Effective release edges: 5 • unique plotted points: 6 • source: release_effective_edges.csv

# 13/88  Canes Venatici (CVn) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

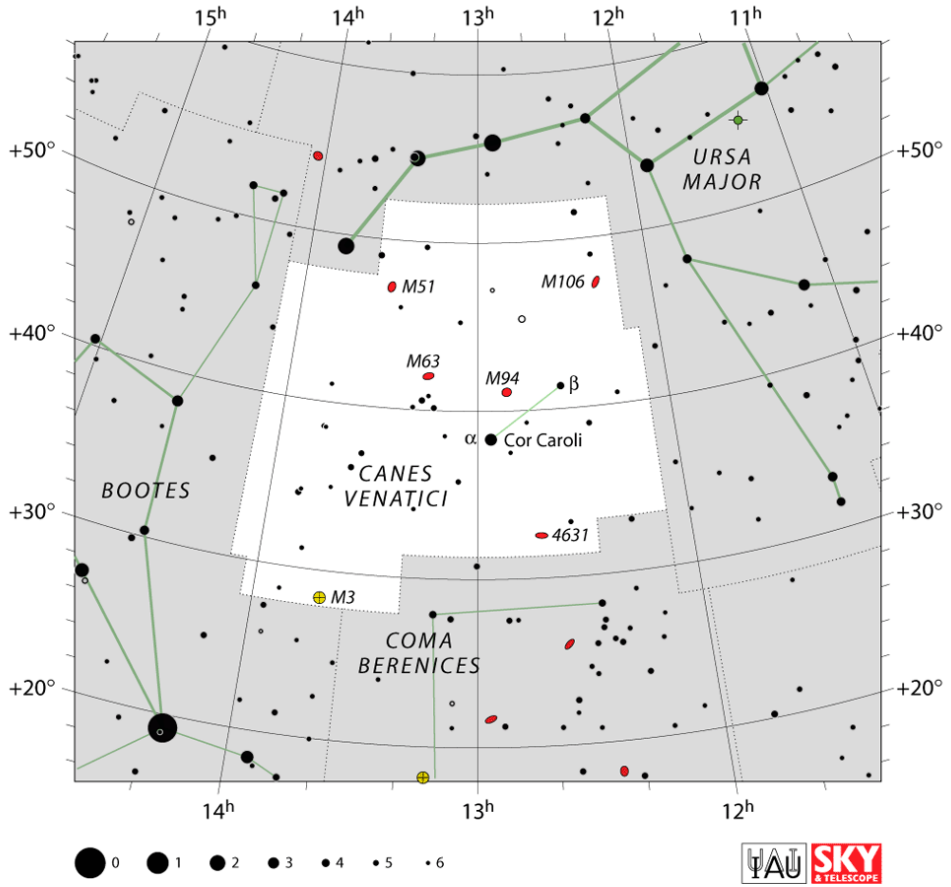
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review:  [ ] Correct  [ ] Incorrect  [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 13/88  Canes Venatici (CVn) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 61317 | 188.437875 | +41.356778 | HIP 63125 |
| HIP 63125 | 194.007667 | +38.318250 | HIP 61317 |

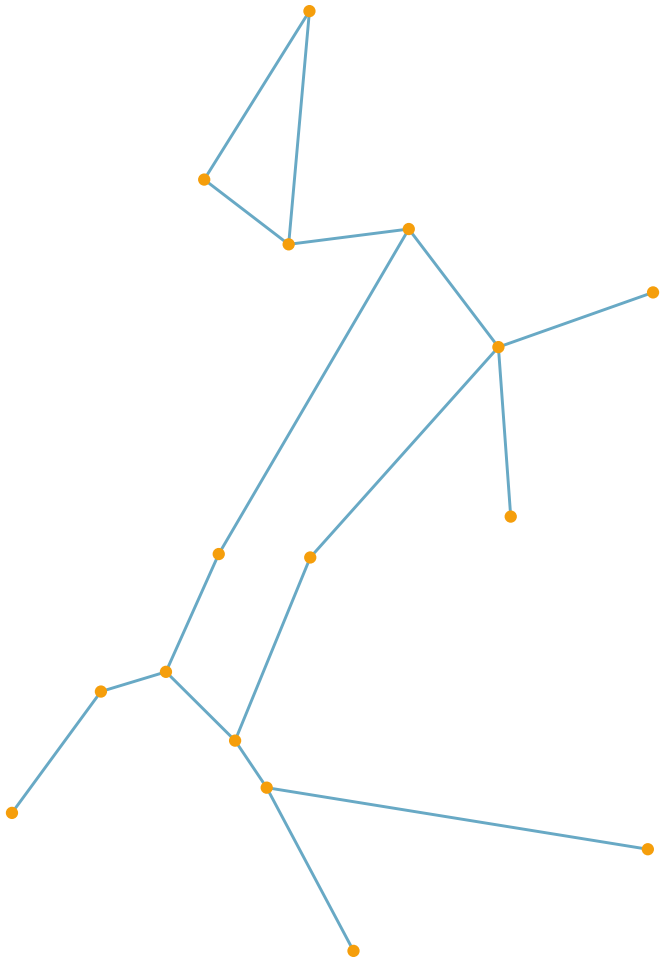
Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv

# 14/88 Canis Major (CMa) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

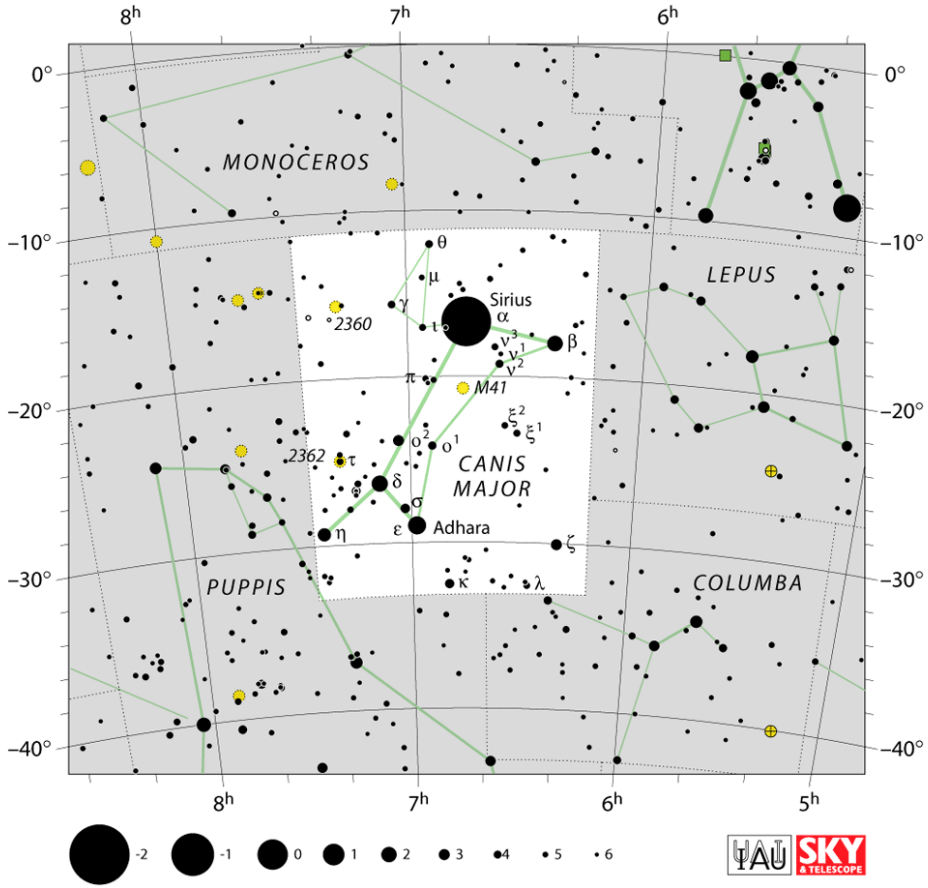
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 14/88  Canis Major (CMa) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 33160 | 103.547833 | -12.038583 | HIP 34045, HIP 33347 |
| HIP 34045 | 105.939542 | -15.633250 | HIP 33160, HIP 33347 |
| HIP 33347 | 104.034292 | -17.054250 | HIP 34045, HIP 32349, HIP 33160 |
| HIP 32349 | 101.288542 | -16.713139 | HIP 33347, HIP 33977, HIP 31592 |
| HIP 33977 | 105.756125 | -23.833306 | HIP 32349, HIP 34444 |
| HIP 34444 | 107.097875 | -26.393194 | HIP 33977, HIP 35037, HIP 33856 |
| HIP 35037 | 108.702750 | -26.772694 | HIP 34444, HIP 35904 |
| HIP 35904 | 111.023792 | -29.303111 | HIP 35037 |
| HIP 33579 | 104.656458 | -28.972083 | HIP 33856, HIP 32759, HIP 30122 |
| HIP 33856 | 105.429792 | -27.934833 | HIP 33579, HIP 34444, HIP 33165 |
| HIP 33165 | 103.554375 | -23.928361 | HIP 33856, HIP 31592 |
| HIP 31592 | 99.170833 | -19.255722 | HIP 33165, HIP 31416, HIP 30324, HIP 32349 |
| HIP 31416 | 98.764083 | -22.964833 | HIP 31592 |
| HIP 30324 | 95.674958 | -17.955917 | HIP 31592 |
| HIP 32759 | 102.460292 | -32.508500 | HIP 33579 |
| HIP 30122 | 95.078292 | -30.063389 | HIP 33579 |

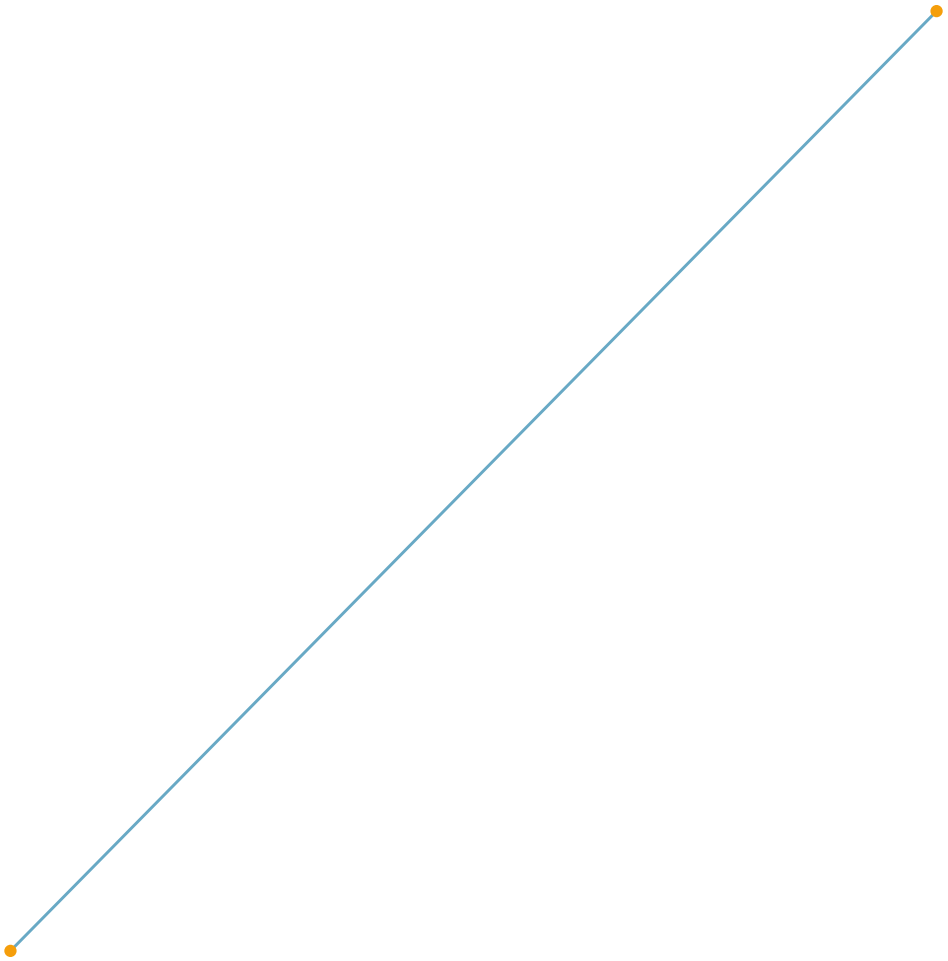
Effective release edges: 17 • unique plotted points: 16 • source: release_effective_edges.csv

# 15/88 Canis Minor (CMi) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

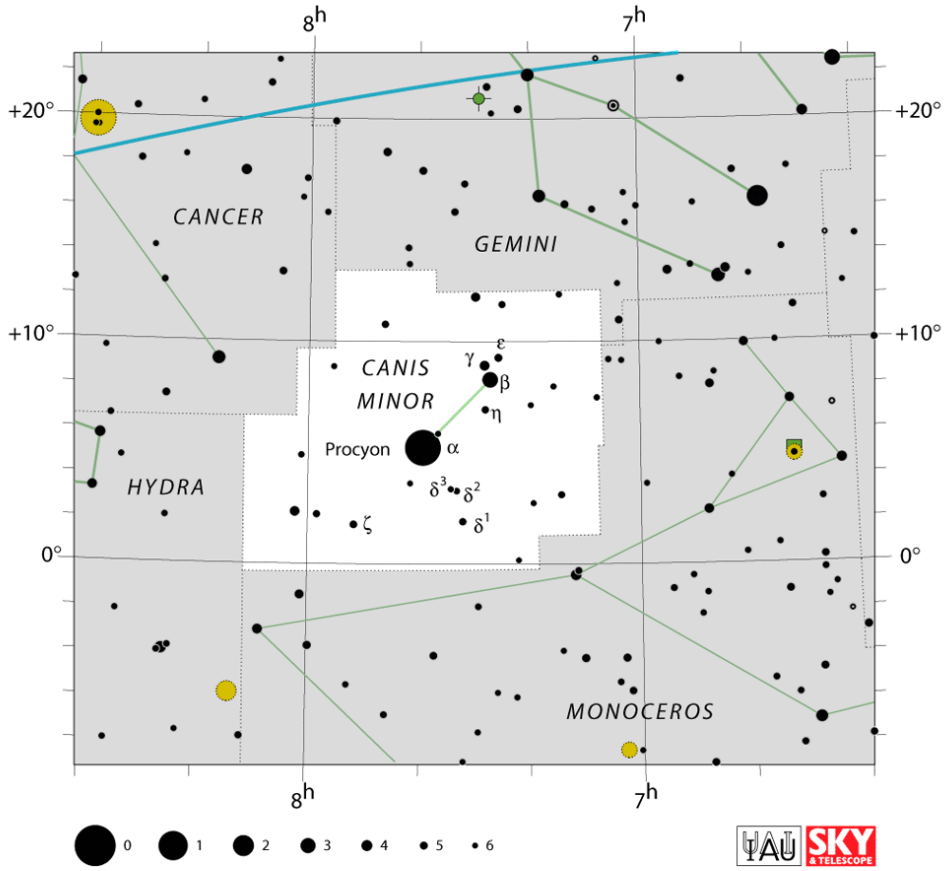
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 15/88  Canis Minor (CMi) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 37279 | 114.827250 | +5.227500 | HIP 36188 |
| HIP 36188 | 111.787792 | +8.289417 | HIP 37279 |

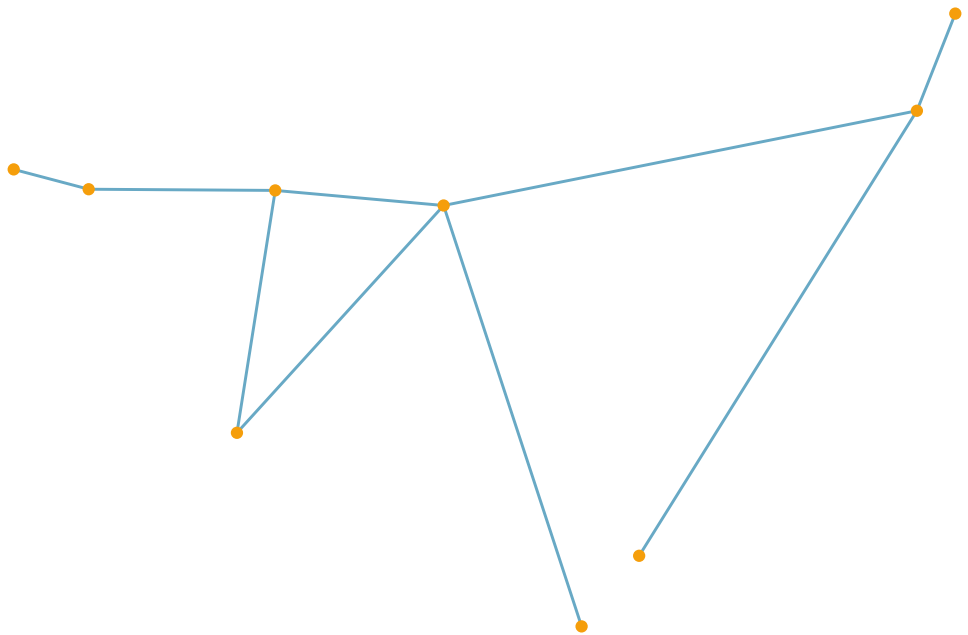
Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv

# 16/88 Capricornus (Cap) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

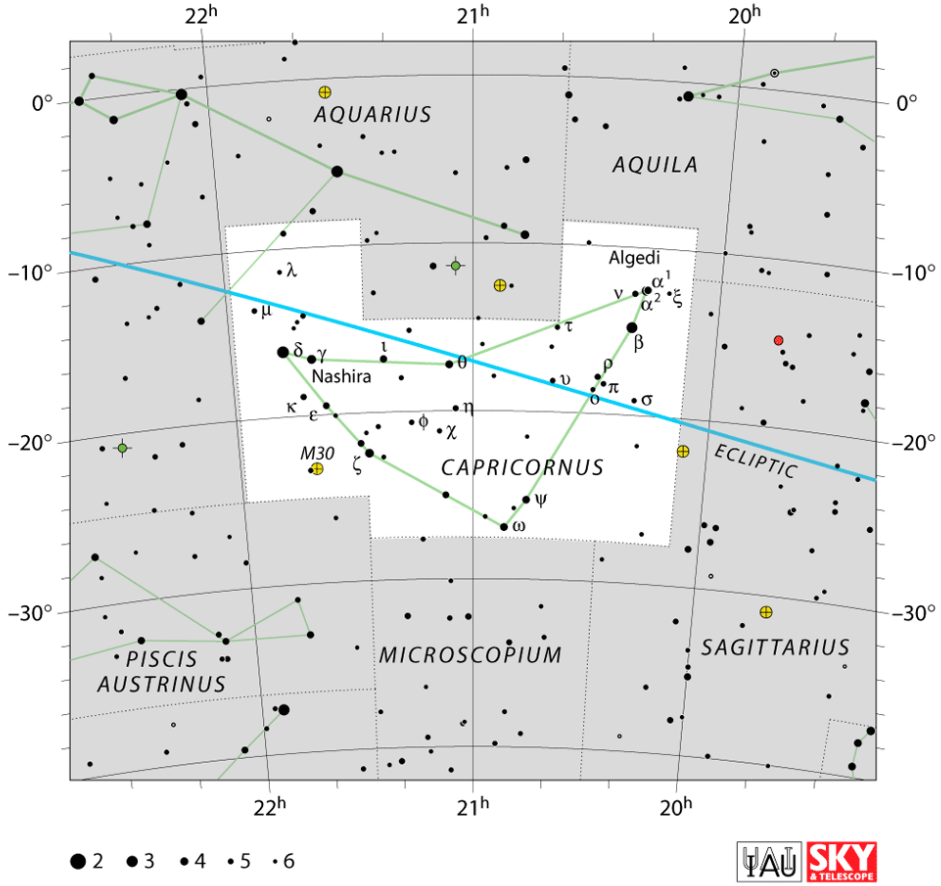
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 16/88  Capricornus (Cap) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 100064 | 304.513417 | -12.544861 | HIP 100345 |
| HIP 100345 | 305.252708 | -14.781389 | HIP 100064, HIP 104139, HIP 102485 |
| HIP 104139 | 316.486583 | -17.232722 | HIP 100345, HIP 105515, HIP 105881, HIP 102978 |
| HIP 105515 | 320.561583 | -16.834556 | HIP 104139, HIP 106985, HIP 105881 |
| HIP 106985 | 325.022250 | -16.662250 | HIP 105515, HIP 107556 |
| HIP 107556 | 326.759542 | -16.126556 | HIP 106985 |
| HIP 105881 | 321.666792 | -22.411389 | HIP 105515, HIP 104139 |
| HIP 102485 | 311.524042 | -25.270528 | HIP 100345 |
| HIP 102978 | 312.955417 | -26.919139 | HIP 104139 |

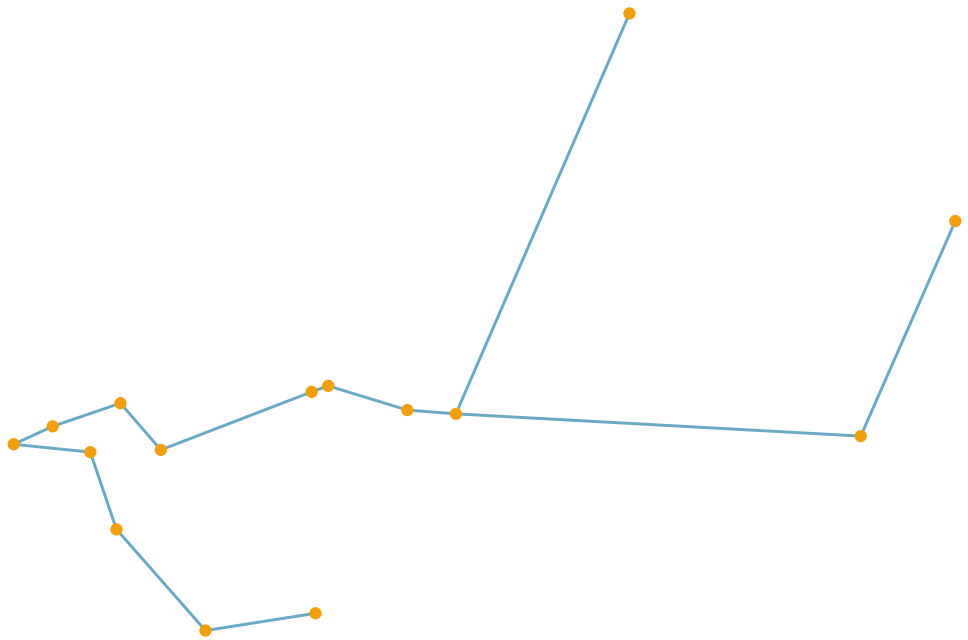
Effective release edges: 9 • unique plotted points: 9 • source: release_effective_edges.csv

# 17/88 Carina (Car) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

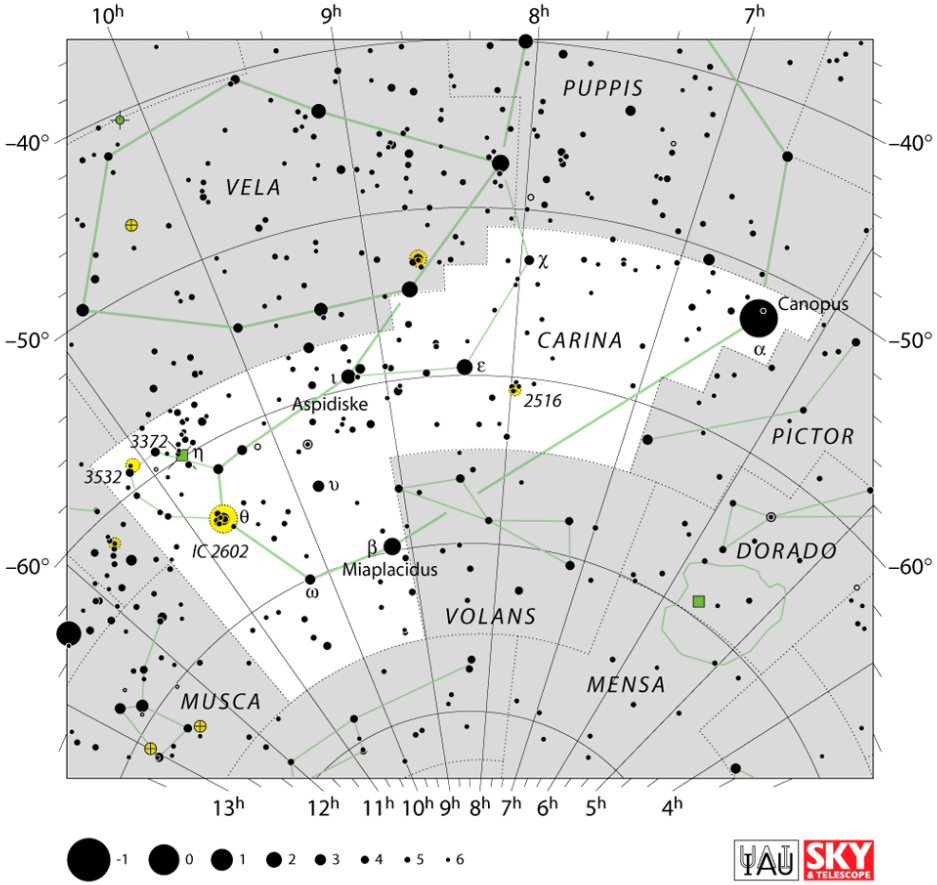
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 17/88  Carina (Car) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 45238 | 138.301000 | -69.717472 | HIP 50099 |
| HIP 50099 | 153.434500 | -70.037917 | HIP 45238, HIP 52419 |
| HIP 52419 | 160.739292 | -64.394472 | HIP 50099, HIP 52468 |
| HIP 52468 | 160.884625 | -60.566639 | HIP 52419, HIP 54463 |
| HIP 54463 | 167.147500 | -58.975056 | HIP 52468, HIP 53253 |
| HIP 53253 | 163.373208 | -58.853278 | HIP 54463, HIP 51232 |
| HIP 51232 | 156.969792 | -58.739417 | HIP 53253, HIP 50371 |
| HIP 50371 | 154.270875 | -61.332333 | HIP 51232, HIP 45556 |
| HIP 45556 | 139.272625 | -59.275250 | HIP 50371, HIP 45080 |
| HIP 42568 | 130.154333 | -59.761028 | HIP 41037, HIP 45080 |
| HIP 41037 | 125.628583 | -59.509528 | HIP 42568, HIP 30438, HIP 39429 |
| HIP 30438 | 95.987875 | -52.695722 | HIP 41037, HIP 31685 |
| HIP 45080 | 137.742125 | -58.966917 | HIP 45556, HIP 42568 |
| HIP 31685 | 99.440292 | -43.195917 | HIP 30438 |
| HIP 39429 | 120.896125 | -40.003194 | HIP 41037 |

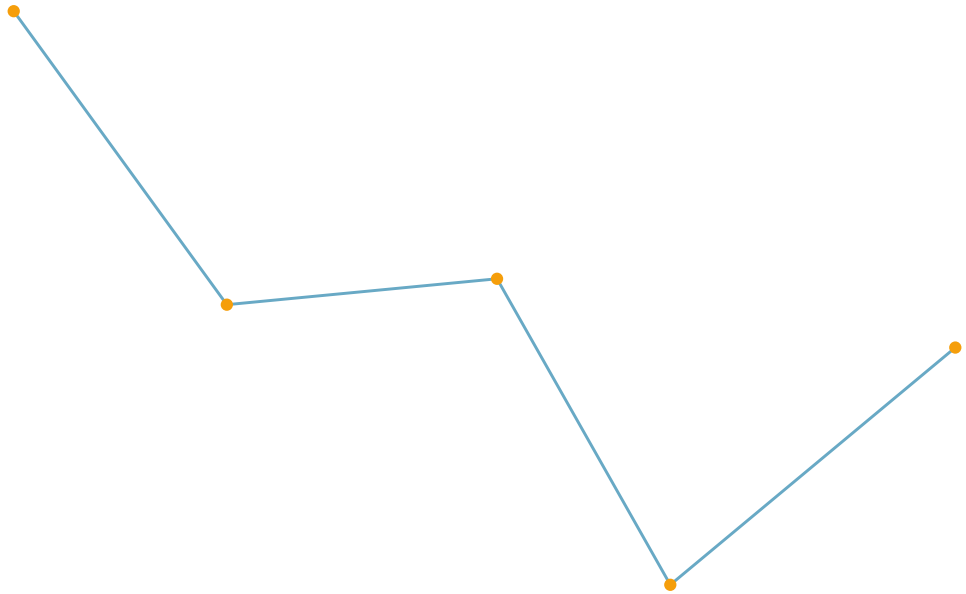
Effective release edges: 14 • unique plotted points: 15 • source: release_effective_edges.csv

# 18/88 Cassiopeia (Cas) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

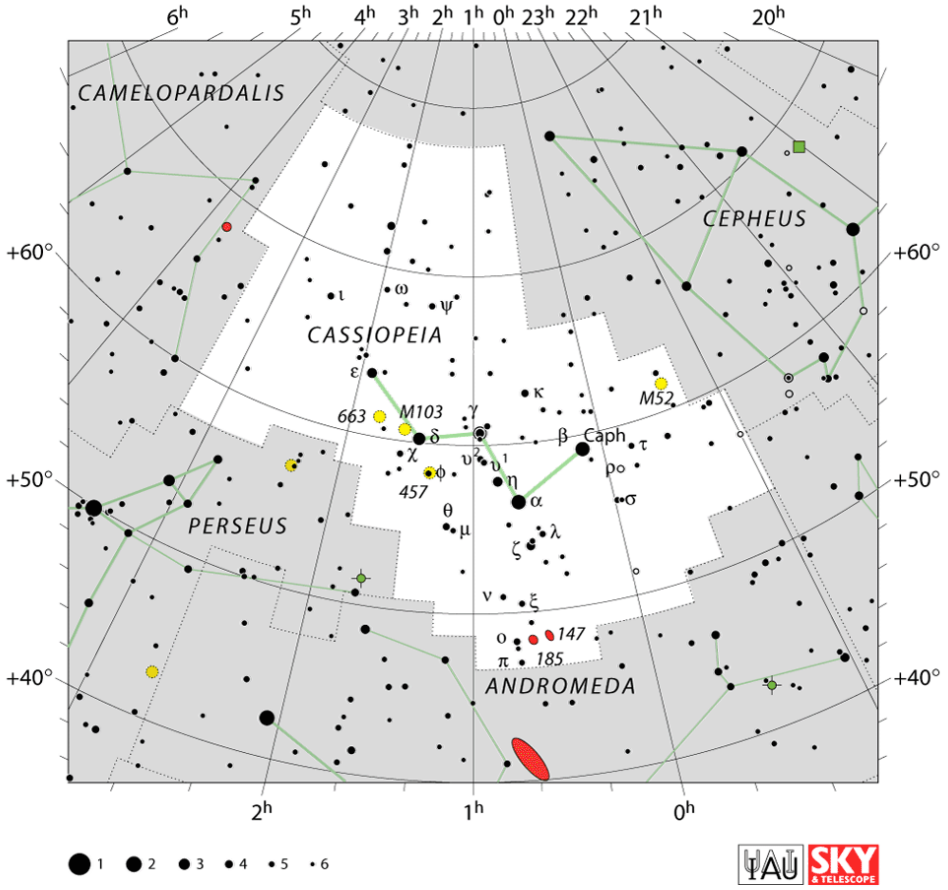
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 18/88  Cassiopeia (Cas) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 8886 | 28.598667 | +63.670139 | HIP 6686 |
| HIP 6686 | 21.452500 | +60.235417 | HIP 8886, HIP 4427 |
| HIP 4427 | 14.177083 | +60.716750 | HIP 6686, HIP 3179 |
| HIP 3179 | 10.126625 | +56.537417 | HIP 4427, HIP 746 |
| HIP 746 | 2.292042 | +59.150222 | HIP 3179 |

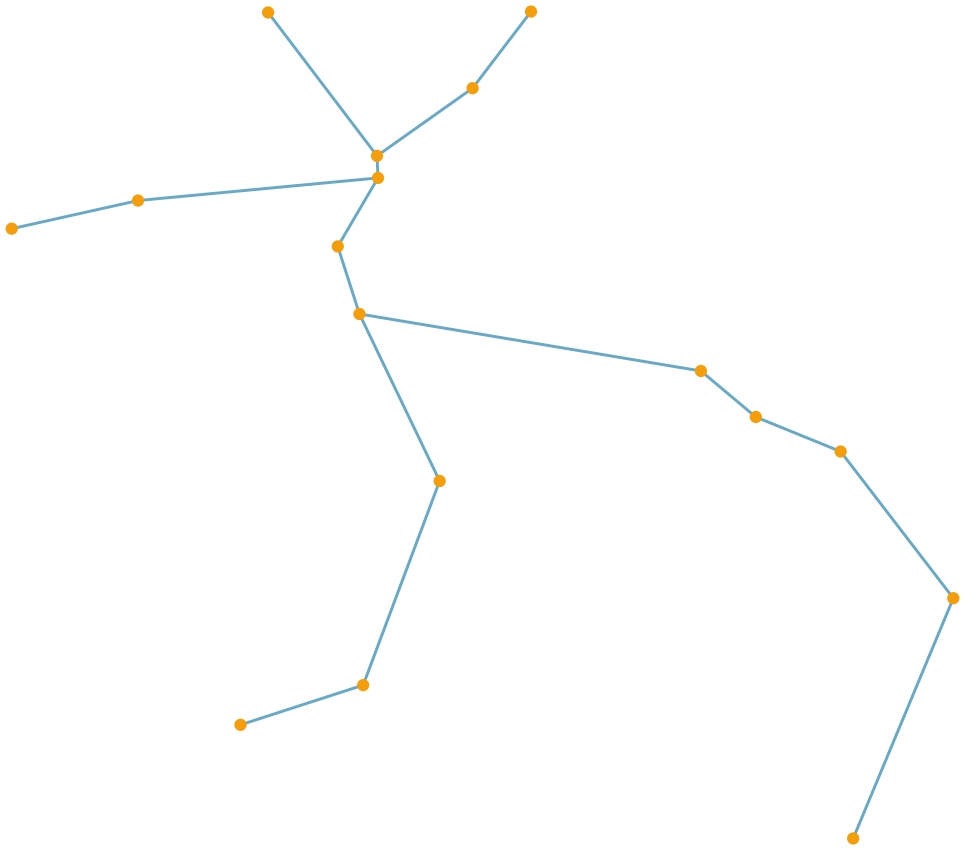
Effective release edges: 4 • unique plotted points: 5 • source: release_effective_edges.csv

# 19/88 Centaurus (Cen) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

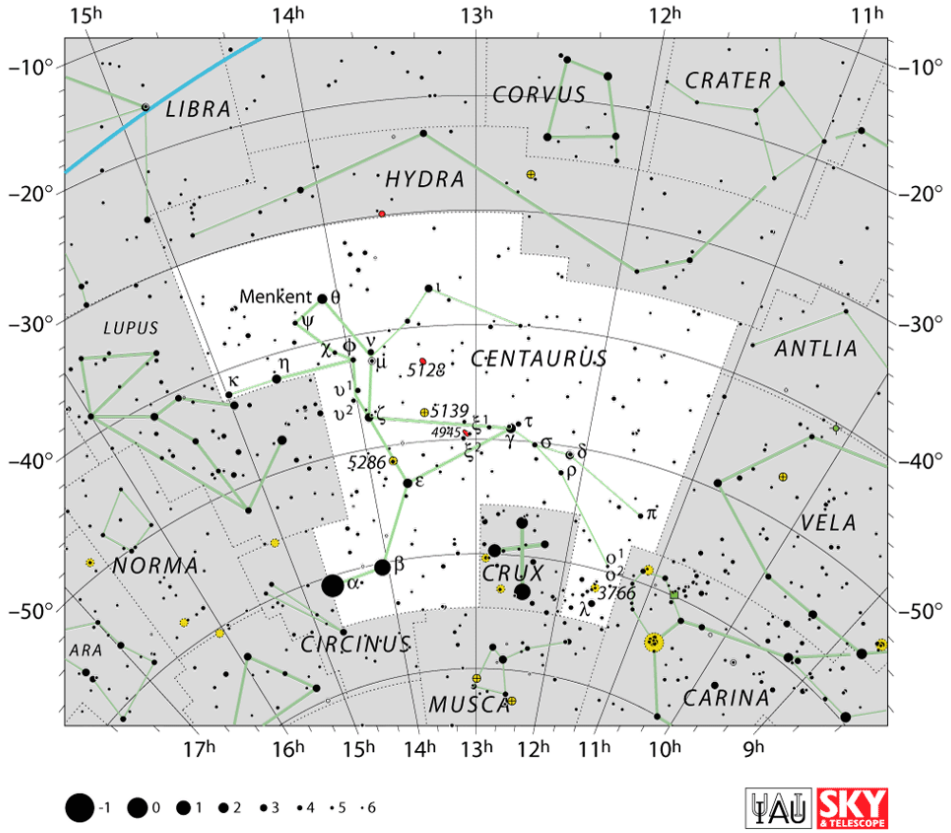
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 19/88  Centaurus (Cen) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 71683 | 219.920417 | -60.835139 | HIP 68702 |
| HIP 68702 | 210.956000 | -60.372972 | HIP 71683, HIP 66657 |
| HIP 66657 | 204.971958 | -53.466361 | HIP 68702, HIP 68002 |
| HIP 68002 | 208.885125 | -47.288278 | HIP 66657, HIP 68282, HIP 61932 |
| HIP 68282 | 209.669875 | -44.803528 | HIP 68002, HIP 67472 |
| HIP 67472 | 207.404208 | -42.473694 | HIP 68282, HIP 67464, HIP 71352 |
| HIP 67464 | 207.376250 | -41.687667 | HIP 67472, HIP 65936, HIP 68933 |
| HIP 65936 | 202.761125 | -39.407278 | HIP 67464, HIP 65109 |
| HIP 65109 | 200.150292 | -36.712083 | HIP 65936 |
| HIP 68933 | 211.672167 | -36.368694 | HIP 67464 |
| HIP 71352 | 218.876875 | -42.157750 | HIP 67472, HIP 73334 |
| HIP 73334 | 224.790417 | -42.104139 | HIP 71352 |
| HIP 61932 | 190.380000 | -48.959889 | HIP 68002, HIP 60823 |
| HIP 60823 | 187.010042 | -50.230611 | HIP 61932, HIP 59196 |
| HIP 59196 | 182.089750 | -50.722417 | HIP 60823, HIP 56480 |
| HIP 56480 | 173.690458 | -54.264139 | HIP 59196, HIP 56561 |
| HIP 56561 | 173.945542 | -63.019833 | HIP 56480 |

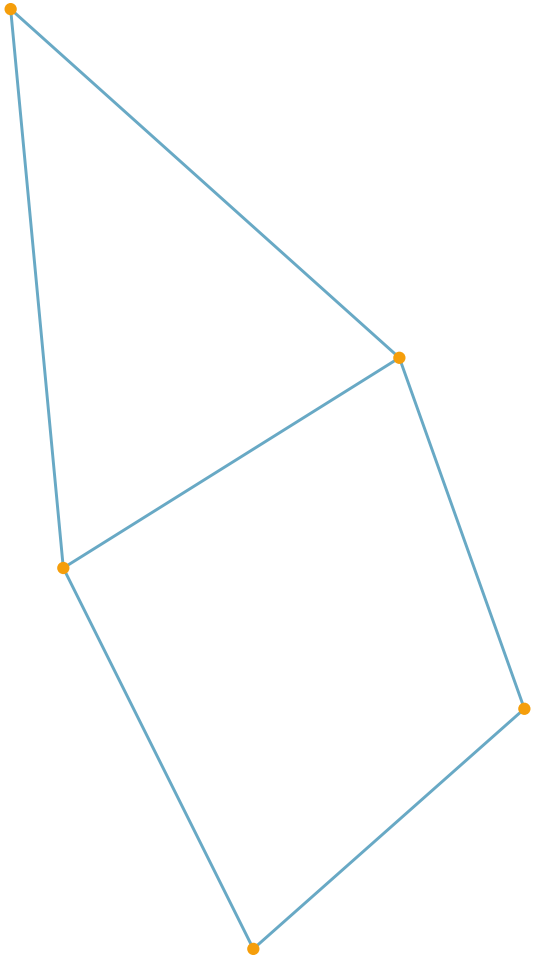
Effective release edges: 16 • unique plotted points: 17 • source: release_effective_edges.csv

# 20/88 Cepheus (Cep) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

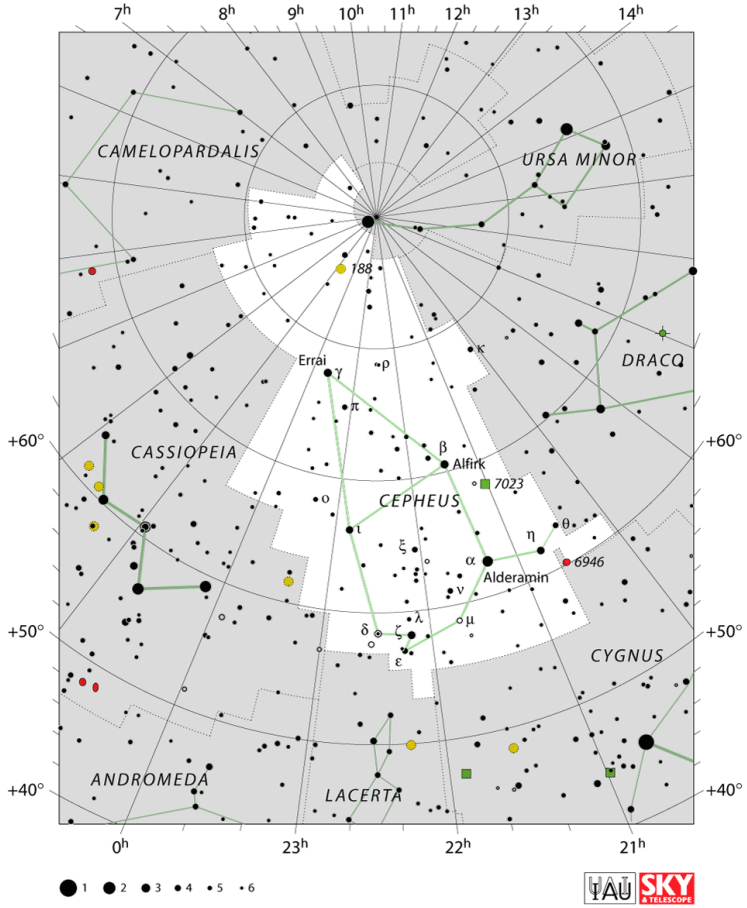
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 20/88  Cepheus (Cep) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 109492 | 332.713583 | +58.201250 | HIP 112724, HIP 105199 |
| HIP 112724 | 342.420458 | +66.200722 | HIP 109492, HIP 106032, HIP 116727 |
| HIP 106032 | 322.164917 | +70.560694 | HIP 112724, HIP 105199, HIP 116727 |
| HIP 105199 | 319.644083 | +62.585444 | HIP 106032, HIP 109492 |
| HIP 116727 | 354.837417 | +77.631972 | HIP 112724, HIP 106032 |

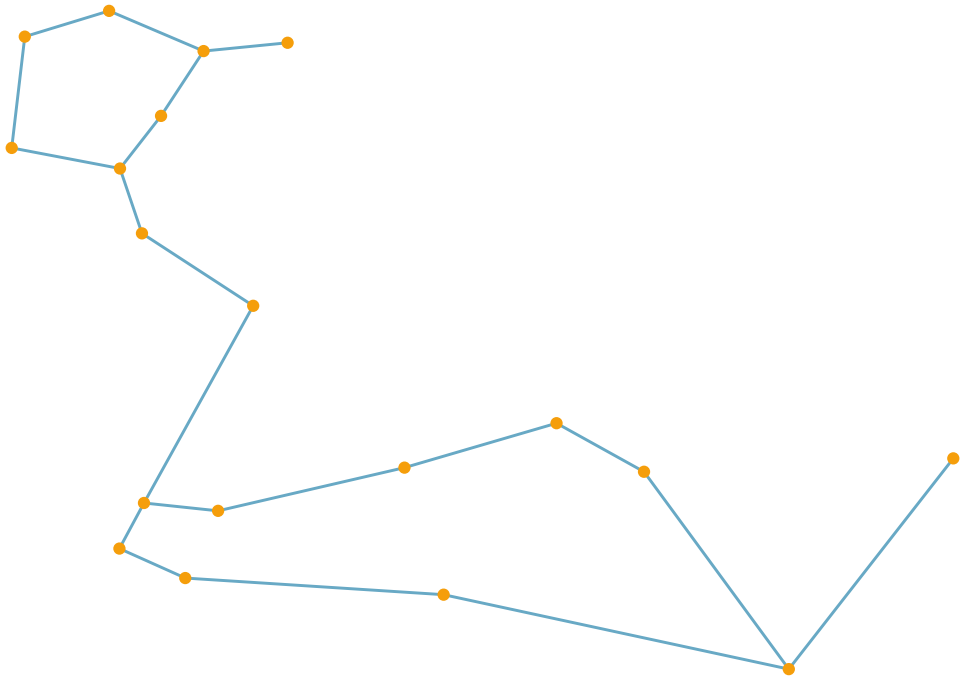
Effective release edges: 6 • unique plotted points: 5 • source: release_effective_edges.csv

# 21/88 Cetus (Cet) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

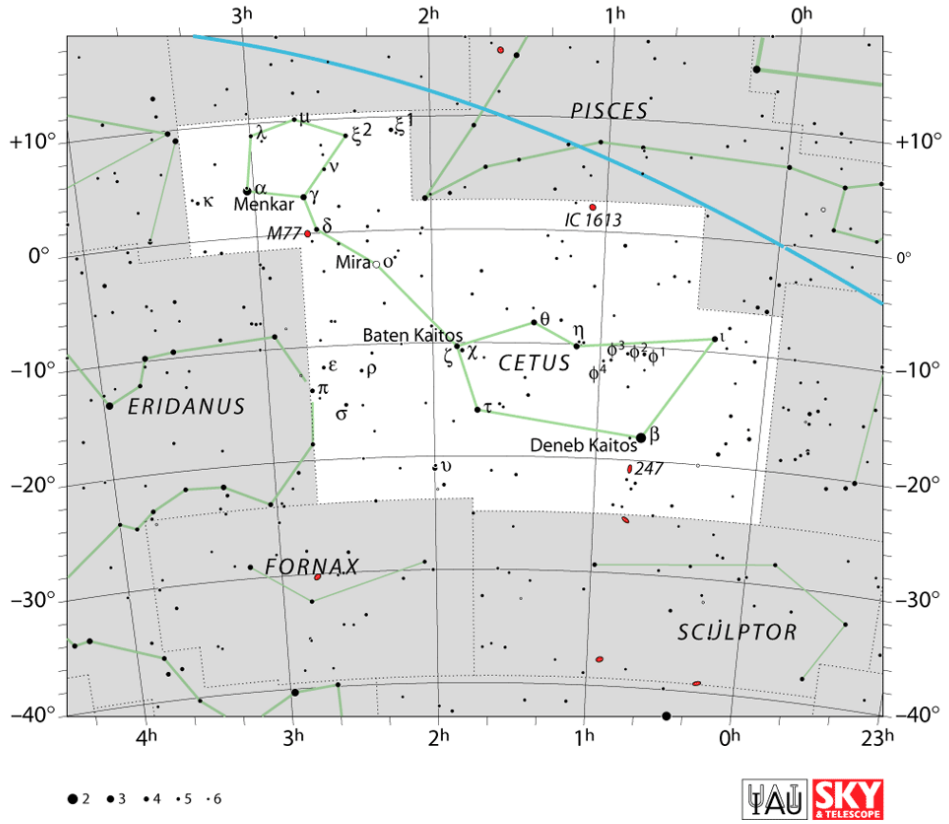
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 21/88  Cetus (Cet) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 10324 | 33.250042 | +8.846750 | HIP 11484 |
| HIP 11484 | 37.039667 | +8.460083 | HIP 10324, HIP 12828, HIP 12093 |
| HIP 8102 | 26.021375 | -15.939556 | HIP 3419, HIP 11783 |
| HIP 3419 | 10.896792 | -17.986694 | HIP 8102, HIP 1562, HIP 5364 |
| HIP 1562 | 4.857000 | -8.823833 | HIP 3419 |
| HIP 5364 | 17.146917 | -10.181917 | HIP 3419, HIP 6537 |
| HIP 6537 | 21.006042 | -8.182750 | HIP 5364, HIP 8645 |
| HIP 8645 | 27.865042 | -10.334944 | HIP 6537, HIP 11345 |
| HIP 11345 | 36.487542 | -12.290444 | HIP 8645, HIP 12390 |
| HIP 12390 | 39.890542 | -11.871583 | HIP 11345, HIP 12770, HIP 10826 |
| HIP 12770 | 41.030625 | -13.858667 | HIP 12390, HIP 11783 |
| HIP 11783 | 38.022000 | -15.244333 | HIP 12770, HIP 8102 |
| HIP 10826 | 34.836625 | -2.977056 | HIP 12390, HIP 12387 |
| HIP 12387 | 39.870625 | +0.328528 | HIP 10826, HIP 12706 |
| HIP 12706 | 40.825500 | +3.236167 | HIP 12387, HIP 14135, HIP 12093 |
| HIP 14135 | 45.569917 | +4.089917 | HIP 12706, HIP 13954 |
| HIP 13954 | 44.928750 | +8.907389 | HIP 14135, HIP 12828 |
| HIP 12828 | 41.234875 | +10.114222 | HIP 13954, HIP 11484 |
| HIP 12093 | 38.968708 | +5.593306 | HIP 11484, HIP 12706 |

Effective release edges: 20 • unique plotted points: 19 • source: release_effective_edges.csv

# 22/88 Chamaeleon (Cha) — side-by-side chart comparison

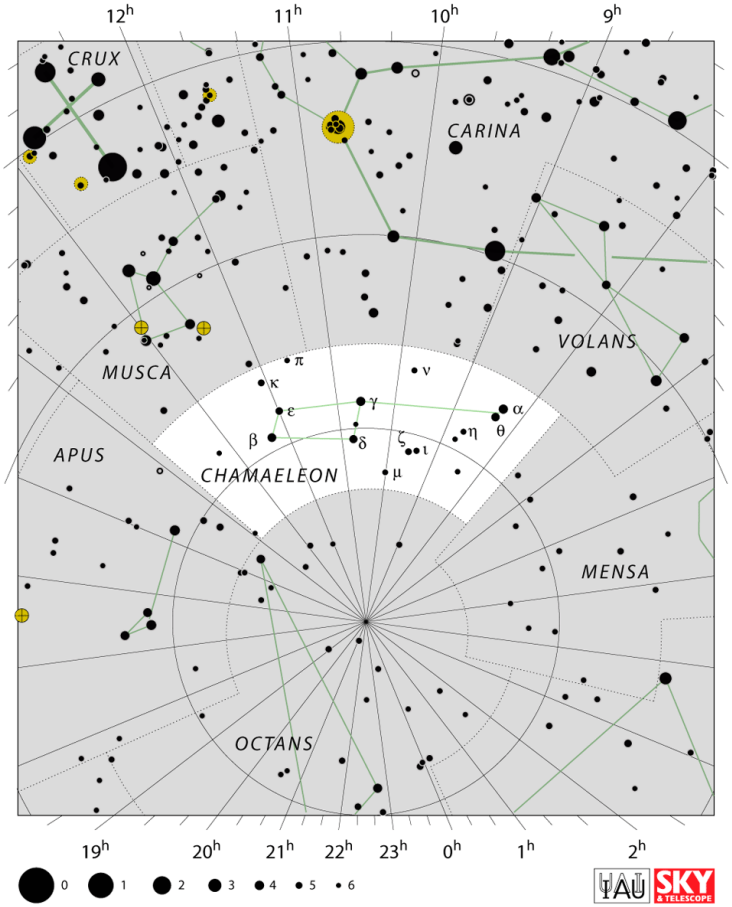
Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV

## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 22/88  Chamaeleon (Cha) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 40702 | 124.630292 | -76.919972 | HIP 51839 |
| HIP 51839 | 158.867583 | -78.607806 | HIP 40702, HIP 60000 |
| HIP 60000 | 184.587250 | -79.312278 | HIP 51839 |

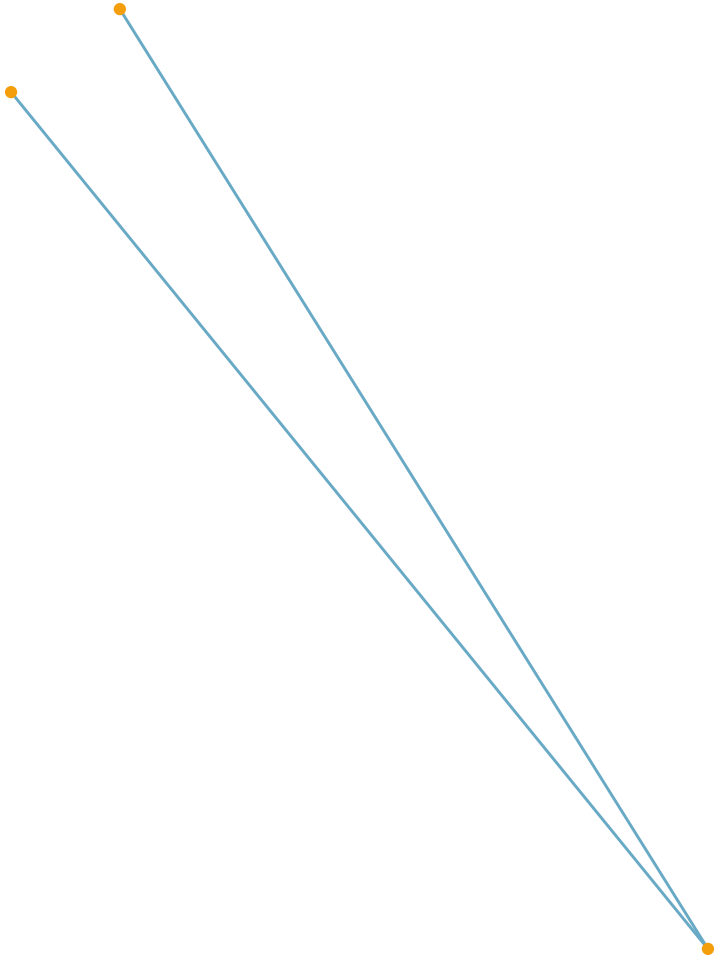
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

# 23/88 Circinus (Cir) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

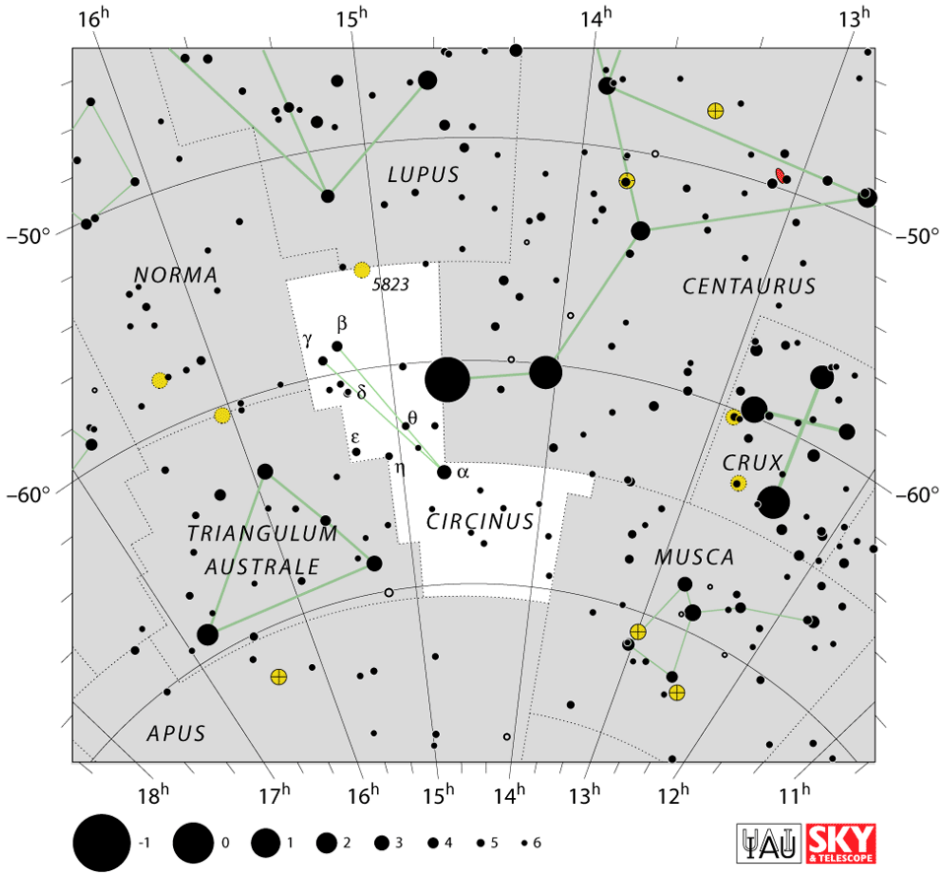
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 23/88  Circinus (Cir) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 71908 | 220.627875 | -64.974583 | HIP 75323, HIP 74824 |
| HIP 75323 | 230.844417 | -59.320694 | HIP 71908 |
| HIP 74824 | 229.379000 | -58.800889 | HIP 71908 |

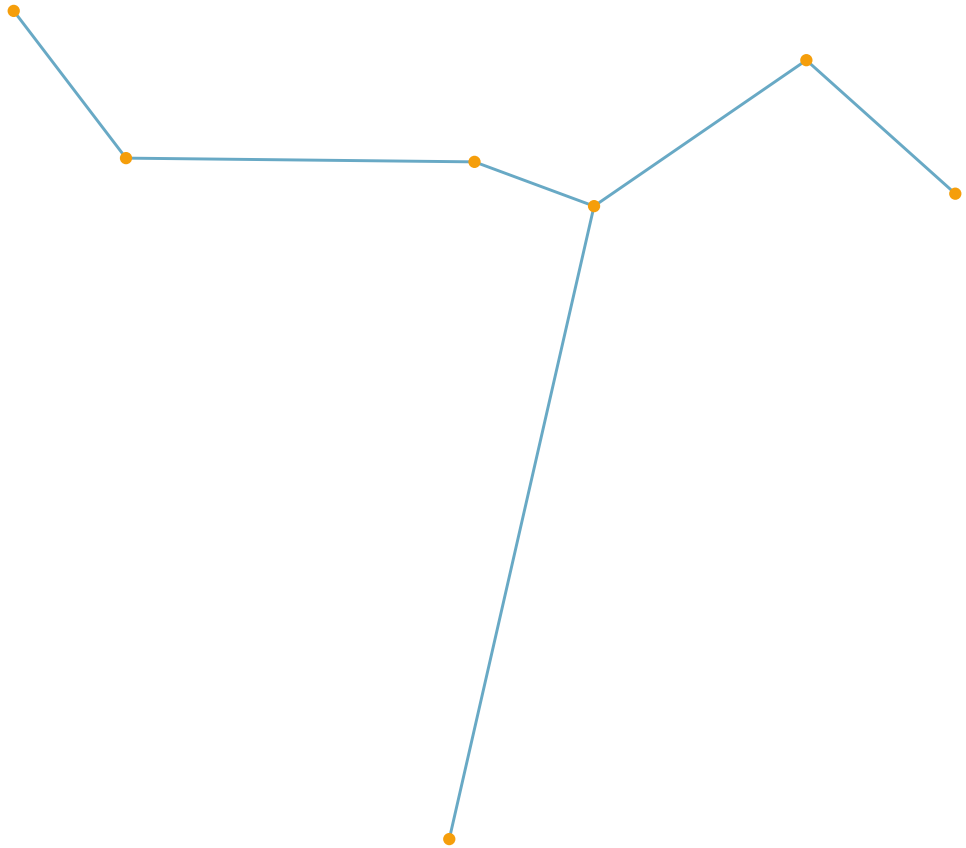
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

# 24/88 Columba (Col) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

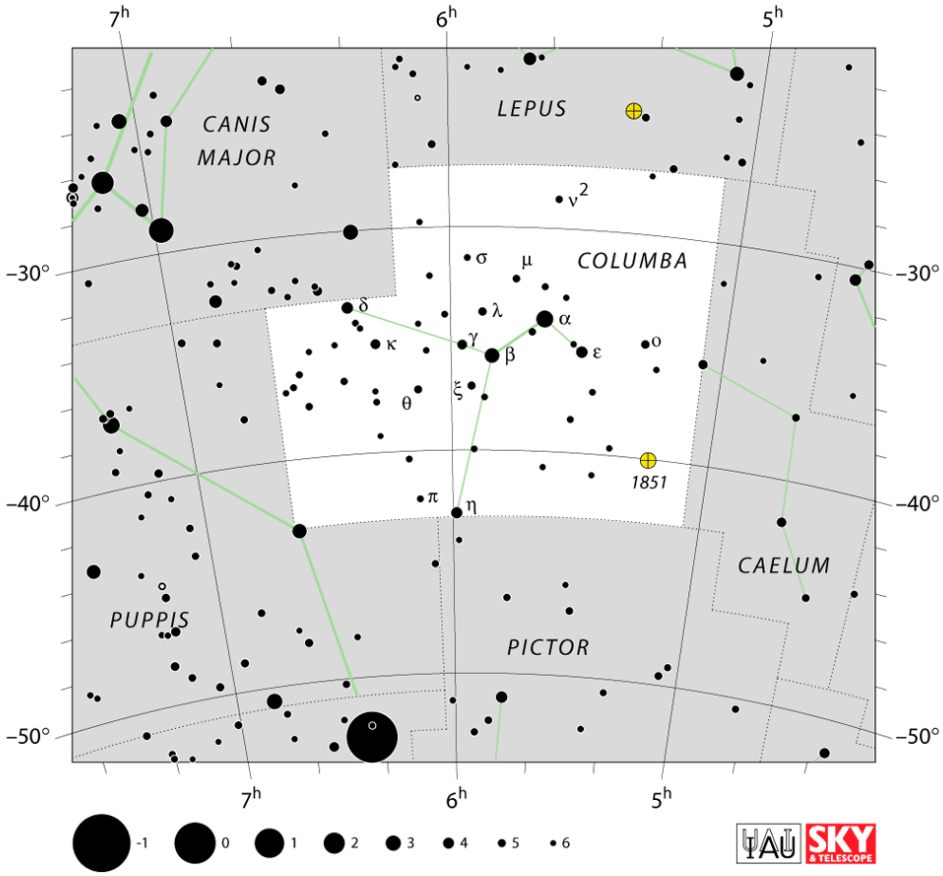
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 24/88  Columba (Col) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 30277 | 95.528542 | -33.436278 | HIP 29807 |
| HIP 29807 | 94.138083 | -35.140722 | HIP 30277, HIP 28199 |
| HIP 28199 | 89.384208 | -35.283306 | HIP 29807, HIP 27628 |
| HIP 27628 | 87.739792 | -35.769306 | HIP 28199, HIP 28328, HIP 26634 |
| HIP 28328 | 89.786625 | -42.815111 | HIP 27628 |
| HIP 26634 | 84.912250 | -34.074056 | HIP 27628, HIP 25859 |
| HIP 25859 | 82.803083 | -35.470444 | HIP 26634 |

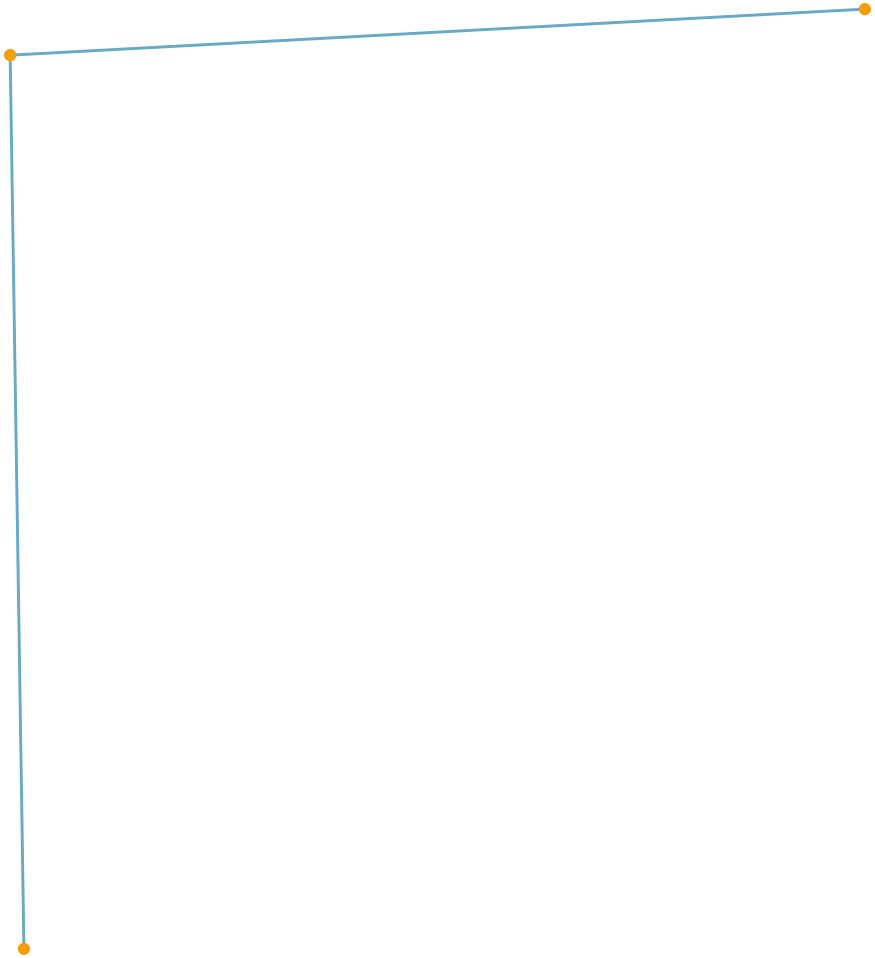
Effective release edges: 6 • unique plotted points: 7 • source: release_effective_edges.csv

# 25/88 Coma Berenices (Com) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

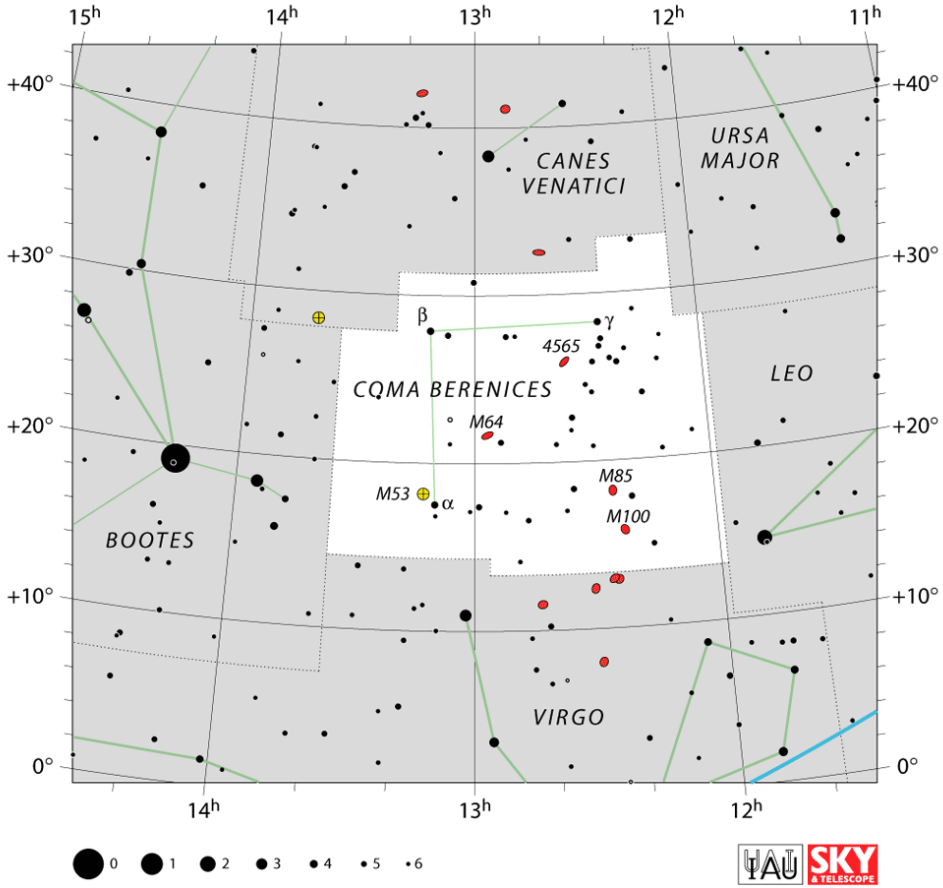
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 25/88  Coma Berenices (Com) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 64241 | 197.498125 | +17.529111 | HIP 64394 |
| HIP 64394 | 197.970500 | +27.876028 | HIP 64241, HIP 60742 |
| HIP 60742 | 186.734708 | +28.268611 | HIP 64394 |

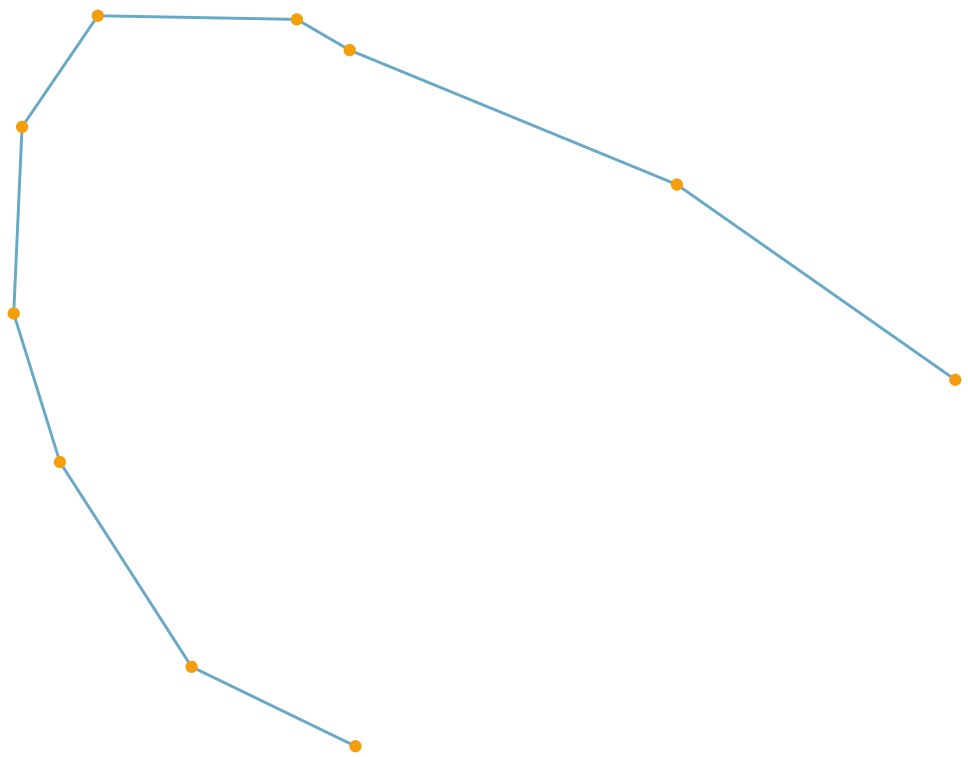
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

# 26/88 Corona Australis (CrA) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

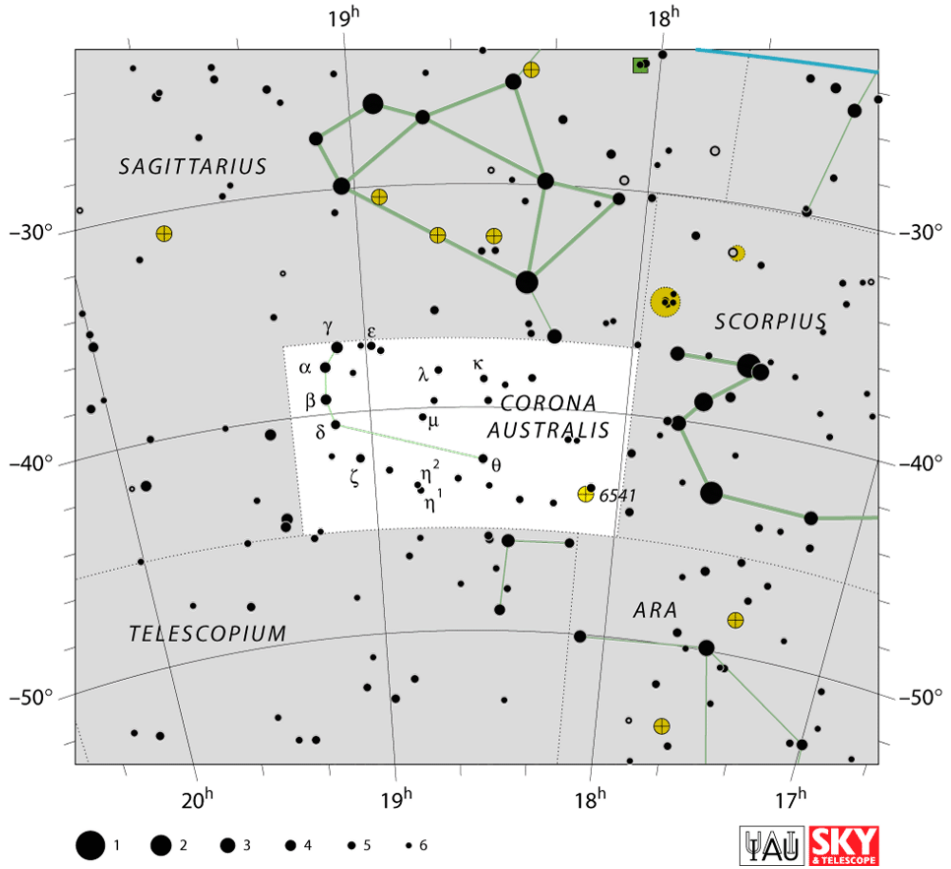
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 26/88  Corona Australis (CrA) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 91875 | 280.945583 | -38.323306 | HIP 92989, HIP 90887 |
| HIP 92989 | 284.168708 | -37.343194 | HIP 91875, HIP 93174 |
| HIP 93174 | 284.681125 | -37.107083 | HIP 92989, HIP 93825 |
| HIP 93825 | 286.604333 | -37.062750 | HIP 93174, HIP 94114 |
| HIP 94114 | 287.367833 | -37.904250 | HIP 93825, HIP 94160 |
| HIP 94160 | 287.507292 | -39.340694 | HIP 94114, HIP 94005 |
| HIP 94005 | 287.087208 | -40.496639 | HIP 94160, HIP 93542 |
| HIP 93542 | 285.778458 | -42.095000 | HIP 94005, HIP 92953 |
| HIP 92953 | 284.070708 | -42.710611 | HIP 93542 |
| HIP 90887 | 278.088792 | -39.703917 | HIP 91875 |

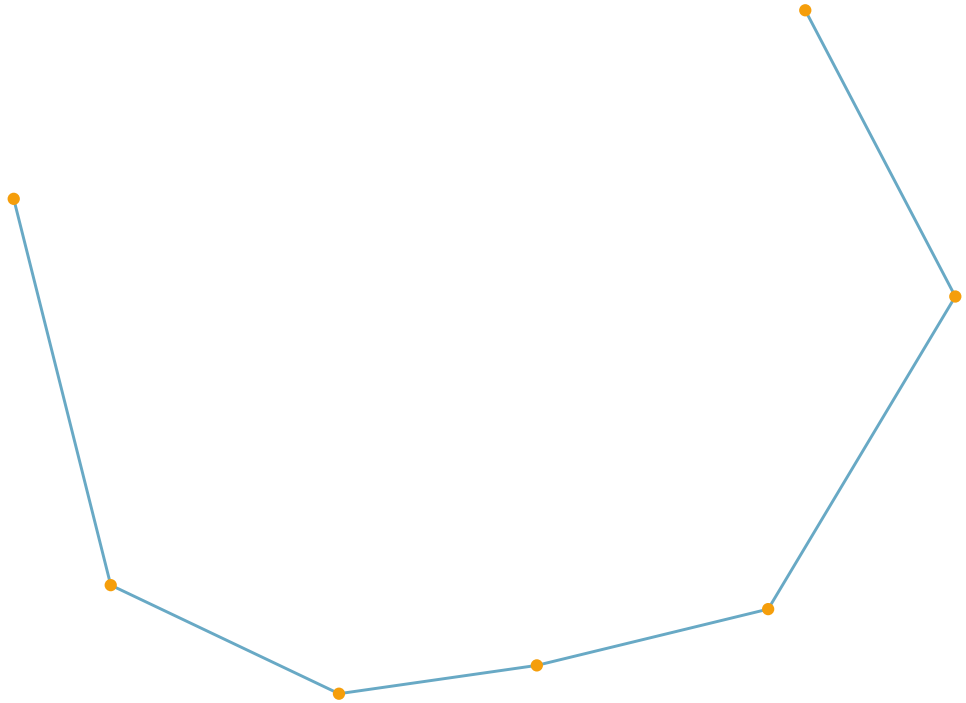
Effective release edges: 9 • unique plotted points: 10 • source: release_effective_edges.csv

# 27/88 Corona Borealis (CrB) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

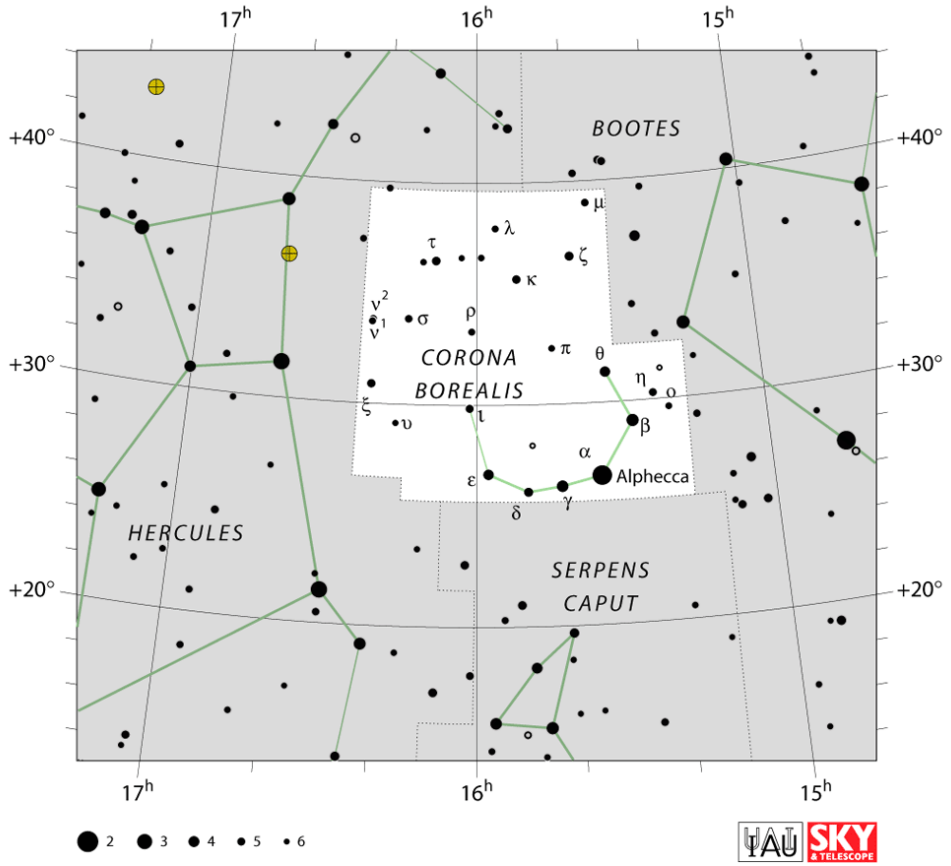
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 27/88  Corona Borealis (CrB) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 76127 | 233.232500 | +31.359167 | HIP 75695 |
| HIP 75695 | 231.957708 | +29.105500 | HIP 76127, HIP 76267 |
| HIP 76267 | 233.671625 | +26.714917 | HIP 75695, HIP 76952 |
| HIP 76952 | 235.686000 | +26.295528 | HIP 76267, HIP 77512 |
| HIP 77512 | 237.398750 | +26.068556 | HIP 76952, HIP 78159 |
| HIP 78159 | 239.397083 | +26.878028 | HIP 77512, HIP 78493 |
| HIP 78493 | 240.360792 | +29.851083 | HIP 78159 |

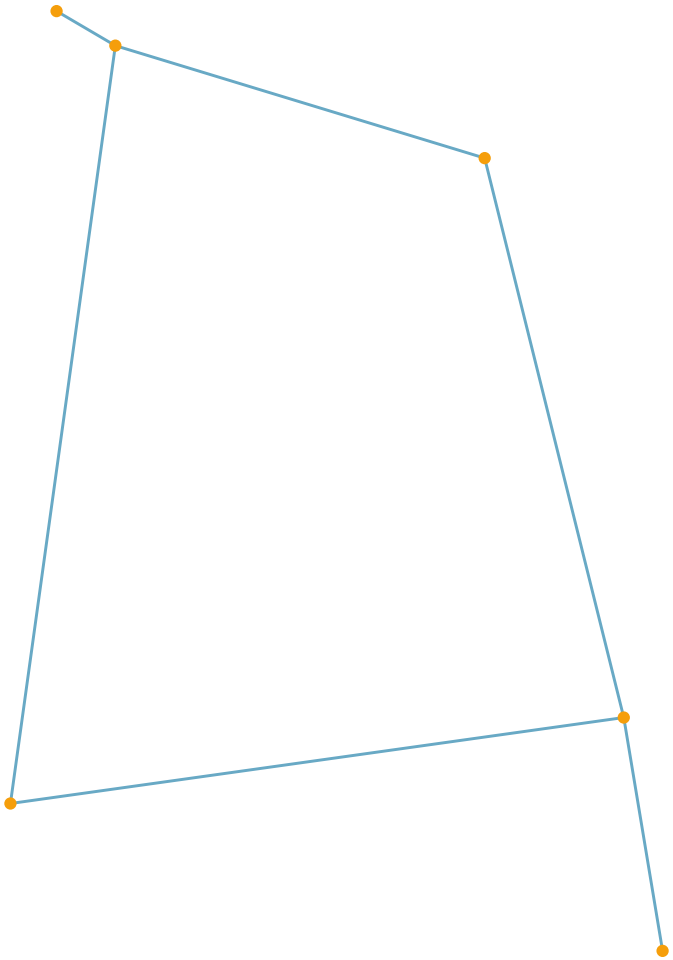
Effective release edges: 6 • unique plotted points: 7 • source: release_effective_edges.csv

# 28/88 Corvus (Crv) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

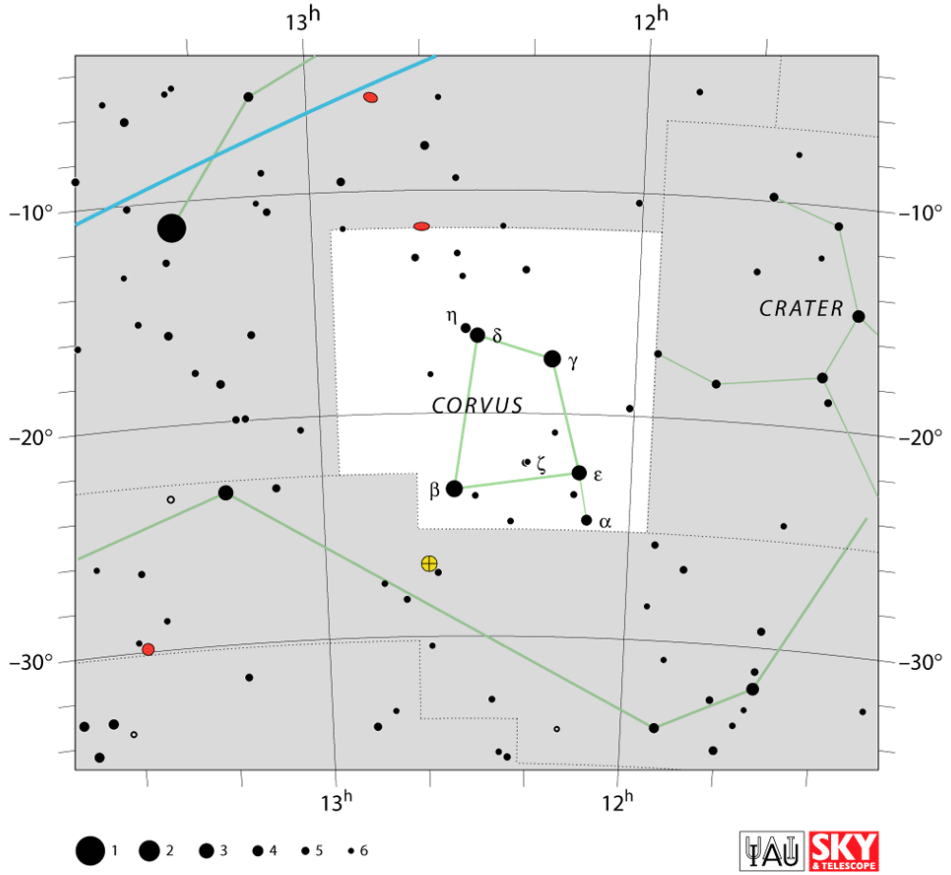
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 28/88  Corvus (Crv) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 61174 | 188.018667 | -16.195861 | HIP 60965 |
| HIP 60965 | 187.466583 | -16.515083 | HIP 61174, HIP 59803, HIP 61359 |
| HIP 59803 | 183.951958 | -17.541972 | HIP 60965, HIP 59316 |
| HIP 59316 | 182.531375 | -22.619806 | HIP 59803, HIP 59199, HIP 61359 |
| HIP 59199 | 182.103125 | -24.728778 | HIP 59316 |
| HIP 61359 | 188.596792 | -23.396611 | HIP 59316, HIP 60965 |

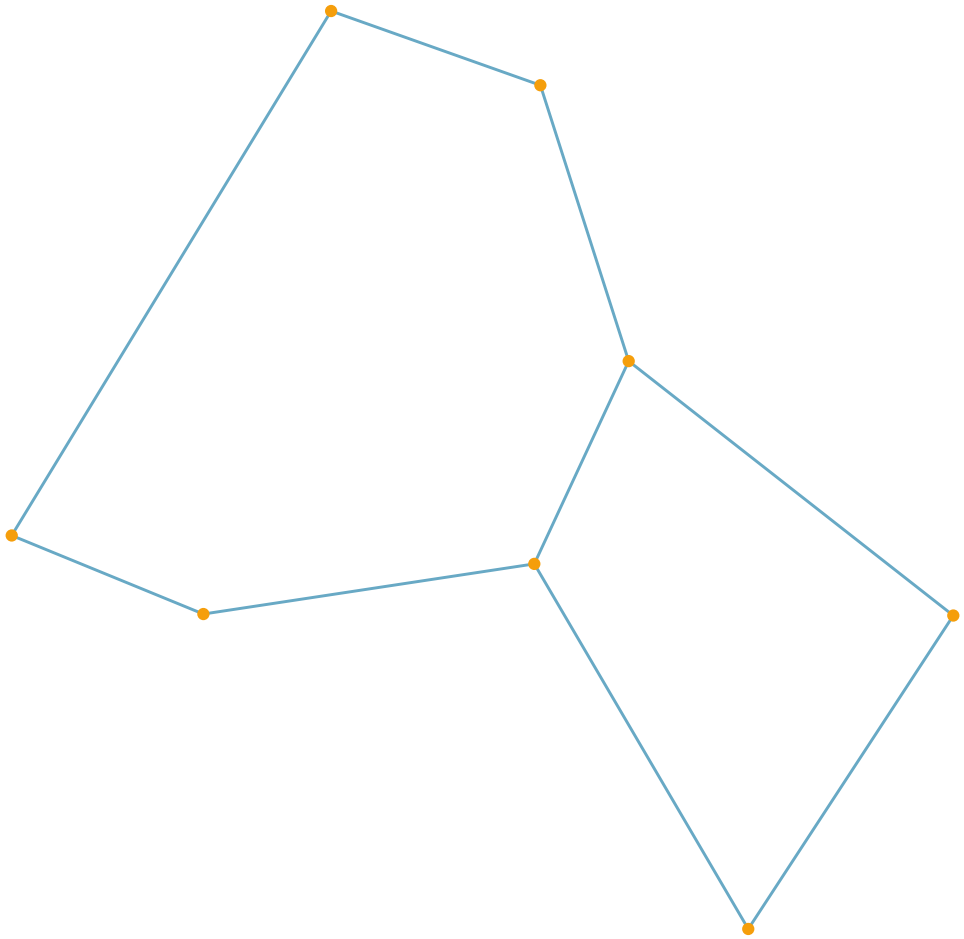
Effective release edges: 6 • unique plotted points: 6 • source: release_effective_edges.csv

# 29/88 Crater (Crt) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

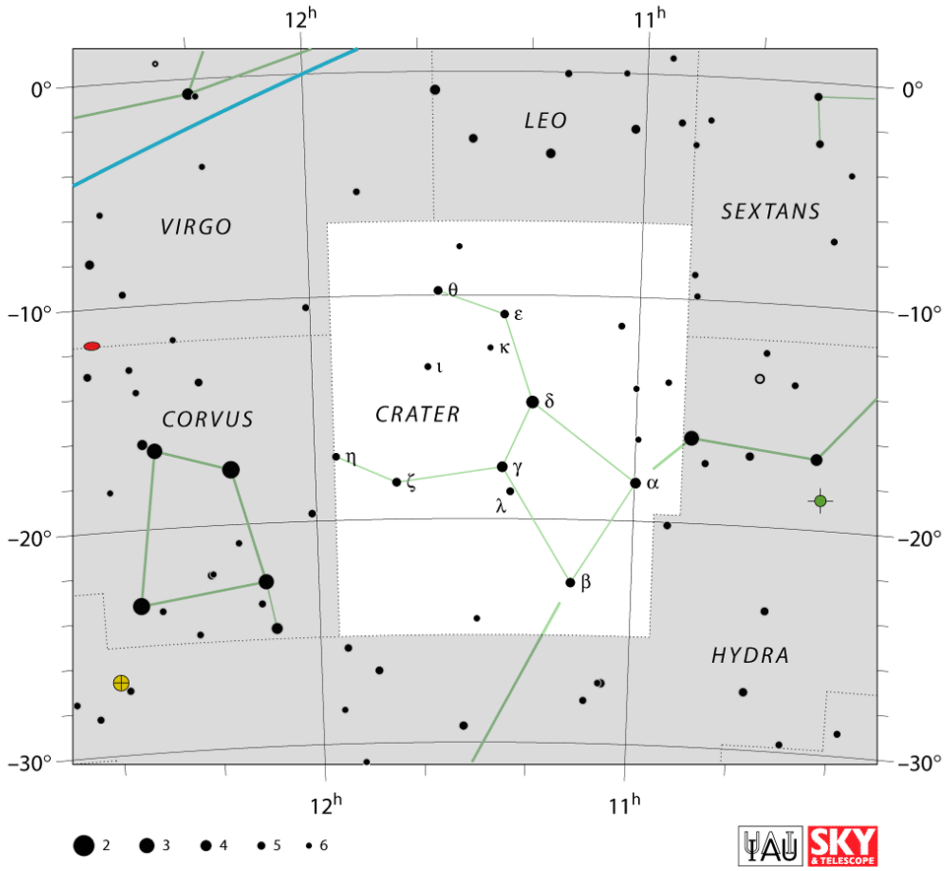
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 29/88  Crater (Crt) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 53740 | 164.944792 | -18.299111 | HIP 54682, HIP 55282 |
| HIP 54682 | 167.914542 | -22.825611 | HIP 53740, HIP 55705 |
| HIP 55705 | 171.220750 | -17.684028 | HIP 54682, HIP 55282, HIP 57283 |
| HIP 55282 | 169.835500 | -14.779056 | HIP 55705, HIP 53740, HIP 55687 |
| HIP 55687 | 171.152542 | -10.859389 | HIP 55282, HIP 56633 |
| HIP 56633 | 174.170625 | -9.802250 | HIP 55687, HIP 58188 |
| HIP 58188 | 179.004083 | -17.150806 | HIP 56633, HIP 57283 |
| HIP 57283 | 176.190667 | -18.350611 | HIP 58188, HIP 55705 |

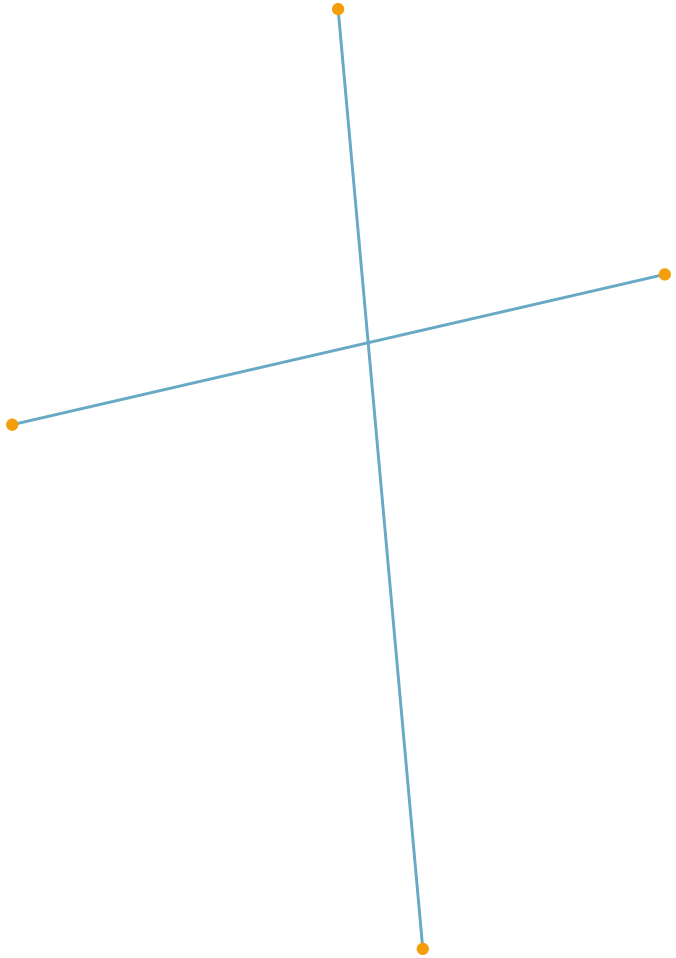
Effective release edges: 9 • unique plotted points: 8 • source: release_effective_edges.csv

# 30/88 Crux (Cru) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

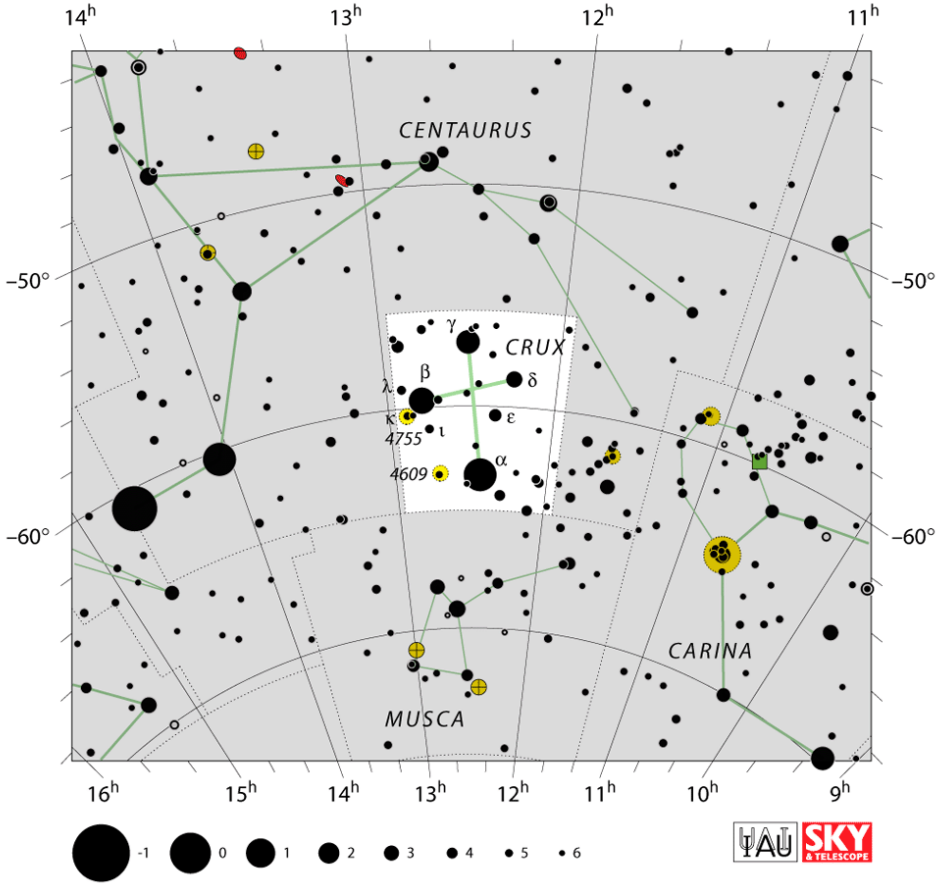
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 30/88  Crux (Cru) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 61084 | 187.791375 | -57.112556 | HIP 60718 |
| HIP 60718 | 186.649750 | -63.099056 | HIP 61084 |
| HIP 62434 | 191.930500 | -59.688722 | HIP 59747 |
| HIP 59747 | 183.786500 | -58.748889 | HIP 62434 |

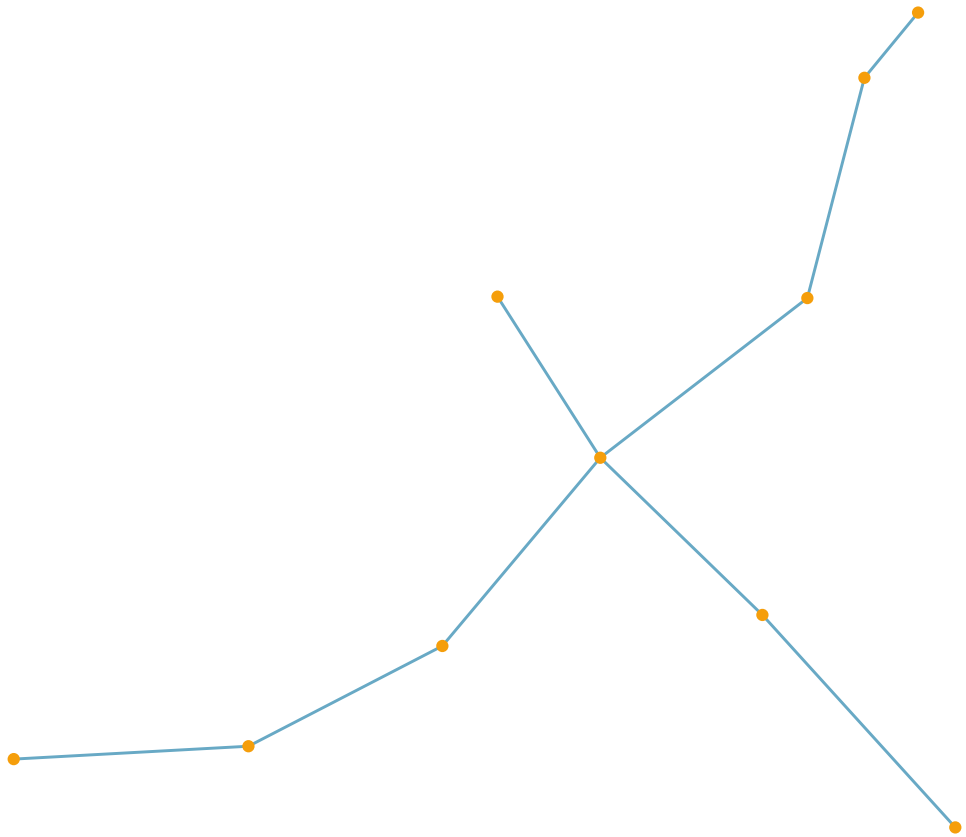
Effective release edges: 2 • unique plotted points: 4 • source: release_effective_edges.csv

# 31/88 Cygnus (Cyg) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

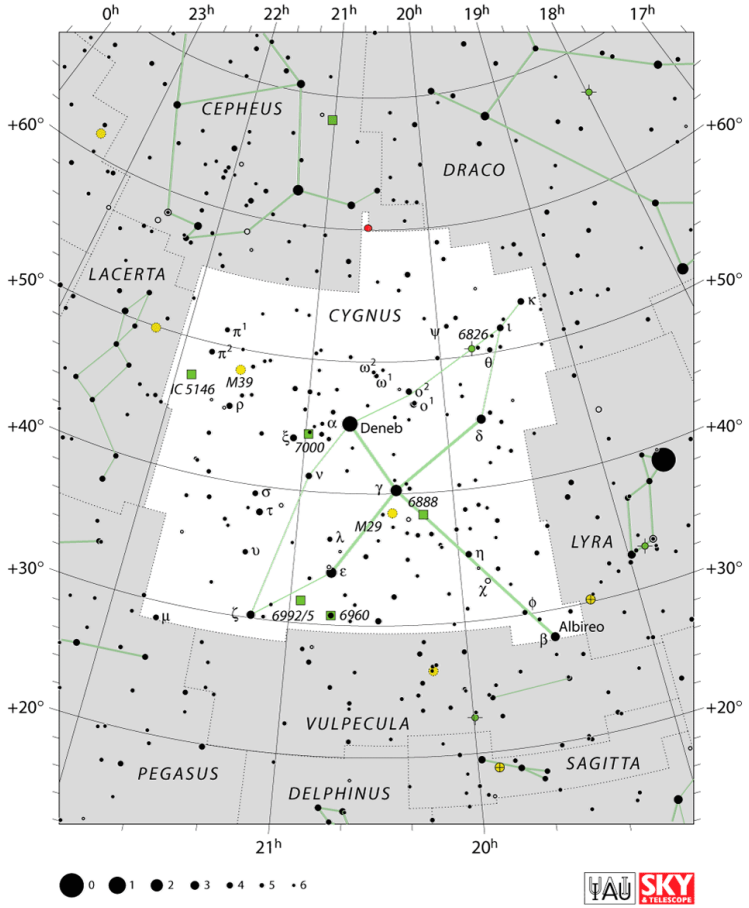
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 31/88  Cygnus (Cyg) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 94779 | 289.275458 | +53.368167 | HIP 95853 |
| HIP 95853 | 292.426417 | +51.729472 | HIP 94779, HIP 97165 |
| HIP 97165 | 296.243500 | +45.130694 | HIP 95853, HIP 100453 |
| HIP 100453 | 305.557083 | +40.256694 | HIP 97165, HIP 102098, HIP 102488, HIP 98110 |
| HIP 102098 | 310.357958 | +45.280333 | HIP 100453 |
| HIP 102488 | 311.551792 | +33.969444 | HIP 100453, HIP 104732 |
| HIP 104732 | 318.234083 | +30.227083 | HIP 102488, HIP 107310 |
| HIP 107310 | 326.035000 | +28.743222 | HIP 104732 |
| HIP 98110 | 299.076667 | +35.083500 | HIP 100453, HIP 95947 |
| HIP 95947 | 292.680375 | +27.959694 | HIP 98110 |

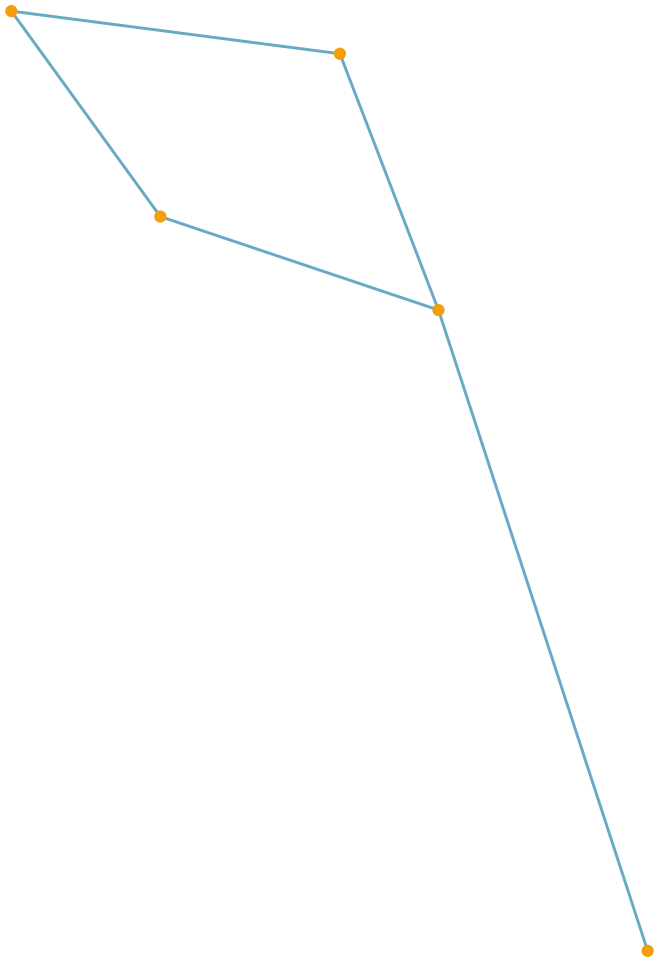
Effective release edges: 9 • unique plotted points: 10 • source: release_effective_edges.csv

# 32/88 Delphinus (Del) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

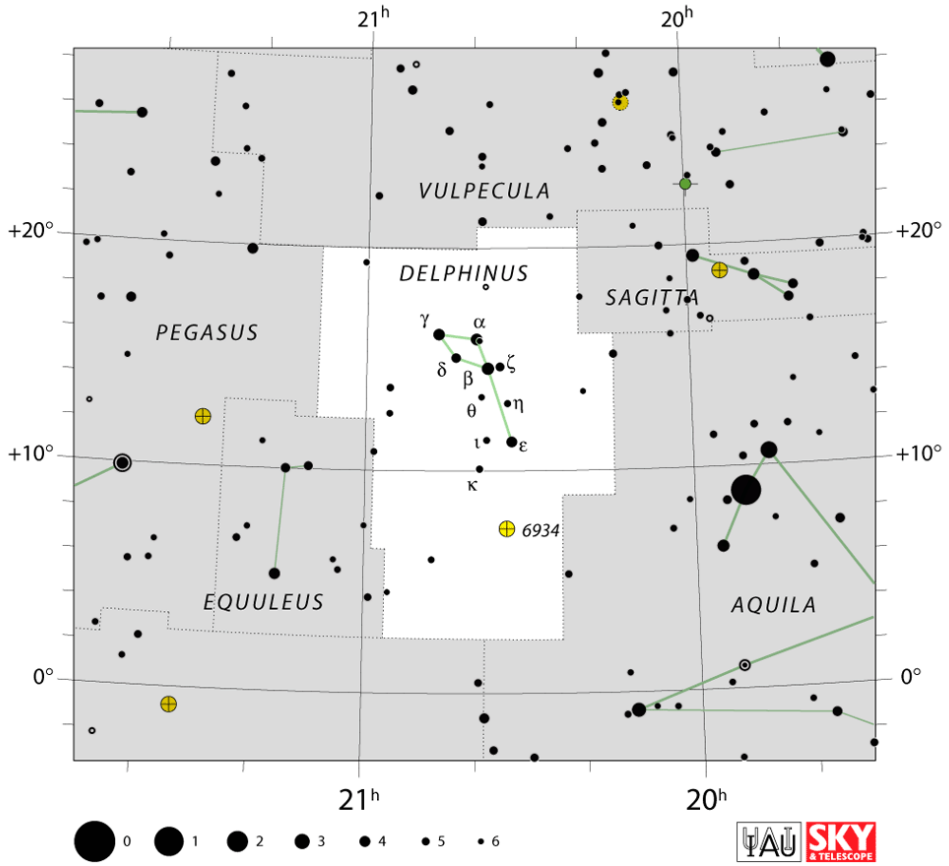
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 32/88  Delphinus (Del) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 101421 | 308.303167 | +11.303333 | HIP 101769 |
| HIP 101769 | 309.386958 | +14.595194 | HIP 101421, HIP 101958, HIP 102281 |
| HIP 101958 | 309.909375 | +15.912056 | HIP 101769, HIP 102532 |
| HIP 102532 | 311.664667 | +16.124778 | HIP 101958, HIP 102281 |
| HIP 102281 | 310.864792 | +15.074694 | HIP 102532, HIP 101769 |

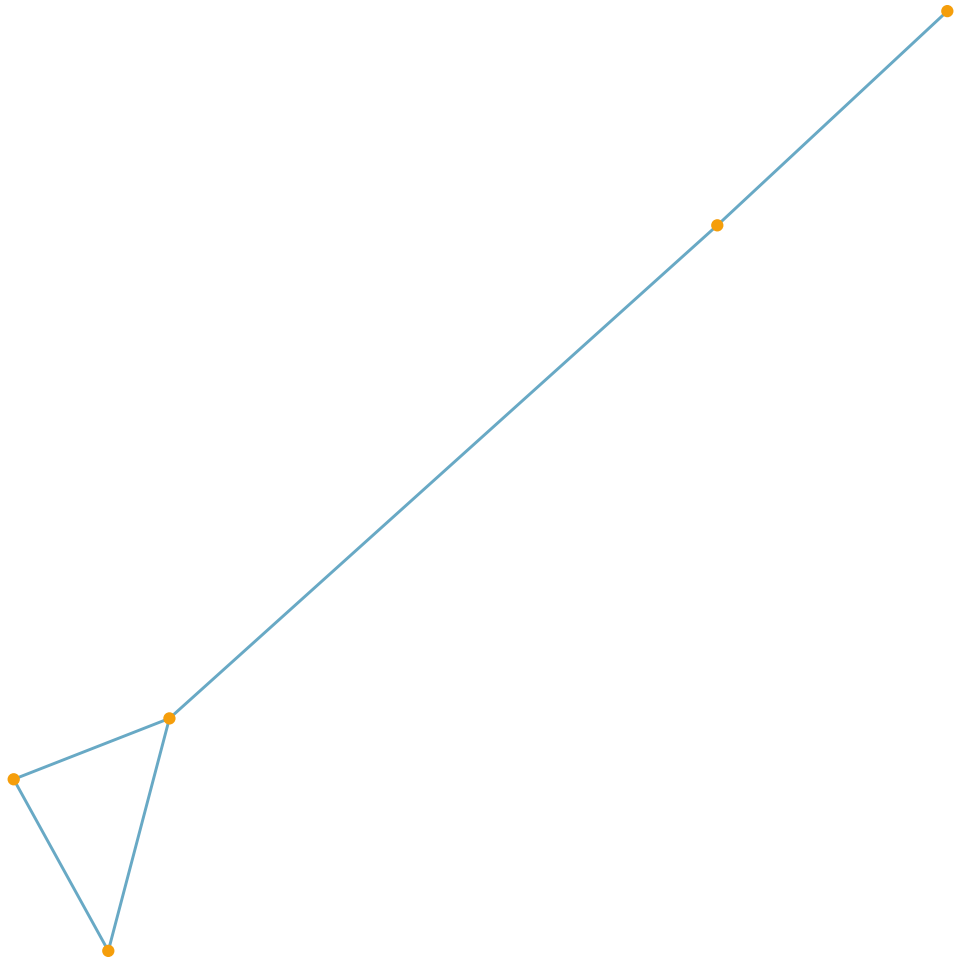
Effective release edges: 5 • unique plotted points: 5 • source: release_effective_edges.csv

# 33/88 Dorado (Dor) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

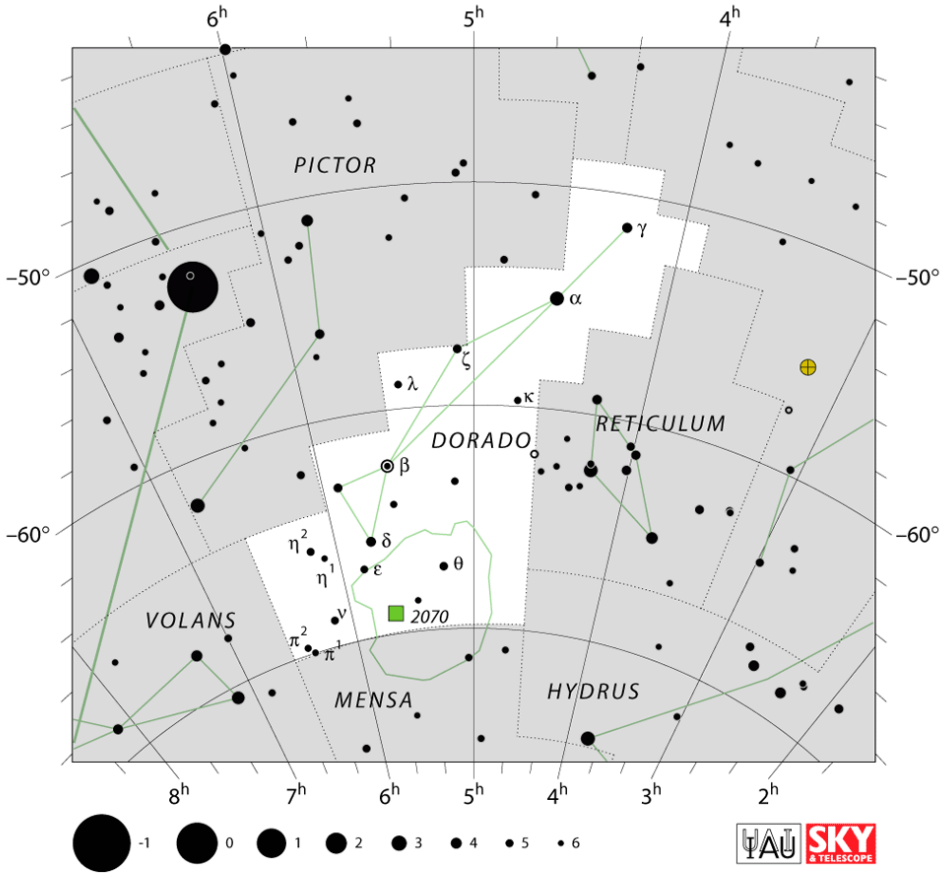
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 33/88  Dorado (Dor) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 27100 | 86.193417 | -65.735528 | HIP 27890, HIP 26069 |
| HIP 27890 | 88.524583 | -63.091028 | HIP 27100, HIP 26069 |
| HIP 26069 | 83.406333 | -62.489861 | HIP 27890, HIP 27100, HIP 21281 |
| HIP 21281 | 68.498833 | -55.045000 | HIP 26069, HIP 19893 |
| HIP 19893 | 64.006208 | -51.487083 | HIP 21281 |

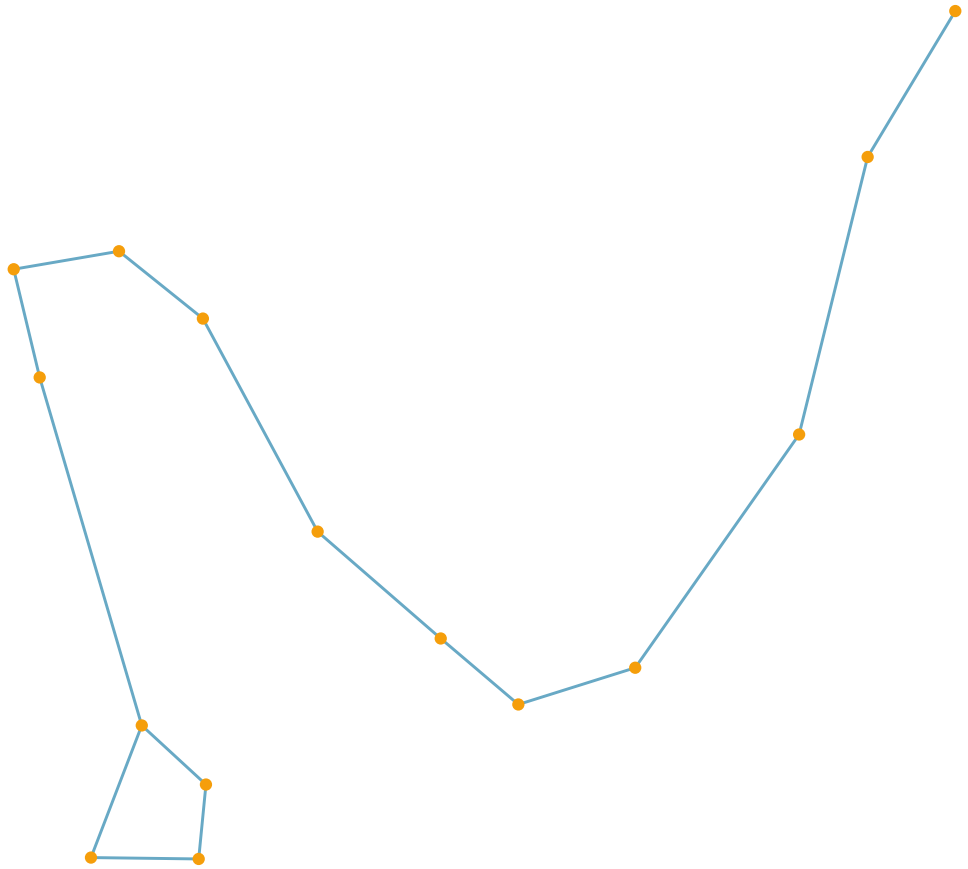
Effective release edges: 5 • unique plotted points: 5 • source: release_effective_edges.csv

# 34/88 Draco (Dra) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

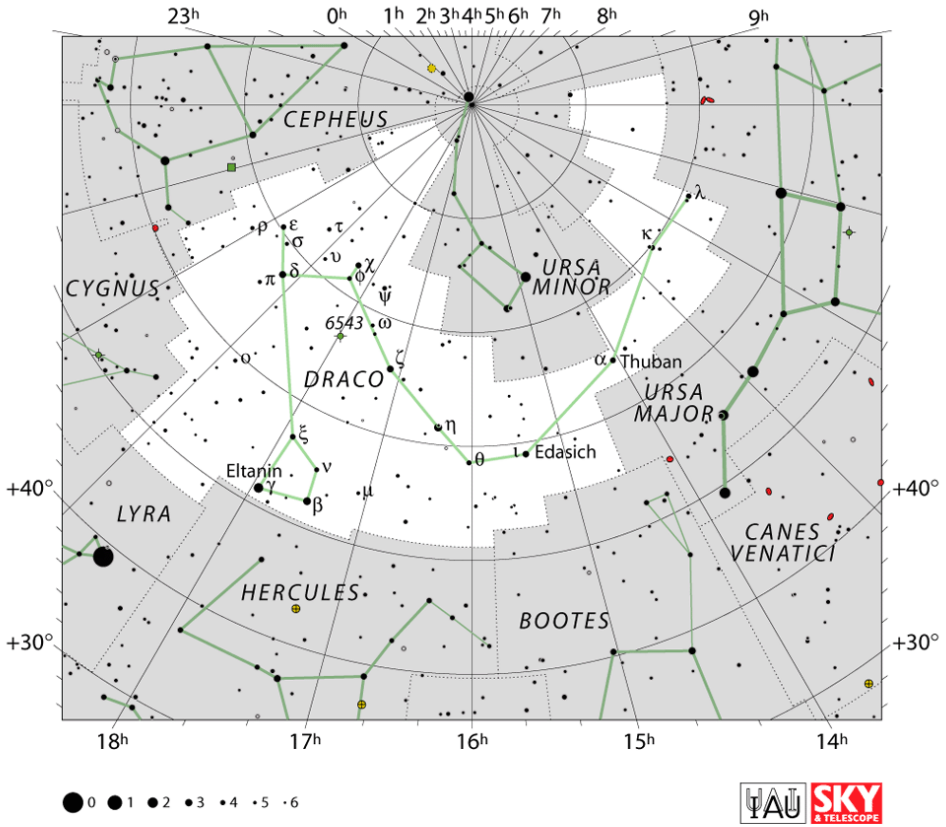
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 34/88  Draco (Dra) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 87585 | 268.381792 | +56.872444 | HIP 87833, HIP 85829, HIP 94376 |
| HIP 87833 | 269.151583 | +51.488944 | HIP 87585, HIP 85670 |
| HIP 85670 | 262.608250 | +52.301361 | HIP 87833, HIP 85829 |
| HIP 85829 | 263.066167 | +55.172806 | HIP 85670, HIP 87585 |
| HIP 94376 | 288.138125 | +67.661306 | HIP 87585, HIP 97433 |
| HIP 97433 | 297.042542 | +70.267833 | HIP 94376, HIP 94648 |
| HIP 94648 | 288.888708 | +73.355222 | HIP 97433, HIP 89937 |
| HIP 89937 | 275.259750 | +72.733694 | HIP 94648, HIP 83895 |
| HIP 83895 | 257.196792 | +65.714639 | HIP 89937, HIP 80331 |
| HIP 80331 | 245.997958 | +61.514083 | HIP 83895, HIP 78527 |
| HIP 78527 | 240.473750 | +58.564444 | HIP 80331, HIP 75458 |
| HIP 75458 | 231.232417 | +58.966028 | HIP 78527, HIP 68756 |
| HIP 68756 | 211.097625 | +64.375806 | HIP 75458, HIP 61281 |
| HIP 61281 | 188.371000 | +69.788222 | HIP 68756, HIP 56211 |
| HIP 56211 | 172.851208 | +69.331111 | HIP 61281 |

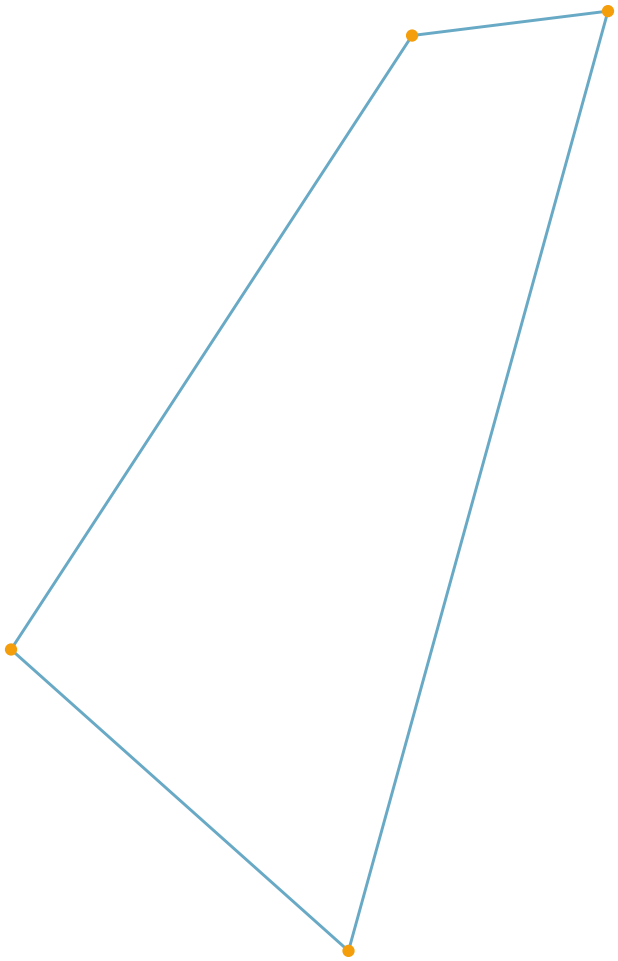
Effective release edges: 15 • unique plotted points: 15 • source: release_effective_edges.csv

# 35/88 Equuleus (Equ) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

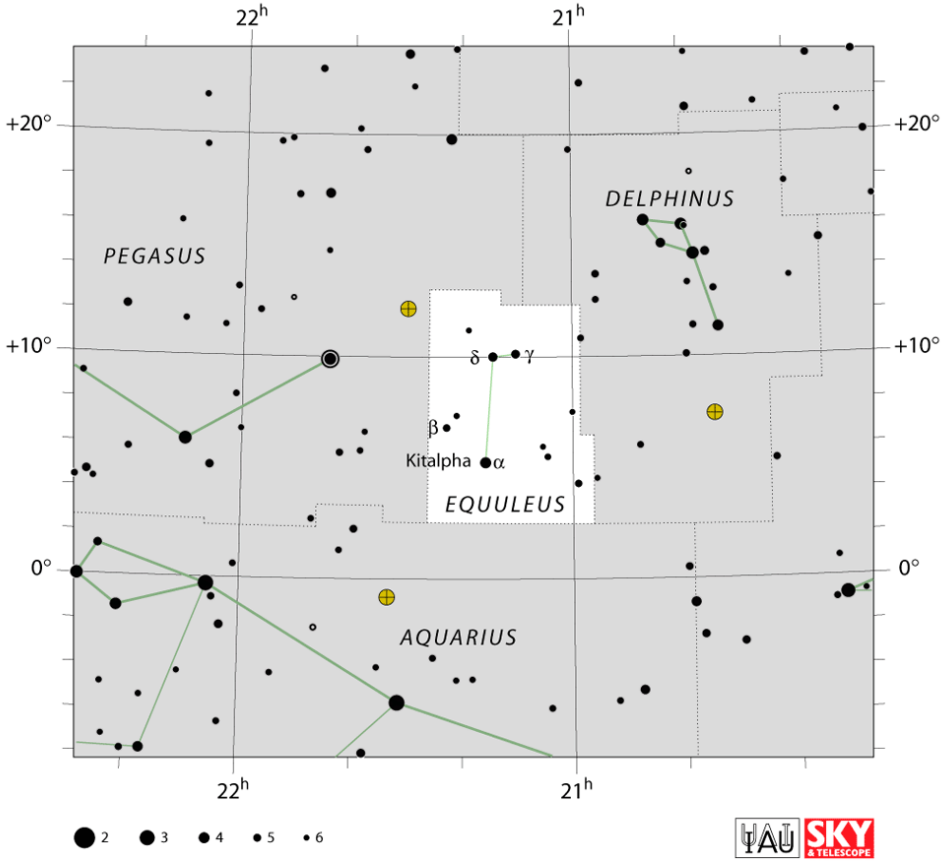
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 35/88  Equuleus (Equ) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 104521 | 317.585292 | +10.131944 | HIP 104858, HIP 104987 |
| HIP 104858 | 318.619958 | +10.007722 | HIP 104521, HIP 105570 |
| HIP 105570 | 320.723250 | +6.811111 | HIP 104858, HIP 104987 |
| HIP 104987 | 318.955833 | +5.248083 | HIP 105570, HIP 104521 |

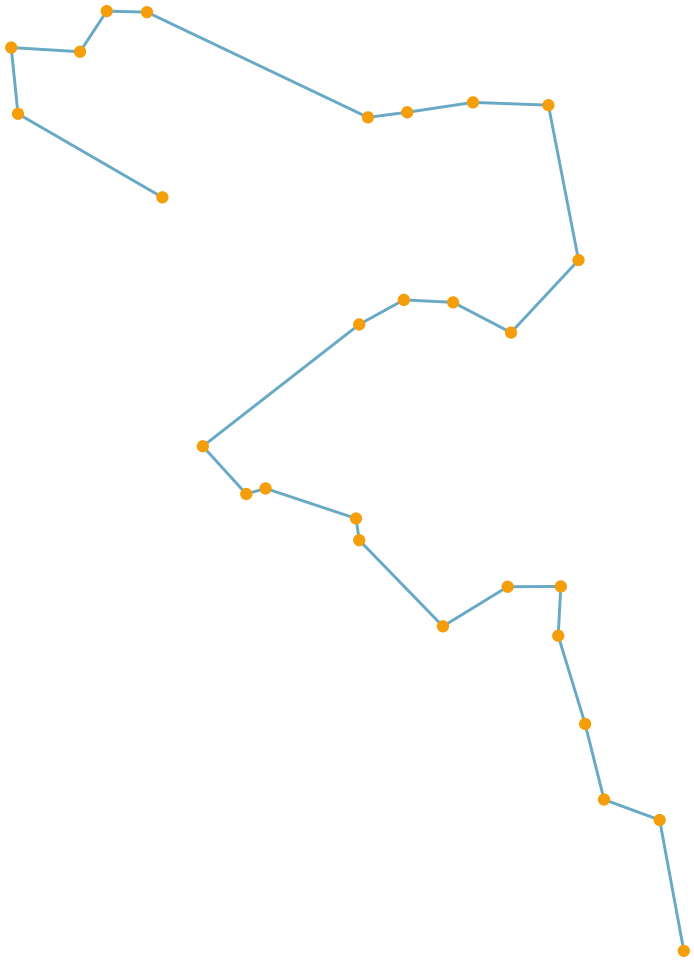
Effective release edges: 4 • unique plotted points: 4 • source: release_effective_edges.csv

# 36/88 Eridanus (Eri) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

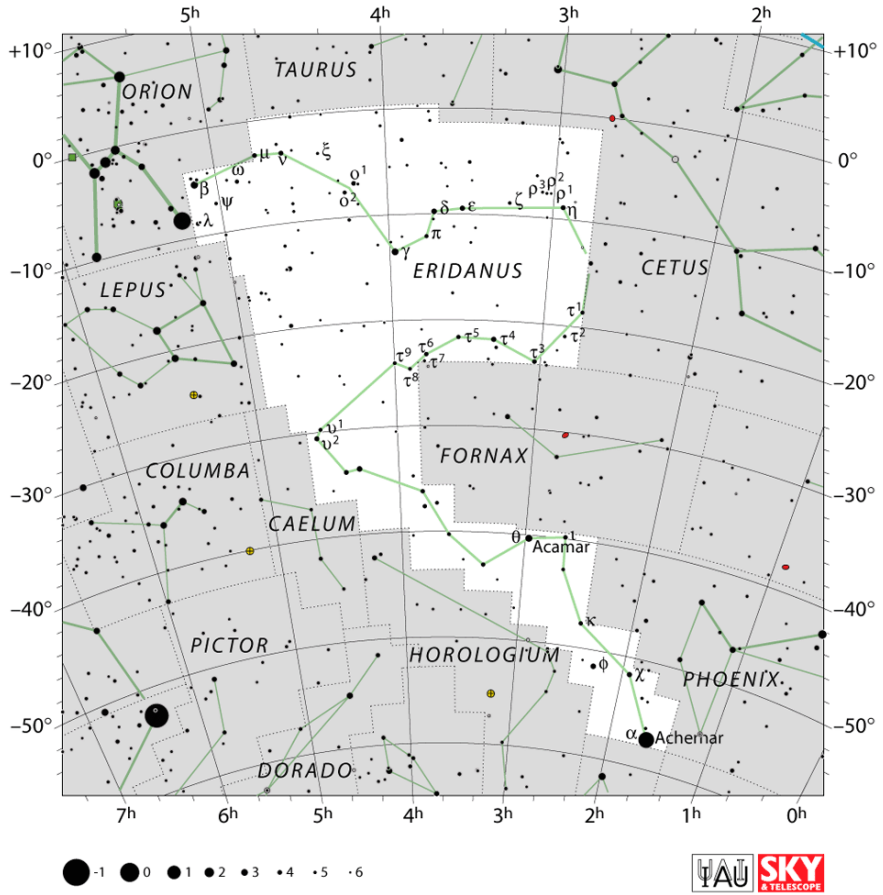
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 36/88  Eridanus (Eri) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 7588 | 24.428125 | -57.236667 | HIP 9007 |
| HIP 9007 | 28.986792 | -51.609583 | HIP 7588, HIP 10602 |
| HIP 10602 | 34.127083 | -51.512111 | HIP 9007, HIP 11407 |
| HIP 11407 | 36.746250 | -47.703833 | HIP 10602, HIP 12413 |
| HIP 12413 | 39.949667 | -42.891639 | HIP 11407, HIP 12486 |
| HIP 12486 | 40.166375 | -39.855306 | HIP 12413, HIP 13847 |
| HIP 13847 | 44.565500 | -40.304722 | HIP 12486, HIP 15510 |
| HIP 15510 | 49.971750 | -43.071556 | HIP 13847, HIP 17797 |
| HIP 17797 | 57.149250 | -37.620139 | HIP 15510, HIP 17874 |
| HIP 17874 | 57.363667 | -36.200111 | HIP 17797, HIP 20042 |
| HIP 20042 | 64.473417 | -33.798333 | HIP 17874, HIP 20535 |
| HIP 20535 | 66.009042 | -34.017000 | HIP 20042, HIP 21393 |
| HIP 21393 | 68.887792 | -30.562306 | HIP 20535, HIP 17651 |
| HIP 17651 | 56.712458 | -23.248444 | HIP 21393, HIP 16611 |
| HIP 16611 | 53.446875 | -21.632806 | HIP 17651, HIP 15474 |
| HIP 15474 | 49.879042 | -21.757944 | HIP 16611, HIP 14146 |
| HIP 14146 | 45.598292 | -23.624333 | HIP 15474, HIP 12843 |
| HIP 12843 | 41.274917 | -18.572639 | HIP 14146, HIP 13701 |
| HIP 13701 | 44.106667 | -8.897611 | HIP 12843, HIP 15197 |
| HIP 15197 | 48.958458 | -8.819833 | HIP 13701, HIP 16537 |
| HIP 16537 | 53.235083 | -9.458306 | HIP 15197, HIP 17378 |
| HIP 17378 | 55.812333 | -9.765194 | HIP 16537, HIP 21444 |
| HIP 21444 | 69.079750 | -3.352444 | HIP 17378, HIP 22109 |
| HIP 22109 | 71.375583 | -3.254611 | HIP 21444, HIP 22701 |
| HIP 22701 | 73.223667 | -5.452750 | HIP 22109, HIP 23875 |
| HIP 23875 | 76.962625 | -5.086250 | HIP 22701, HIP 23972 |
| HIP 23972 | 77.286583 | -8.754083 | HIP 23875, HIP 21594 |
| HIP 21594 | 69.545292 | -14.303583 | HIP 23972 |

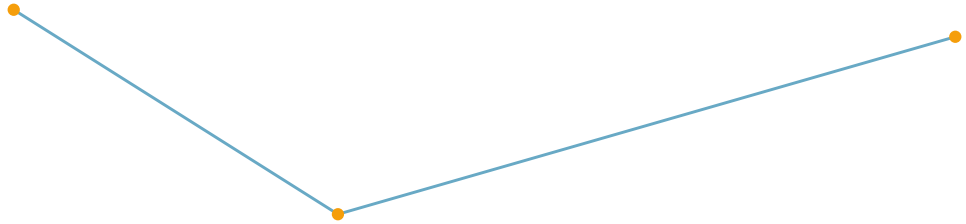
Effective release edges: 27 • unique plotted points: 28 • source: release_effective_edges.csv

# 37/88 Fornax (For) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

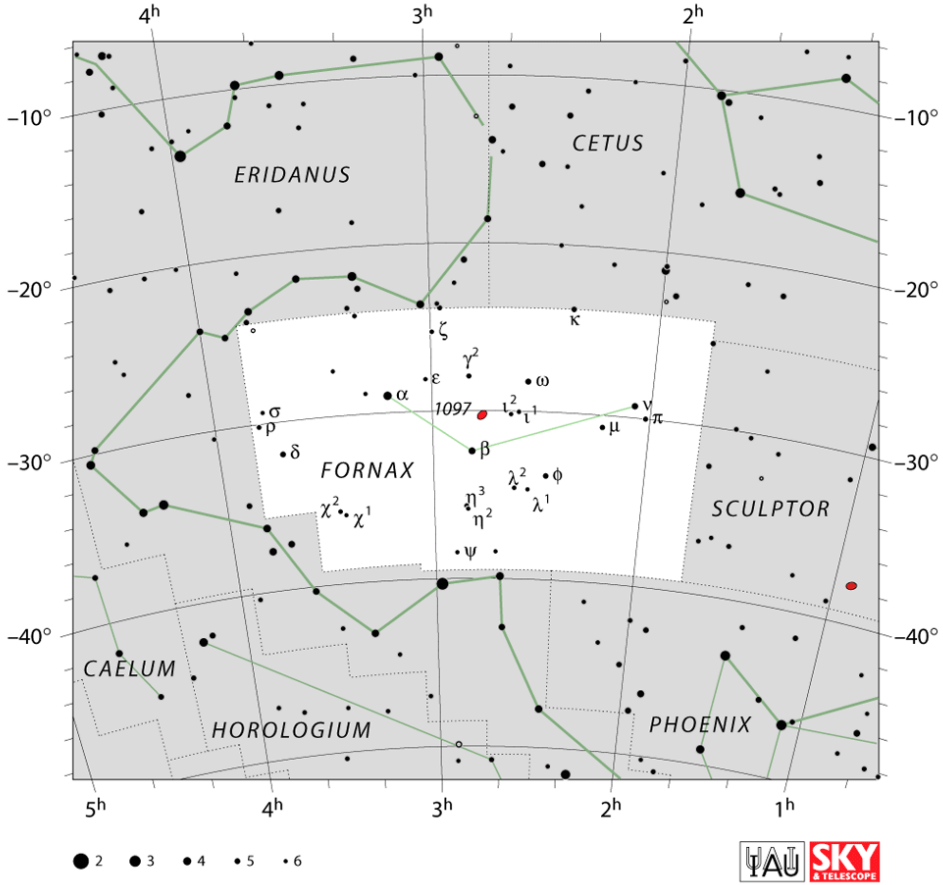
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 37/88  Fornax (For) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| Dalim (Alpha Fornacis) | 48.018875 | -28.987611 | Beta Fornacis |
| Beta Fornacis | 42.272583 | -32.405889 | Dalim (Alpha Fornacis), Nu Fornacis |
| Nu Fornacis | 31.122667 | -29.296806 | Beta Fornacis |

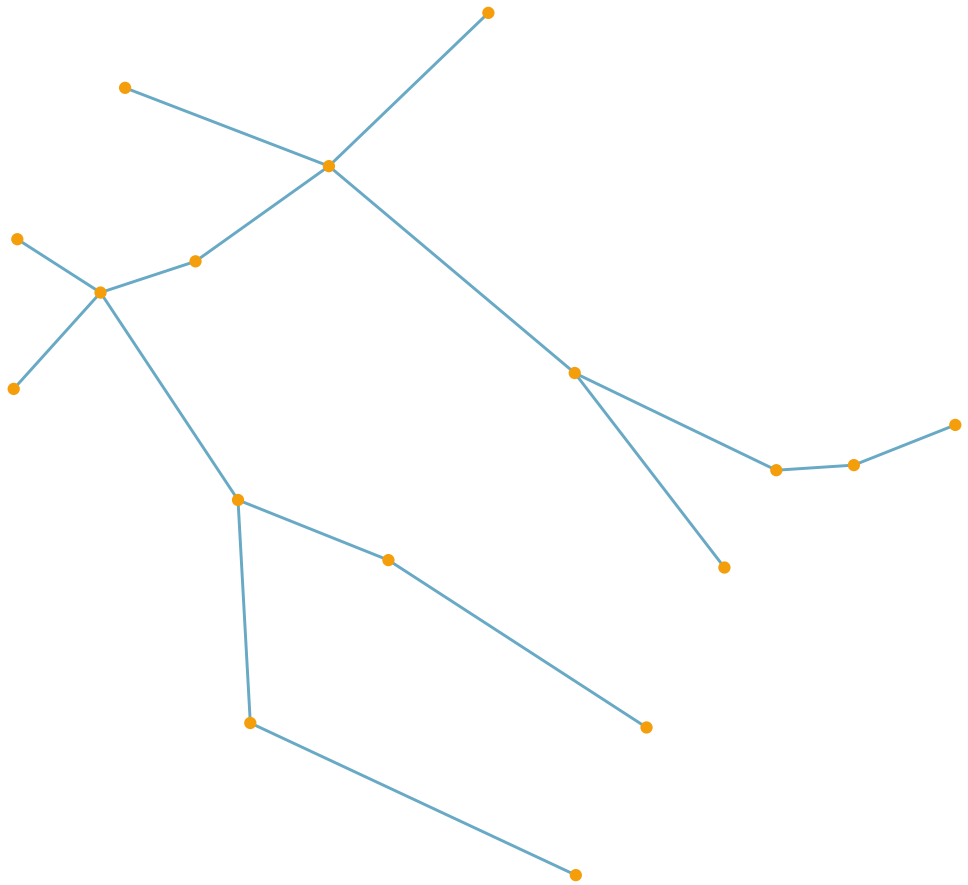
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

# 38/88 Gemini (Gem) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

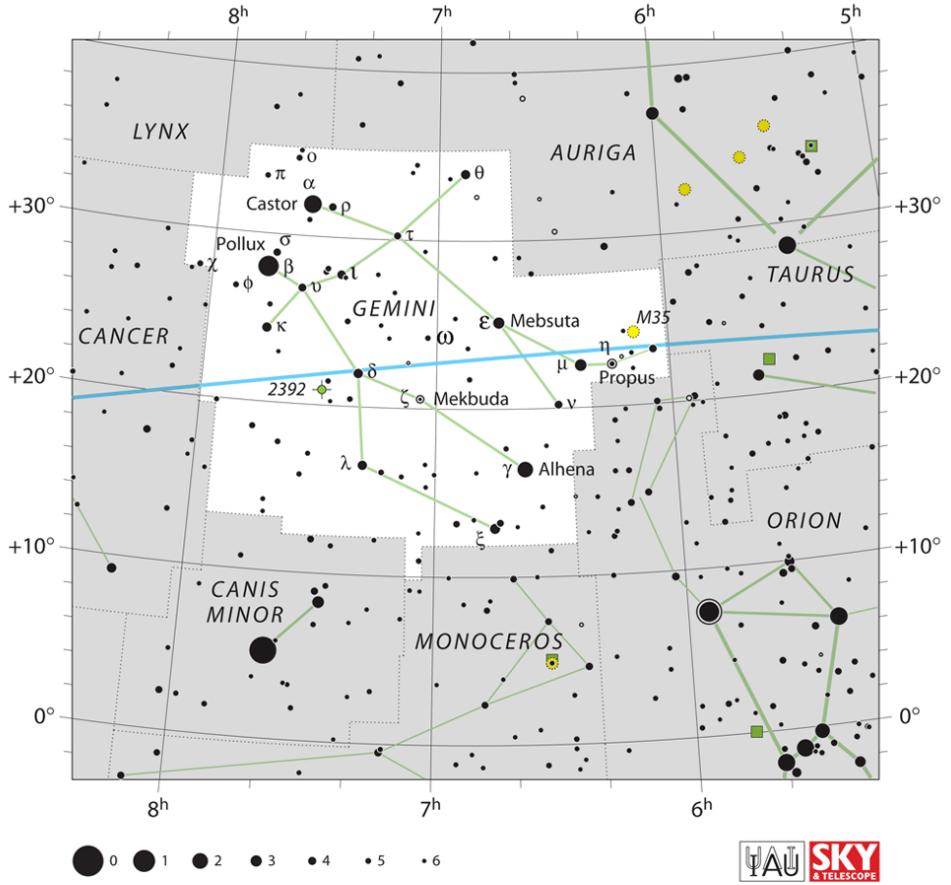
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 38/88  Gemini (Gem) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 31681 | 99.427917 | +16.399417 | HIP 34088 |
| HIP 34088 | 106.027250 | +20.570306 | HIP 31681, HIP 35550 |
| HIP 35550 | 110.030792 | +21.982333 | HIP 34088, HIP 35350, HIP 36962 |
| HIP 35350 | 109.523375 | +16.540472 | HIP 35550, HIP 32362 |
| HIP 32362 | 101.322625 | +12.896056 | HIP 35350 |
| HIP 36962 | 113.980708 | +26.896000 | HIP 35550, HIP 37740, HIP 37826, HIP 36046 |
| HIP 37740 | 116.111958 | +24.398139 | HIP 36962 |
| HIP 37826 | 116.330667 | +28.026306 | HIP 36962 |
| HIP 36046 | 111.432000 | +27.798278 | HIP 36962, HIP 34693 |
| HIP 34693 | 107.784958 | +30.245278 | HIP 36046, HIP 36850, HIP 33018, HIP 32246 |
| HIP 36850 | 113.650000 | +31.888639 | HIP 34693 |
| HIP 33018 | 103.197250 | +33.961361 | HIP 34693 |
| HIP 32246 | 100.983042 | +25.131167 | HIP 34693, HIP 30883, HIP 30343 |
| HIP 30883 | 97.240792 | +20.212167 | HIP 32246 |
| HIP 30343 | 95.739958 | +22.513861 | HIP 32246, HIP 29655 |
| HIP 29655 | 93.719583 | +22.506833 | HIP 30343, HIP 28734 |
| HIP 28734 | 91.030083 | +23.263639 | HIP 29655 |

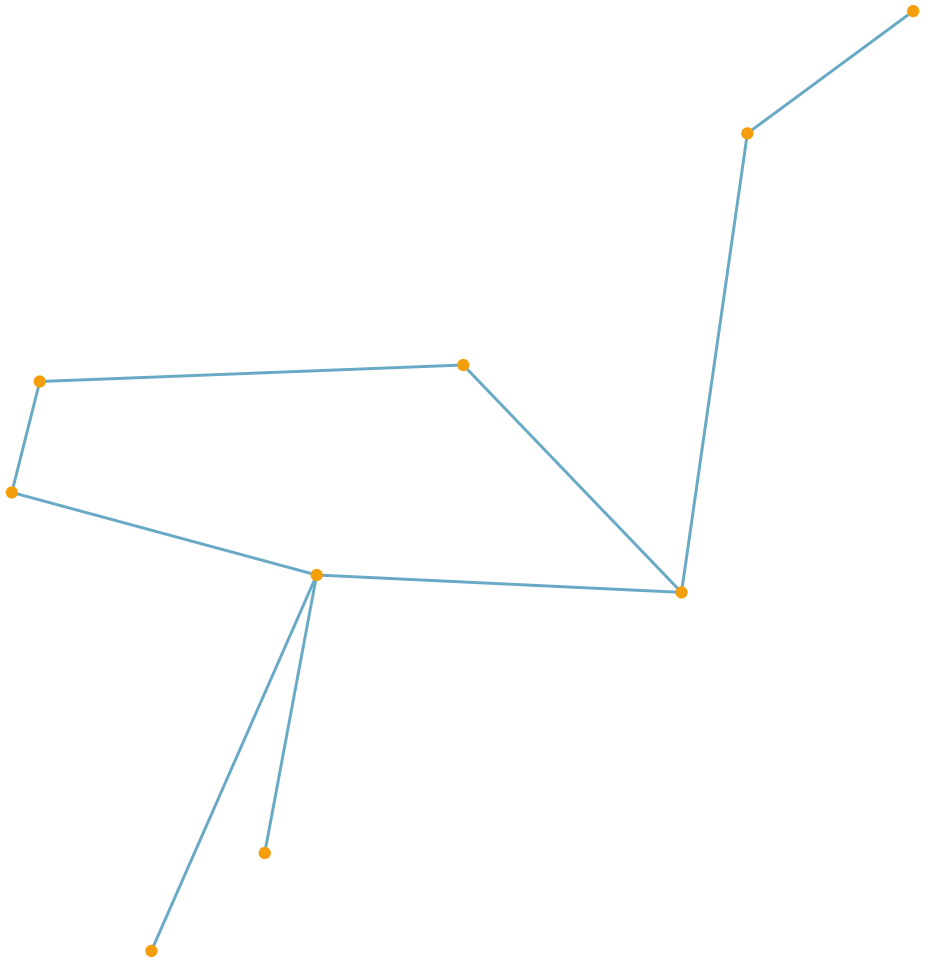
Effective release edges: 16 • unique plotted points: 17 • source: release_effective_edges.csv

# 39/88 Grus (Gru) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

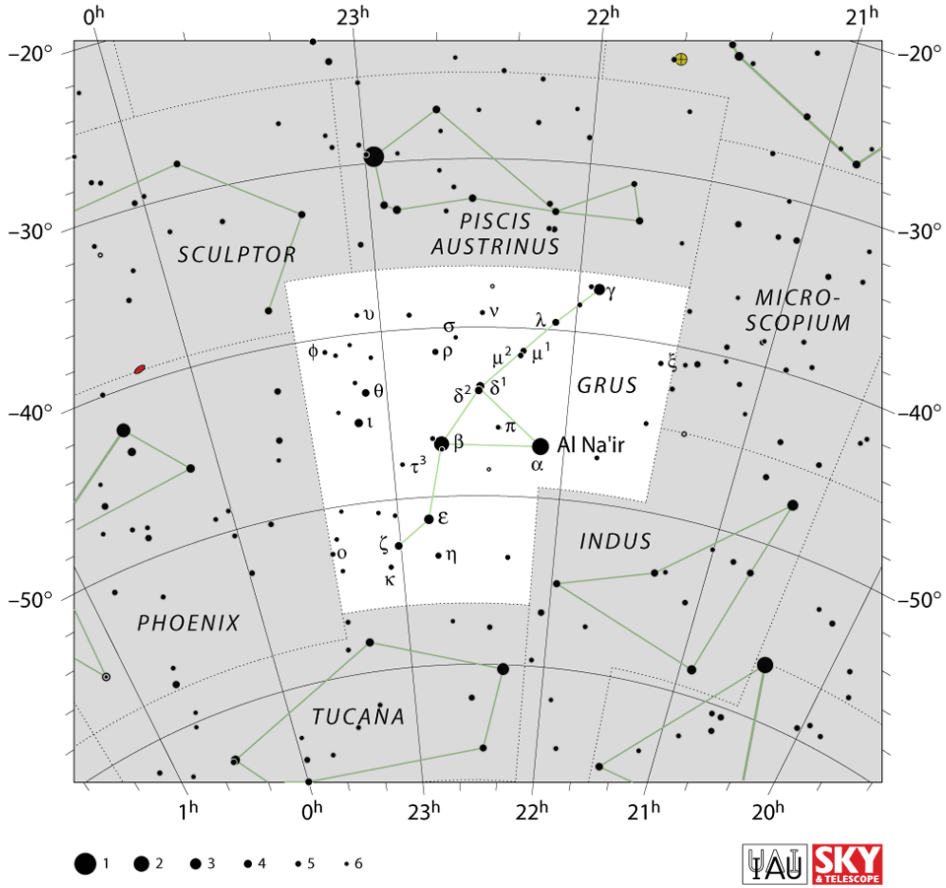
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 39/88  Grus (Gru) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 114131 | 346.719875 | -43.520333 | HIP 110997, HIP 114421 |
| HIP 110997 | 337.317292 | -43.495556 | HIP 114131, HIP 109268 |
| HIP 109268 | 332.057833 | -46.960611 | HIP 110997, HIP 112122, HIP 109111 |
| HIP 112122 | 340.666375 | -46.884556 | HIP 109268, HIP 114421, HIP 113638, HIP 112623 |
| HIP 114421 | 347.589292 | -45.246639 | HIP 112122, HIP 114131 |
| HIP 113638 | 345.220292 | -52.754111 | HIP 112122 |
| HIP 112623 | 342.138333 | -51.316694 | HIP 112122 |
| HIP 109111 | 331.528750 | -39.543056 | HIP 109268, HIP 108085 |
| HIP 108085 | 328.481875 | -37.364833 | HIP 109111 |

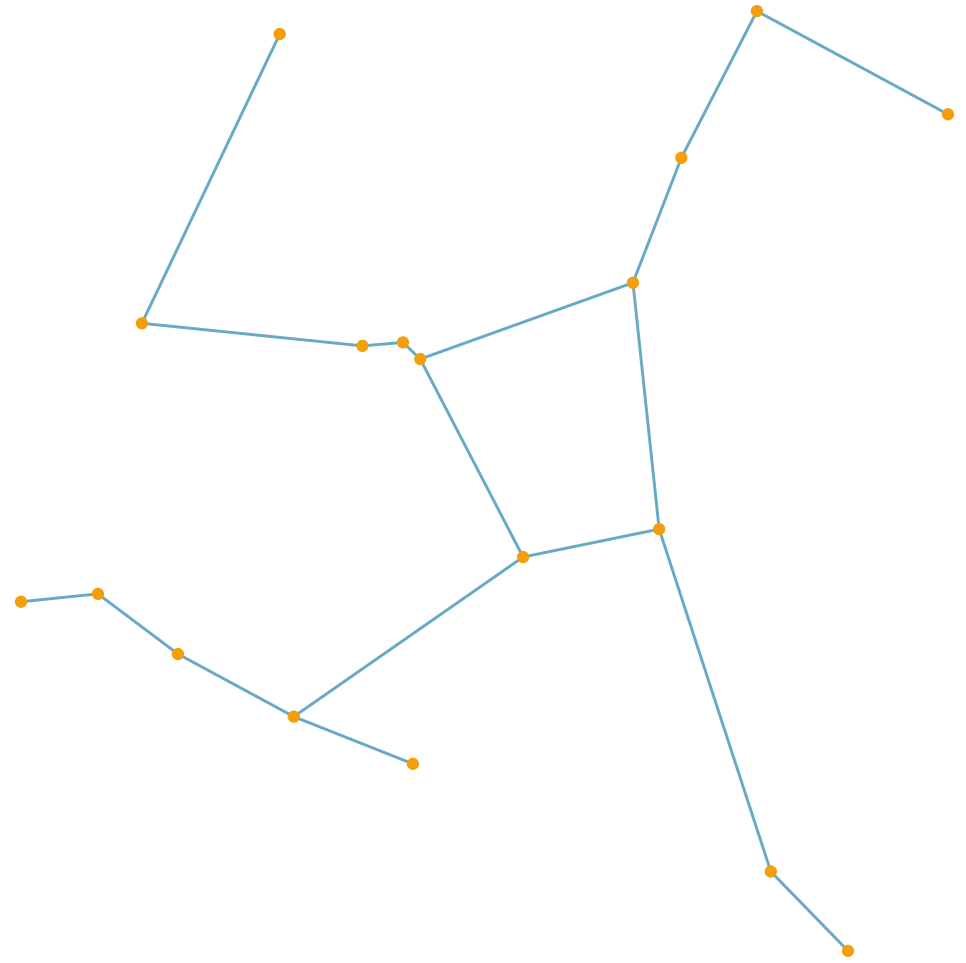
Effective release edges: 9 • unique plotted points: 9 • source: release_effective_edges.csv

# 40/88 Hercules (Her) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

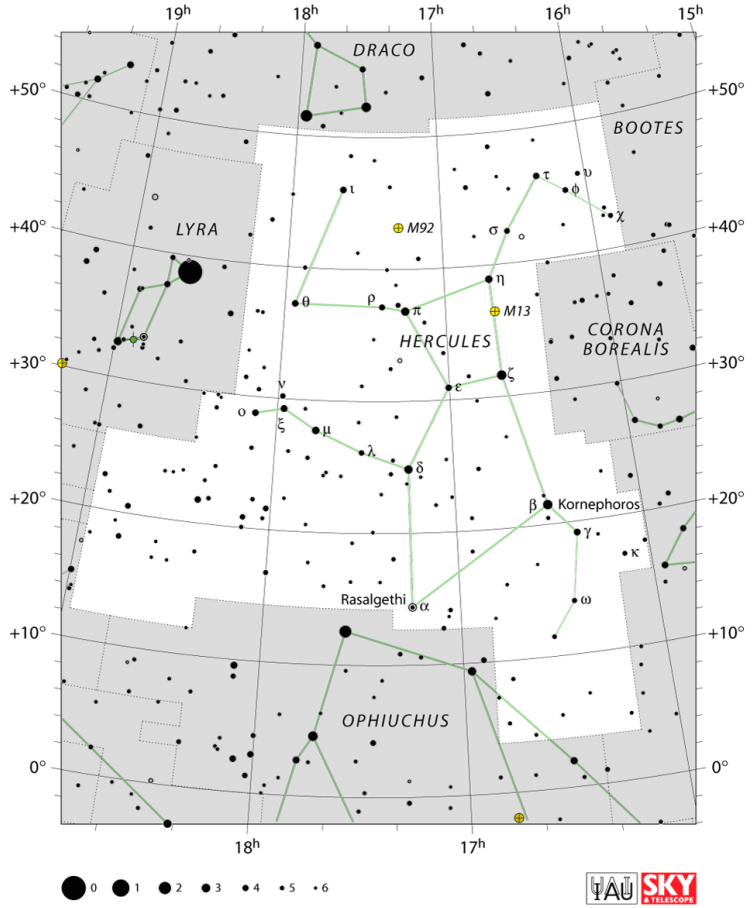
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 40/88  Hercules (Her) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 86414 | 264.866208 | +46.006333 | HIP 87808 |
| HIP 87808 | 269.063250 | +37.250528 | HIP 86414, HIP 85112 |
| HIP 85112 | 260.920708 | +37.145917 | HIP 87808, HIP 84606 |
| HIP 84606 | 259.417875 | +37.291333 | HIP 85112, HIP 84380 |
| HIP 84380 | 258.761875 | +36.809167 | HIP 84606, HIP 81833, HIP 83207 |
| HIP 81833 | 250.723917 | +38.922472 | HIP 84380, HIP 81126, HIP 81693 |
| HIP 81126 | 248.525792 | +42.436889 | HIP 81833, HIP 79992 |
| HIP 79992 | 244.935208 | +46.313278 | HIP 81126, HIP 77760 |
| HIP 77760 | 238.167458 | +42.450000 | HIP 79992 |
| HIP 81693 | 250.322833 | +31.601889 | HIP 81833, HIP 80816, HIP 83207 |
| HIP 80816 | 247.555250 | +21.489639 | HIP 81693, HIP 80170 |
| HIP 80170 | 245.480167 | +19.153028 | HIP 80816 |
| HIP 83207 | 255.072542 | +30.926333 | HIP 81693, HIP 85693, HIP 84380 |
| HIP 85693 | 262.684583 | +26.110611 | HIP 83207, HIP 84379, HIP 86974 |
| HIP 84379 | 258.758000 | +24.839583 | HIP 85693 |
| HIP 86974 | 266.615500 | +27.722500 | HIP 87933, HIP 85693 |
| HIP 87933 | 269.440958 | +29.247917 | HIP 86974, HIP 88794 |
| HIP 88794 | 271.885625 | +28.762472 | HIP 87933 |

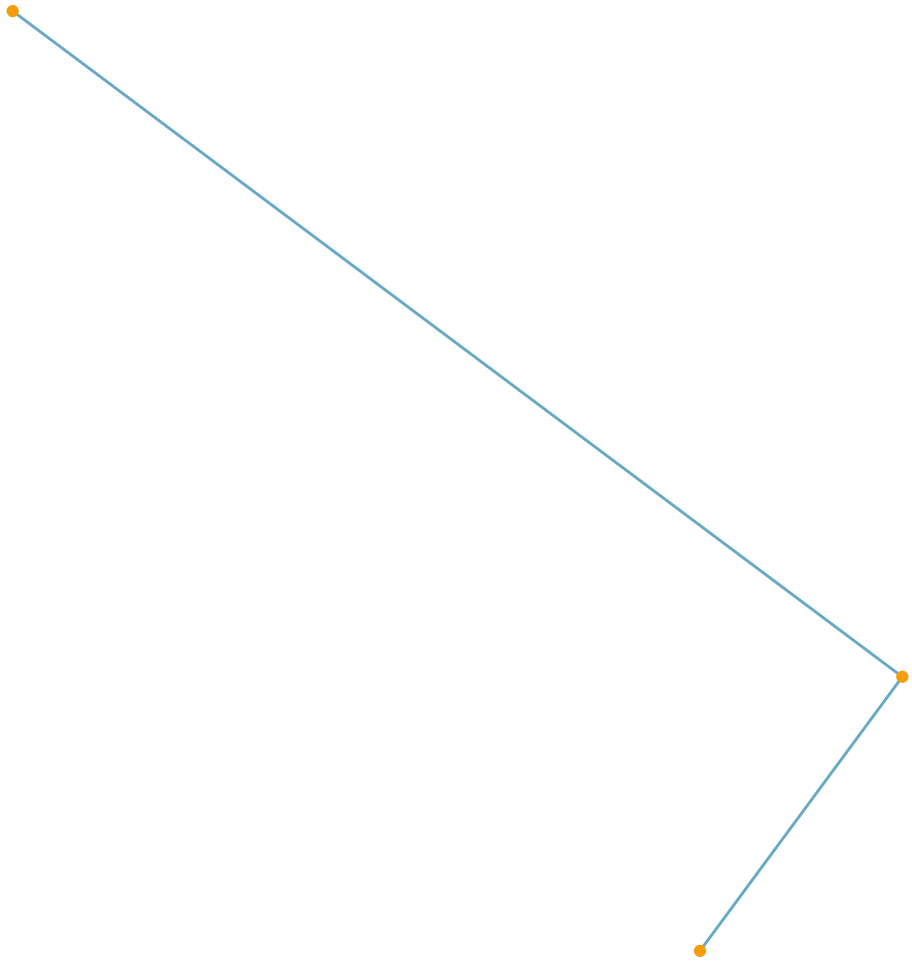
Effective release edges: 18 • unique plotted points: 18 • source: release_effective_edges.csv

# 41/88 Horologium (Hor) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

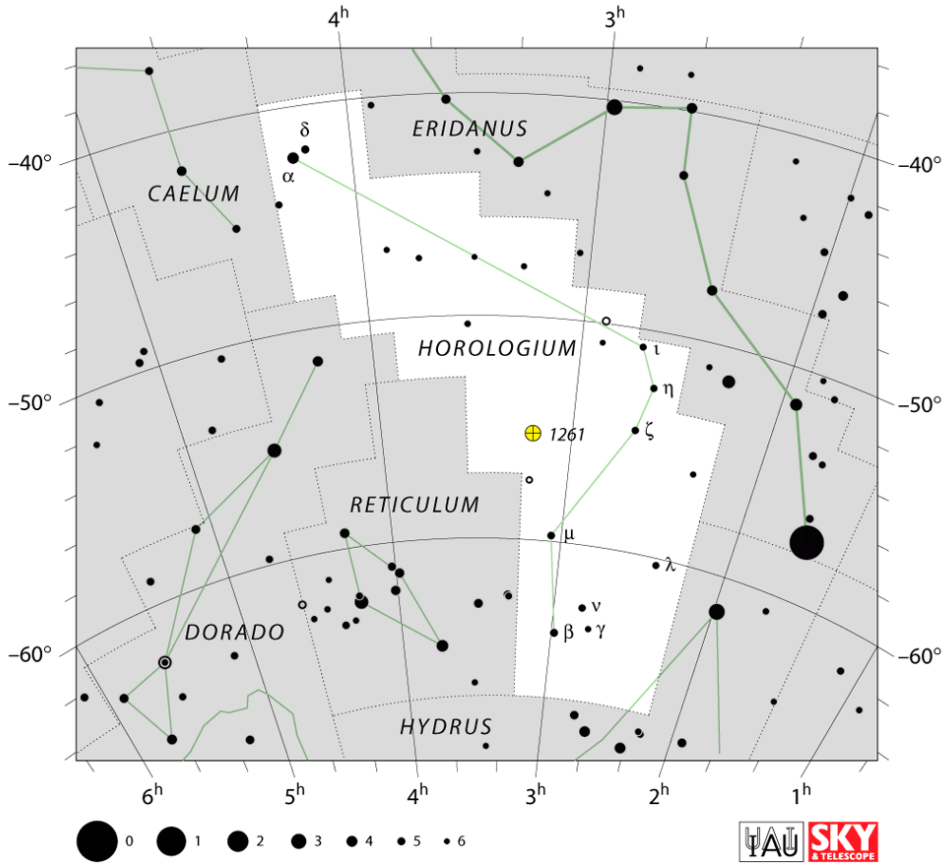
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 41/88  Horologium (Hor) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 19747 | 63.500333 | -42.293861 | HIP 12484 |
| HIP 12484 | 40.164917 | -54.549917 | HIP 19747, HIP 14240 |
| HIP 14240 | 45.903750 | -59.737611 | HIP 12484 |

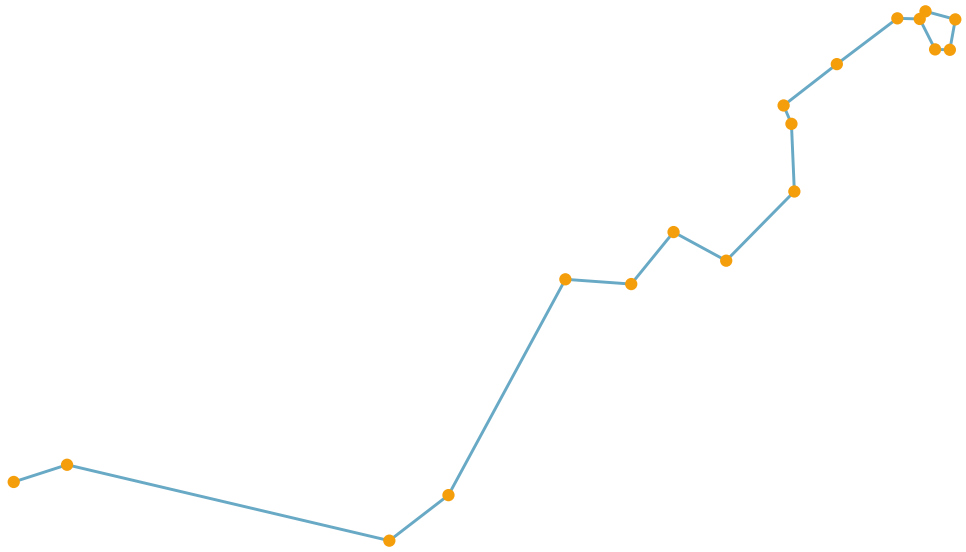
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

# 42/88 Hydra (Hya) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

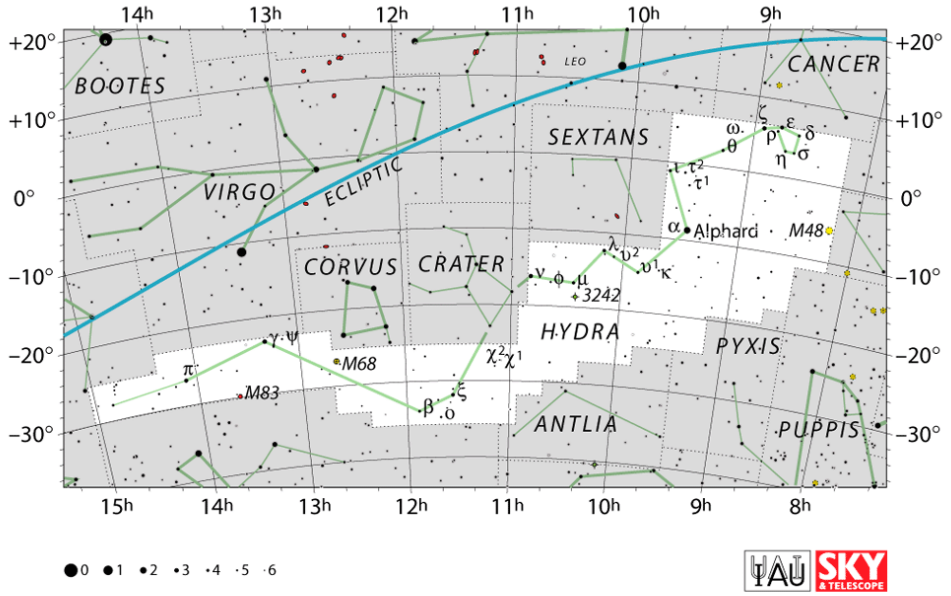
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 42/88  Hydra (Hya) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 42799 | 130.806208 | +3.398667 | HIP 42402, HIP 43234 |
| HIP 42402 | 129.689375 | +3.341472 | HIP 42799, HIP 42313 |
| HIP 42313 | 129.414208 | +5.703806 | HIP 42402, HIP 43109 |
| HIP 43109 | 131.694375 | +6.418917 | HIP 42313, HIP 43234 |
| HIP 43234 | 132.108250 | +5.837889 | HIP 43109, HIP 42799, HIP 43813 |
| HIP 43813 | 133.848667 | +5.945528 | HIP 43234, HIP 45336 |
| HIP 45336 | 138.590792 | +2.315028 | HIP 43813, HIP 46776 |
| HIP 46776 | 142.995542 | -1.184667 | HIP 45336, HIP 46509 |
| HIP 46509 | 142.286833 | -2.768944 | HIP 46776, HIP 46390 |
| HIP 46390 | 141.896875 | -8.658694 | HIP 46509, HIP 48356 |
| HIP 48356 | 147.869500 | -14.846556 | HIP 46390, HIP 49841 |
| HIP 49841 | 152.647500 | -12.353833 | HIP 48356, HIP 51069 |
| HIP 51069 | 156.522958 | -16.836083 | HIP 49841, HIP 52943 |
| HIP 52943 | 162.405958 | -16.194139 | HIP 51069, HIP 56343 |
| HIP 56343 | 173.251083 | -31.857528 | HIP 52943, HIP 57936 |
| HIP 57936 | 178.227333 | -33.908139 | HIP 56343, HIP 64166 |
| HIP 64166 | 197.263667 | -23.117972 | HIP 57936, HIP 64962 |
| HIP 64962 | 199.730208 | -23.171417 | HIP 64166 |

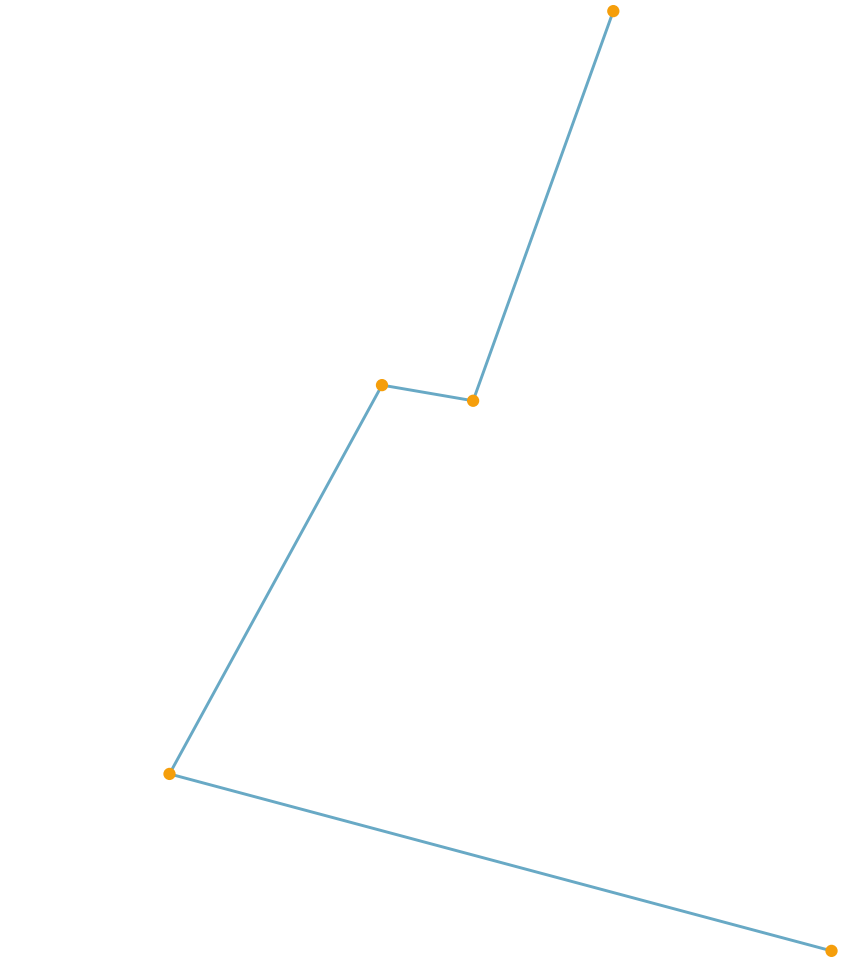
Effective release edges: 18 • unique plotted points: 18 • source: release_effective_edges.csv

# 43/88 Hydrus (Hyi) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

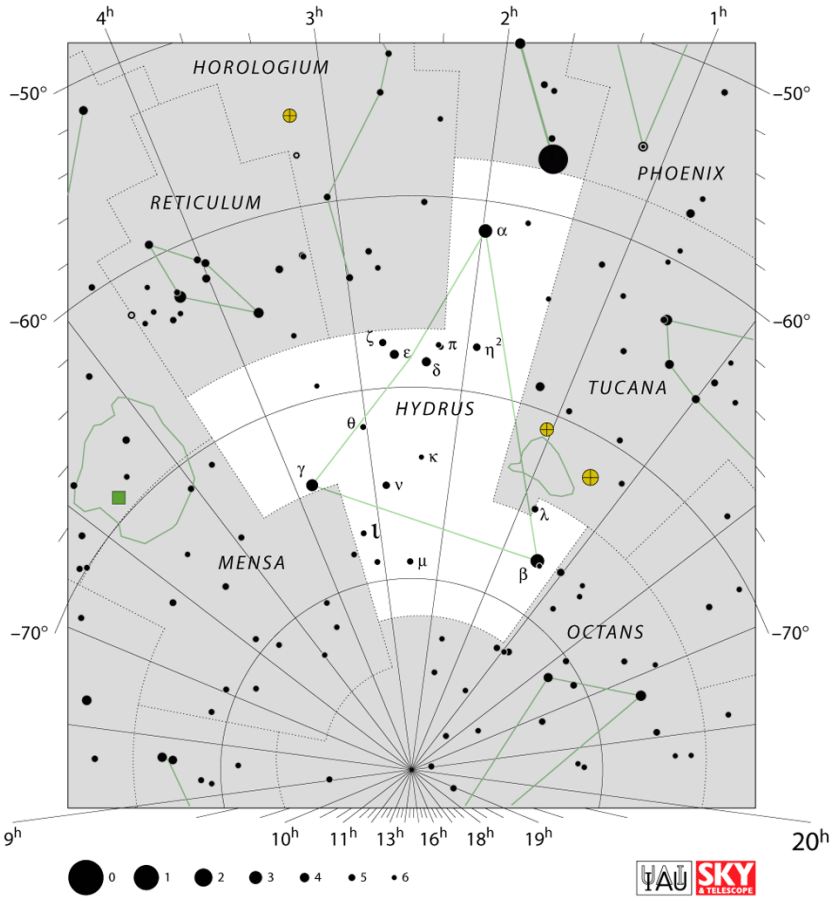
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 43/88  Hydrus (Hyi) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 2021 | 6.413333 | -77.255028 | HIP 17678 |
| HIP 17678 | 56.809292 | -74.239250 | HIP 2021, HIP 12394 |
| HIP 12394 | 39.896750 | -68.266944 | HIP 17678, HIP 11001 |
| HIP 11001 | 35.437583 | -68.659417 | HIP 12394, HIP 9236 |
| HIP 9236 | 29.691125 | -61.569917 | HIP 11001 |

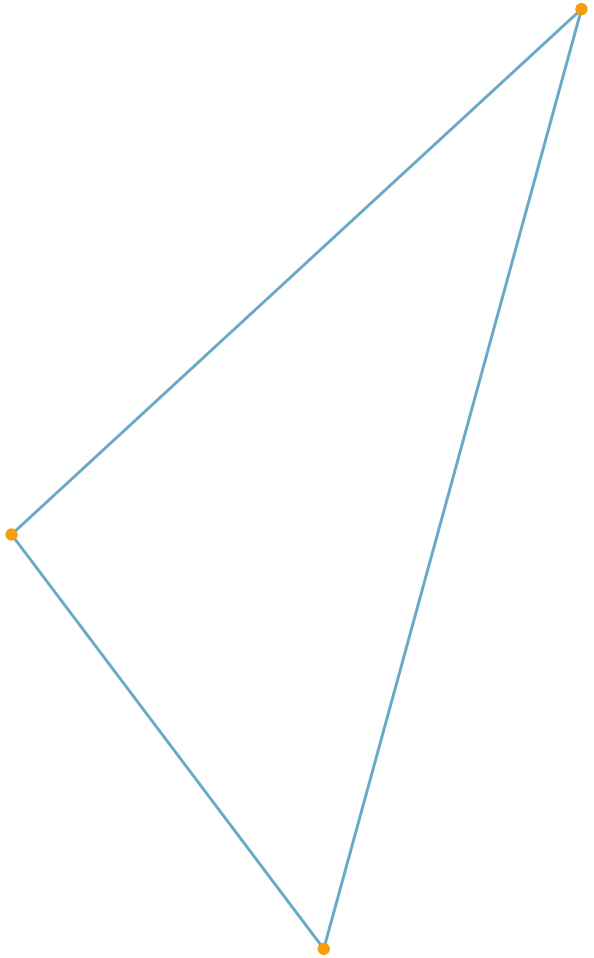
Effective release edges: 4 • unique plotted points: 5 • source: release_effective_edges.csv

# 44/88 Indus (Ind) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

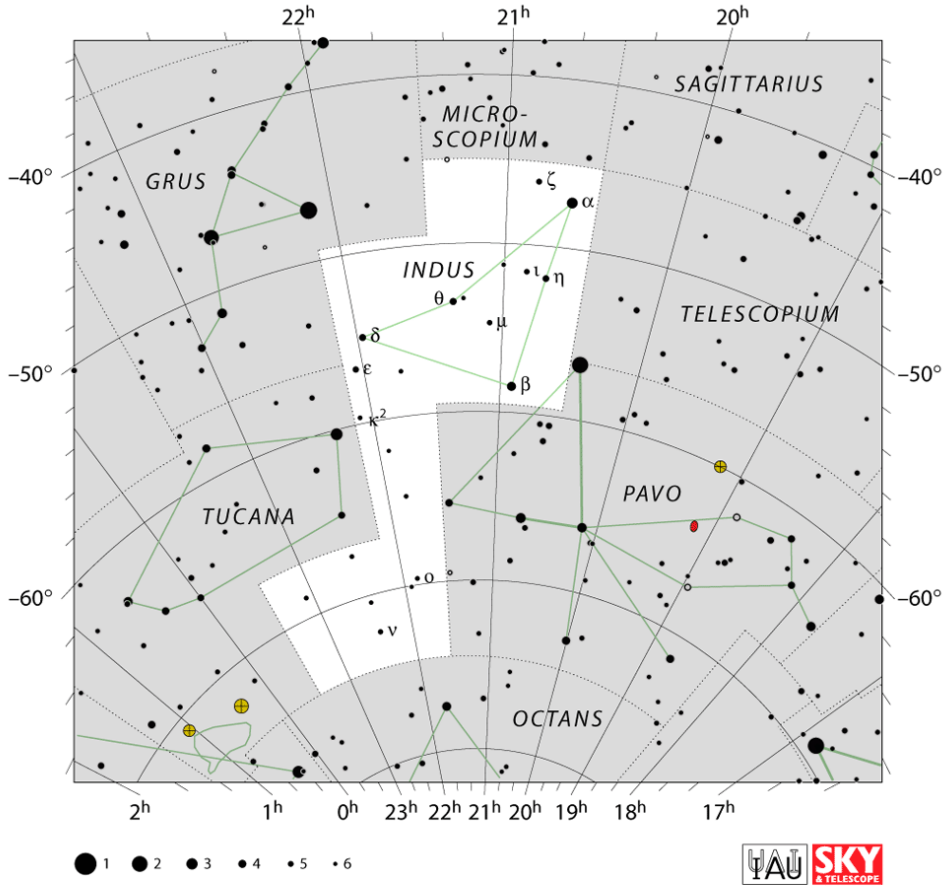
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 44/88  Indus (Ind) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 105319 | 319.966167 | -53.449278 | HIP 101772, HIP 103227 |
| HIP 101772 | 309.391625 | -47.291667 | HIP 105319, HIP 103227 |
| HIP 103227 | 313.702417 | -58.454083 | HIP 101772, HIP 105319 |

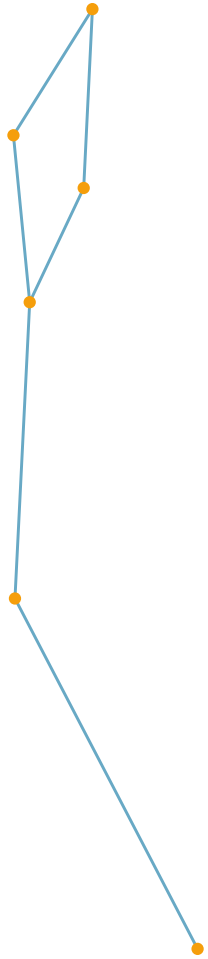
Effective release edges: 3 • unique plotted points: 3 • source: release_effective_edges.csv

# 45/88 Lacerta (Lac) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

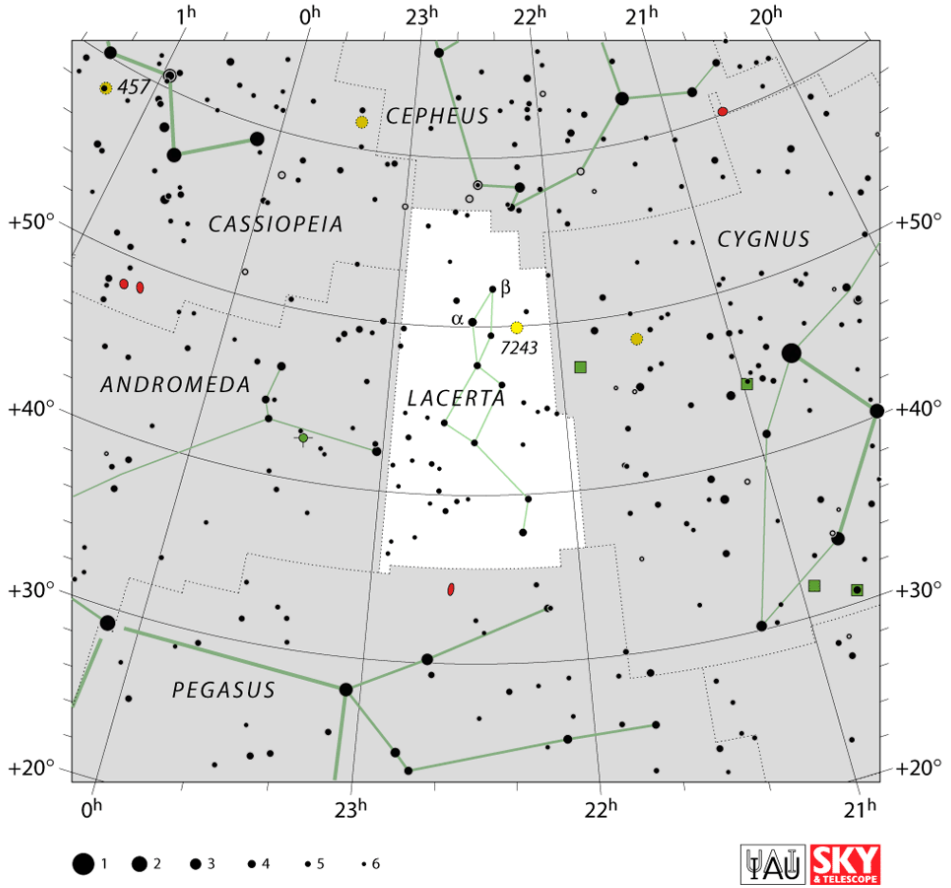
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 45/88  Lacerta (Lac) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 109937 | 333.992375 | +37.748722 | HIP 111104 |
| HIP 111104 | 337.621917 | +43.123389 | HIP 109937, HIP 111022 |
| HIP 111022 | 337.382583 | +47.706889 | HIP 111104, HIP 110609, HIP 111169 |
| HIP 110609 | 336.129167 | +49.476389 | HIP 111022, HIP 110538 |
| HIP 110538 | 335.890167 | +52.229500 | HIP 110609, HIP 111169 |
| HIP 111169 | 337.822417 | +50.282444 | HIP 110538, HIP 111022 |

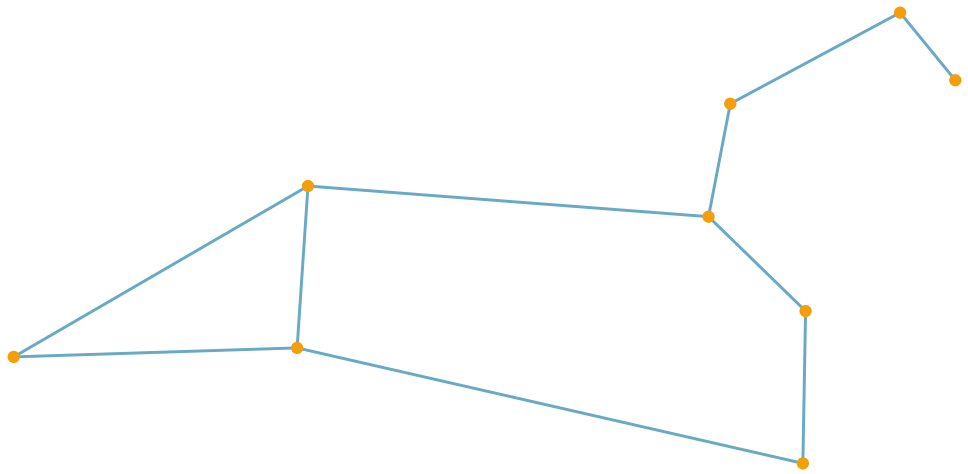
Effective release edges: 6 • unique plotted points: 6 • source: release_effective_edges.csv

# 46/88 Leo (Leo) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

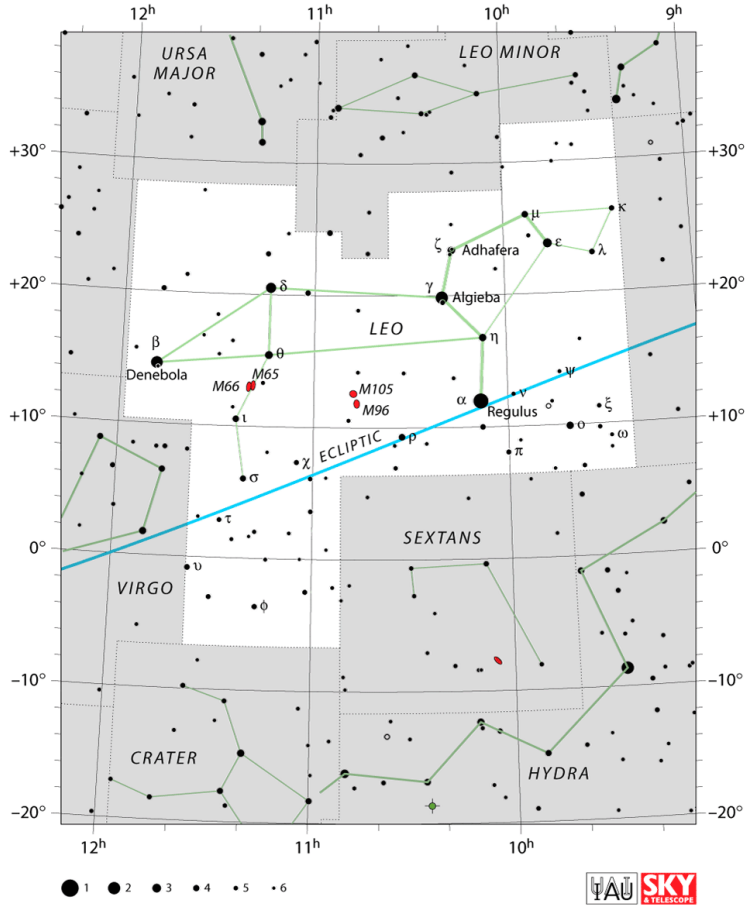
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 46/88  Leo (Leo) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 57632 | 177.266167 | +14.572333 | HIP 54879, HIP 54872 |
| HIP 54879 | 168.560167 | +15.429750 | HIP 57632, HIP 49669, HIP 54872 |
| HIP 49669 | 152.093583 | +11.967194 | HIP 54879, HIP 49583 |
| HIP 49583 | 151.833125 | +16.762667 | HIP 49669, HIP 50583 |
| HIP 50583 | 154.992333 | +19.841861 | HIP 49583, HIP 54872, HIP 50335 |
| HIP 54872 | 168.526708 | +20.524028 | HIP 50583, HIP 57632, HIP 54879 |
| HIP 50335 | 154.172500 | +23.417333 | HIP 50583, HIP 48455 |
| HIP 48455 | 148.191500 | +26.007083 | HIP 50335, HIP 47908 |
| HIP 47908 | 146.462917 | +23.774278 | HIP 48455 |

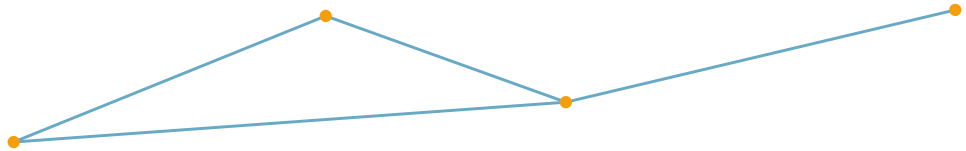
Effective release edges: 10 • unique plotted points: 9 • source: release_effective_edges.csv

# 47/88 Leo Minor (LMi) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

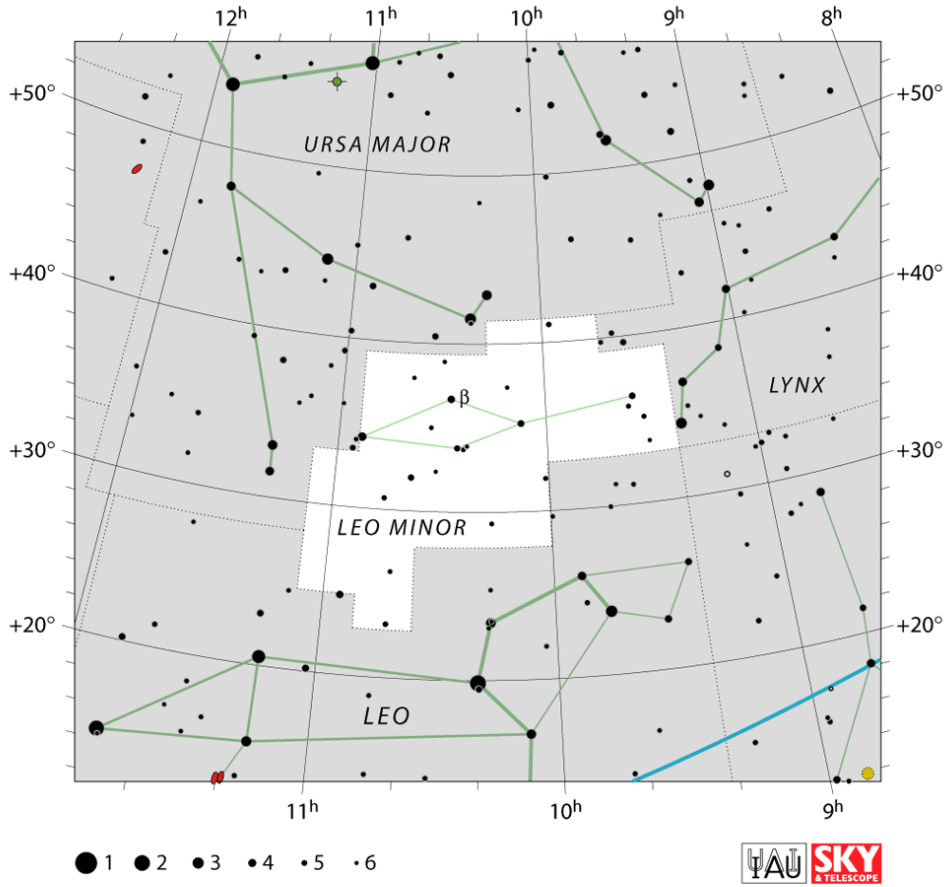
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 47/88  Leo Minor (LMi) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 53229 | 163.327667 | +34.215556 | HIP 51233, HIP 49593 |
| HIP 51233 | 156.971208 | +36.707472 | HIP 53229, HIP 49593 |
| HIP 49593 | 151.857208 | +35.244694 | HIP 51233, HIP 46952, HIP 53229 |
| HIP 46952 | 143.555750 | +36.397611 | HIP 49593 |

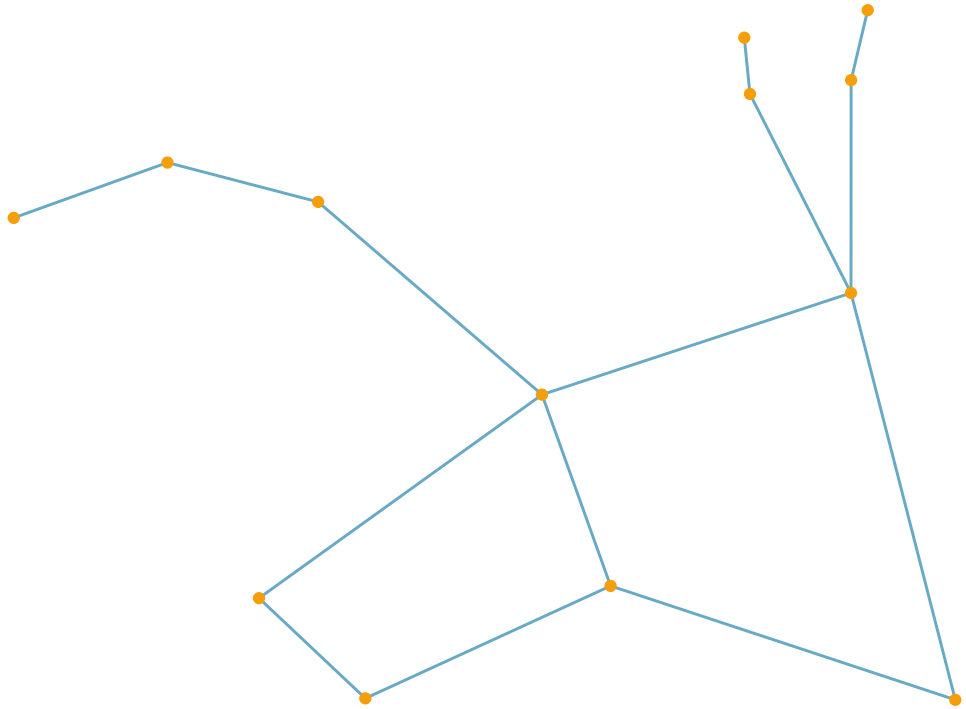
Effective release edges: 4 • unique plotted points: 4 • source: release_effective_edges.csv

# 48/88 Lepus (Lep) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

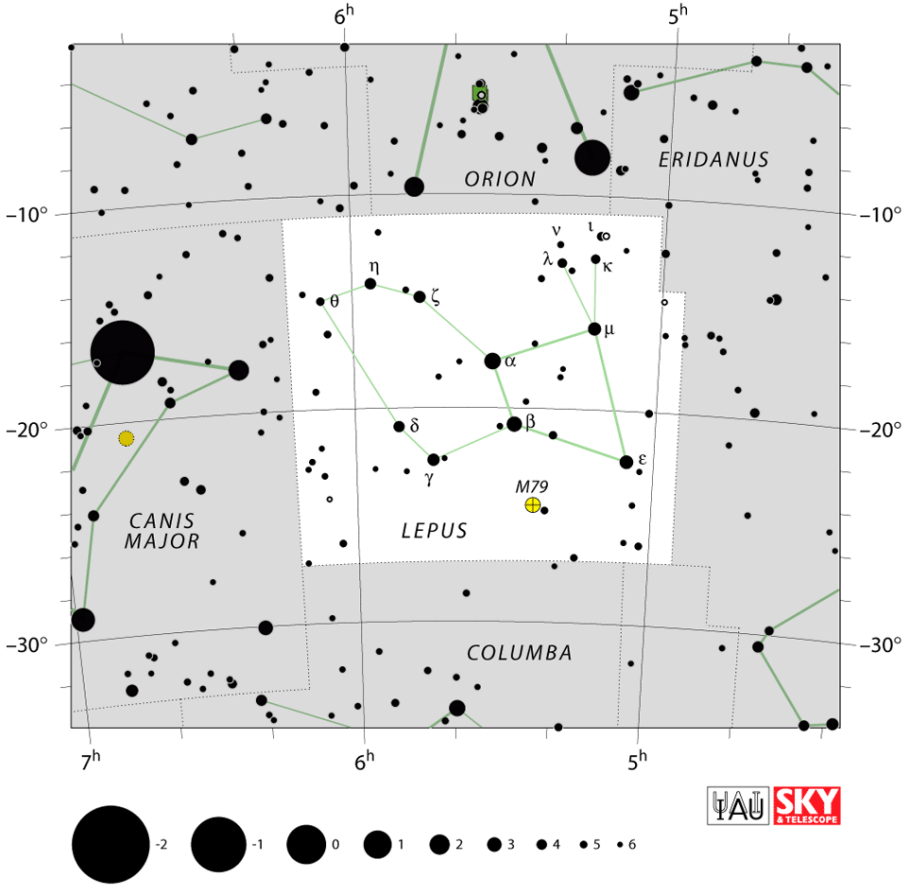
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 48/88  Lepus (Lep) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 28910 | 91.538875 | -14.935278 | HIP 28103 |
| HIP 28103 | 89.101333 | -14.168028 | HIP 28910, HIP 27288 |
| HIP 27288 | 86.738958 | -14.821944 | HIP 28103, HIP 25985 |
| HIP 25985 | 83.182542 | -17.822306 | HIP 27288, HIP 24305, HIP 27654, HIP 25606 |
| HIP 24305 | 78.232792 | -16.205417 | HIP 25985, HIP 24845, HIP 24327, HIP 23685 |
| HIP 27654 | 87.829792 | -20.877500 | HIP 25985, HIP 27072 |
| HIP 27072 | 86.116542 | -22.447500 | HIP 27654, HIP 25606 |
| HIP 25606 | 82.061375 | -20.759222 | HIP 27072, HIP 23685, HIP 25985 |
| HIP 23685 | 76.365208 | -22.370861 | HIP 25606, HIP 24305 |
| HIP 24845 | 79.893875 | -13.176778 | HIP 24305, HIP 24873 |
| HIP 24327 | 78.307875 | -12.941278 | HIP 24305, HIP 24244 |
| HIP 24244 | 78.074542 | -11.869139 | HIP 24327 |
| HIP 24873 | 79.995958 | -12.315611 | HIP 24845 |

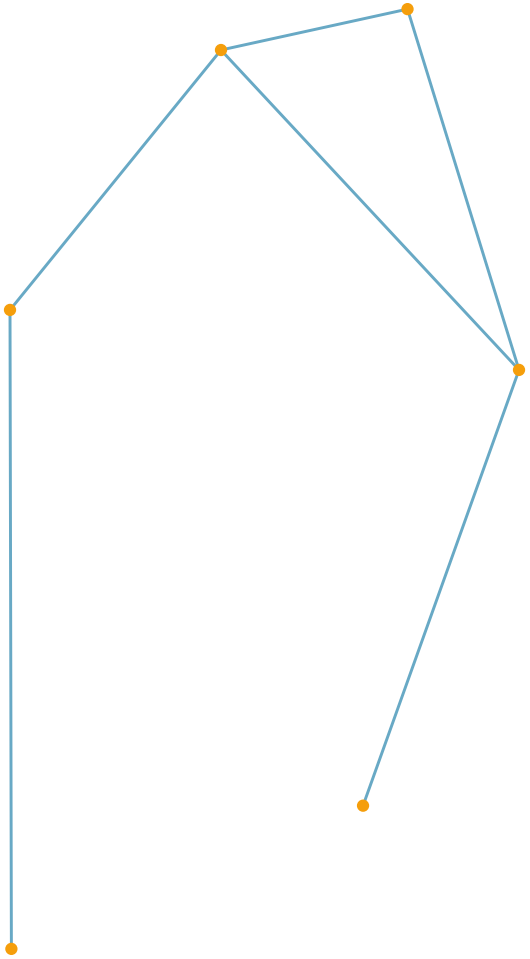
Effective release edges: 14 • unique plotted points: 13 • source: release_effective_edges.csv

# 49/88 Libra (Lib) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

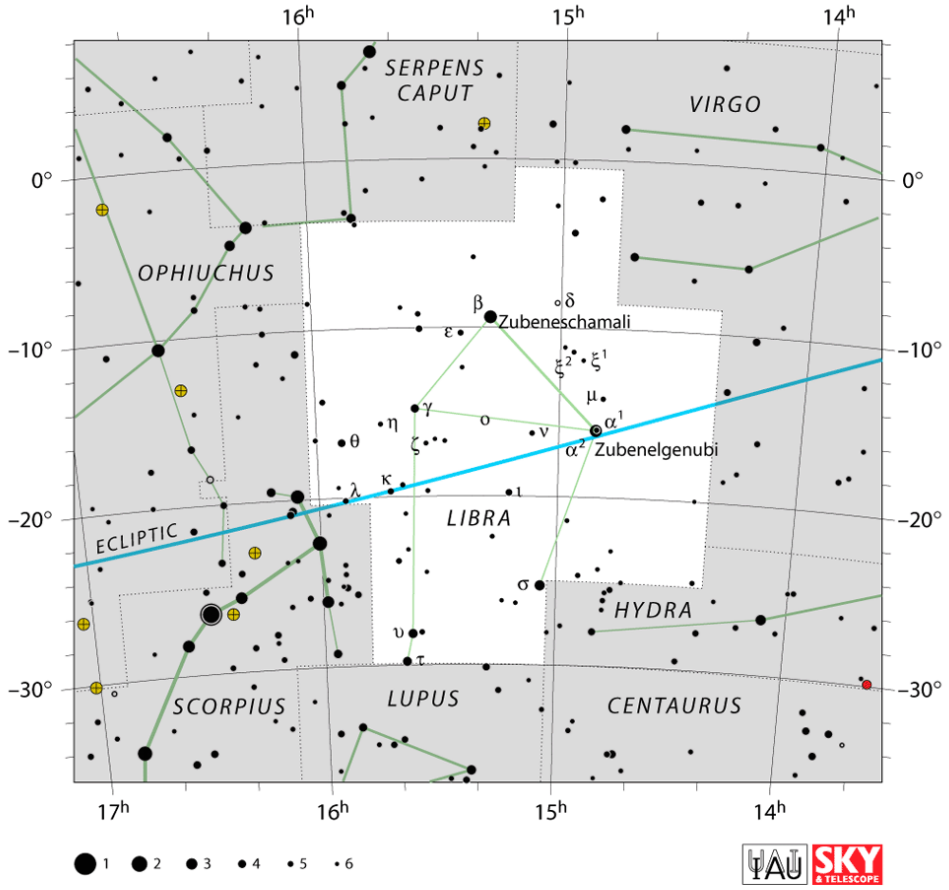
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 49/88  Libra (Lib) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| Delta Librae | 225.296700 | -8.519700 | Beta Librae, Alpha Librae |
| Alpha Librae | 222.719600 | -16.041800 | Delta Librae, Beta Librae, Sigma Librae |
| Sigma Librae | 226.017600 | -25.281900 | Alpha Librae |
| Gamma Librae | 233.881600 | -14.789500 | Beta Librae, Upsilon Librae |
| Upsilon Librae | 234.256100 | -28.135100 | Gamma Librae |
| Beta Librae | 229.251700 | -9.382900 | Delta Librae, Alpha Librae, Gamma Librae |

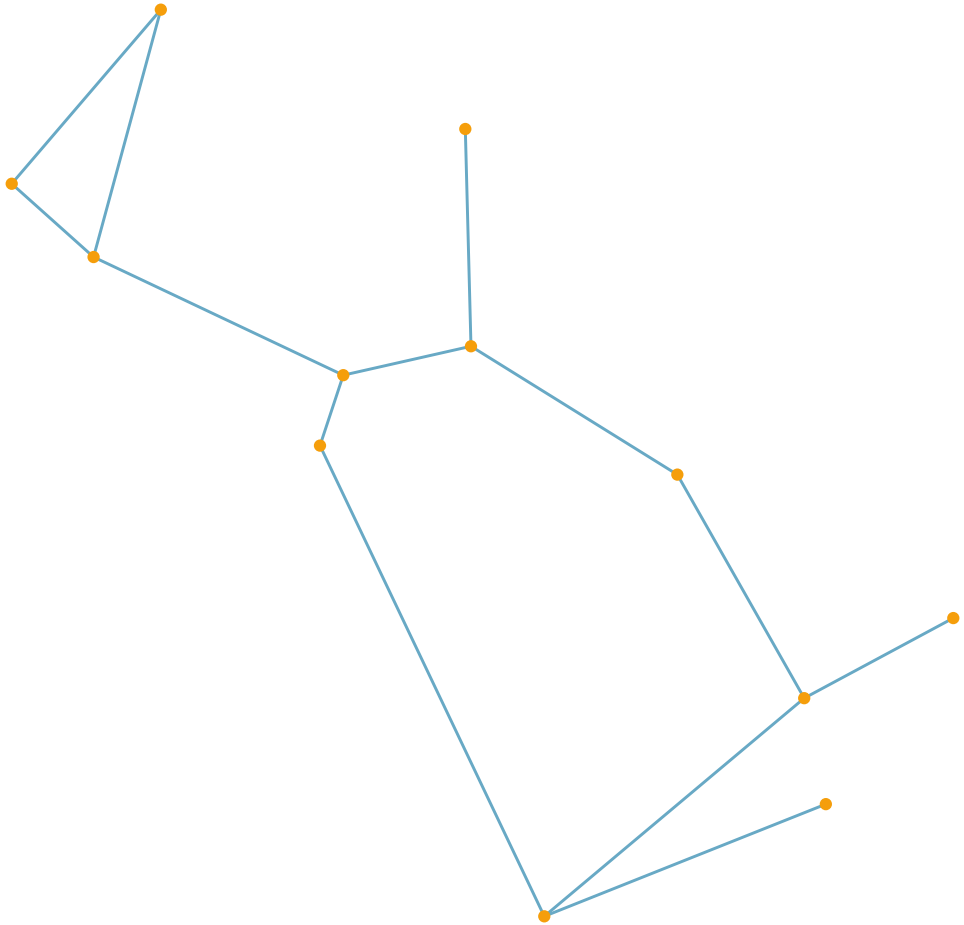
Effective release edges: 6 • unique plotted points: 6 • source: release_effective_edges.csv

# 50/88 Lupus (Lup) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

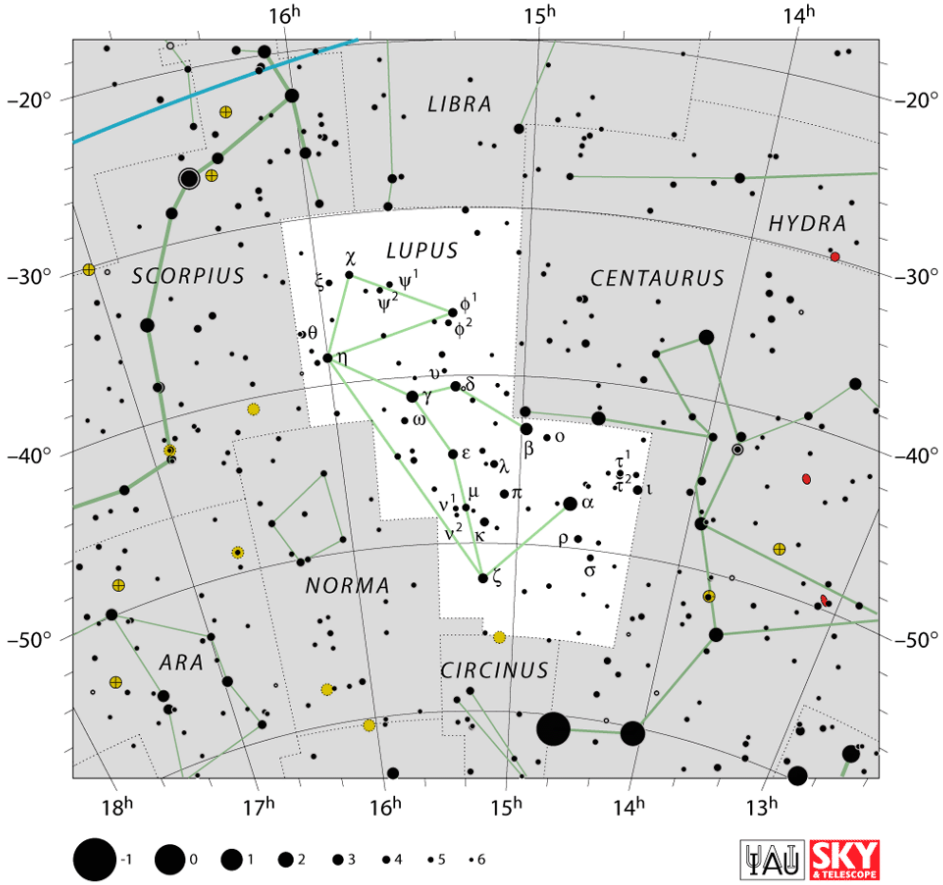
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 50/88  Lupus (Lup) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 77634 | 237.739750 | -33.627111 | HIP 78970, HIP 78384 |
| HIP 78970 | 241.817375 | -36.755528 | HIP 77634, HIP 78384 |
| HIP 78384 | 240.030583 | -38.396639 | HIP 78970, HIP 77634, HIP 76297 |
| HIP 76297 | 233.785250 | -41.166694 | HIP 78384, HIP 75141, HIP 76552 |
| HIP 75141 | 230.343083 | -40.647472 | HIP 76297, HIP 75177, HIP 73273 |
| HIP 75177 | 230.451833 | -36.261167 | HIP 75141 |
| HIP 73273 | 224.633125 | -43.133861 | HIP 75141, HIP 71860 |
| HIP 76552 | 234.513833 | -42.567472 | HIP 76297, HIP 74395 |
| HIP 74395 | 228.071667 | -52.099083 | HIP 76552, HIP 71860, HIP 71536 |
| HIP 71860 | 220.482375 | -47.388139 | HIP 74395, HIP 70576, HIP 73273 |
| HIP 71536 | 219.471875 | -49.425750 | HIP 74395 |
| HIP 70576 | 216.545000 | -45.379250 | HIP 71860 |

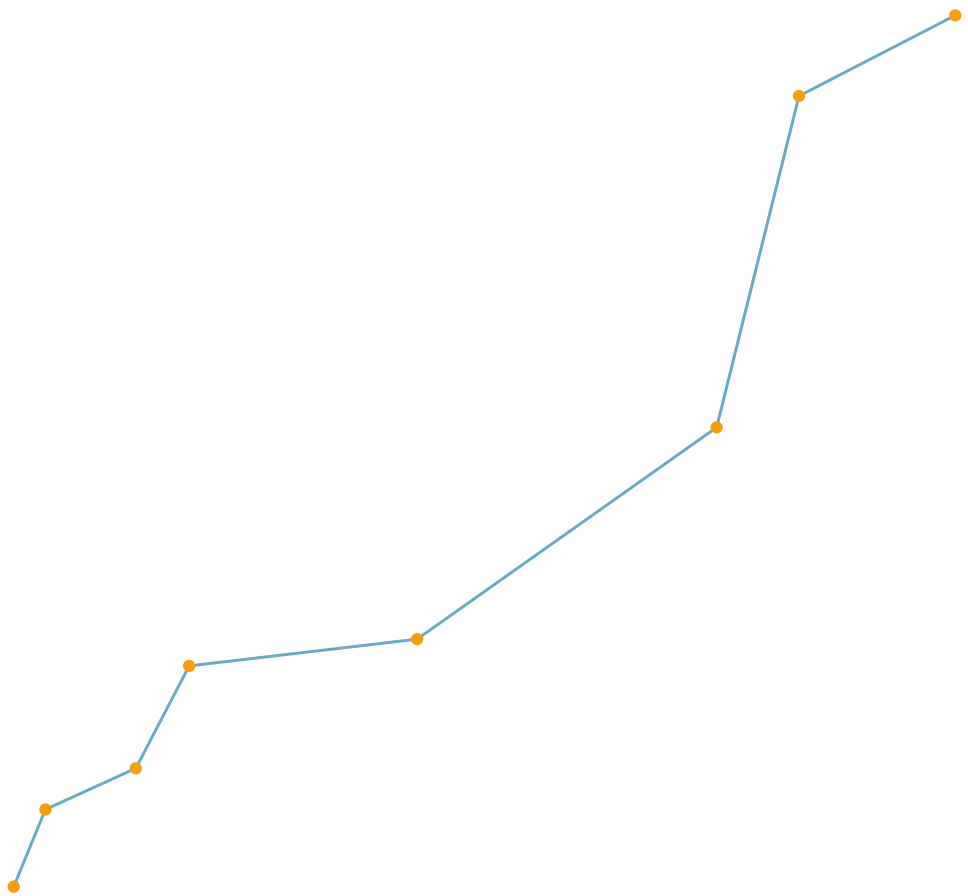
Effective release edges: 13 • unique plotted points: 12 • source: release_effective_edges.csv

# 51/88 Lynx (Lyn) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

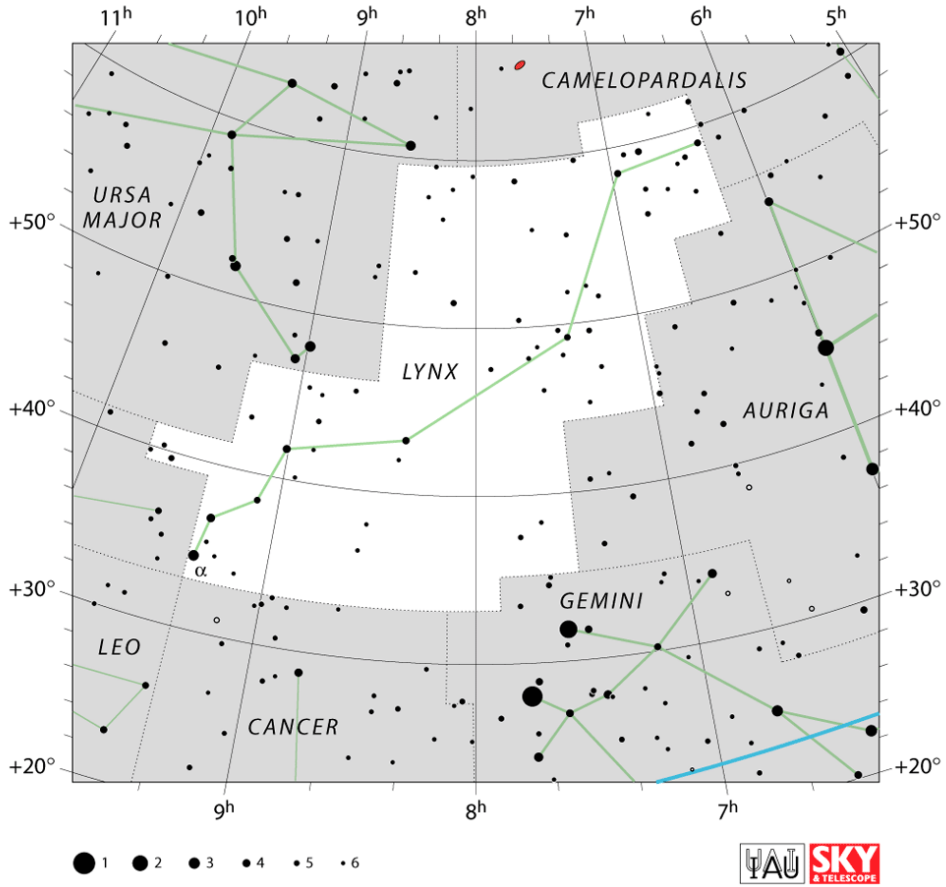
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 51/88  Lynx (Lyn) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 45860 | 140.264417 | +34.392528 | HIP 45688 |
| HIP 45688 | 139.711125 | +36.802889 | HIP 45860, HIP 44700 |
| HIP 44700 | 136.632458 | +38.452250 | HIP 45688, HIP 44248 |
| HIP 44248 | 135.161458 | +41.783444 | HIP 44700, HIP 41075 |
| HIP 41075 | 125.708875 | +43.188361 | HIP 44248, HIP 36145 |
| HIP 36145 | 111.678583 | +49.211639 | HIP 41075, HIP 33449 |
| HIP 33449 | 104.319167 | +58.423056 | HIP 36145, HIP 30060 |
| HIP 30060 | 94.905792 | +59.010917 | HIP 33449 |

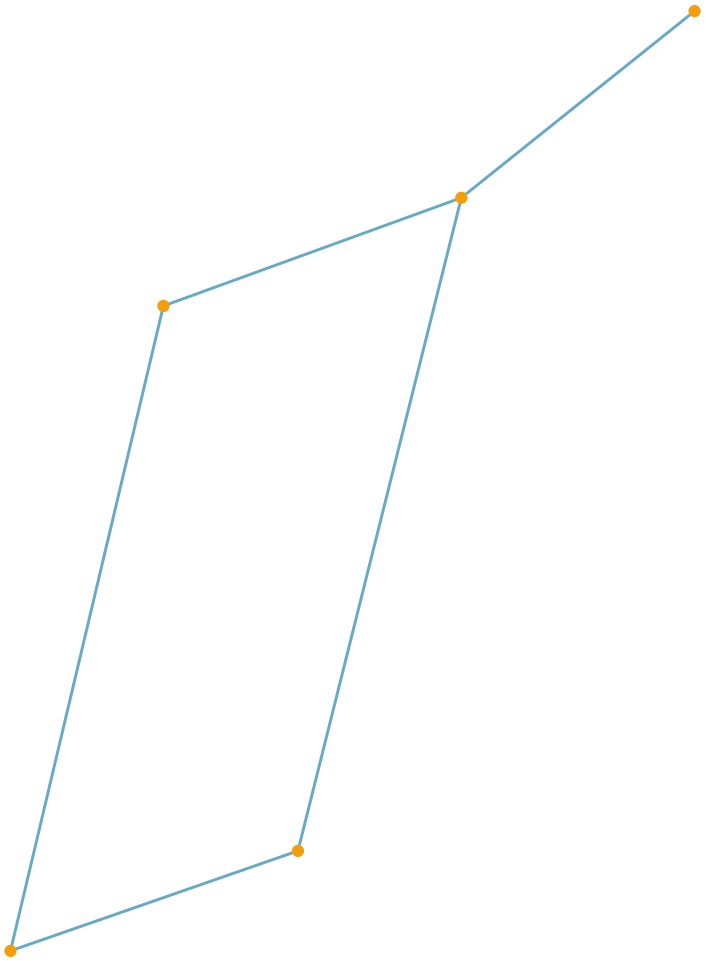
Effective release edges: 7 • unique plotted points: 8 • source: release_effective_edges.csv

# 52/88 Lyra (Lyr) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

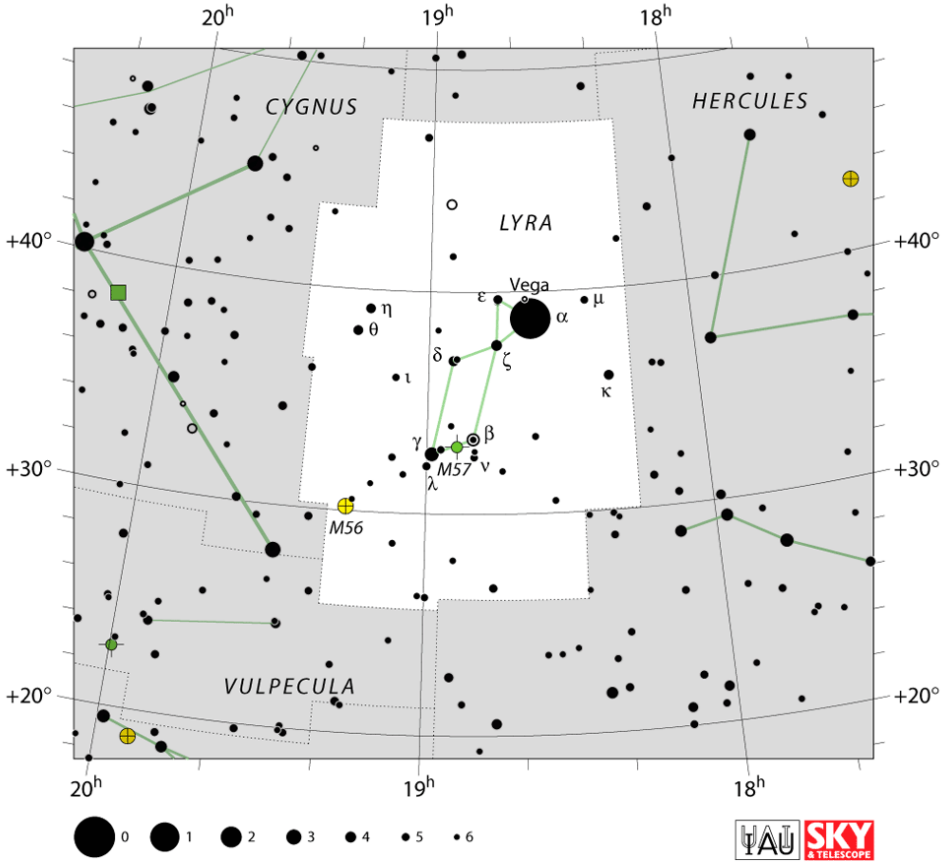
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 52/88  Lyra (Lyr) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 91262 | 279.234125 | +38.783000 | HIP 91971 |
| HIP 91971 | 281.193083 | +37.605056 | HIP 91262, HIP 92420, HIP 92791 |
| HIP 92420 | 282.519958 | +33.362667 | HIP 91971, HIP 93194 |
| HIP 93194 | 284.735917 | +32.689556 | HIP 92420, HIP 92791 |
| HIP 92791 | 283.626208 | +36.898611 | HIP 93194, HIP 91971 |

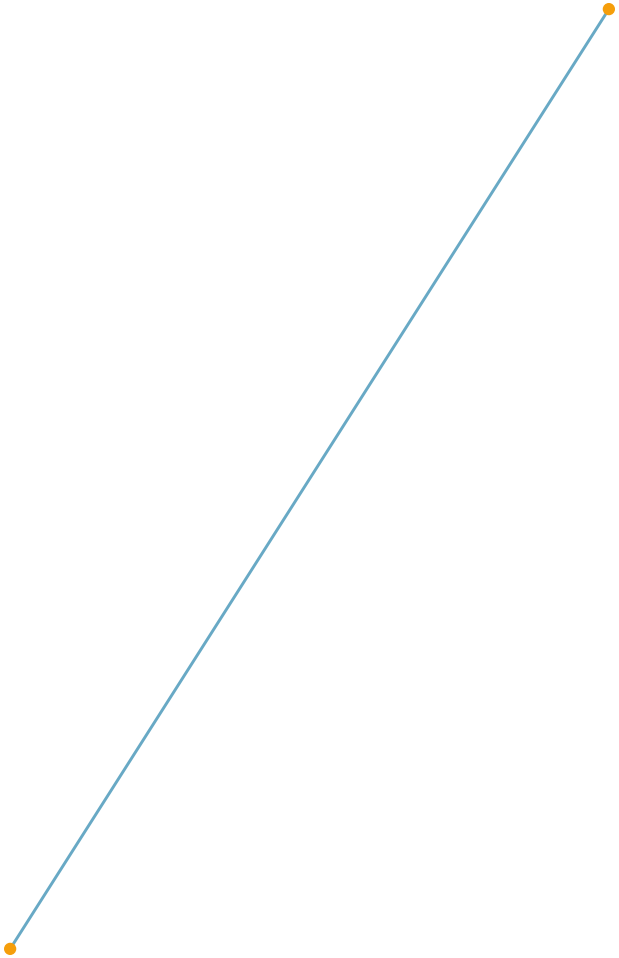
Effective release edges: 5 • unique plotted points: 5 • source: release_effective_edges.csv

# 53/88  Mensa (Men) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

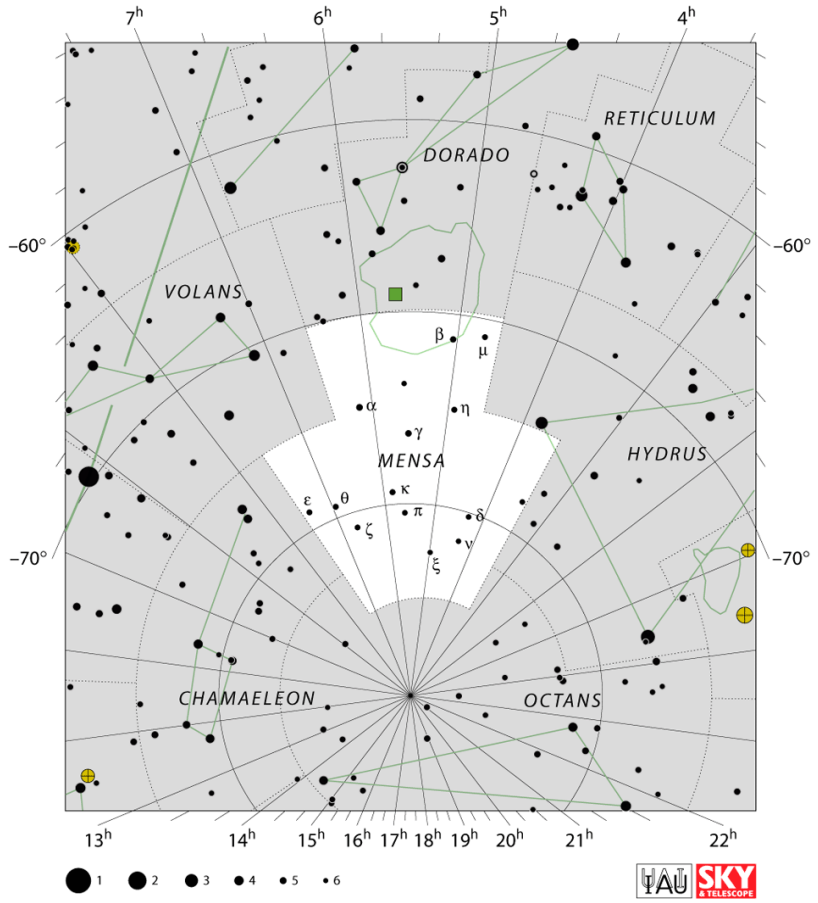
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review:  [ ] Correct  [ ] Incorrect  [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 53/88  Mensa (Men) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 25918 | 82.969417 | -76.341667 | HIP 21949 |
| HIP 21949 | 70.766458 | -70.931111 | HIP 25918 |

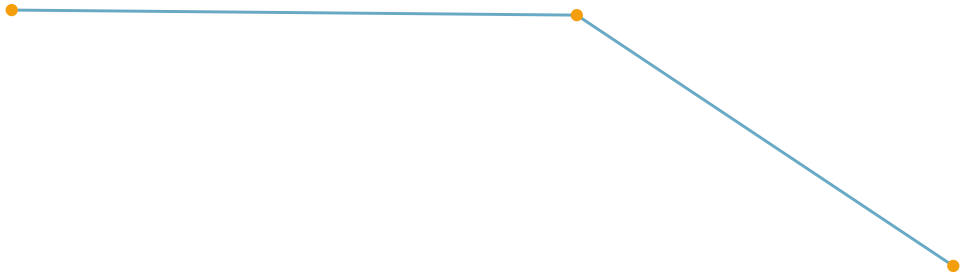
Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv

# 54/88 Microscopium (Mic) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

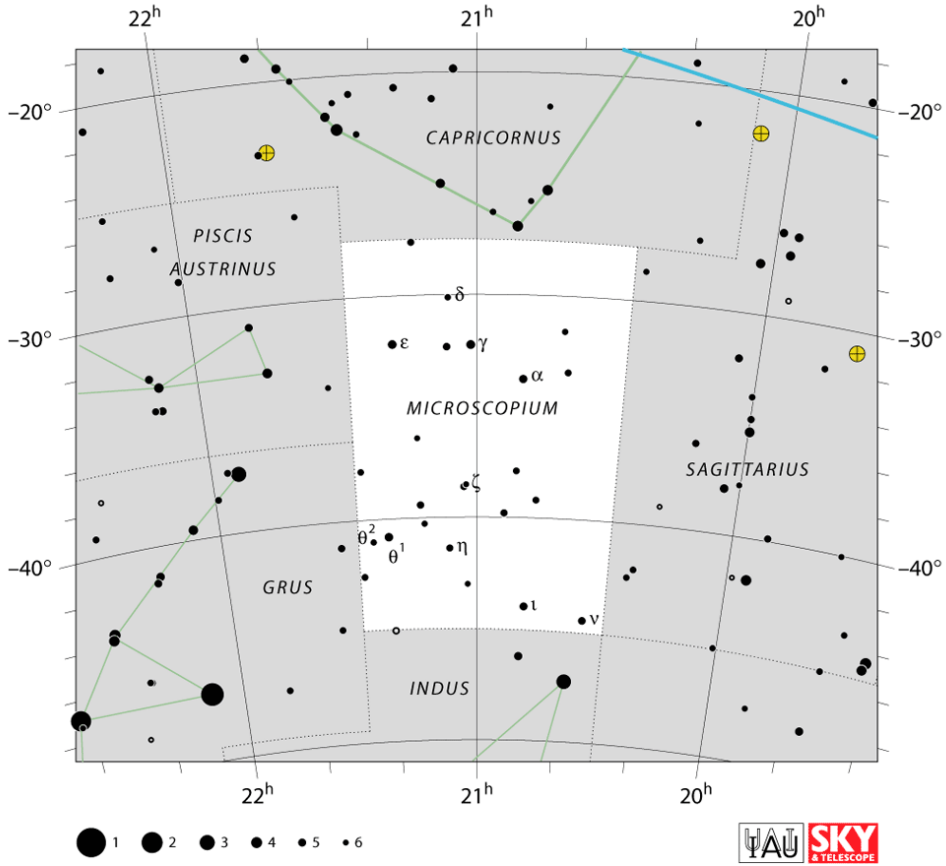
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 54/88  Microscopium (Mic) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 105140 | 319.484375 | -32.172472 | HIP 103738 |
| HIP 103738 | 315.322750 | -32.257778 | HIP 105140, HIP 102831 |
| HIP 102831 | 312.492000 | -33.779667 | HIP 103738 |

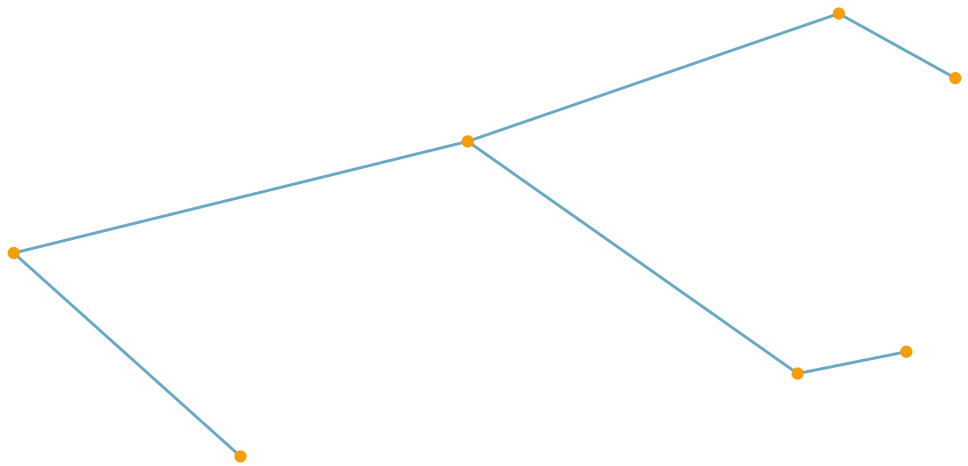
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

# 55/88 Monoceros (Mon) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

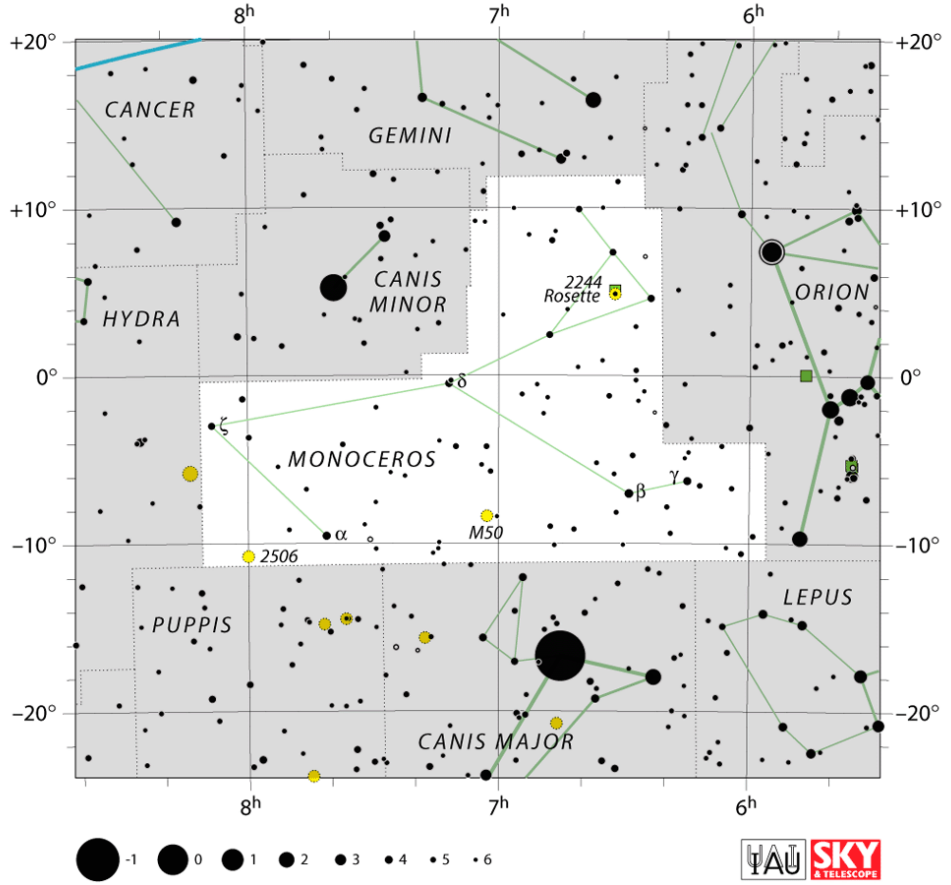
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 55/88  Monoceros (Mon) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 29651 | 93.713917 | -6.274722 | HIP 30867 |
| HIP 30867 | 97.204458 | -7.033056 | HIP 29651, HIP 34769 |
| HIP 34769 | 107.966083 | +0.492778 | HIP 30867, HIP 30419, HIP 39863 |
| HIP 30419 | 95.942083 | +4.592833 | HIP 34769, HIP 29151 |
| HIP 29151 | 92.241250 | +2.499722 | HIP 30419 |
| HIP 39863 | 122.148583 | -2.983778 | HIP 34769, HIP 37447 |
| HIP 37447 | 115.312000 | -9.551083 | HIP 39863 |

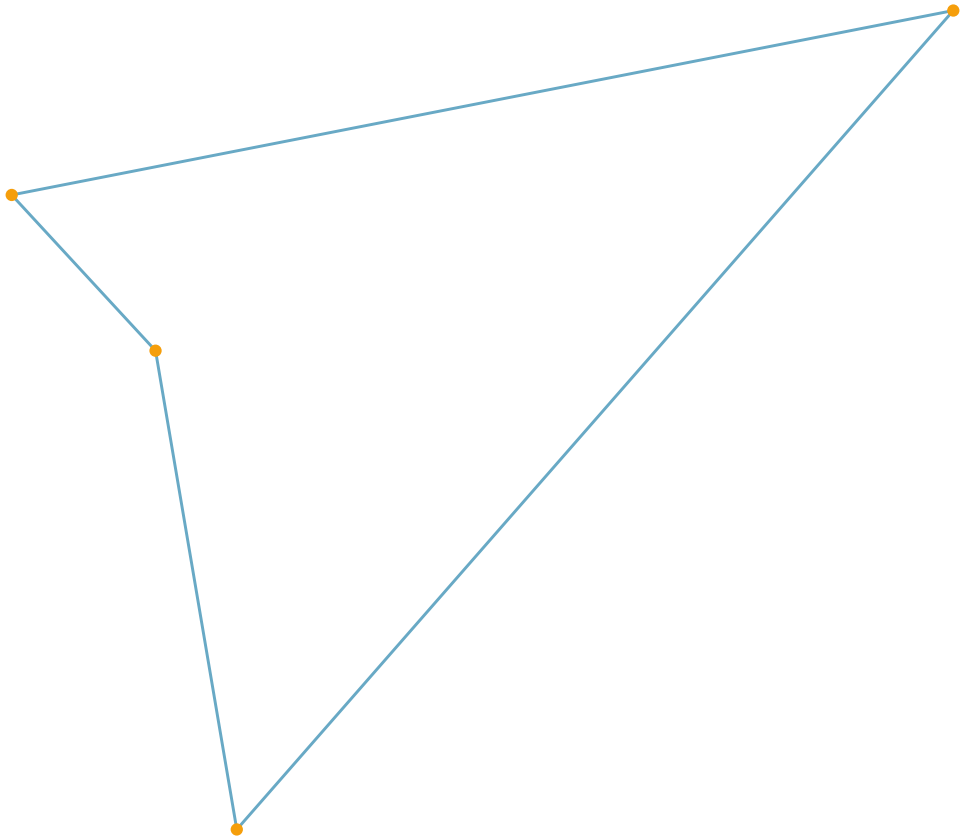
Effective release edges: 6 • unique plotted points: 7 • source: release_effective_edges.csv

# 56/88 Musca (Mus) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

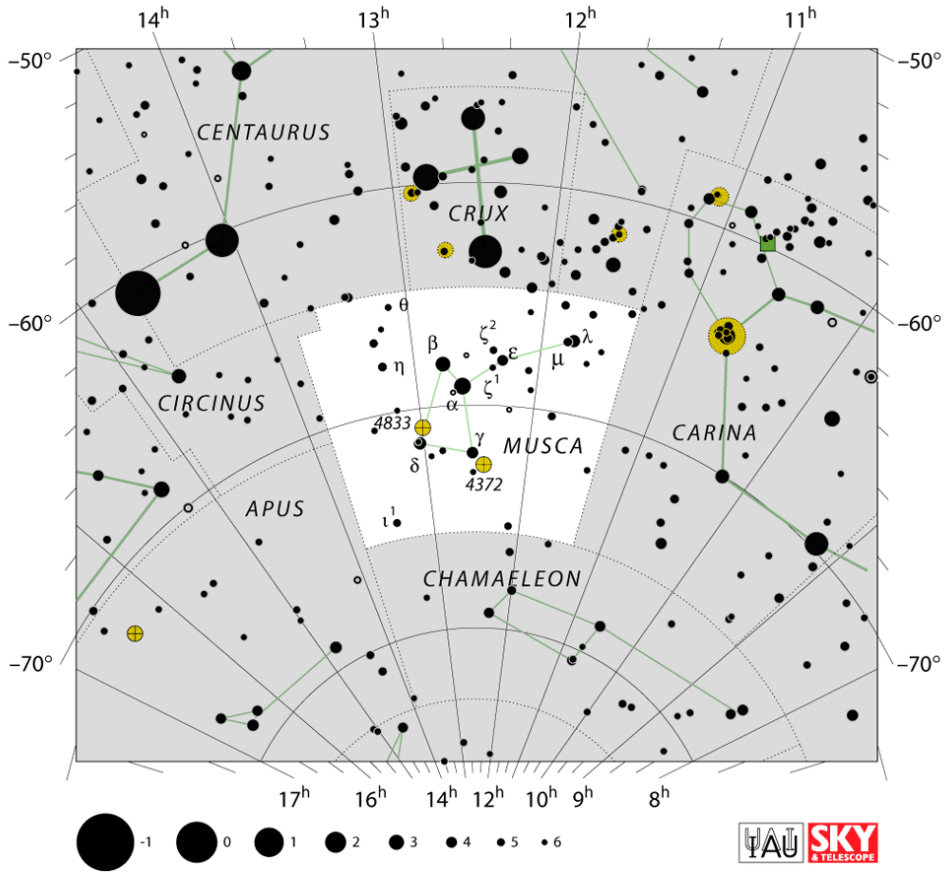
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 56/88  Musca (Mus) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 62322 | 191.570292 | -68.108083 | HIP 57363, HIP 61585 |
| HIP 57363 | 176.402375 | -66.728833 | HIP 62322, HIP 61199 |
| HIP 61199 | 188.117125 | -72.132972 | HIP 57363, HIP 61585 |
| HIP 61585 | 189.296167 | -69.135528 | HIP 61199, HIP 62322 |

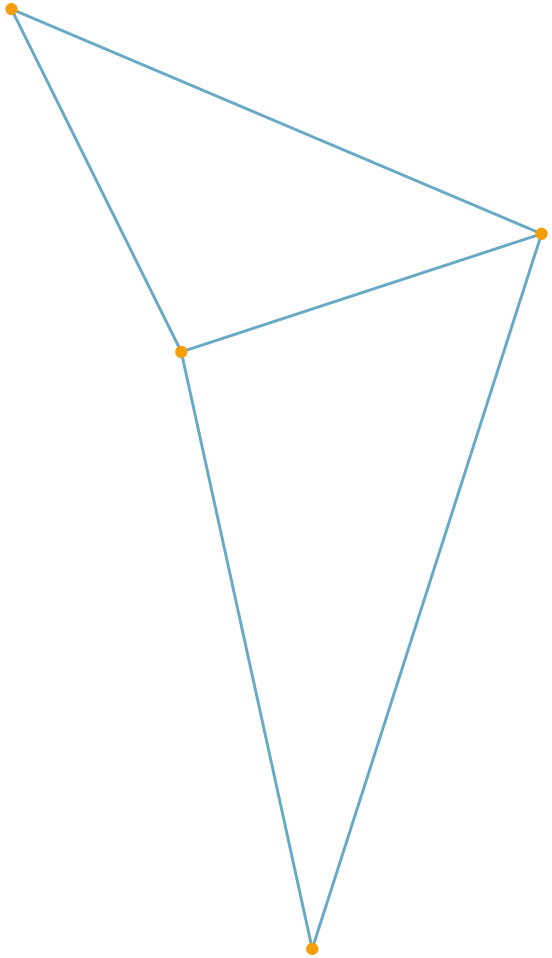
Effective release edges: 4 • unique plotted points: 4 • source: release_effective_edges.csv

# 57/88 Norma (Nor) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

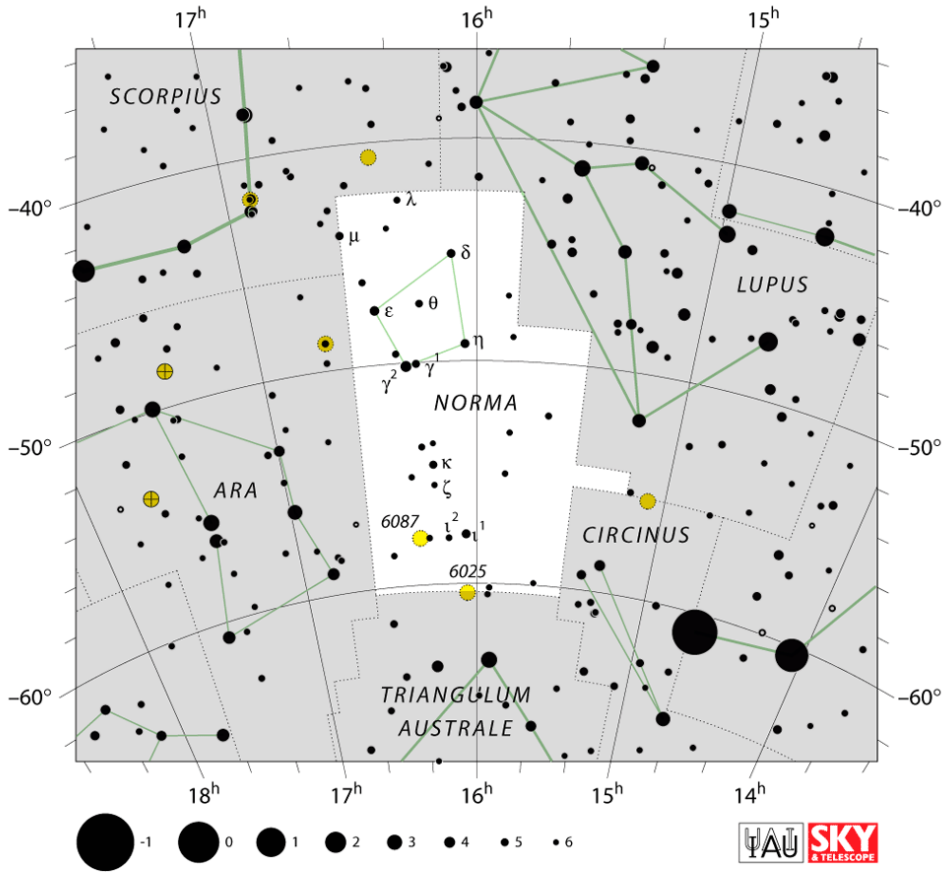
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 57/88  Norma (Nor) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 79509 | 243.369708 | -54.630417 | HIP 80000, HIP 78639 |
| HIP 80000 | 244.960708 | -50.155389 | HIP 79509, HIP 80582, HIP 78639 |
| HIP 80582 | 246.796042 | -47.554722 | HIP 80000, HIP 78639 |
| HIP 78639 | 240.803583 | -49.229722 | HIP 80582, HIP 80000, HIP 79509 |

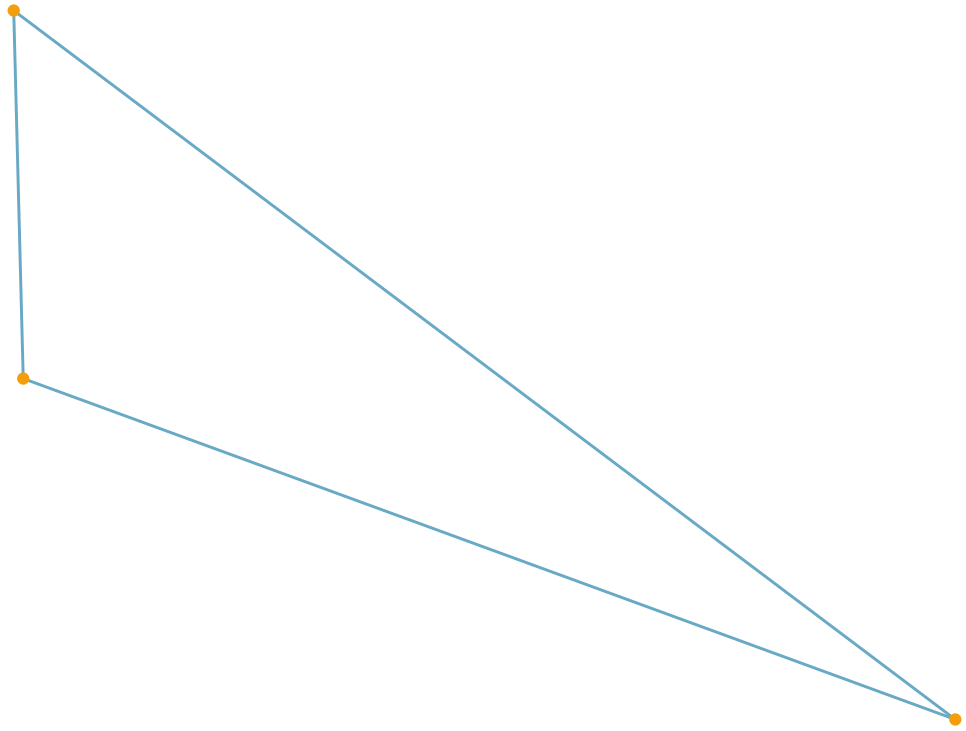
Effective release edges: 5 • unique plotted points: 4 • source: release_effective_edges.csv

# 58/88 Octans (Oct) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

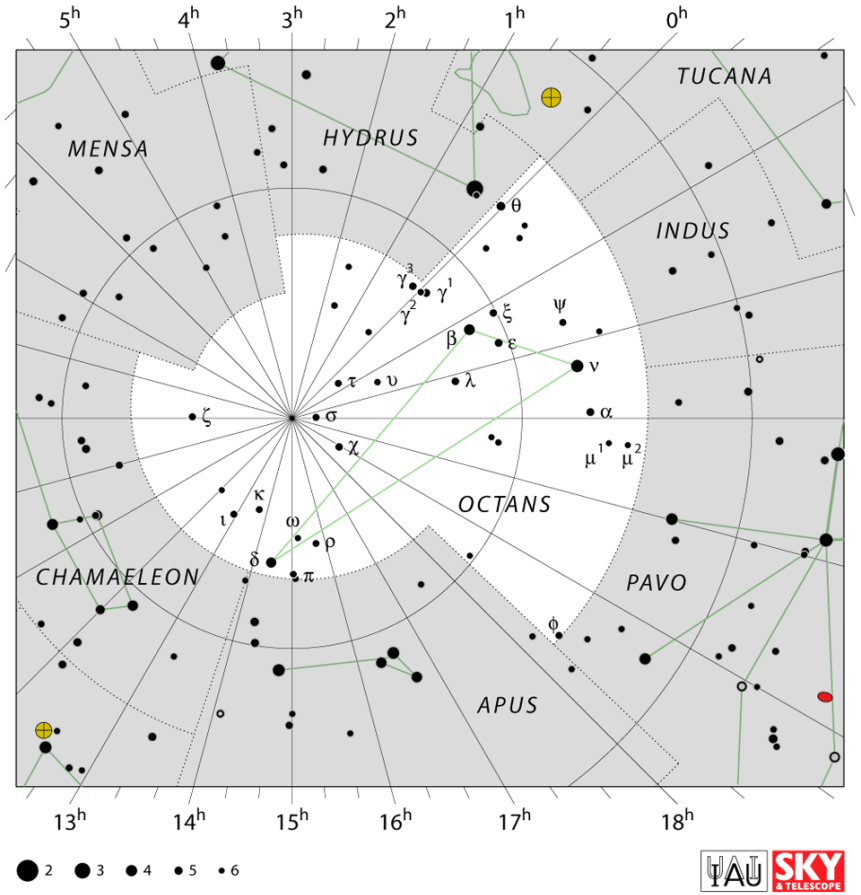
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 58/88  Octans (Oct) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 107089 | 325.368625 | -77.389472 | HIP 112405, HIP 70638 |
| HIP 112405 | 341.515500 | -81.381611 | HIP 107089, HIP 70638 |
| HIP 70638 | 216.732250 | -83.667861 | HIP 112405, HIP 107089 |

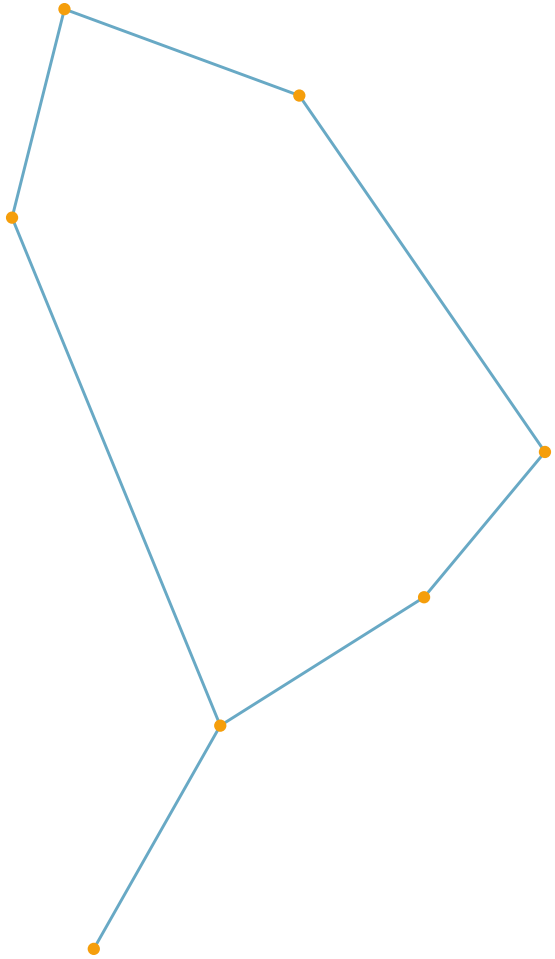
Effective release edges: 3 • unique plotted points: 3 • source: release_effective_edges.csv

# 59/88 Ophiuchus (Oph) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

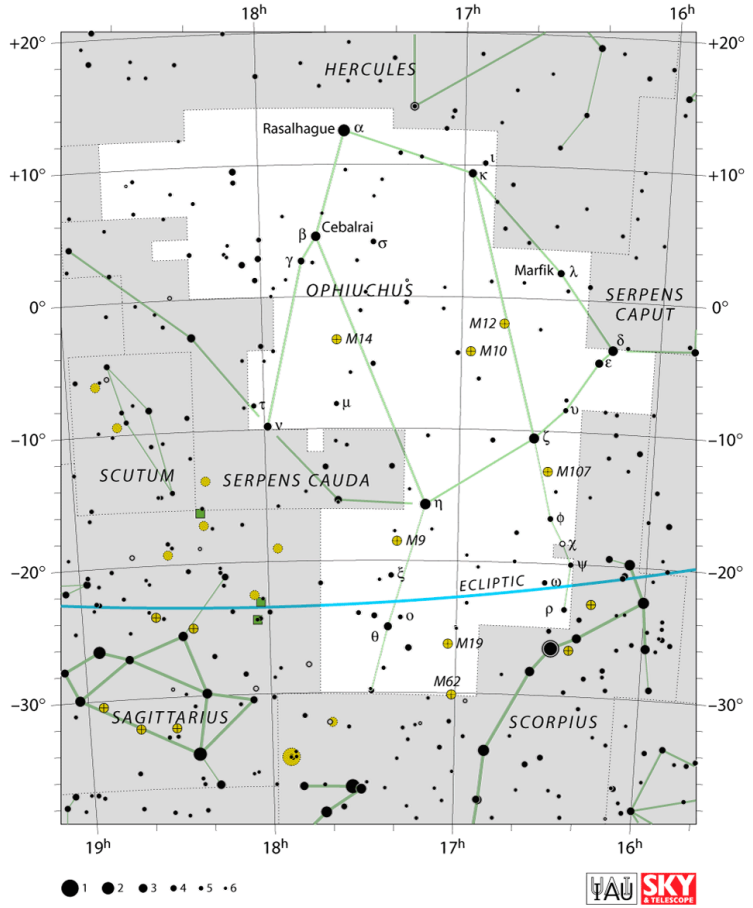
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 59/88  Ophiuchus (Oph) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 86032 | 263.733333 | +12.560583 | HIP 86742, HIP 83000 |
| HIP 86742 | 265.868250 | +4.566917 | HIP 86032, HIP 84012 |
| HIP 84012 | 257.594417 | -15.725139 | HIP 86742, HIP 81377, HIP 85755 |
| HIP 83000 | 254.417792 | +9.375056 | HIP 86032, HIP 79882 |
| HIP 79882 | 244.580167 | -4.692611 | HIP 83000, HIP 81377 |
| HIP 81377 | 249.289708 | -10.567139 | HIP 79882, HIP 84012 |
| HIP 85755 | 262.853958 | -23.962583 | HIP 84012 |

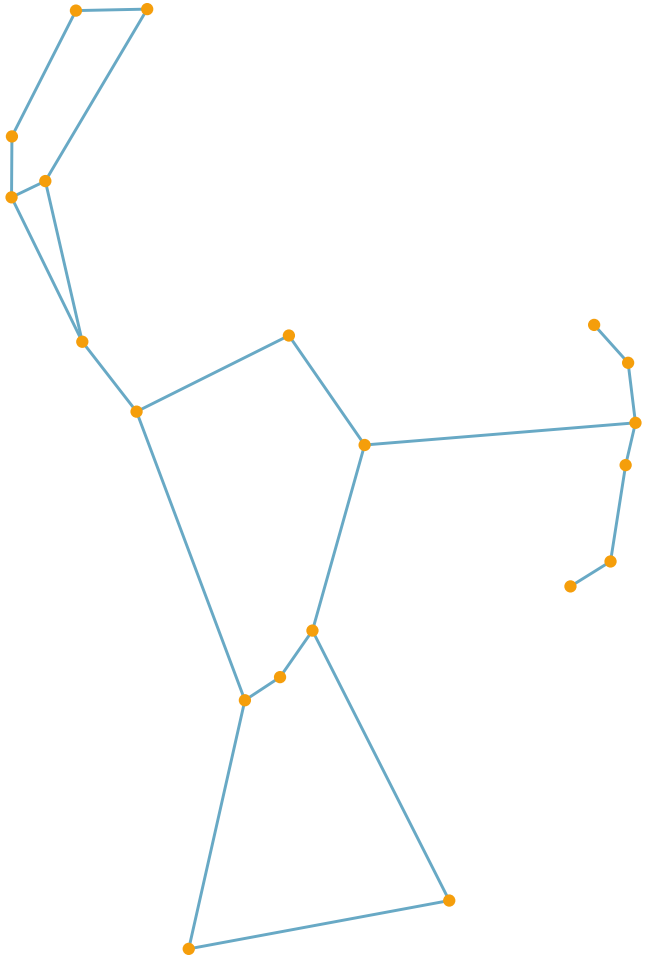
Effective release edges: 7 • unique plotted points: 7 • source: release_effective_edges.csv

# 60/88 Orion (Ori) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

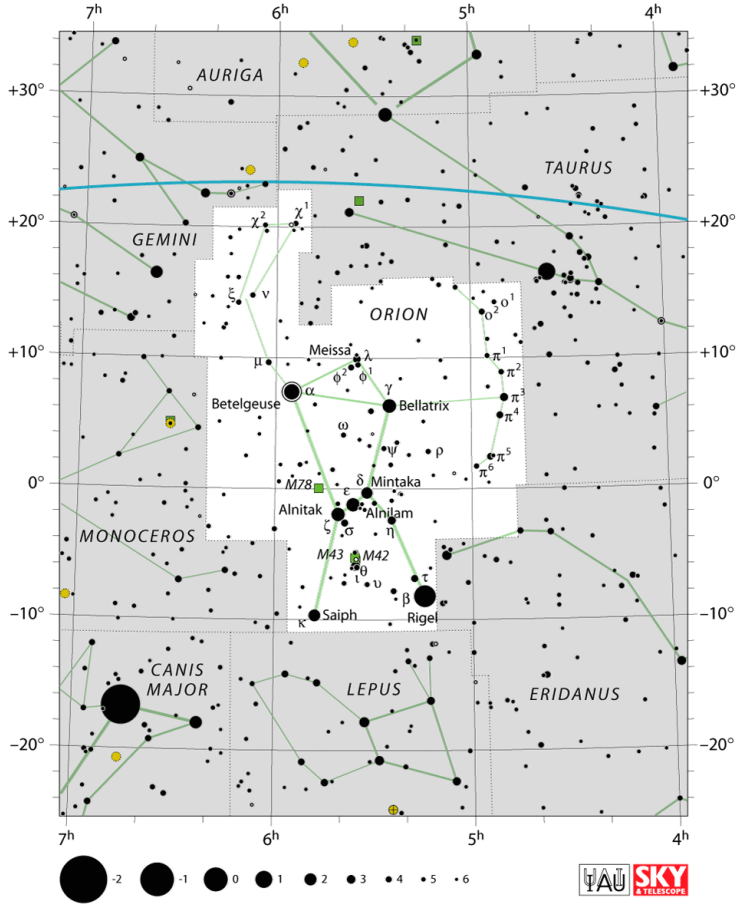
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 60/88  Orion (Ori) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 26727 | 85.189667 | -1.942583 | HIP 26311, HIP 27989, HIP 27366 |
| HIP 26311 | 84.053375 | -1.201917 | HIP 26727, HIP 25930 |
| HIP 25930 | 83.001667 | +0.299083 | HIP 26311, HIP 24436, HIP 25336 |
| HIP 29434 | 93.013667 | +16.130444 | HIP 29426, HIP 28716 |
| HIP 29426 | 92.985000 | +14.208806 | HIP 29434, HIP 29038, HIP 28614 |
| HIP 28716 | 90.979917 | +20.138472 | HIP 29434, HIP 27913 |
| HIP 27913 | 88.596167 | +20.276417 | HIP 28716, HIP 29038 |
| HIP 29038 | 91.893000 | +14.768528 | HIP 29426, HIP 27913, HIP 28614 |
| HIP 28614 | 90.595792 | +9.647361 | HIP 29426, HIP 27989, HIP 29038 |
| HIP 27989 | 88.792875 | +7.407028 | HIP 28614, HIP 26727, HIP 26207 |
| HIP 27366 | 86.939125 | -9.669611 | HIP 26727, HIP 24436 |
| HIP 24436 | 78.634458 | -8.201639 | HIP 27366, HIP 25930 |
| HIP 25336 | 81.282792 | +6.349722 | HIP 25930, HIP 26207, HIP 22449 |
| HIP 26207 | 83.784500 | +9.934167 | HIP 25336, HIP 27989 |
| HIP 22449 | 72.458917 | +6.961250 | HIP 25336, HIP 22549, HIP 22509 |
| HIP 22549 | 72.801542 | +5.605111 | HIP 22449, HIP 22730 |
| HIP 22730 | 73.344833 | +2.508278 | HIP 22549, HIP 23123 |
| HIP 23123 | 74.637083 | +1.714028 | HIP 22730 |
| HIP 22509 | 72.653000 | +8.900250 | HIP 22449, HIP 22845 |
| HIP 22845 | 73.723750 | +10.151139 | HIP 22509 |

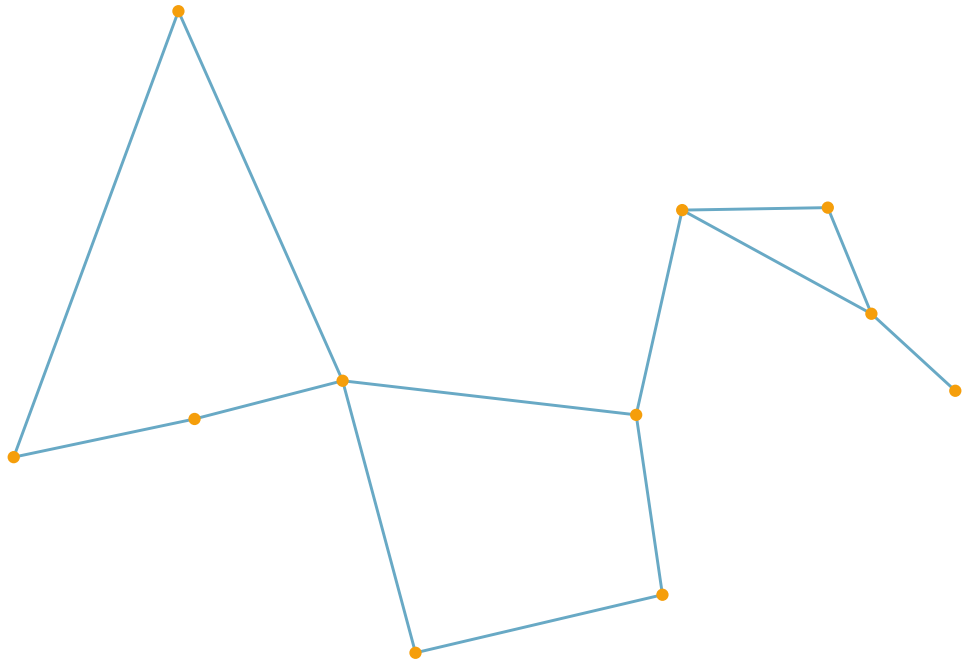
Effective release edges: 23 • unique plotted points: 20 • source: release_effective_edges.csv

# 61/88  Pavo (Pav) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

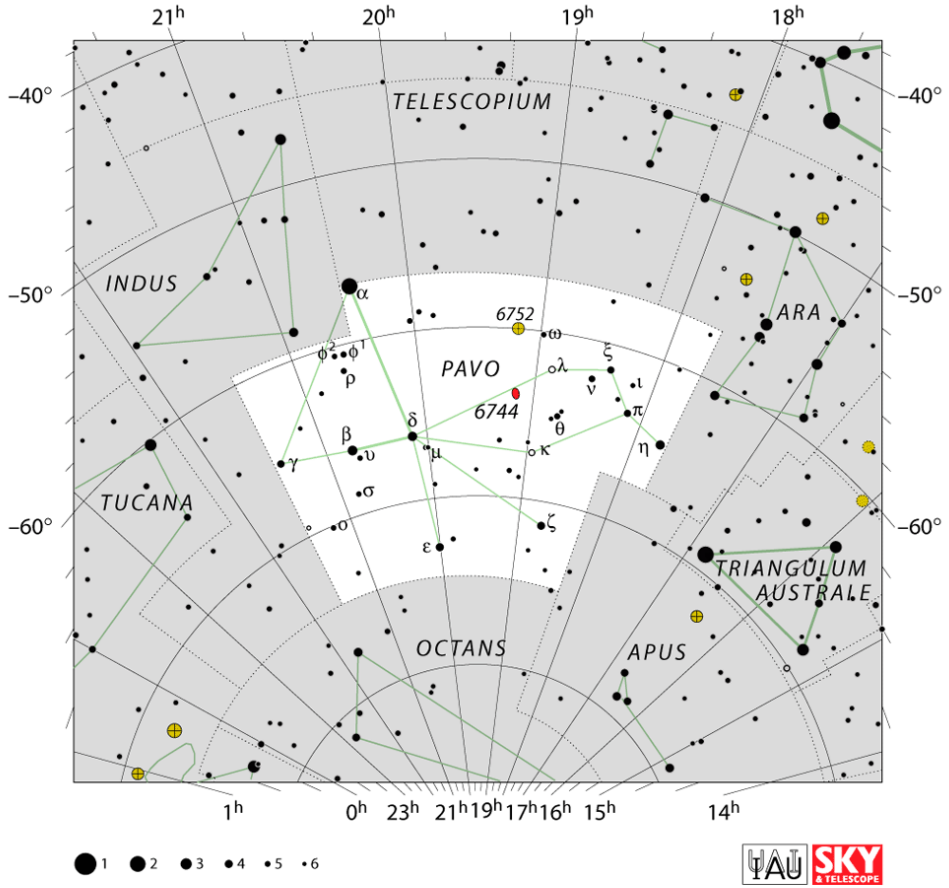
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review:  [ ] Correct  [ ] Incorrect  [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 61/88  Pavo (Pav) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 100751 | 306.411875 | -56.734889 | HIP 105858, HIP 99240 |
| HIP 105858 | 321.610375 | -65.368139 | HIP 100751, HIP 102395 |
| HIP 102395 | 311.239833 | -66.203250 | HIP 105858, HIP 99240 |
| HIP 99240 | 302.174417 | -66.179333 | HIP 102395, HIP 100751, HIP 98495, HIP 93015 |
| HIP 98495 | 300.147458 | -72.910194 | HIP 99240, HIP 91792 |
| HIP 91792 | 280.758875 | -71.427722 | HIP 98495, HIP 93015 |
| HIP 93015 | 284.237667 | -67.233528 | HIP 91792, HIP 99240, HIP 92609 |
| HIP 92609 | 283.054333 | -62.187556 | HIP 93015, HIP 90098, HIP 88866 |
| HIP 90098 | 275.806750 | -61.493917 | HIP 92609, HIP 88866 |
| HIP 88866 | 272.144958 | -63.668056 | HIP 90098, HIP 92609, HIP 86929 |
| HIP 86929 | 266.433333 | -64.723722 | HIP 88866 |

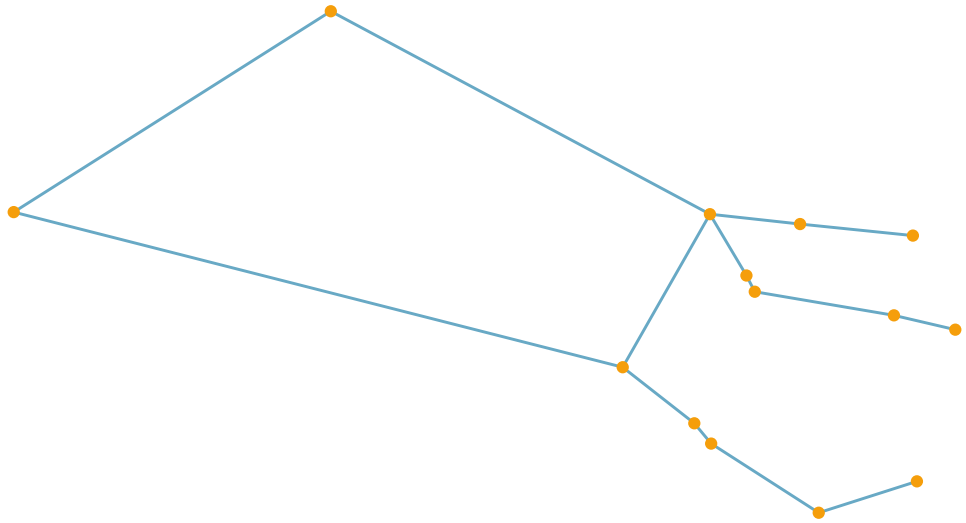
Effective release edges: 13 • unique plotted points: 11 • source: release_effective_edges.csv

# 62/88 Pegasus (Peg) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

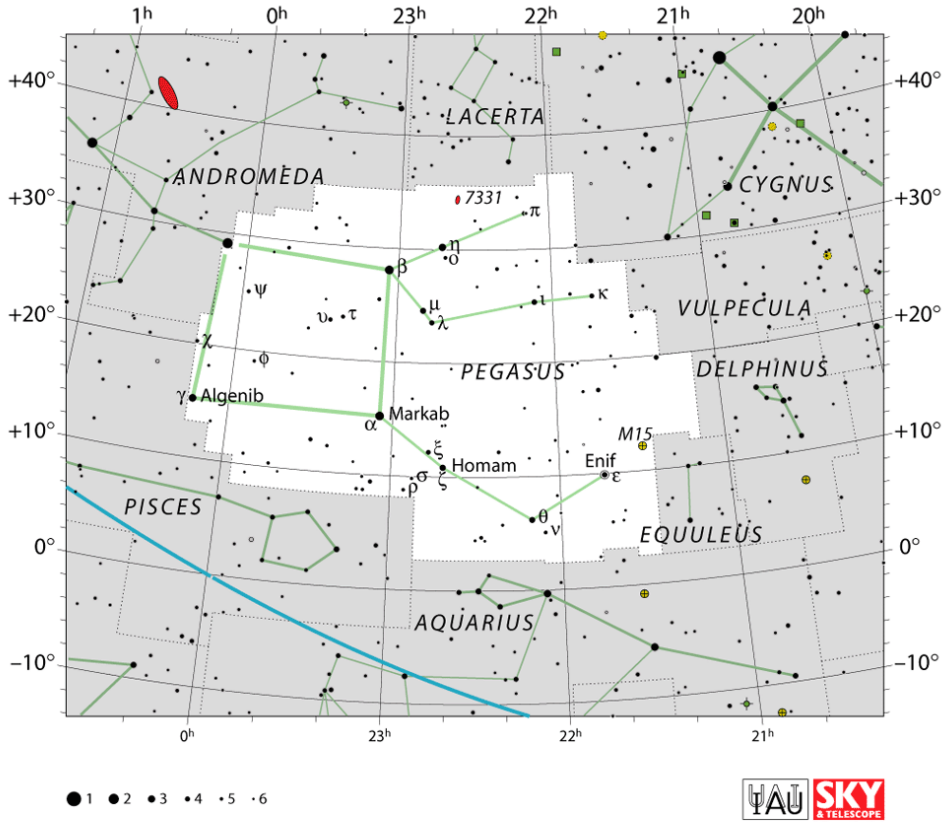
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 62/88  Pegasus (Peg) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 1067 | 3.308958 | +15.183611 | HIP 113963, HIP 677 |
| HIP 113963 | 346.190083 | +15.205361 | HIP 1067, HIP 112447, HIP 113881 |
| HIP 113881 | 345.943042 | +28.082444 | HIP 112158, HIP 112748, HIP 677, HIP 113963 |
| HIP 112158 | 340.750542 | +30.221306 | HIP 113881, HIP 109352 |
| HIP 109352 | 332.307000 | +33.172500 | HIP 112158 |
| HIP 112748 | 342.500417 | +24.601694 | HIP 113881, HIP 112440 |
| HIP 112440 | 341.632667 | +23.565667 | HIP 112748, HIP 109176 |
| HIP 109176 | 331.751958 | +25.345056 | HIP 112440, HIP 107354 |
| HIP 107354 | 326.161250 | +25.645000 | HIP 109176 |
| HIP 112447 | 341.672667 | +12.174083 | HIP 113963, HIP 112029 |
| HIP 112029 | 340.365292 | +10.831389 | HIP 112447, HIP 109427 |
| HIP 109427 | 332.549250 | +6.197778 | HIP 112029, HIP 107315 |
| HIP 107315 | 326.046417 | +9.875000 | HIP 109427 |
| HIP 677 | 2.096542 | +29.090833 | HIP 113881, HIP 1067 |

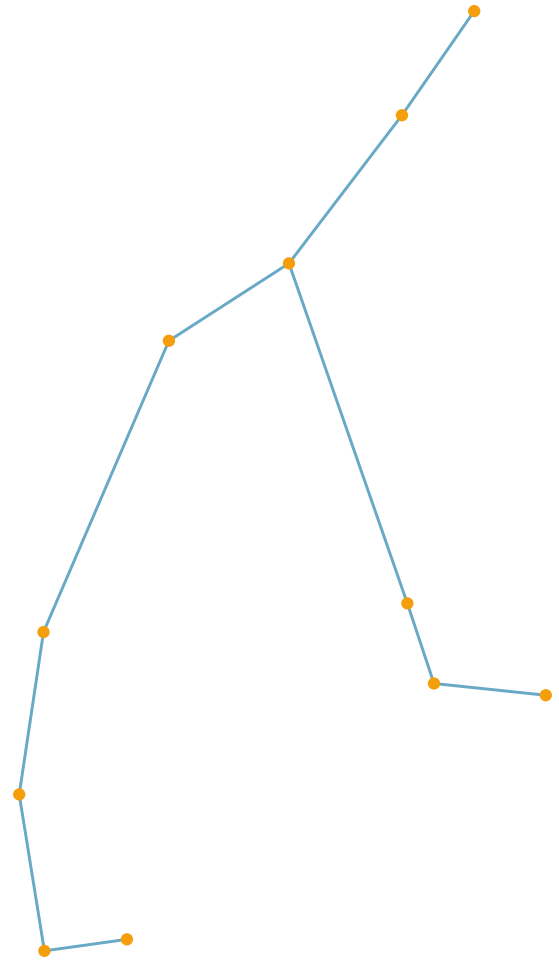
Effective release edges: 14 • unique plotted points: 14 • source: release_effective_edges.csv

# 63/88 Perseus (Per) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

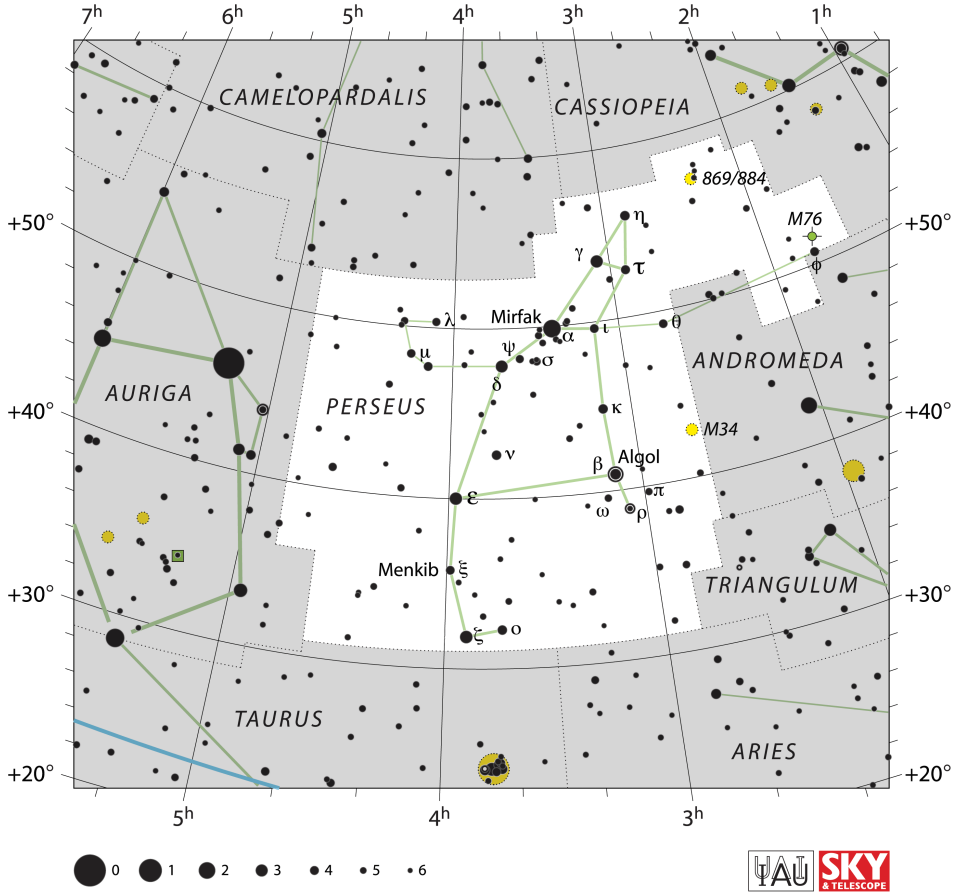
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 63/88  Perseus (Per) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 17448 | 56.079708 | +32.288278 | HIP 18246 |
| HIP 18246 | 58.533000 | +31.883667 | HIP 17448, HIP 18614 |
| HIP 18614 | 59.741250 | +35.791028 | HIP 18246, HIP 18532 |
| HIP 18532 | 59.463417 | +40.010278 | HIP 18614, HIP 17358 |
| HIP 17358 | 55.731167 | +47.787667 | HIP 18532, HIP 15863 |
| HIP 15863 | 51.080625 | +49.861250 | HIP 17358, HIP 14328, HIP 14576 |
| HIP 14328 | 46.199125 | +53.506444 | HIP 15863, HIP 13268 |
| HIP 13268 | 42.674125 | +55.895528 | HIP 14328 |
| HIP 14576 | 47.042208 | +40.955639 | HIP 15863, HIP 14354 |
| HIP 14354 | 46.293750 | +38.840528 | HIP 14576, HIP 13254 |
| HIP 13254 | 42.645458 | +38.318917 | HIP 14354 |

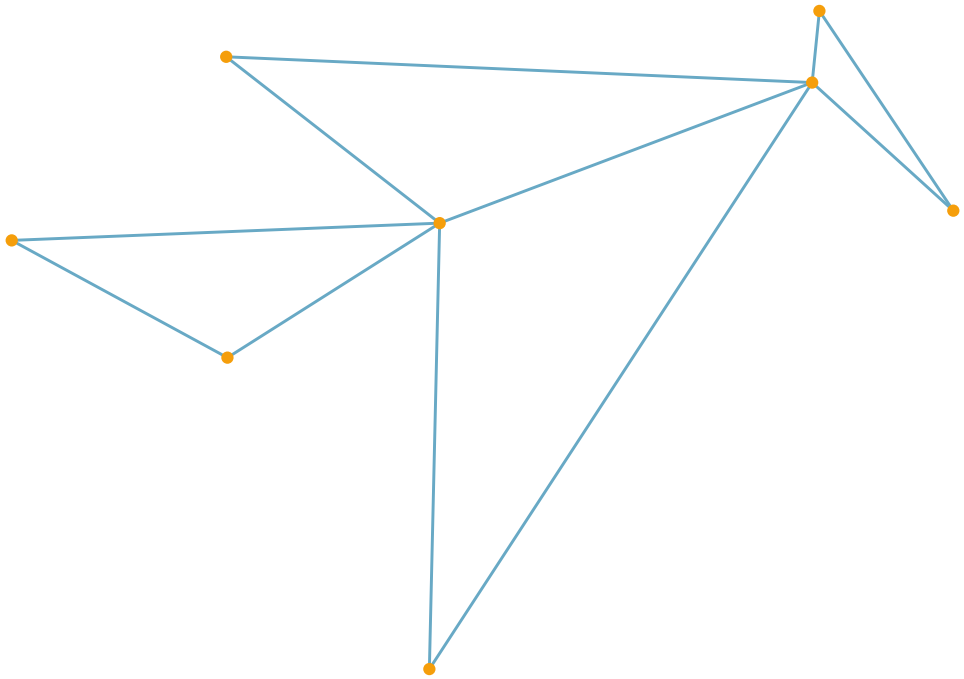
Effective release edges: 10 • unique plotted points: 11 • source: release_effective_edges.csv

# 64/88 Phoenix (Phe) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

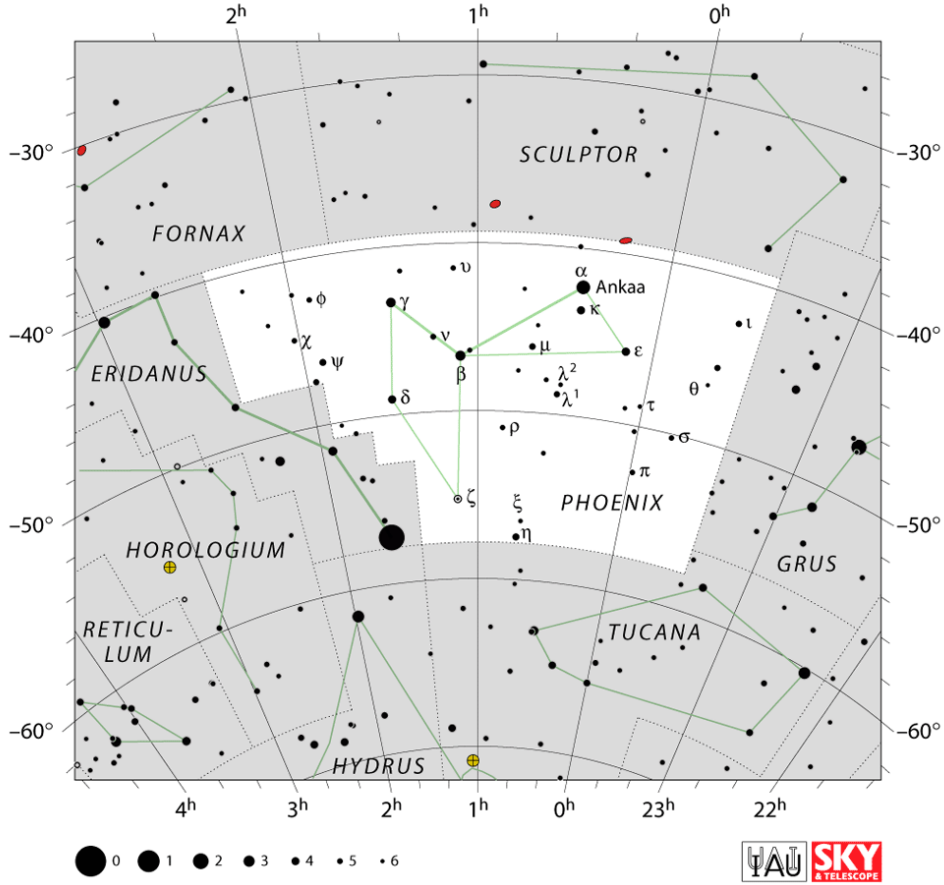
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 64/88  Phoenix (Phe) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 5348 | 17.096083 | -55.245833 | HIP 5165, HIP 2072 |
| HIP 5165 | 16.521292 | -46.718500 | HIP 5348, HIP 2072, HIP 7083, HIP 8837, HIP 6867 |
| HIP 2072 | 6.550500 | -43.679917 | HIP 5165, HIP 5348, HIP 6867, HIP 2081, HIP 765 |
| HIP 7083 | 22.812417 | -49.073083 | HIP 5165, HIP 8837 |
| HIP 8837 | 28.411750 | -46.302444 | HIP 7083, HIP 5165 |
| HIP 6867 | 22.091417 | -43.317722 | HIP 5165, HIP 2072 |
| HIP 2081 | 6.570292 | -42.305111 | HIP 2072, HIP 765 |
| HIP 765 | 2.352250 | -45.747000 | HIP 2081, HIP 2072 |

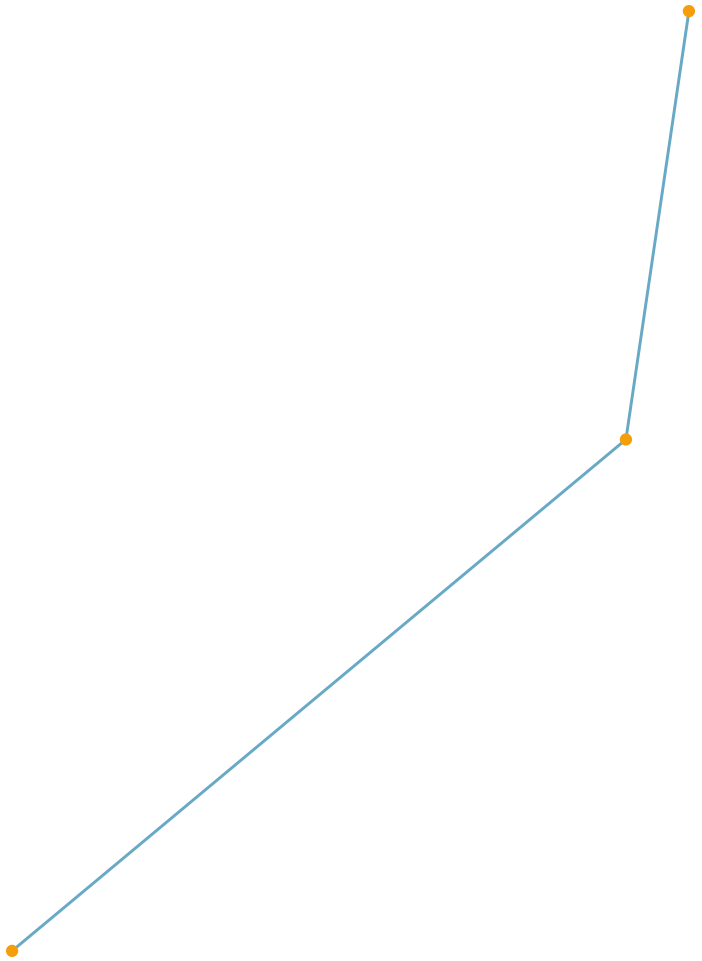
Effective release edges: 11 • unique plotted points: 8 • source: release_effective_edges.csv

# 65/88 Pictor (Pic) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

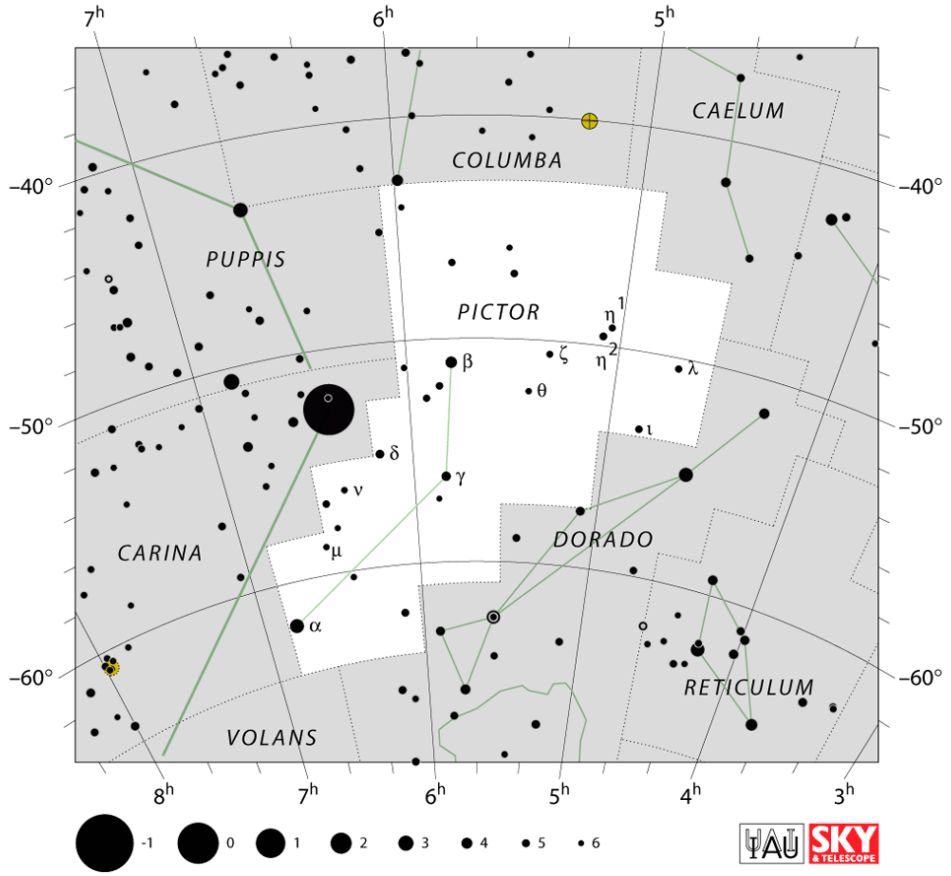
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 65/88  Pictor (Pic) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 32607 | 102.048083 | -61.941972 | HIP 27530 |
| HIP 27530 | 87.456583 | -56.166500 | HIP 32607, HIP 27321 |
| HIP 27321 | 86.821167 | -51.066722 | HIP 27530 |

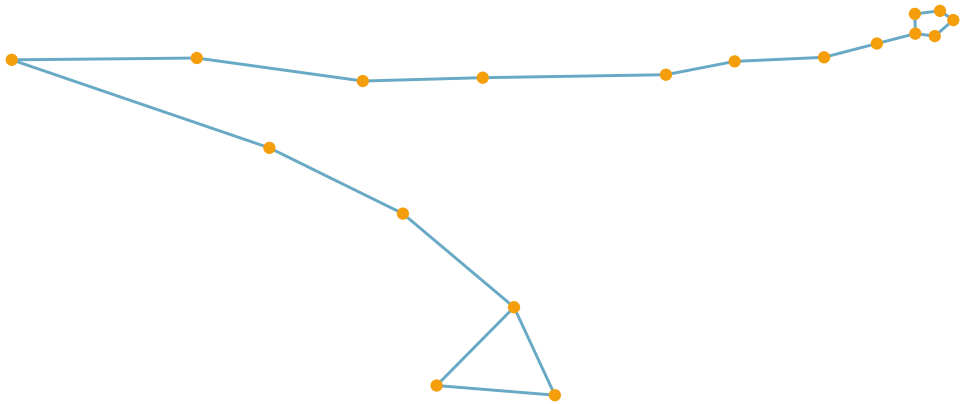
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

# 66/88 Pisces (Psc) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

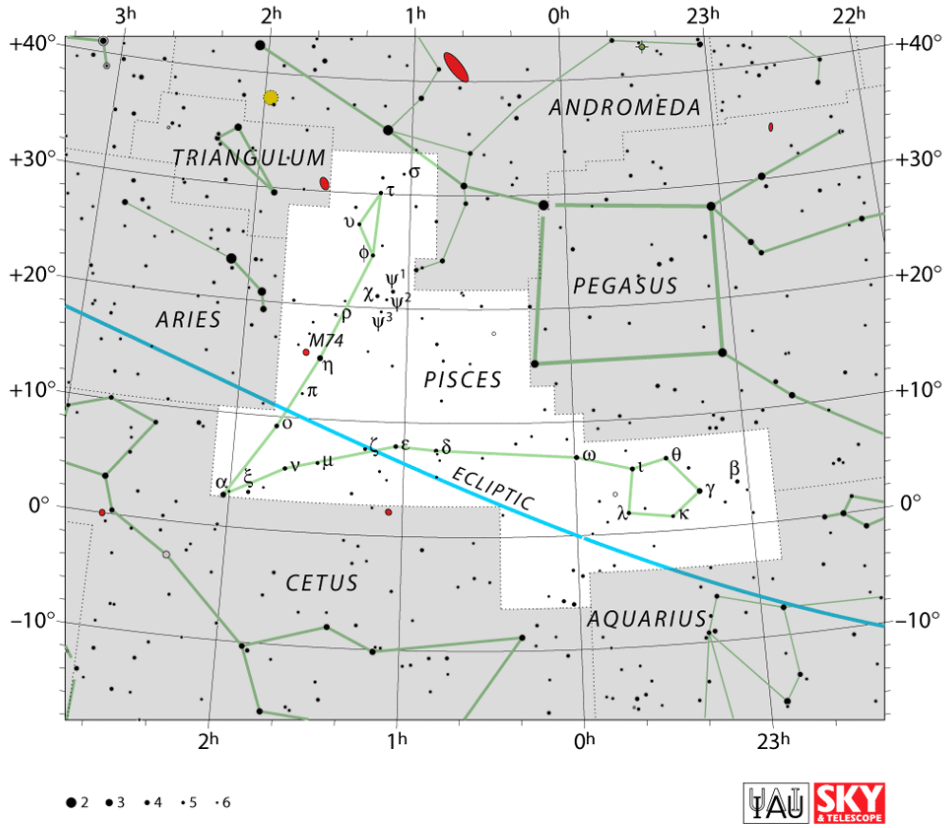
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 66/88  Pisces (Psc) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 4889 | 15.704542 | +31.804333 | HIP 5742, HIP 6193 |
| HIP 5742 | 18.437250 | +24.583778 | HIP 4889, HIP 6193, HIP 7097 |
| HIP 6193 | 19.866583 | +27.264083 | HIP 4889, HIP 5742 |
| HIP 7097 | 22.870792 | +15.345833 | HIP 5742, HIP 8198 |
| HIP 8198 | 26.348292 | +9.157639 | HIP 7097, HIP 9487 |
| HIP 9487 | 30.511667 | +2.763750 | HIP 8198, HIP 8833 |
| HIP 8833 | 28.388917 | +3.187472 | HIP 9487, HIP 7884 |
| HIP 7884 | 25.357958 | +5.487611 | HIP 8833, HIP 7007 |
| HIP 7007 | 22.545583 | +6.143944 | HIP 7884, HIP 4906 |
| HIP 4906 | 15.736083 | +7.890083 | HIP 7007, HIP 3760 |
| HIP 3760 | 12.072250 | +7.299917 | HIP 4906, HIP 1645 |
| HIP 1645 | 5.149417 | +8.190250 | HIP 3760, HIP 118268 |
| HIP 118268 | 359.827500 | +6.863583 | HIP 1645, HIP 116771 |
| HIP 116771 | 354.986750 | +5.627361 | HIP 118268, HIP 116928, HIP 115830 |
| HIP 116928 | 355.512000 | +1.780417 | HIP 116771, HIP 115738 |
| HIP 115738 | 351.732958 | +1.255833 | HIP 116928, HIP 114971 |
| HIP 114971 | 349.289542 | +3.282250 | HIP 115738, HIP 115830 |
| HIP 115830 | 351.992375 | +6.379111 | HIP 114971, HIP 116771 |

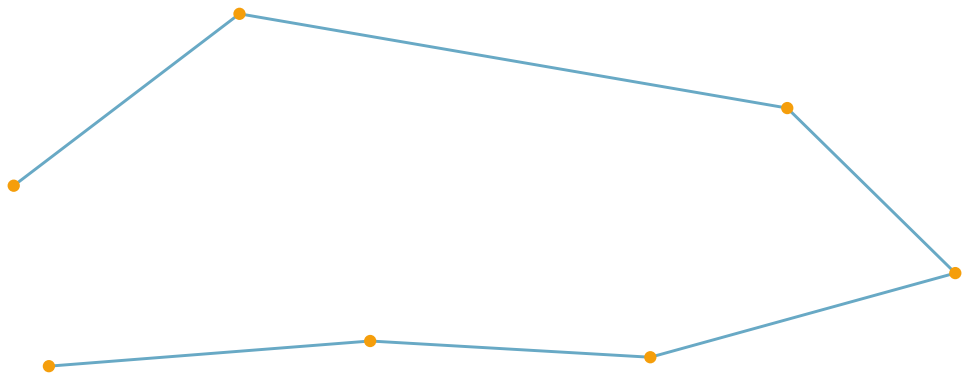
Effective release edges: 19 • unique plotted points: 18 • source: release_effective_edges.csv

# 67/88 Piscis Austrinus (PsA) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

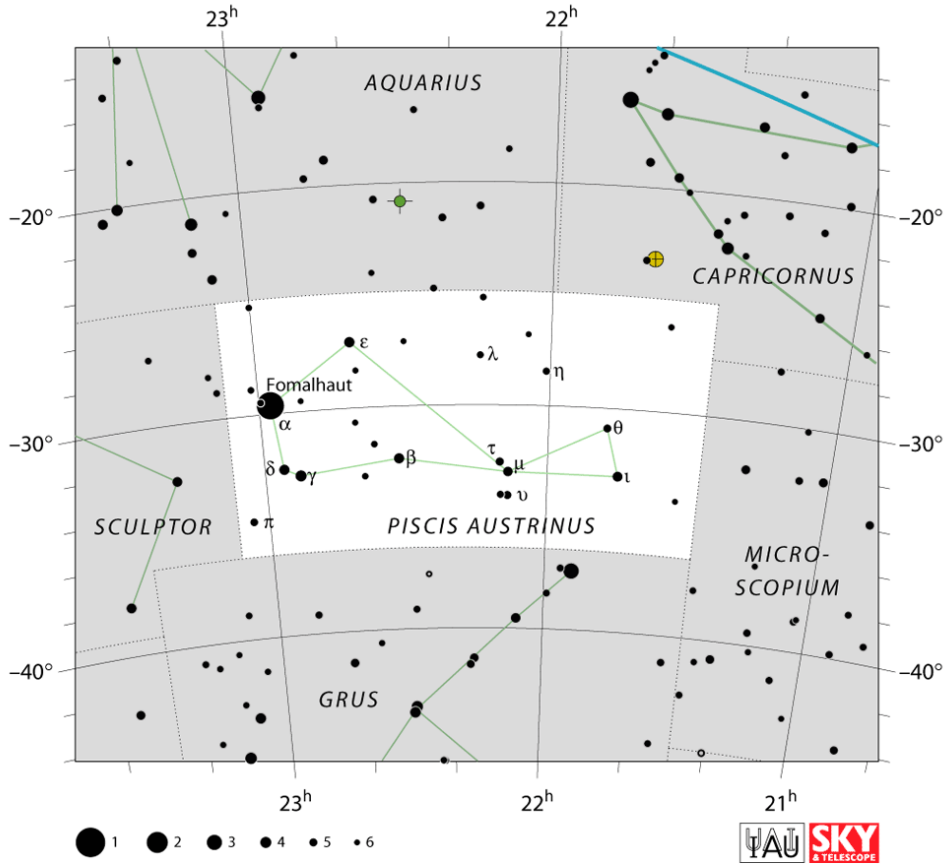
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 67/88  Piscis Austrinus (PsA) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 113368 | 344.411792 | -29.621833 | HIP 111954 |
| HIP 111954 | 340.163875 | -27.043611 | HIP 113368, HIP 108661 |
| HIP 108661 | 330.209250 | -28.453750 | HIP 111954, HIP 107608 |
| HIP 107608 | 326.934042 | -30.898306 | HIP 108661, HIP 109422 |
| HIP 109422 | 332.535333 | -32.548444 | HIP 107608, HIP 111188 |
| HIP 111188 | 337.876208 | -32.346028 | HIP 109422, HIP 113246 |
| HIP 113246 | 343.987042 | -32.539694 | HIP 111188 |

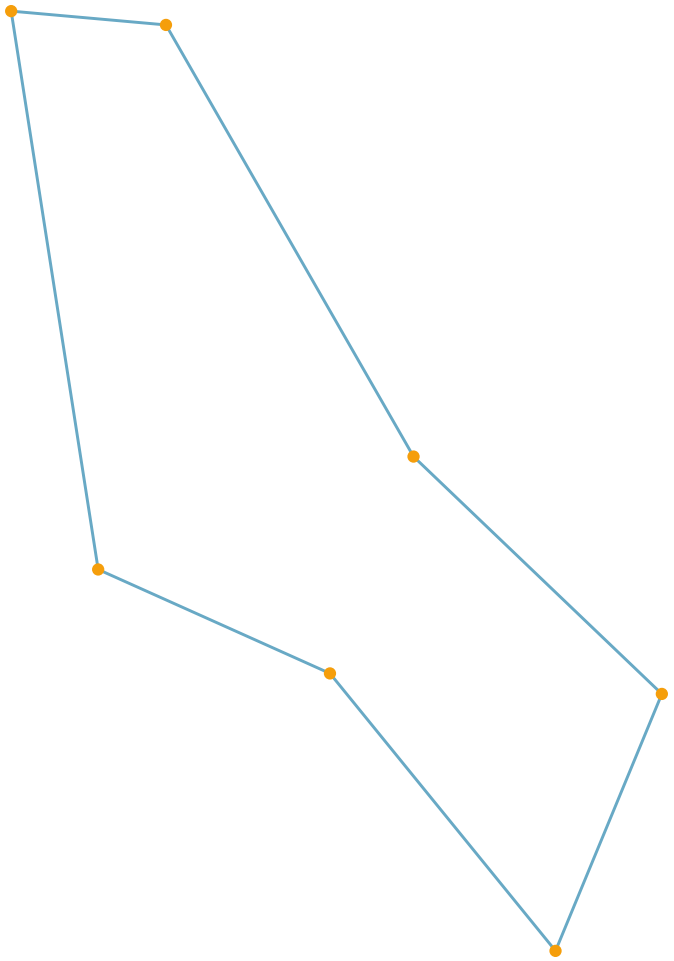
Effective release edges: 6 • unique plotted points: 7 • source: release_effective_edges.csv

# 68/88 Puppis (Pup) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

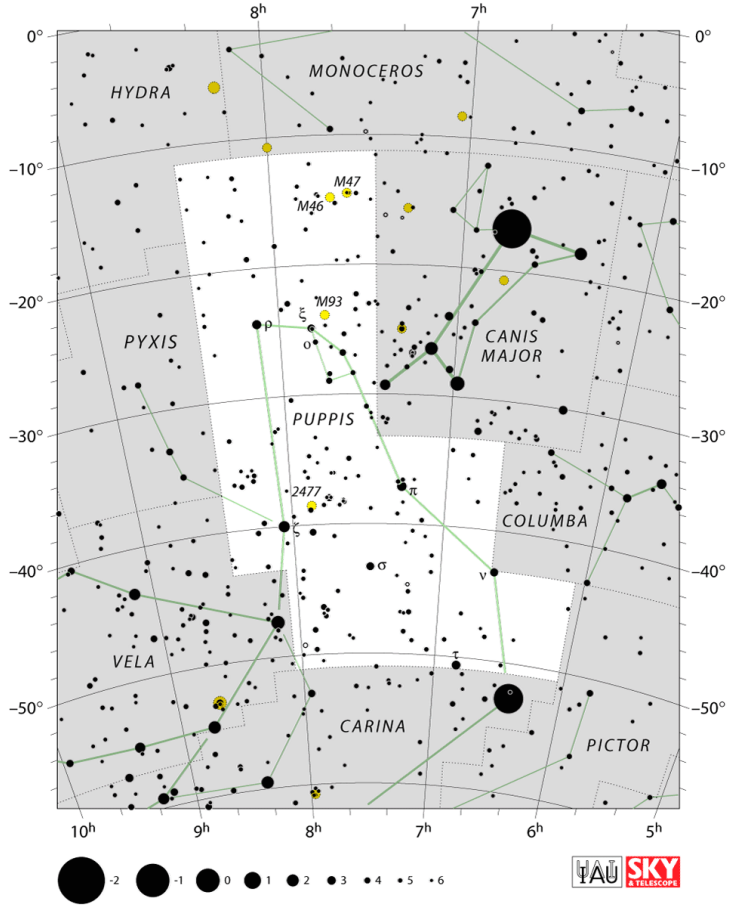
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 68/88  Puppis (Pup) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 39757 | 121.886250 | -24.304444 | HIP 38146, HIP 39429 |
| HIP 38146 | 117.257083 | -24.912250 | HIP 39757, HIP 35264 |
| HIP 35264 | 109.285667 | -37.097500 | HIP 38146, HIP 31685 |
| HIP 31685 | 99.440292 | -43.195917 | HIP 35264, HIP 32768 |
| HIP 32768 | 102.483917 | -50.614389 | HIP 31685, HIP 36377 |
| HIP 36377 | 112.307833 | -43.301889 | HIP 32768, HIP 39429 |
| HIP 39429 | 120.896125 | -40.003194 | HIP 36377, HIP 39757 |

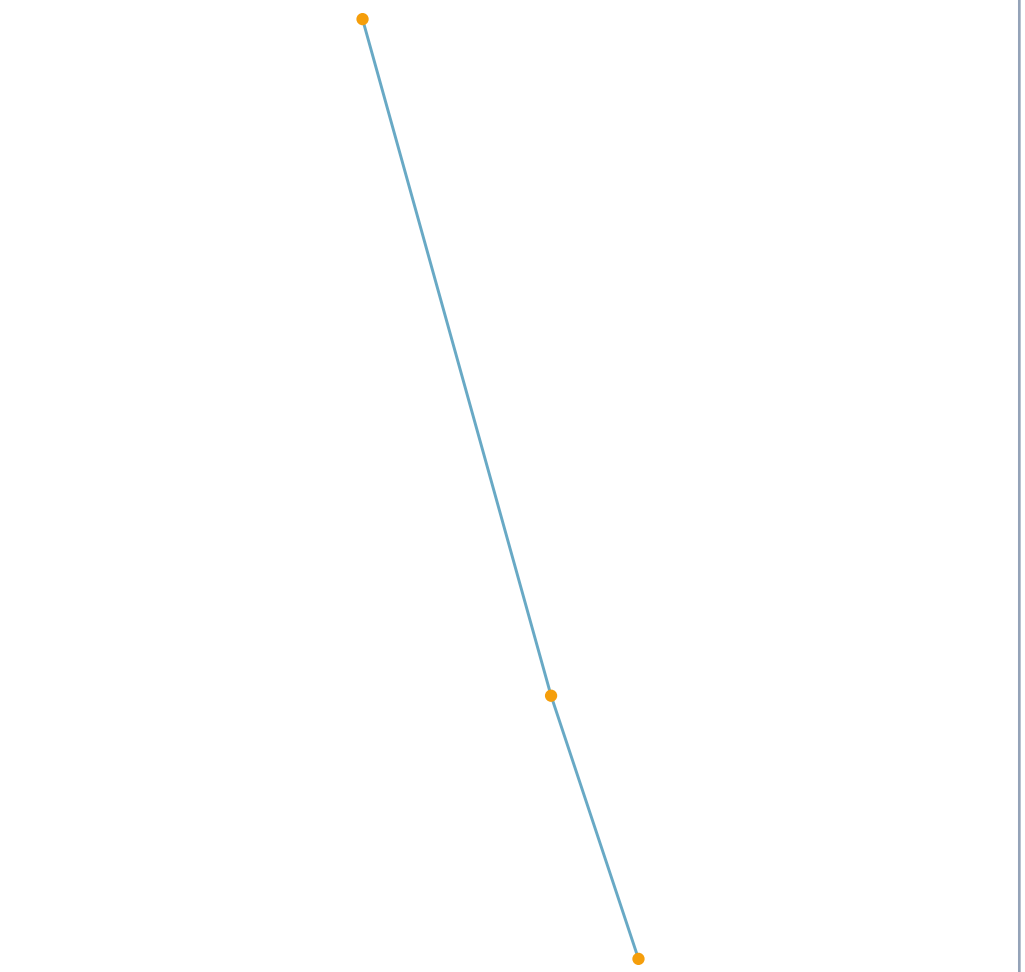
Effective release edges: 7 • unique plotted points: 7 • source: release_effective_edges.csv

# 69/88  Pyxis (Pyx) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

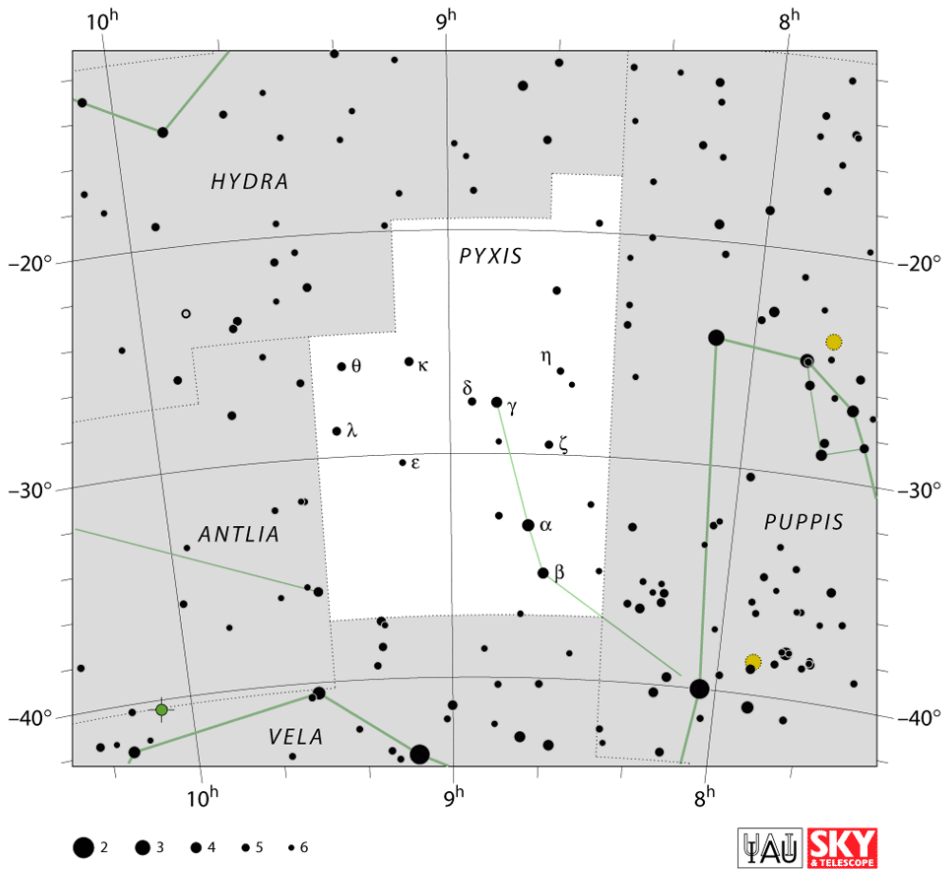
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review:  [ ] Correct  [ ] Incorrect  [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 69/88  Pyxis (Pyx) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 42515 | 130.025583 | -35.308306 | HIP 42828 |
| HIP 42828 | 130.898125 | -33.186417 | HIP 42515, HIP 43409 |
| HIP 43409 | 132.633375 | -27.710056 | HIP 42828 |

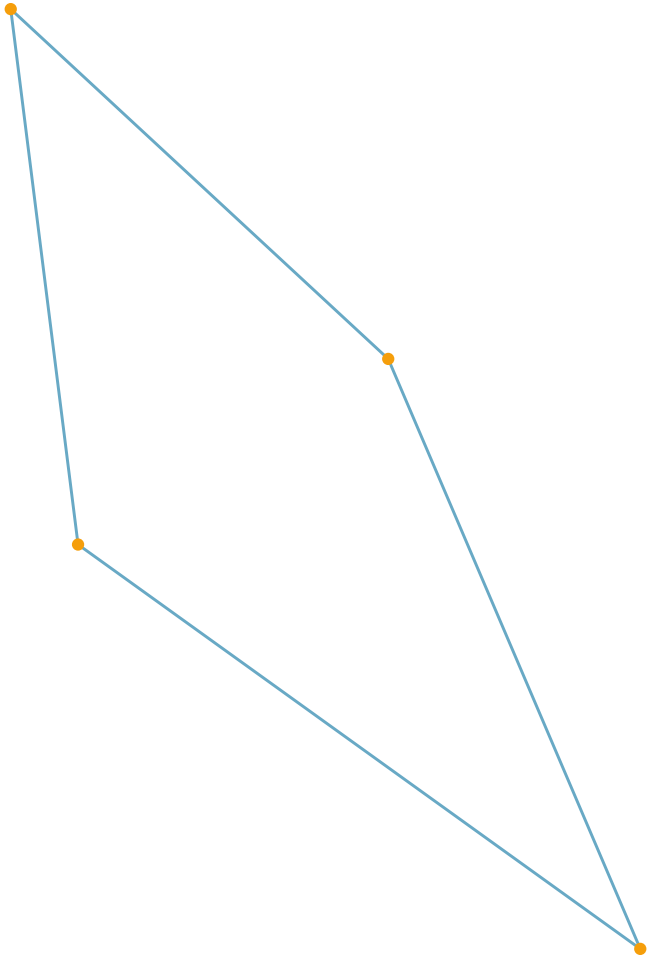
Effective release edges: 2 • unique plotted points: 3 • source: release_effective_edges.csv

# 70/88 Reticulum (Ret) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

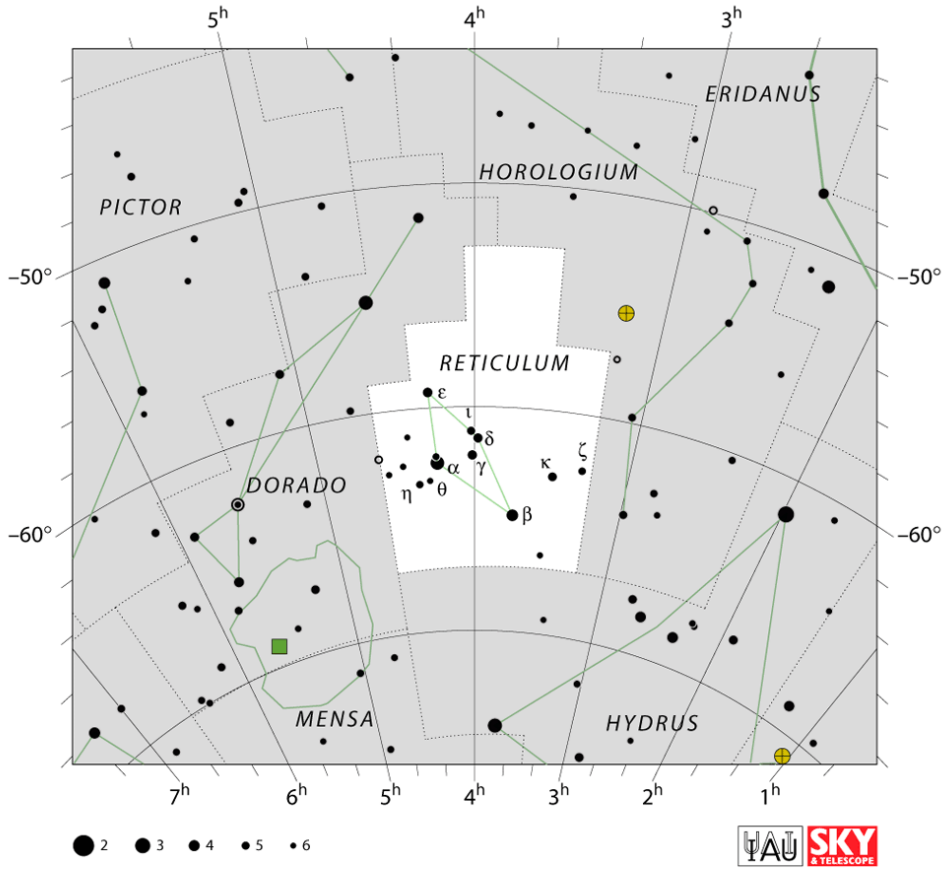
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 70/88  Reticulum (Ret) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 19780 | 63.605958 | -62.473972 | HIP 19921, HIP 17440 |
| HIP 19921 | 64.121167 | -59.301750 | HIP 19780, HIP 18597 |
| HIP 18597 | 59.686417 | -61.400139 | HIP 19921, HIP 17440 |
| HIP 17440 | 56.048125 | -64.807083 | HIP 18597, HIP 19780 |

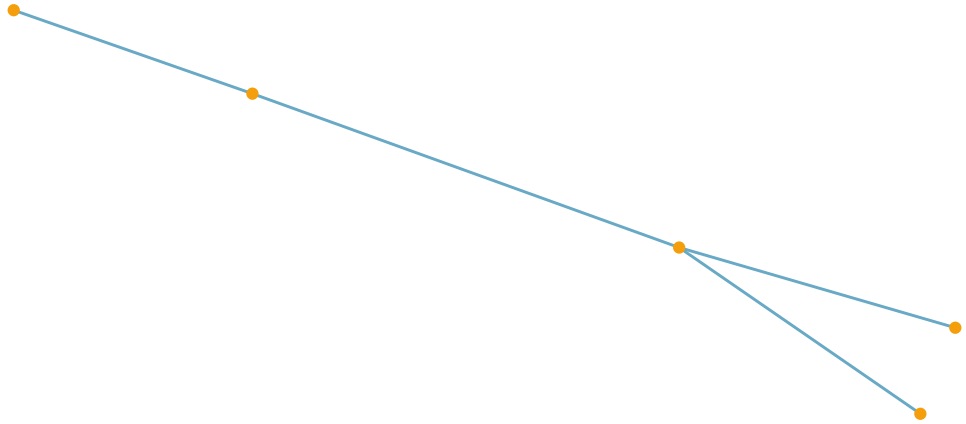
Effective release edges: 4 • unique plotted points: 4 • source: release_effective_edges.csv

# 71/88  Sagitta (Sge) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

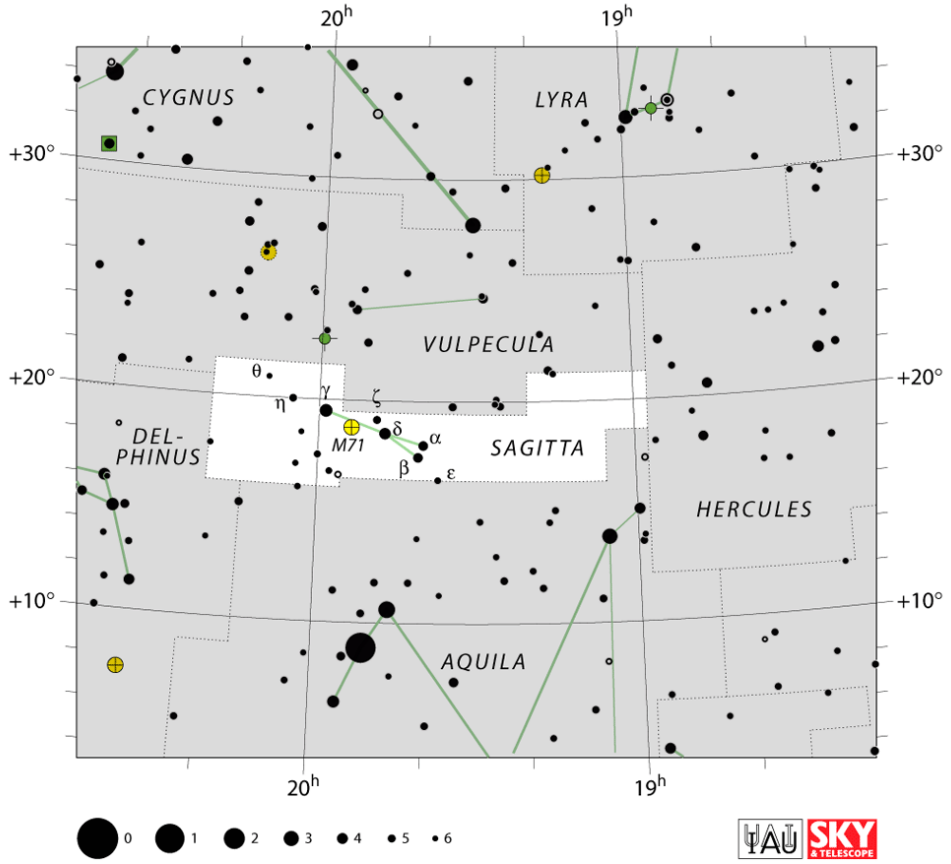
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review:  [ ] Correct  [ ] Incorrect  [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 71/88  Sagitta (Sge) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 96837 | 295.262208 | +17.476111 | HIP 97365 |
| HIP 97365 | 296.846958 | +18.534250 | HIP 96837, HIP 96757, HIP 98337 |
| HIP 96757 | 295.024083 | +18.013944 | HIP 97365 |
| HIP 98337 | 299.689125 | +19.492083 | HIP 97365, HIP 98920 |
| HIP 98920 | 301.289458 | +19.990889 | HIP 98337 |

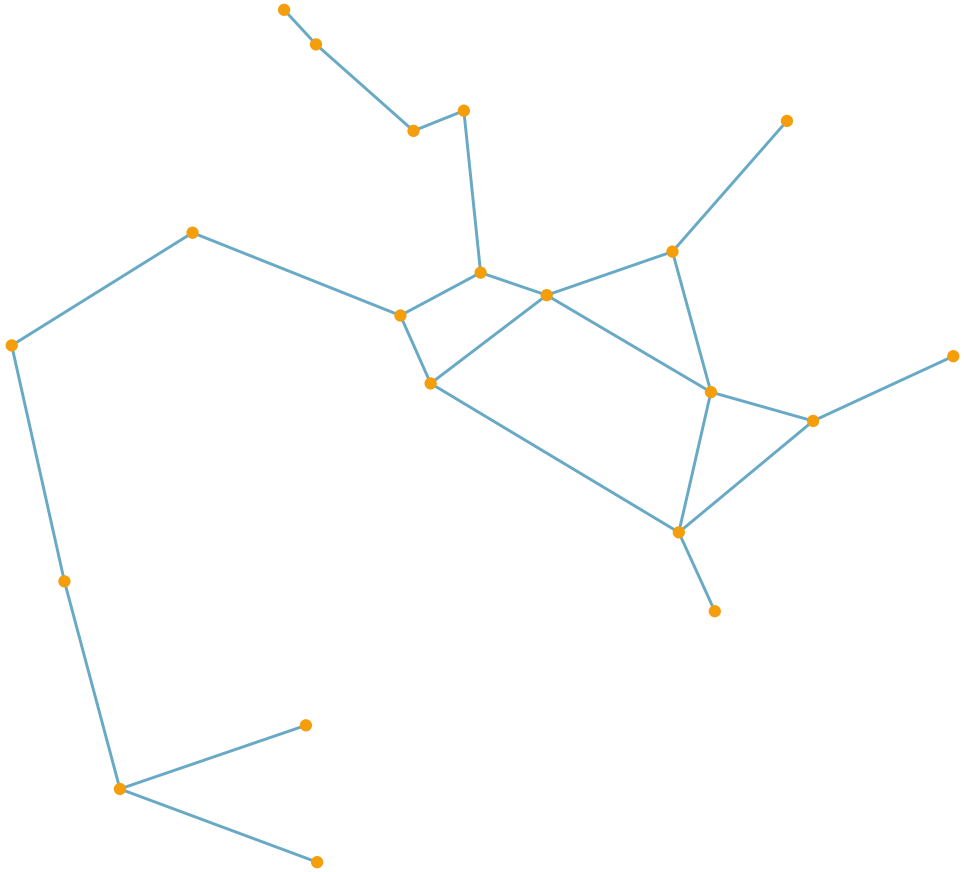
Effective release edges: 4 • unique plotted points: 5 • source: release_effective_edges.csv

# 72/88 Sagittarius (Sgr) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

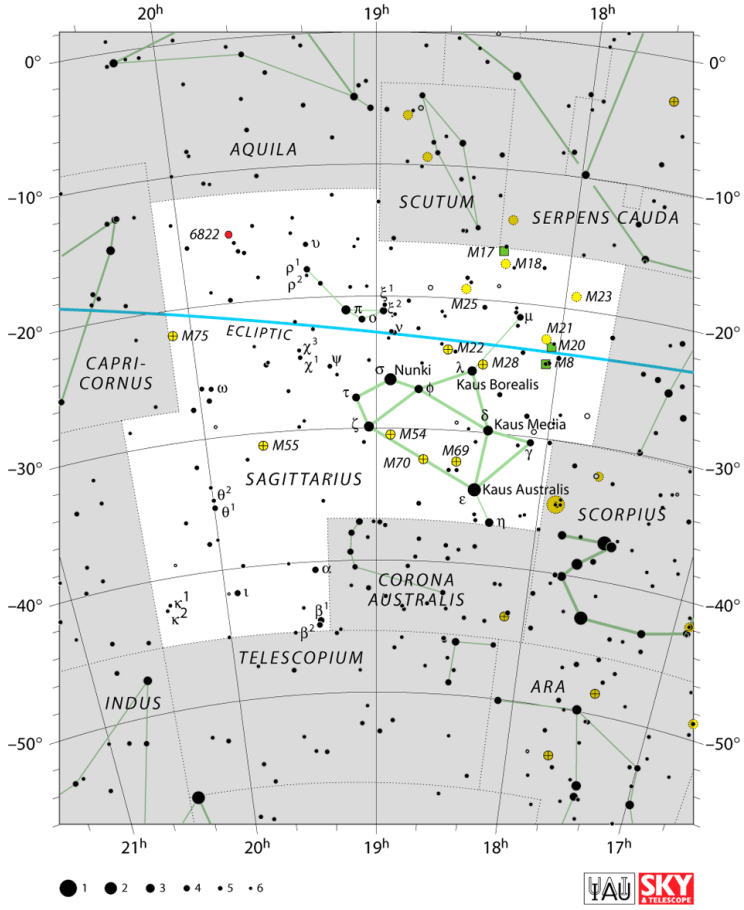
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 72/88  Sagittarius (Sgr) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 89931 | 275.248417 | -29.828028 | HIP 90496, HIP 88635, HIP 90185, HIP 92041 |
| HIP 90496 | 276.992792 | -25.421250 | HIP 89931, HIP 92041, HIP 89341 |
| HIP 89642 | 274.407208 | -36.761278 | HIP 90185 |
| HIP 90185 | 276.043125 | -34.384306 | HIP 89642, HIP 88635, HIP 89931, HIP 93506 |
| HIP 88635 | 271.452167 | -30.423639 | HIP 90185, HIP 87072, HIP 89931 |
| HIP 87072 | 266.890125 | -27.830750 | HIP 88635 |
| HIP 93506 | 285.653000 | -29.880111 | HIP 90185, HIP 92041, HIP 93864 |
| HIP 92041 | 281.413958 | -26.990778 | HIP 93506, HIP 89931, HIP 90496, HIP 92855 |
| HIP 89341 | 273.440875 | -21.058833 | HIP 90496 |
| HIP 93864 | 286.735167 | -27.669806 | HIP 93506, HIP 92855, HIP 96406 |
| HIP 92855 | 283.816333 | -26.296583 | HIP 93864, HIP 92041, HIP 93085 |
| HIP 93085 | 284.432417 | -21.106611 | HIP 92855, HIP 93683 |
| HIP 93683 | 286.170542 | -21.741361 | HIP 93085, HIP 94820 |
| HIP 94820 | 289.408708 | -18.952889 | HIP 93683, HIP 95168 |
| HIP 95168 | 290.418250 | -17.847250 | HIP 94820 |
| HIP 96406 | 294.006875 | -24.719028 | HIP 93864, HIP 98688 |
| HIP 98688 | 300.664417 | -27.709889 | HIP 96406, HIP 98412 |
| HIP 98412 | 299.934042 | -35.276250 | HIP 98688, HIP 98032 |
| HIP 98032 | 298.815333 | -41.868417 | HIP 98412, HIP 95347, HIP 95294 |
| HIP 95347 | 290.971458 | -40.615639 | HIP 98032 |
| HIP 95294 | 290.804417 | -44.799639 | HIP 98032 |

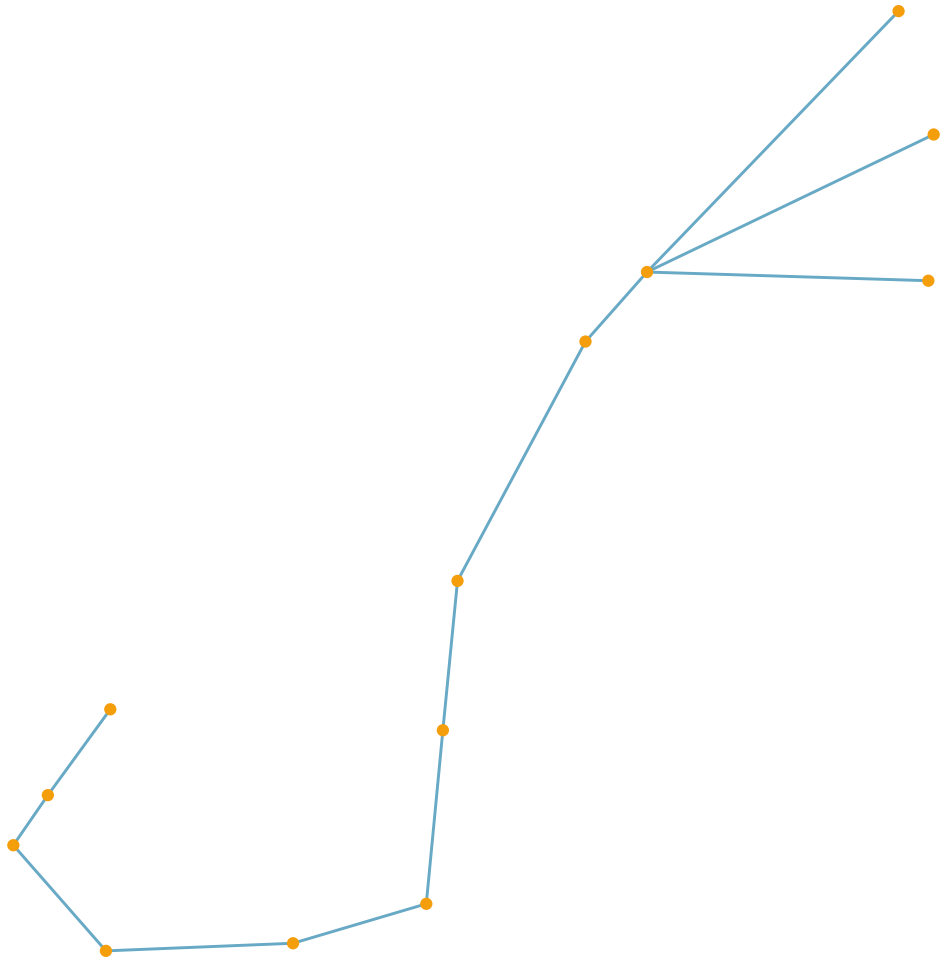
Effective release edges: 24 • unique plotted points: 21 • source: release_effective_edges.csv

# 73/88  Scorpius (Sco) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

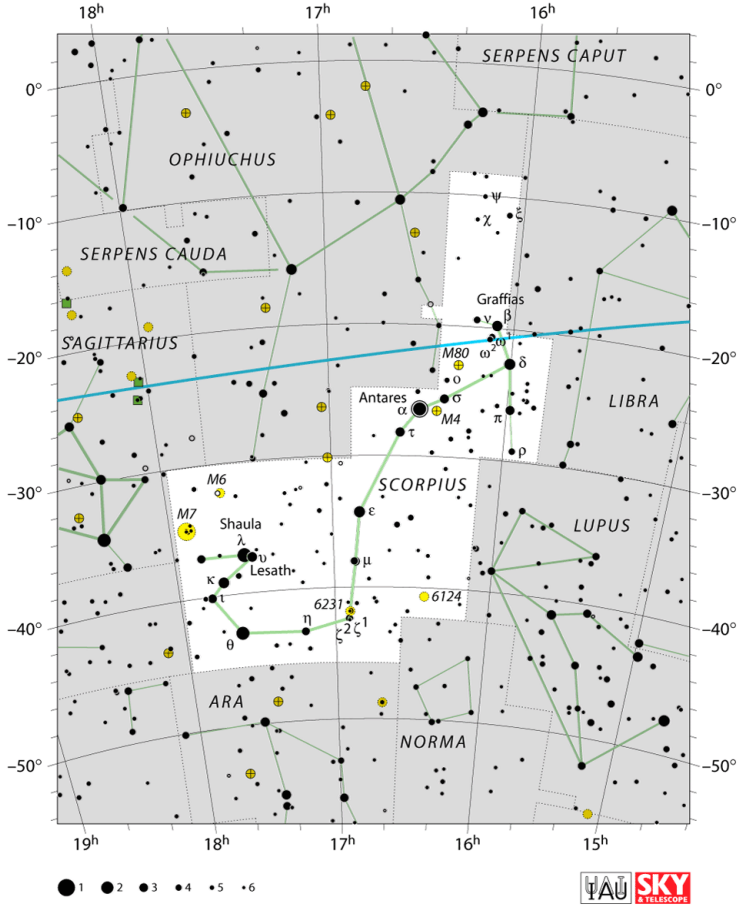
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review:  [ ] Correct  [ ] Incorrect  [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 73/88  Scorpius (Sco) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 85927 | 263.402208 | -37.103750 | HIP 86670 |
| HIP 86670 | 265.622000 | -39.029917 | HIP 85927, HIP 87073 |
| HIP 87073 | 266.896167 | -40.126972 | HIP 86670, HIP 86228 |
| HIP 86228 | 264.329708 | -42.997833 | HIP 87073, HIP 84143 |
| HIP 84143 | 258.038250 | -43.238500 | HIP 86228, HIP 82671 |
| HIP 82671 | 253.498875 | -42.362028 | HIP 84143, HIP 82514 |
| HIP 82514 | 252.967667 | -38.047333 | HIP 82671, HIP 82396 |
| HIP 82396 | 252.542667 | -34.292611 | HIP 82514, HIP 81266 |
| HIP 81266 | 248.970667 | -28.215972 | HIP 82396, HIP 80763 |
| HIP 80763 | 247.351958 | -26.431944 | HIP 81266, HIP 78401, HIP 78265, HIP 78820 |
| HIP 78401 | 240.083375 | -22.621611 | HIP 80763 |
| HIP 78265 | 239.713000 | -26.114056 | HIP 80763 |
| HIP 78820 | 241.359292 | -19.805389 | HIP 80763 |

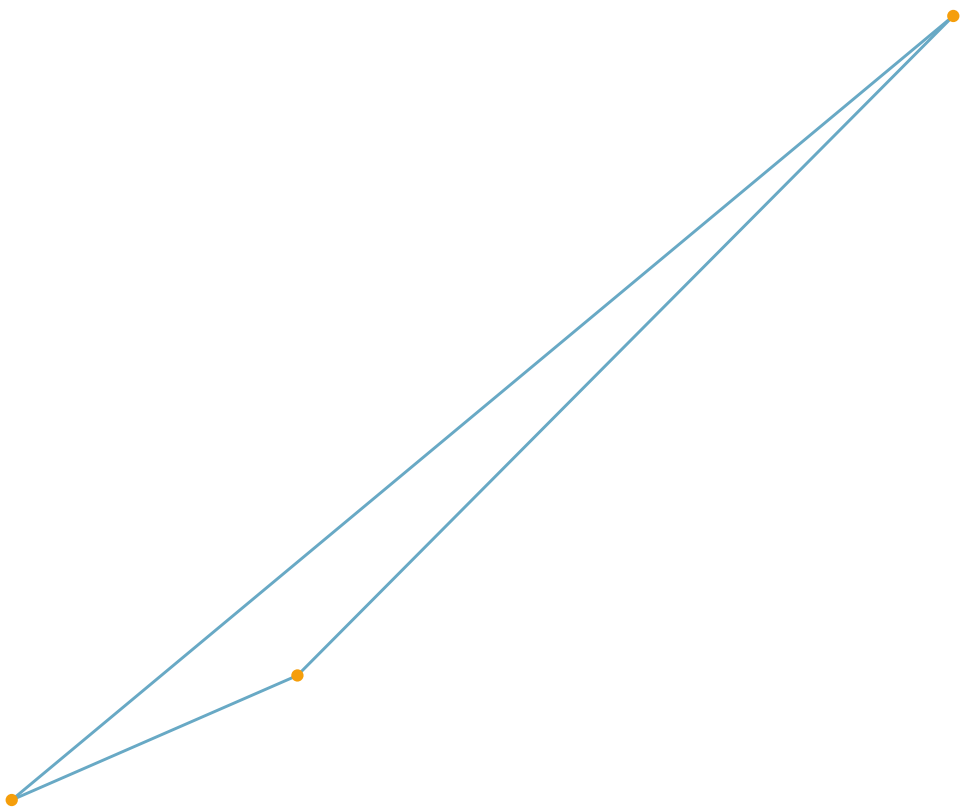
Effective release edges: 12 • unique plotted points: 13 • source: release_effective_edges.csv

# 74/88 Sculptor (Scl) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

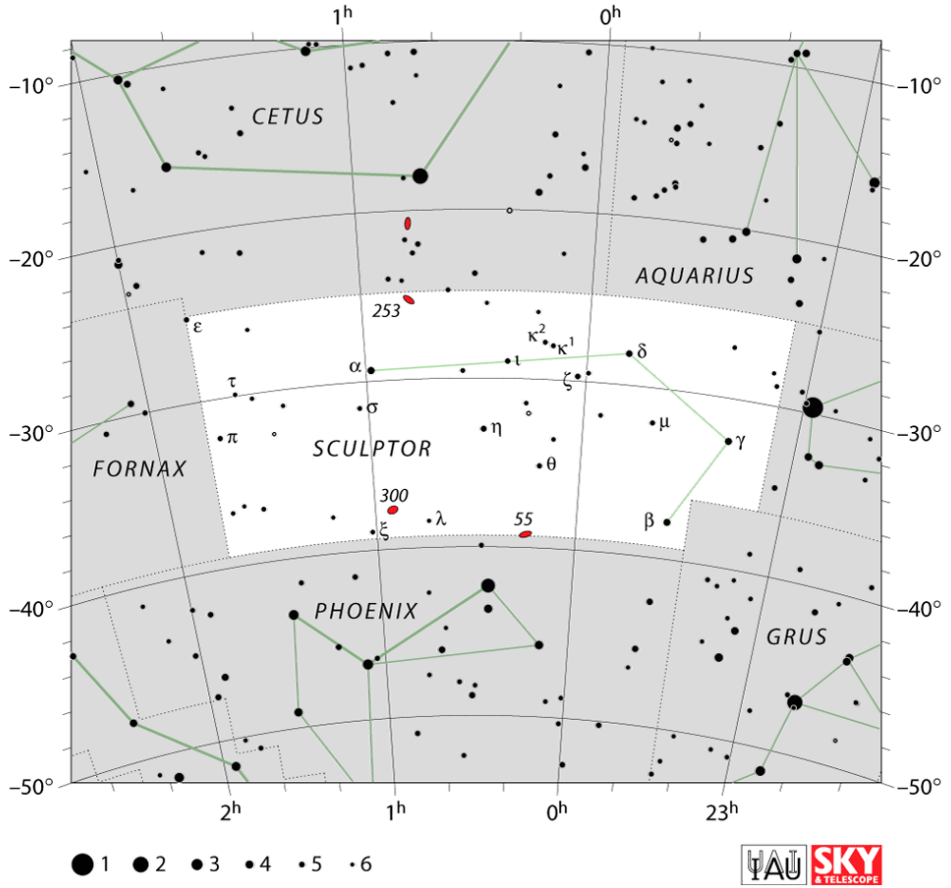
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 74/88  Sculptor (Scl) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 116231 | 353.242458 | -37.818361 | HIP 4577, HIP 115102 |
| HIP 4577 | 14.651458 | -29.357472 | HIP 116231, HIP 115102 |
| HIP 115102 | 349.705958 | -32.531833 | HIP 4577, HIP 116231 |

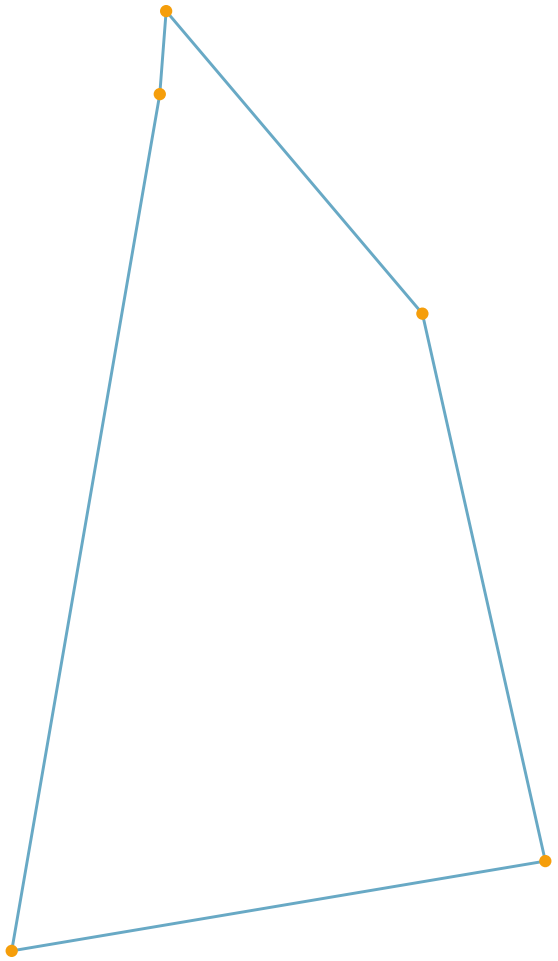
Effective release edges: 3 • unique plotted points: 3 • source: release_effective_edges.csv

# 75/88 Scutum (Sct) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

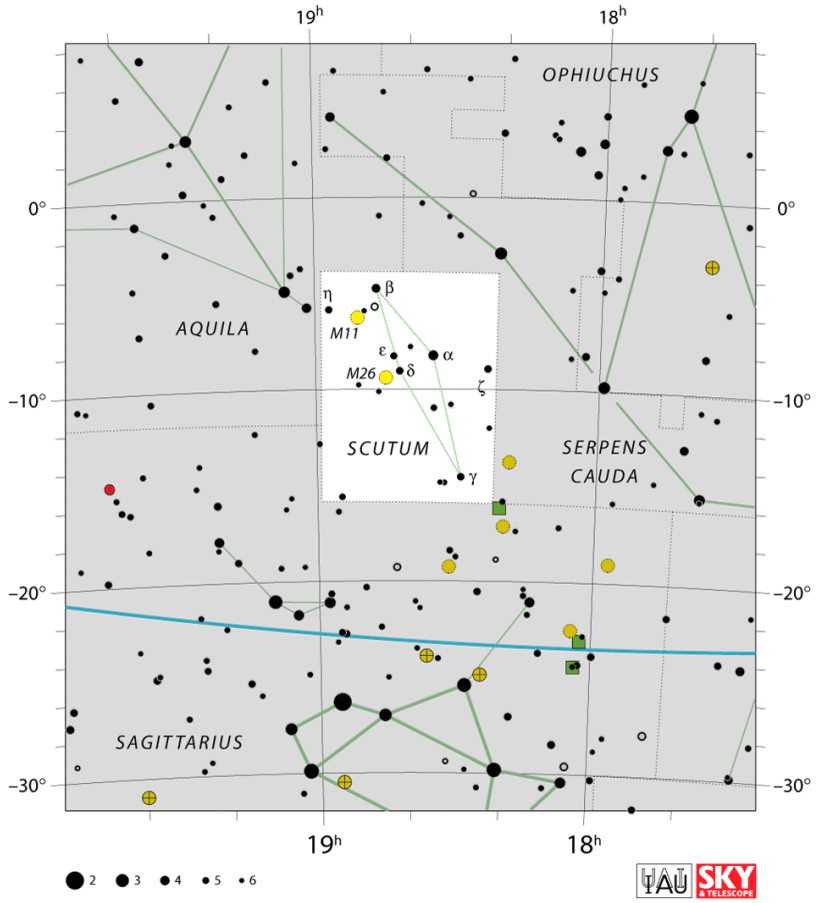
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 75/88  Scutum (Sct) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 92175 | 281.793667 | -4.747833 | HIP 92202, HIP 91117 |
| HIP 92202 | 281.870750 | -5.705083 | HIP 92175, HIP 92814 |
| HIP 92814 | 283.679667 | -15.603028 | HIP 92202, HIP 90595 |
| HIP 90595 | 277.299375 | -14.565806 | HIP 92814, HIP 91117 |
| HIP 91117 | 278.801833 | -8.243306 | HIP 90595, HIP 92175 |

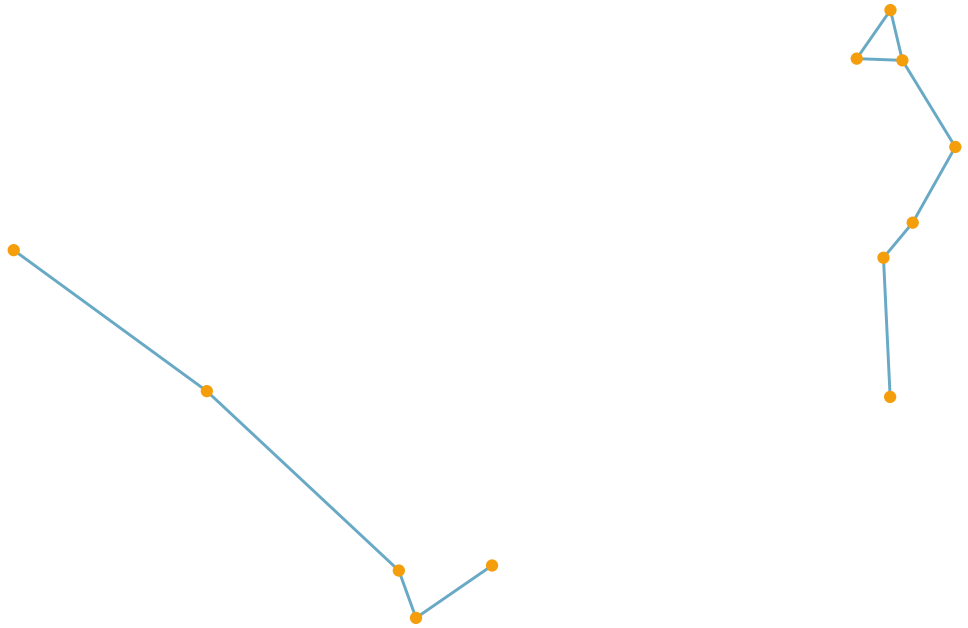
Effective release edges: 5 • unique plotted points: 5 • source: release_effective_edges.csv

# 76/88  Serpens (Ser) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

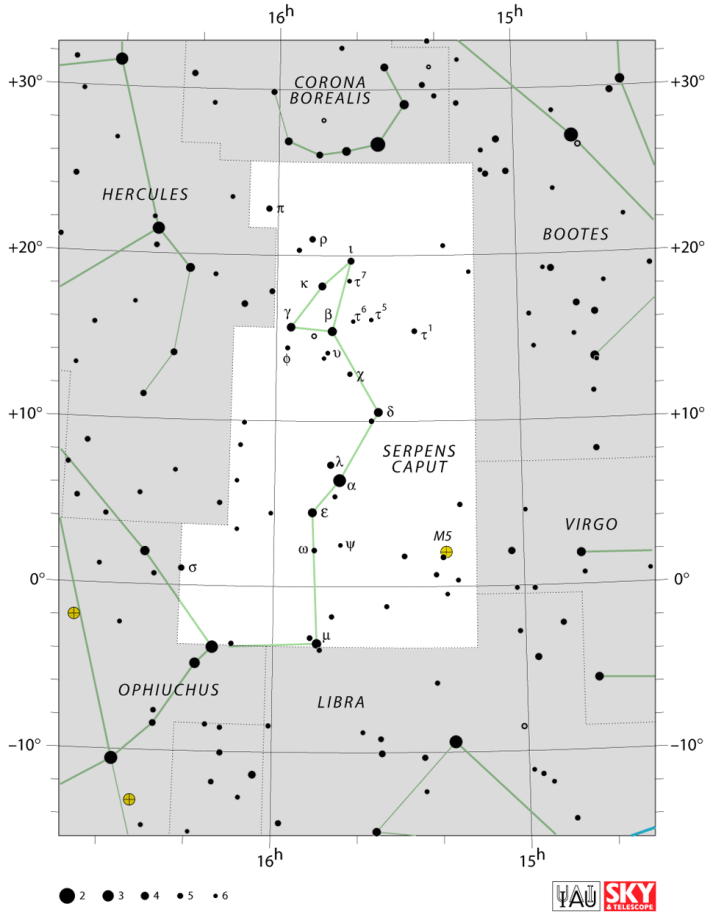
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review:  [ ] Correct  [ ] Incorrect  [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 76/88  Serpens (Ser) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 77516 | 237.405292 | -3.430139 | HIP 77622 |
| HIP 77622 | 237.703708 | +4.477583 | HIP 77516, HIP 77070 |
| HIP 77070 | 236.066667 | +6.425528 | HIP 77622, HIP 76276 |
| HIP 76276 | 233.700792 | +10.538861 | HIP 77070, HIP 77233 |
| HIP 77233 | 236.546708 | +15.421917 | HIP 76276, HIP 78072, HIP 77450 |
| HIP 78072 | 239.112458 | +15.664722 | HIP 77233, HIP 77450 |
| HIP 77450 | 237.185042 | +18.141778 | HIP 78072, HIP 77233 |
| HIP 92946 | 284.054833 | +4.203528 | HIP 89962 |
| HIP 89962 | 275.328833 | -2.897111 | HIP 92946, HIP 86565 |
| HIP 86565 | 265.353833 | -12.875167 | HIP 89962, HIP 86263 |
| HIP 86263 | 264.396792 | -15.398417 | HIP 86565, HIP 84880 |
| HIP 84880 | 260.206833 | -12.846889 | HIP 86263 |

Effective release edges: 11 • unique plotted points: 12 • source: release_effective_edges.csv

# 77/88 Sextans (Sex) — side-by-side chart comparison

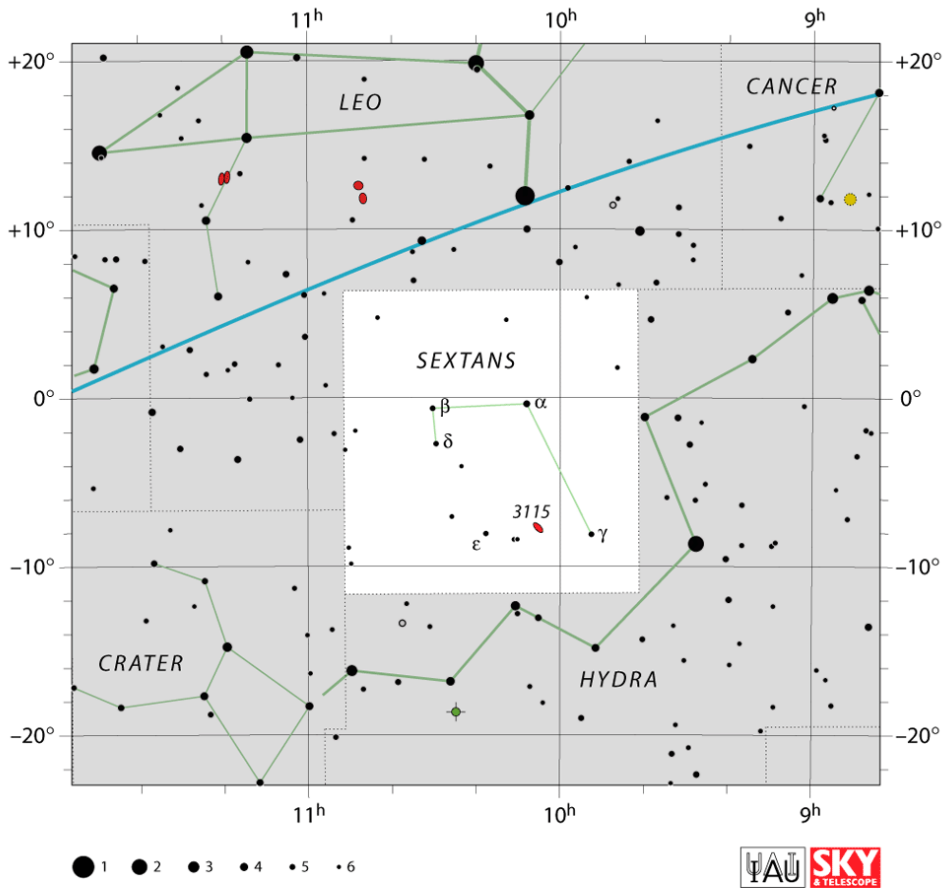
Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV

## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 77/88  Sextans (Sex) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 51437 | 157.572917 | +0.636972 | HIP 49641 |
| HIP 49641 | 151.984583 | +0.371639 | HIP 51437 |

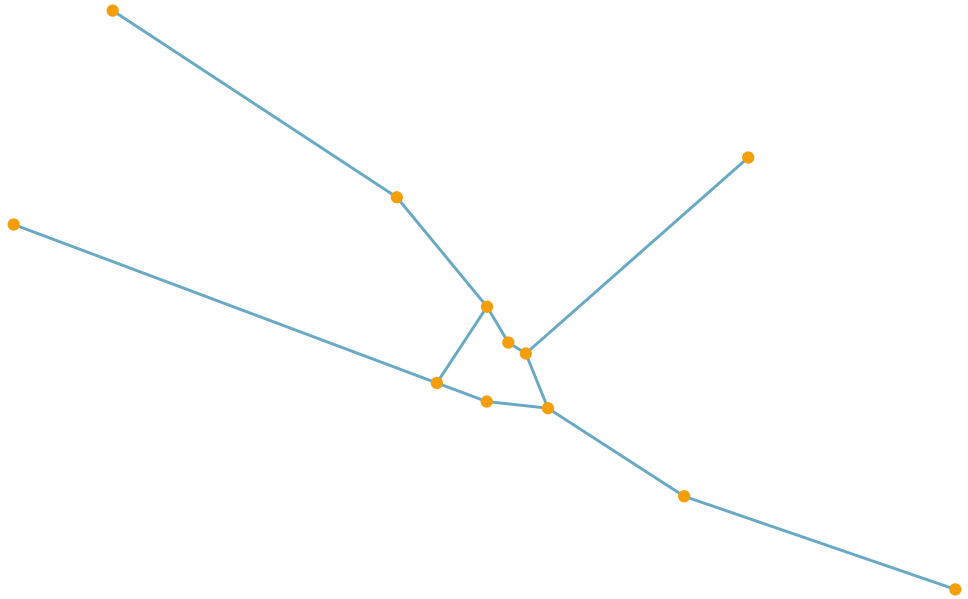
Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv

# 78/88 Taurus (Tau) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

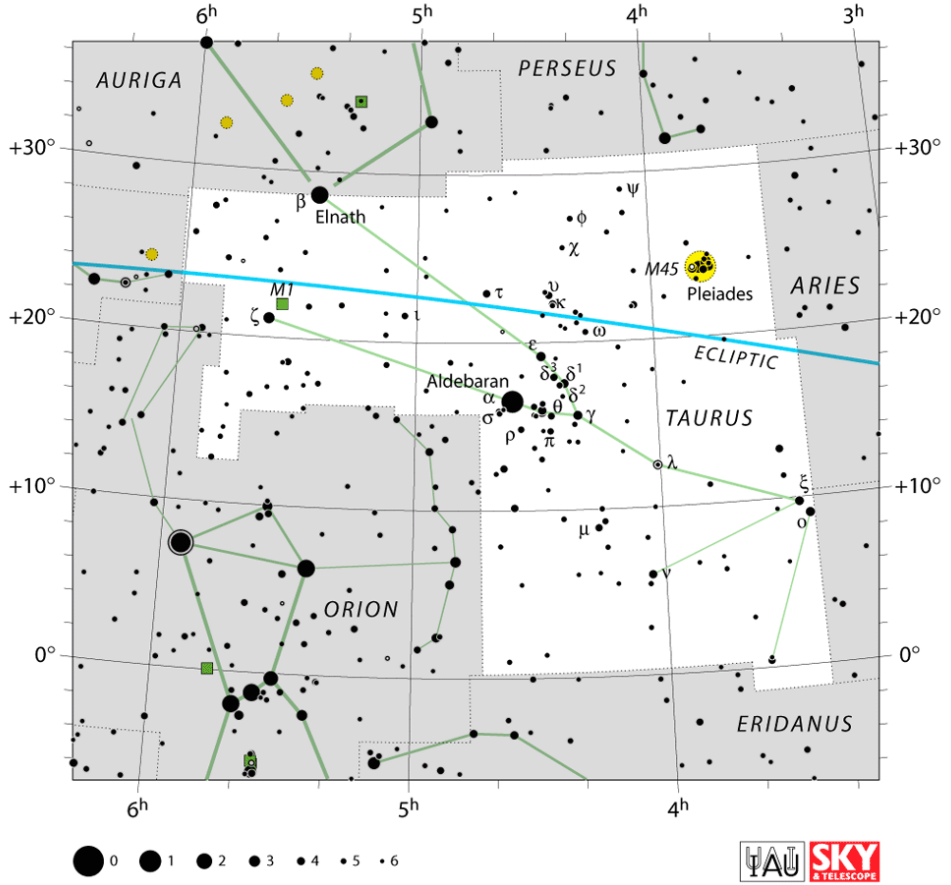
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 78/88  Taurus (Tau) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 25428 | 81.572917 | +28.607861 | HIP 21881 |
| HIP 21881 | 70.561250 | +22.956972 | HIP 25428, HIP 20889 |
| HIP 20889 | 67.153875 | +19.180528 | HIP 21881, HIP 21421, HIP 20648 |
| HIP 21421 | 68.980000 | +16.509750 | HIP 26451, HIP 20889, HIP 20894 |
| HIP 26451 | 84.411167 | +21.142583 | HIP 21421 |
| HIP 20205 | 64.948042 | +15.627694 | HIP 20455, HIP 18724, HIP 20894 |
| HIP 20455 | 65.733458 | +17.542583 | HIP 20205, HIP 20648, HIP 17847 |
| HIP 18724 | 60.170083 | +12.490389 | HIP 20205, HIP 15900 |
| HIP 15900 | 51.203500 | +9.029056 | HIP 18724 |
| HIP 20894 | 67.165292 | +15.870944 | HIP 21421, HIP 20205 |
| HIP 20648 | 66.372167 | +17.928000 | HIP 20889, HIP 20455 |
| HIP 17847 | 57.290542 | +24.053528 | HIP 20455 |

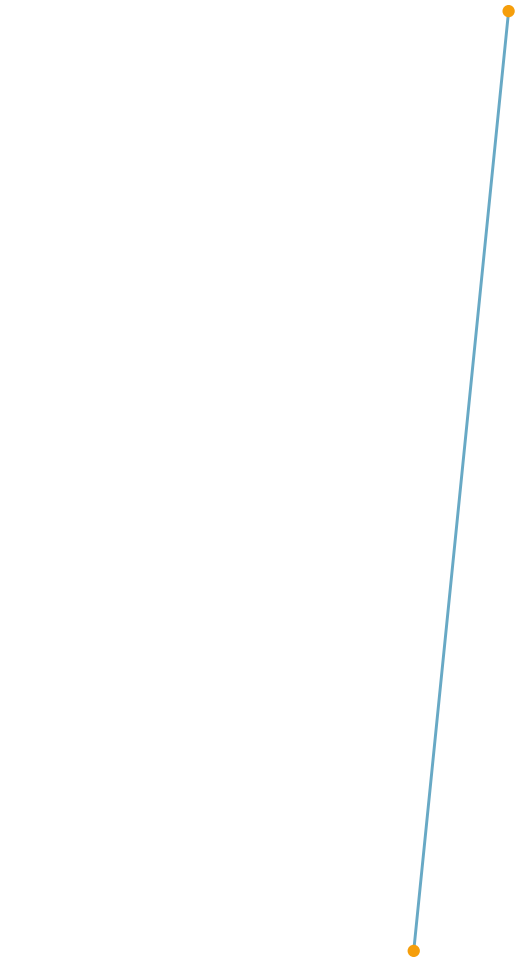
Effective release edges: 12 • unique plotted points: 12 • source: release_effective_edges.csv

# 79/88 Telescopium (Tel) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

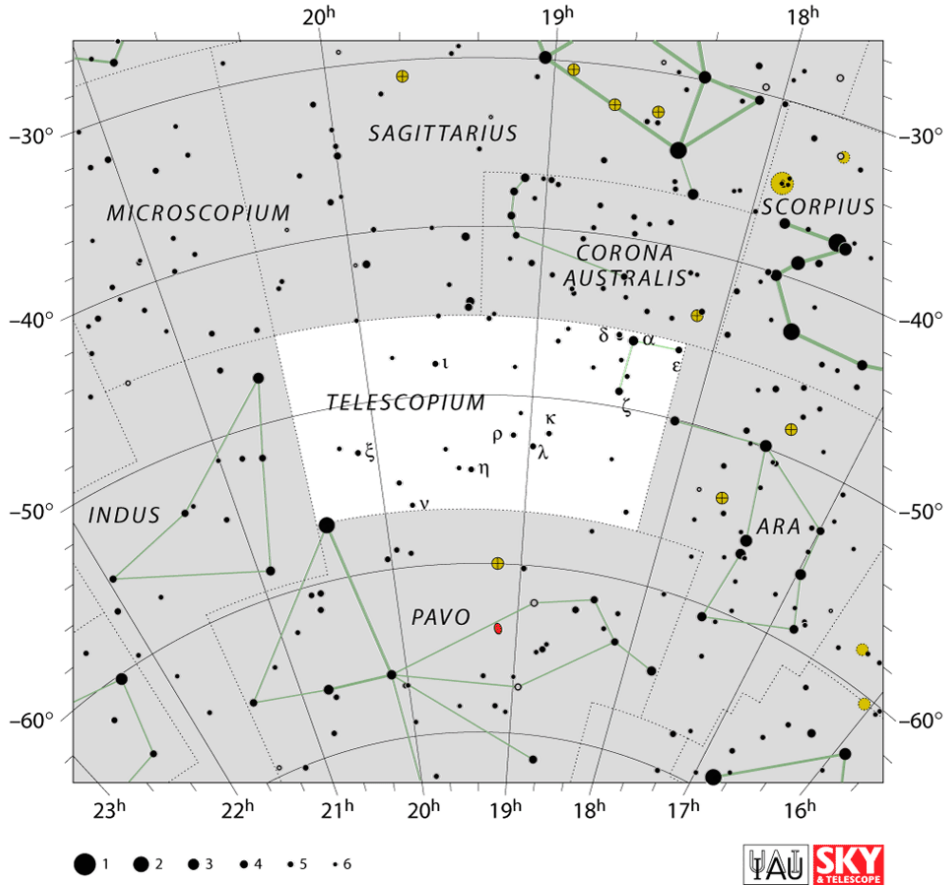
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 79/88  Telescopium (Tel) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 90568 | 277.207250 | -49.070028 | HIP 90422 |
| HIP 90422 | 276.743458 | -45.968333 | HIP 90568 |

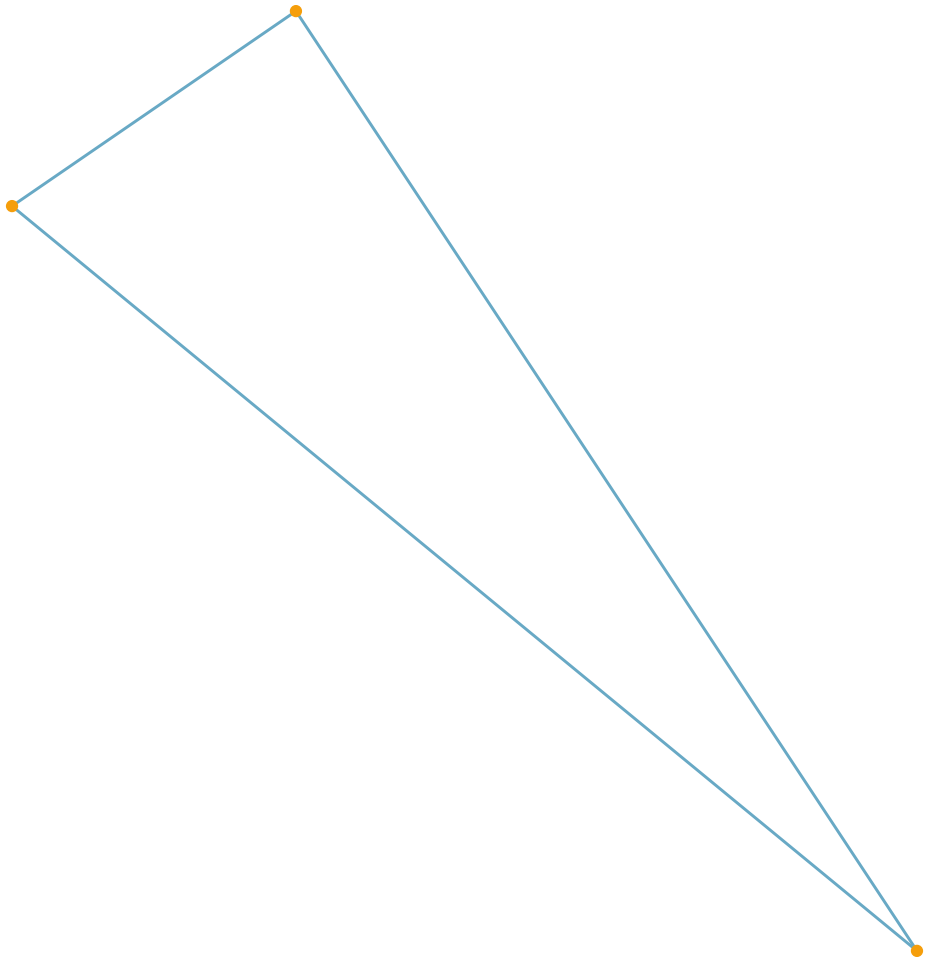
Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv

# 80/88  Triangulum (Tri) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

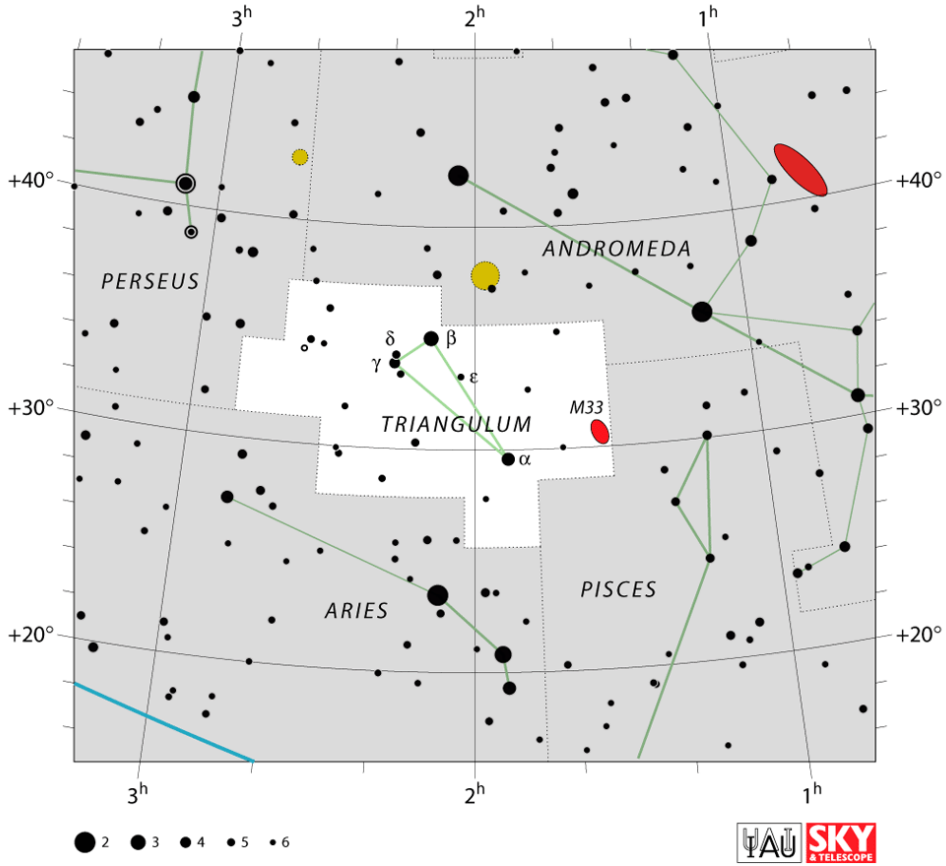
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review:  [ ] Correct  [ ] Incorrect  [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 80/88  Triangulum (Tri) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 10670 | 34.328500 | +33.847333 | HIP 10064, HIP 8796 |
| HIP 10064 | 32.385500 | +34.987389 | HIP 10670, HIP 8796 |
| HIP 8796 | 28.270417 | +29.579389 | HIP 10064, HIP 10670 |

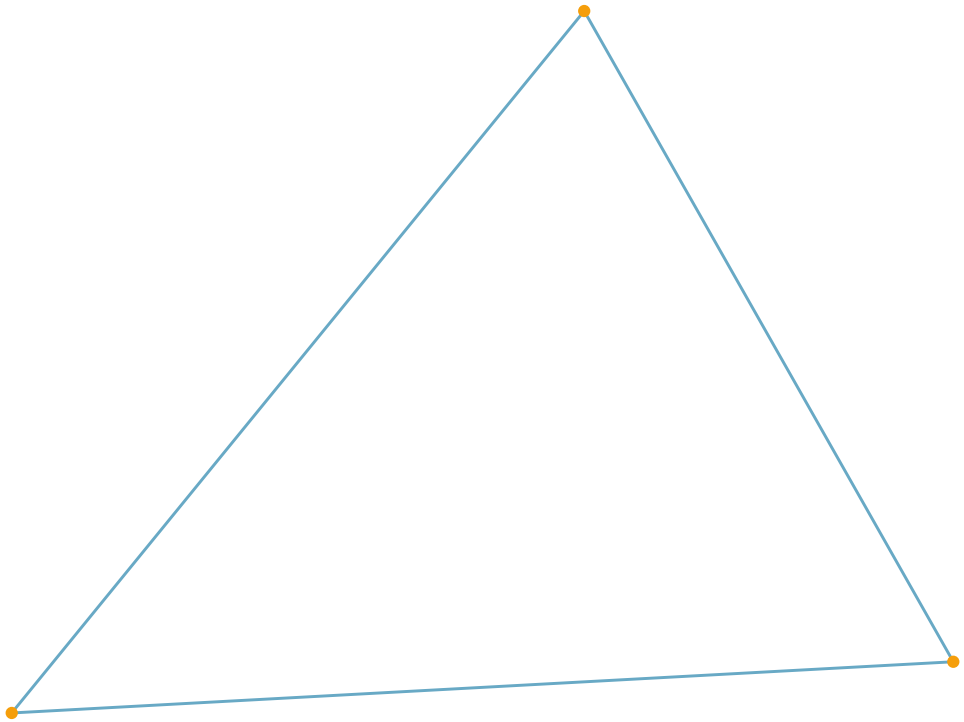
Effective release edges: 3 • unique plotted points: 3 • source: release_effective_edges.csv

# 81/88 Triangulum Australe (TrA) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

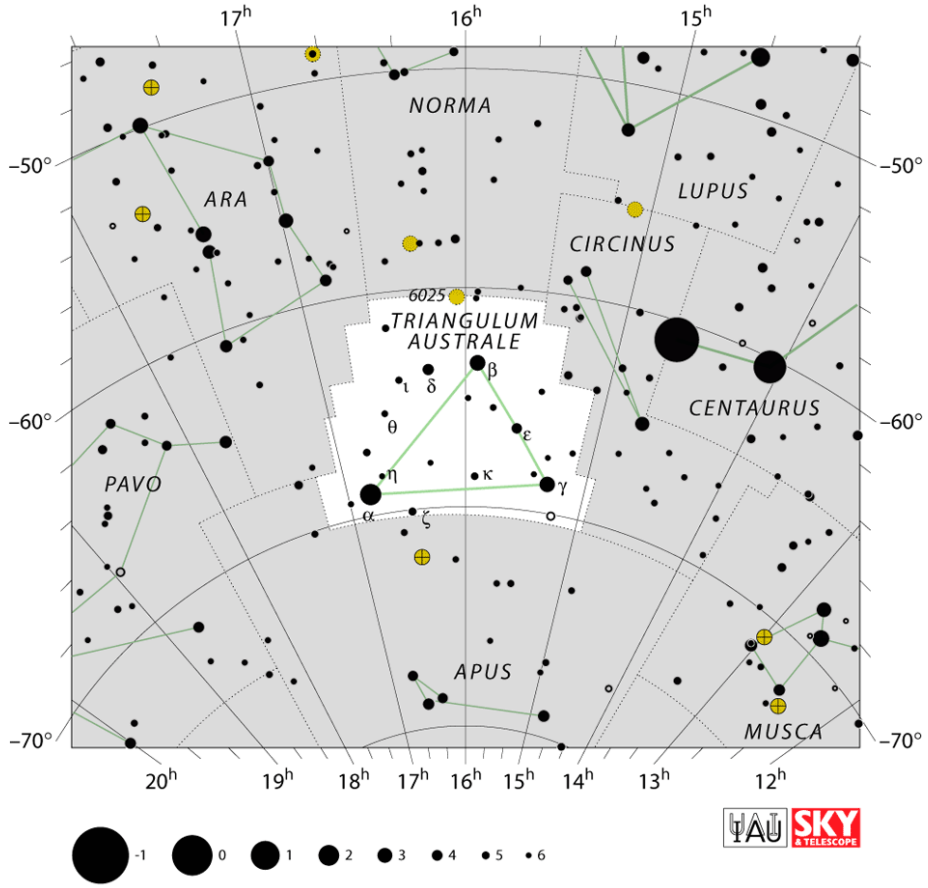
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 81/88  Triangulum Australe (TrA) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 82273 | 252.166125 | -69.027639 | HIP 74946, HIP 77952 |
| HIP 74946 | 229.727875 | -68.679472 | HIP 82273, HIP 77952 |
| HIP 77952 | 238.786708 | -63.429750 | HIP 74946, HIP 82273 |

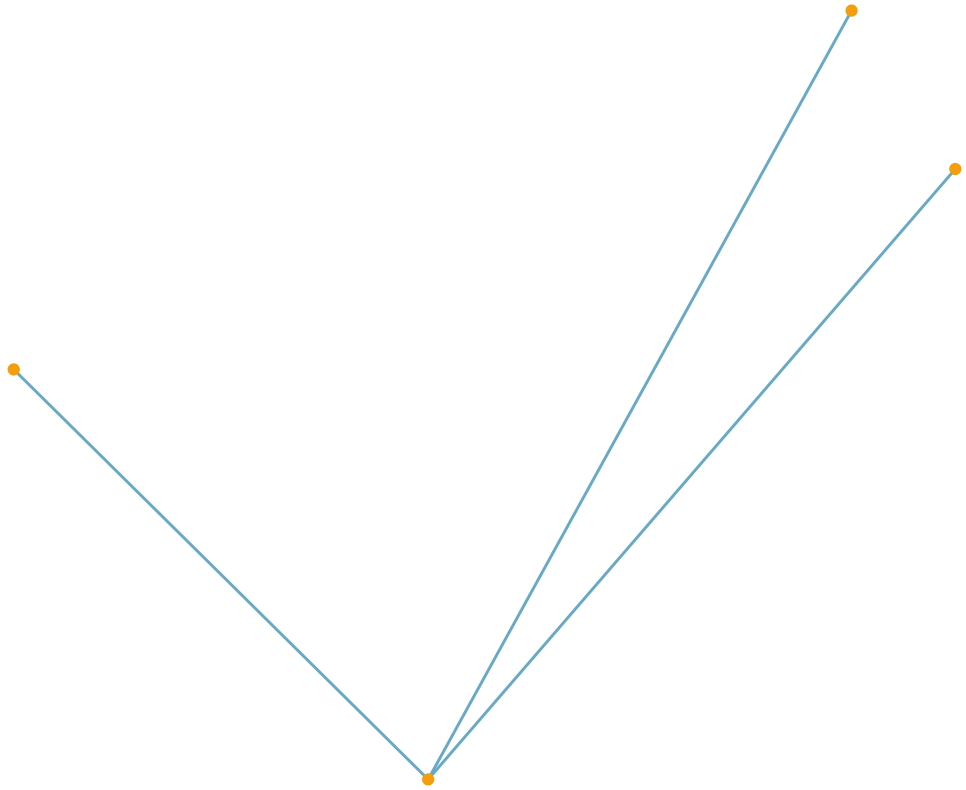
Effective release edges: 3 • unique plotted points: 3 • source: release_effective_edges.csv

# 82/88 Tucana (Tuc) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

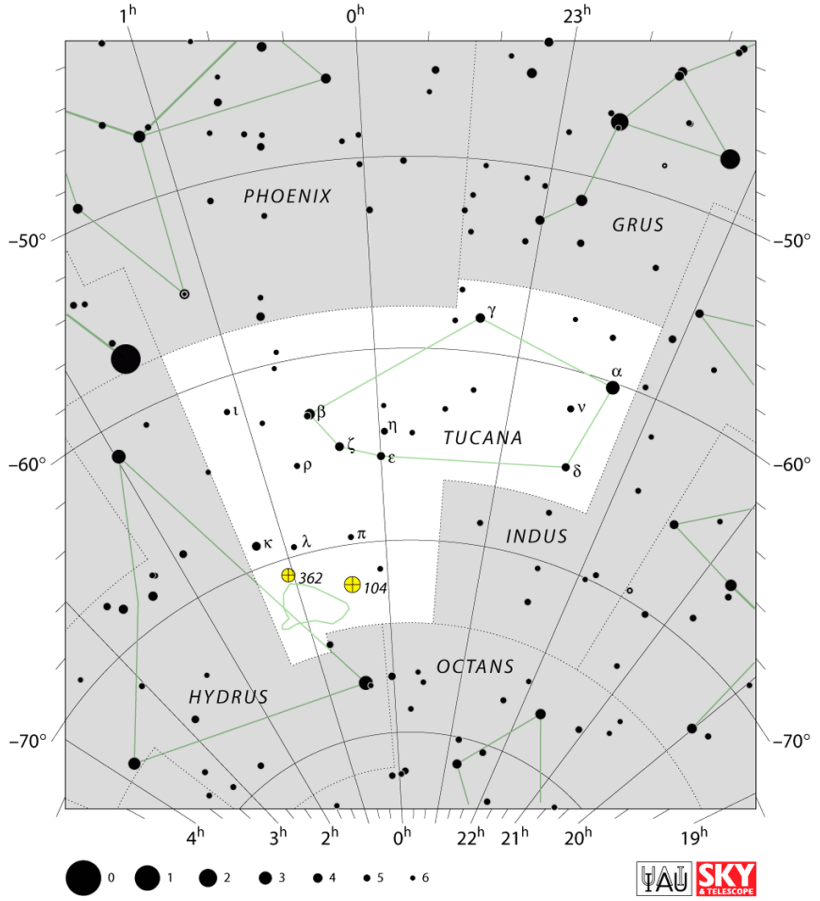
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 82/88  Tucana (Tuc) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 110130 | 334.625750 | -60.259500 | HIP 114996 |
| HIP 114996 | 349.357542 | -58.235917 | HIP 110130, HIP 1599, HIP 2484 |
| HIP 1599 | 5.007958 | -64.877611 | HIP 114996 |
| HIP 2484 | 7.885667 | -62.958083 | HIP 114996 |

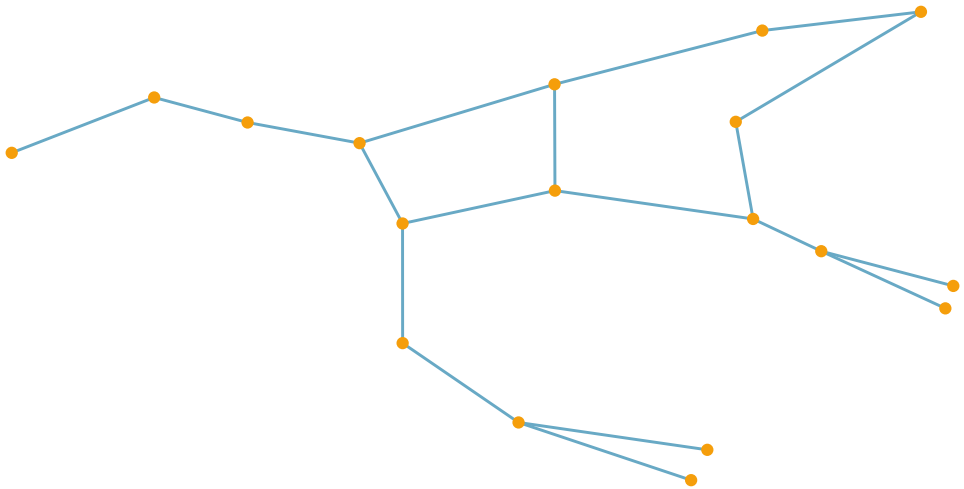
Effective release edges: 3 • unique plotted points: 4 • source: release_effective_edges.csv

# 83/88 Ursa Major (UMa) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

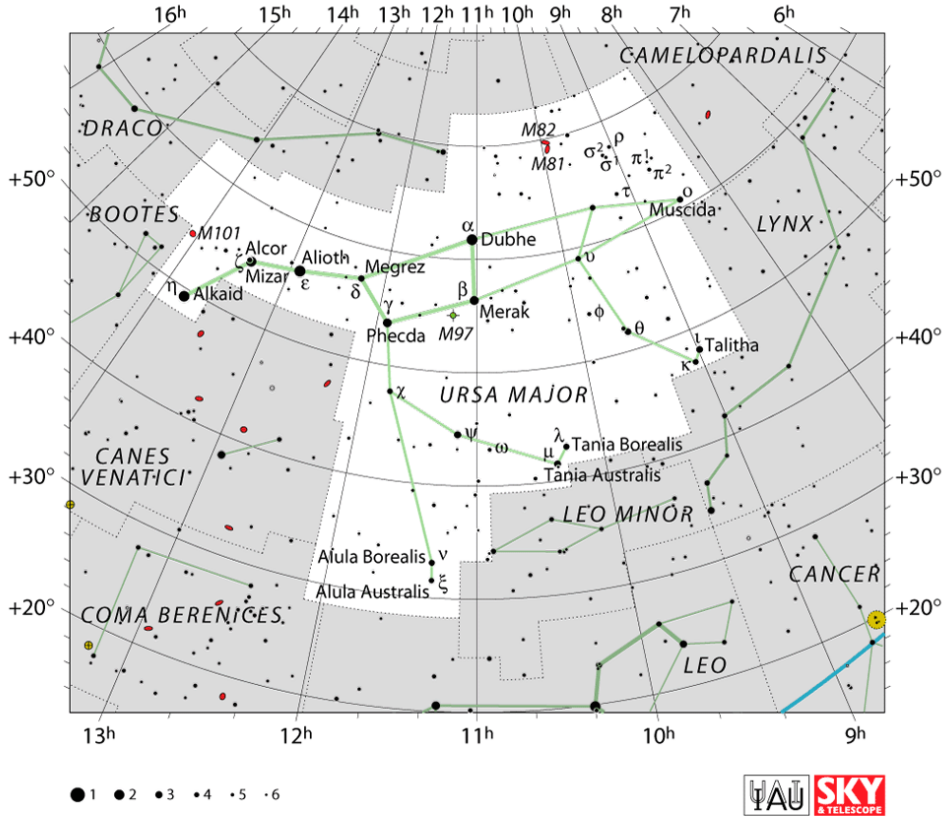
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 83/88  Ursa Major (UMa) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 67301 | 206.885625 | +49.313306 | HIP 65378 |
| HIP 65378 | 200.980917 | +54.925417 | HIP 67301, HIP 62956 |
| HIP 62956 | 193.506792 | +55.959833 | HIP 65378, HIP 59774 |
| HIP 59774 | 183.856042 | +57.032611 | HIP 62956, HIP 54061, HIP 58001 |
| HIP 54061 | 165.932667 | +61.751111 | HIP 59774, HIP 53910, HIP 46733 |
| HIP 53910 | 165.459958 | +56.382333 | HIP 54061, HIP 58001, HIP 48402 |
| HIP 58001 | 178.457250 | +53.694722 | HIP 53910, HIP 59774, HIP 57399 |
| HIP 57399 | 176.513042 | +47.779333 | HIP 58001, HIP 54539 |
| HIP 54539 | 167.416083 | +44.498556 | HIP 57399, HIP 50372, HIP 50801 |
| HIP 50372 | 154.274708 | +42.914472 | HIP 54539 |
| HIP 50801 | 155.582500 | +41.499444 | HIP 54539 |
| HIP 48402 | 148.026500 | +54.064278 | HIP 53910, HIP 46853, HIP 48319 |
| HIP 46853 | 143.218042 | +51.678611 | HIP 48402, HIP 44471, HIP 44127 |
| HIP 44471 | 135.906500 | +47.156667 | HIP 46853 |
| HIP 44127 | 134.803500 | +48.042361 | HIP 46853 |
| HIP 48319 | 147.748708 | +59.039111 | HIP 48402, HIP 41704 |
| HIP 41704 | 127.566792 | +60.718444 | HIP 48319, HIP 46733 |
| HIP 46733 | 142.881542 | +63.061806 | HIP 41704, HIP 54061 |

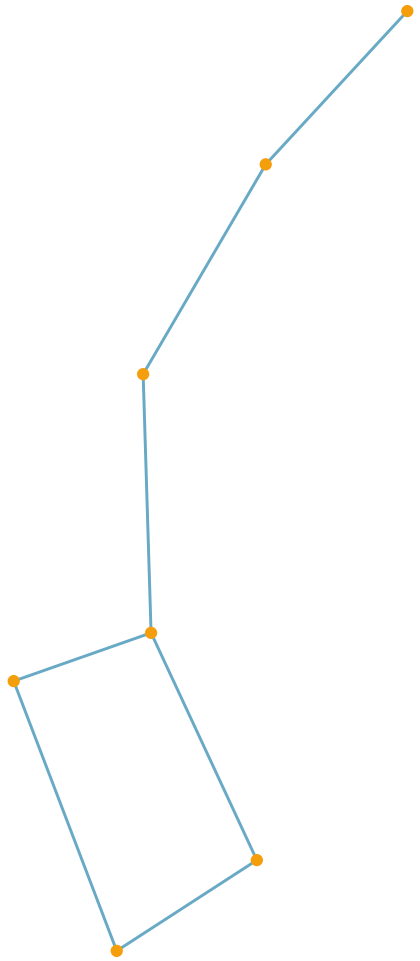
Effective release edges: 19 • unique plotted points: 18 • source: release_effective_edges.csv

# 84/88 Ursa Minor (UMi) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

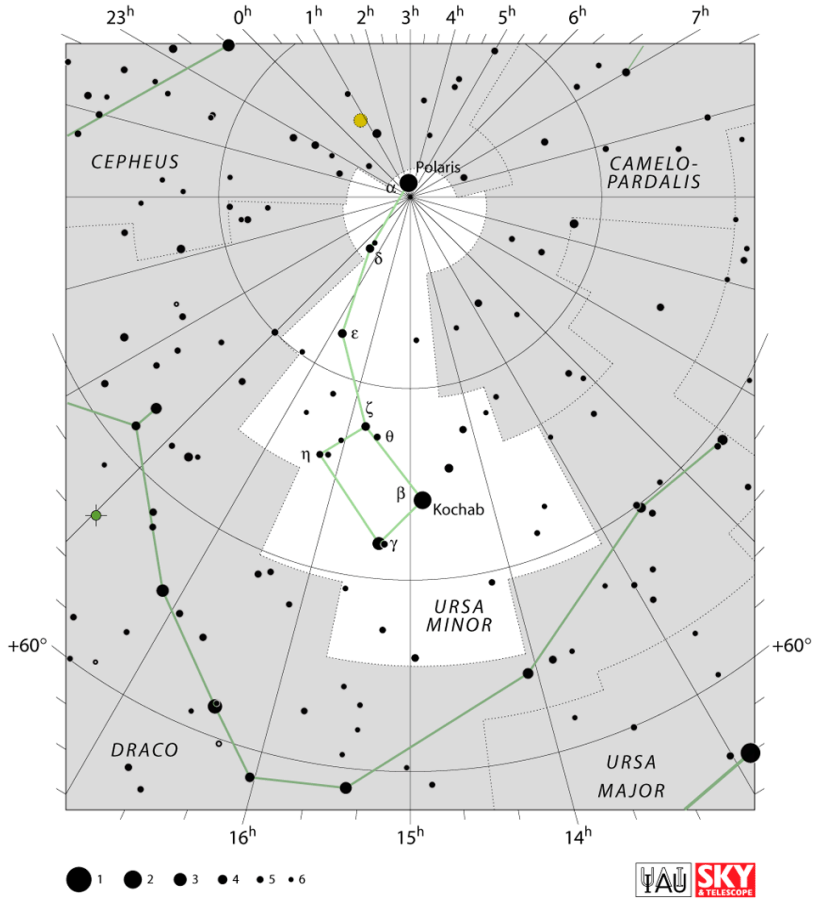
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 84/88  Ursa Minor (UMi) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 11767 | 37.946167 | +89.264139 | HIP 85822 |
| HIP 85822 | 263.053750 | +86.586333 | HIP 11767, HIP 82080 |
| HIP 82080 | 251.492333 | +82.037250 | HIP 85822, HIP 77055 |
| HIP 77055 | 236.014417 | +77.794500 | HIP 82080, HIP 79822, HIP 72607 |
| HIP 79822 | 244.377083 | +75.754694 | HIP 77055, HIP 75097 |
| HIP 75097 | 230.182292 | +71.833972 | HIP 79822, HIP 72607 |
| HIP 72607 | 222.676667 | +74.155472 | HIP 75097, HIP 77055 |

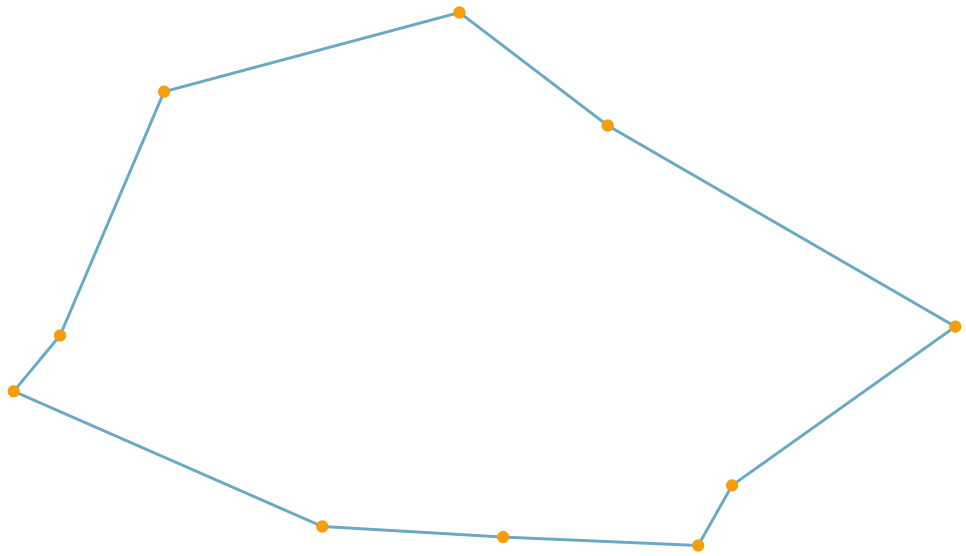
Effective release edges: 7 • unique plotted points: 7 • source: release_effective_edges.csv

# 85/88 Vela (Vel) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

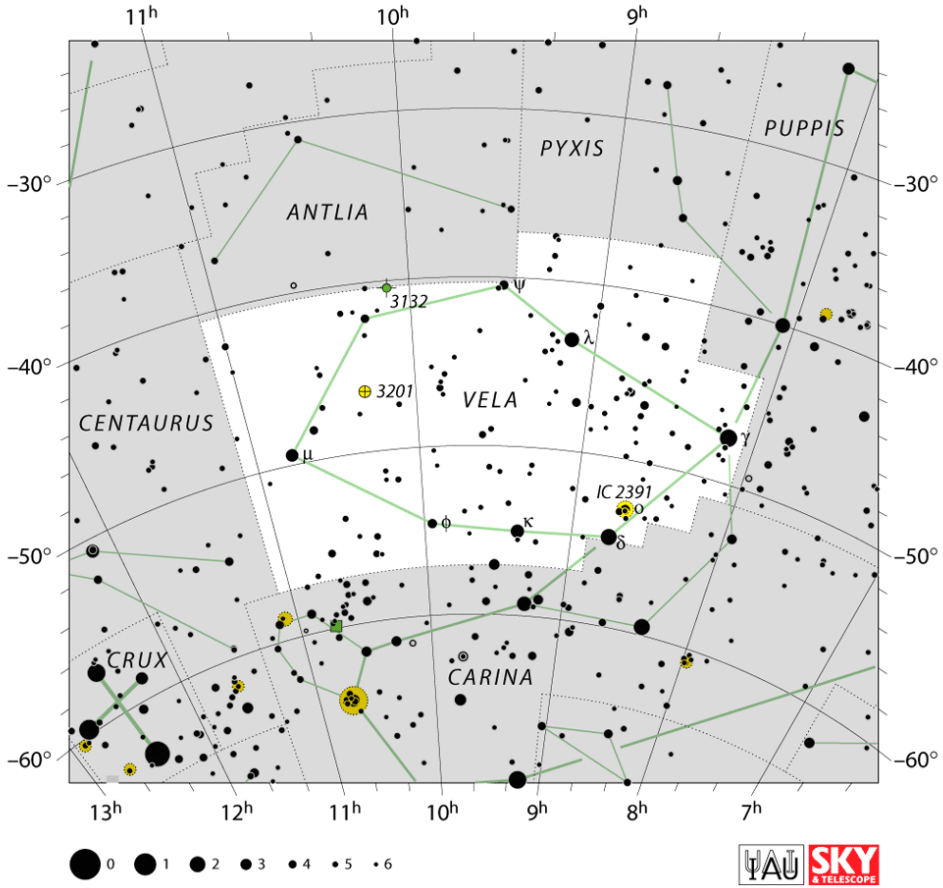
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 85/88  Vela (Vel) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 39953 | 122.383167 | -47.336611 | HIP 42536, HIP 44816 |
| HIP 42536 | 130.073375 | -52.921972 | HIP 39953, HIP 42913 |
| HIP 42913 | 131.175833 | -54.708556 | HIP 42536, HIP 45941 |
| HIP 45941 | 140.528458 | -55.010694 | HIP 42913, HIP 48774 |
| HIP 48774 | 149.215625 | -54.567806 | HIP 45941, HIP 52727 |
| HIP 52727 | 161.692167 | -49.420139 | HIP 48774, HIP 51986 |
| HIP 51986 | 159.326083 | -48.225611 | HIP 52727, HIP 50191 |
| HIP 50191 | 153.684458 | -42.122056 | HIP 51986, HIP 46651 |
| HIP 46651 | 142.675458 | -40.466889 | HIP 50191, HIP 44816 |
| HIP 44816 | 136.999083 | -43.432611 | HIP 46651, HIP 39953 |

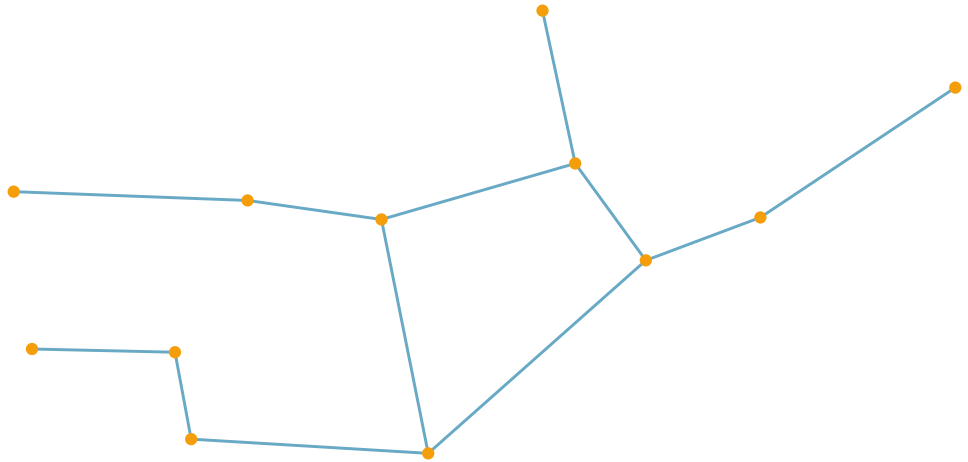
Effective release edges: 10 • unique plotted points: 10 • source: release_effective_edges.csv

# 86/88 Virgo (Vir) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

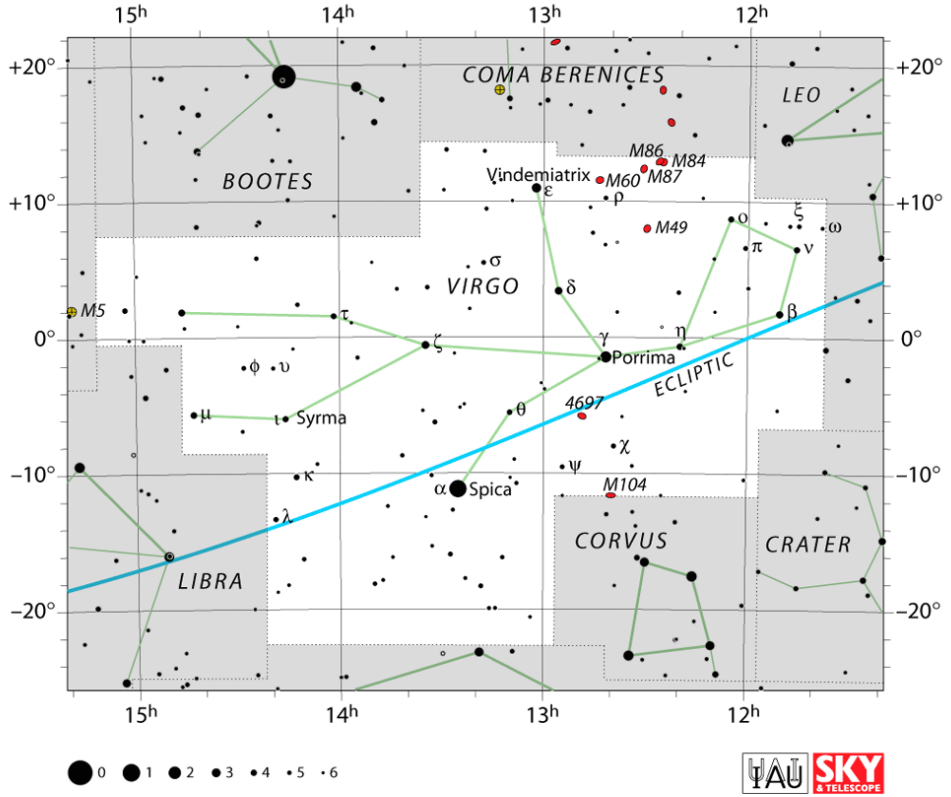
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 86/88  Virgo (Vir) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 57380 | 176.464875 | +6.529806 | HIP 60129 |
| HIP 60129 | 184.976625 | +0.666750 | HIP 57380, HIP 61941 |
| HIP 61941 | 190.416667 | -1.449528 | HIP 60129, HIP 65474, HIP 63090 |
| HIP 65474 | 201.298333 | -11.161250 | HIP 61941, HIP 69427, HIP 66249 |
| HIP 69427 | 213.223917 | -10.274056 | HIP 65474, HIP 69701 |
| HIP 69701 | 214.003667 | -5.999528 | HIP 69427, HIP 71957 |
| HIP 71957 | 220.764833 | -5.657417 | HIP 69701 |
| HIP 66249 | 203.673958 | +0.595944 | HIP 65474, HIP 68520, HIP 63090 |
| HIP 68520 | 210.411583 | +1.544583 | HIP 66249, HIP 72220 |
| HIP 72220 | 221.562458 | +1.892944 | HIP 68520 |
| HIP 63090 | 193.902000 | +3.397611 | HIP 66249, HIP 63608, HIP 61941 |
| HIP 63608 | 195.544833 | +10.959111 | HIP 63090 |

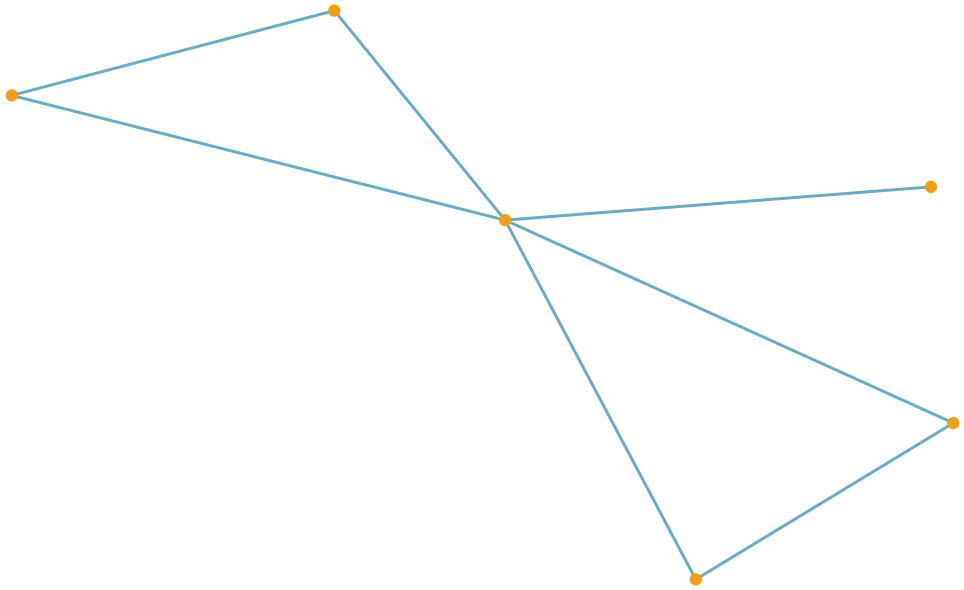
Effective release edges: 12 • unique plotted points: 12 • source: release_effective_edges.csv

# 87/88  Volans (Vol) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

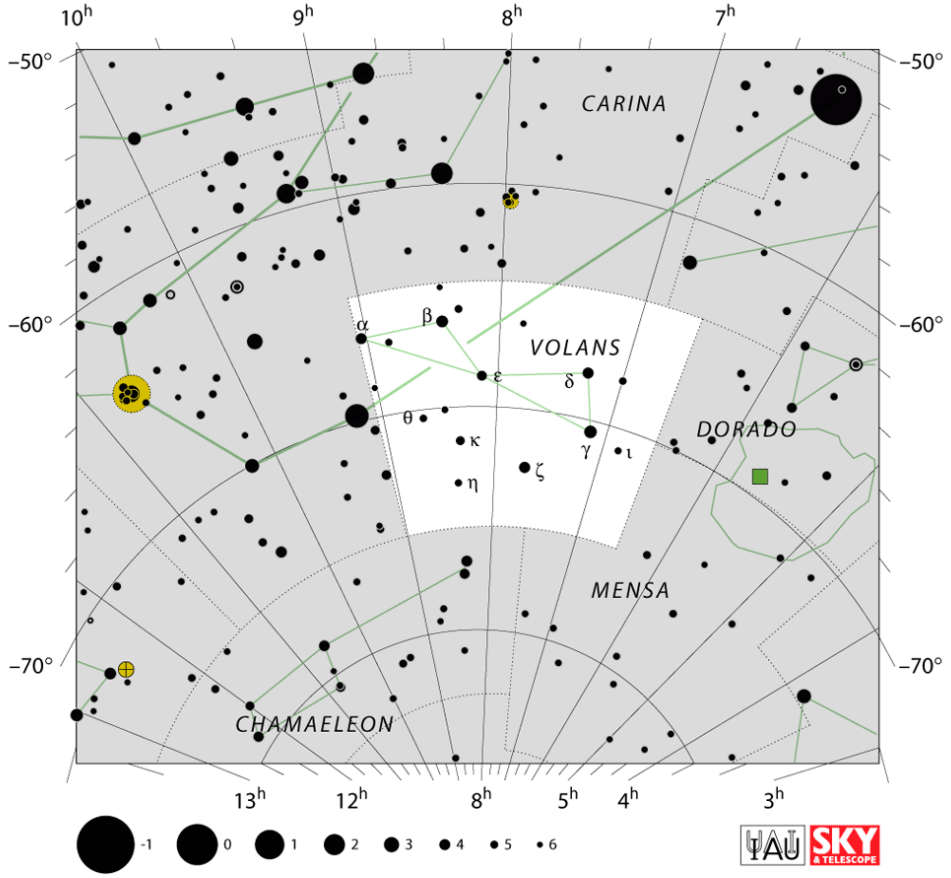
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review:  [ ] Correct  [ ] Incorrect  [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 87/88  Volans (Vol) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 37504 | 115.455000 | -72.606139 | HIP 34481, HIP 39794 |
| HIP 34481 | 107.186750 | -70.499194 | HIP 37504, HIP 39794 |
| HIP 39794 | 121.982667 | -68.617139 | HIP 34481, HIP 37504, HIP 35228, HIP 41312, HIP 44382 |
| HIP 35228 | 109.207625 | -67.957167 | HIP 39794 |
| HIP 41312 | 126.434375 | -66.136528 | HIP 39794, HIP 44382 |
| HIP 44382 | 135.611667 | -66.395833 | HIP 41312, HIP 39794 |

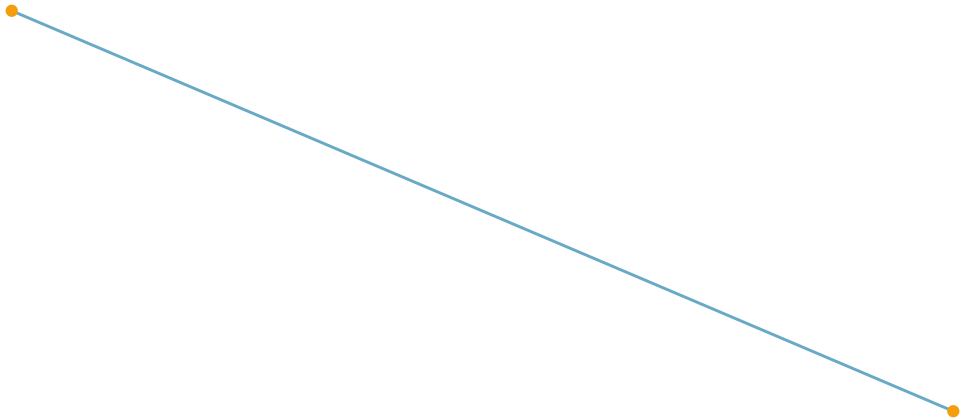
Effective release edges: 7 • unique plotted points: 6 • source: release_effective_edges.csv

# 88/88 Vulpecula (Vul) — side-by-side chart comparison

Left: Galaxy Viewer release JavaScript topology • Right: actual IAU / Sky & Telescope J2000 chart image

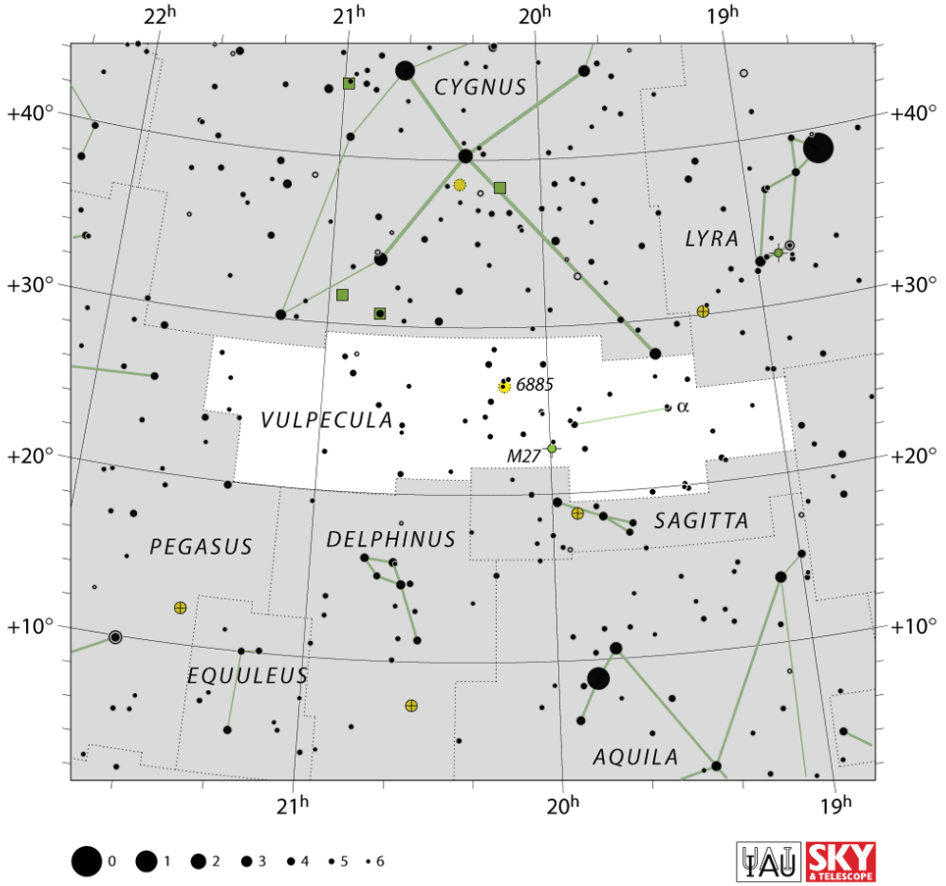
## GALAXY VIEWER — RELEASE JS

Edges parsed from release module + its runtime upstream CSV



## REFERENCE CHART / SOURCE WIRES

IAU / Sky & Telescope J2000 chart • credited mirror, CC BY 4.0




**Review: [ ] Correct [ ] Incorrect [ ] Needs review**

**Missing / extra connections:** _____

Important: the IAU officially standardizes constellation boundaries and names, not one universal stick-figure line pattern.

# 88/88  Vulpecula (Vul) — coordinate and segment register

Coordinates are J2000 equatorial RA/Dec in decimal degrees, parsed from the catalog's HIP source table; custom points are labeled explicitly.

| Point / HIP ID | RA (deg) | Dec (deg) | Connected to (star labels) |
|---|---|---|---|
| HIP 95771 | 292.176708 | +24.665167 | HIP 98543 |
| HIP 98543 | 300.275042 | +27.753556 | HIP 95771 |

Effective release edges: 1 • unique plotted points: 2 • source: release_effective_edges.csv